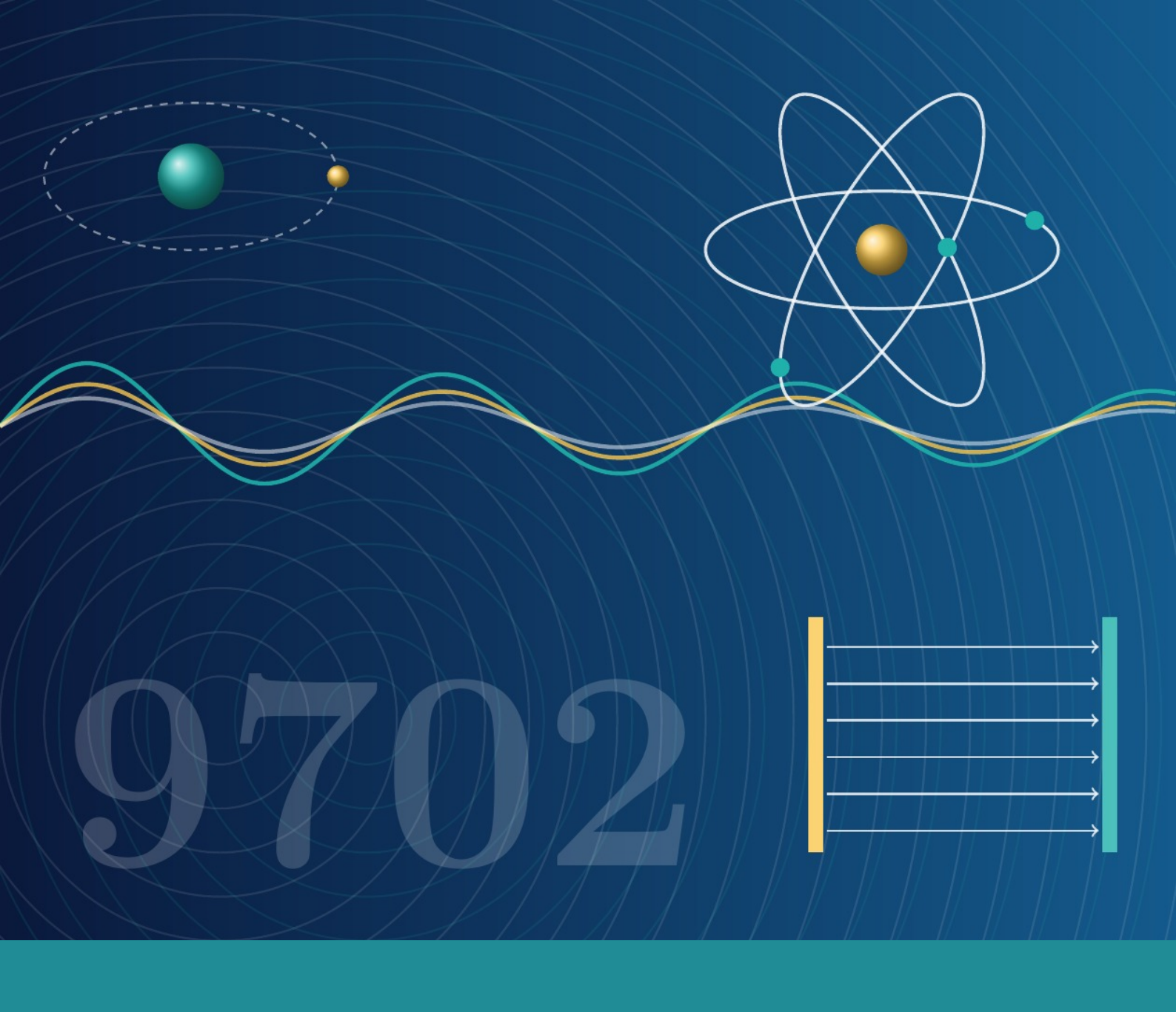


CIE A Level 物理 9702

中英双语精讲讲义

Cambridge International AS & A Level Physics 9702 ——教师版

Teacher's Edition

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组织：考纲 Syllabus 2025–2027

时间：2026 年 10 月

版本：1.0

真题素材：2026 F/M & M/J 9702 试卷

PHYSICS 9702

教师版含全部例题与练习的完整解答、评分要点与教学提示。

前言与使用说明 Preface

本讲义依据 Cambridge International AS & A Level Physics 9702 考纲（2025–2027 考试适用）编写，覆盖考纲全部 25 个主题以及 Paper 3、Paper 5 所需的实验技能。讲义以中文讲解为主，所有考纲条目、定义、定律保留英文原文并配中文释义，真题保留英文原题，帮助学生同时掌握物理概念与英文答题表达。

考试结构 Assessment overview

试卷	内容	时长	分值	A Level 权重
Paper 1	Multiple Choice（AS 内容，40 题）	1 h 15 min	40	15.5%
Paper 2	AS Level Structured Questions	1 h 15 min	60	23%
Paper 3	Advanced Practical Skills	2 h	40	11.5%
Paper 4	A Level Structured Questions（含 AS 知识）	2 h	100	38.5%
Paper 5	Planning, Analysis and Evaluation	1 h 15 min	30	11.5%

章节结构 How each chapter is organised

(1) **考纲要求**：逐条列出该主题的英文学习目标（learning outcomes）及中文说明；
(2) **知识讲解**：定义（绿色 **定义**框）、定律（橙色 **定律**框）、公式（蓝色 **公式**框）、配图、推导、易错提醒与考试技巧；
(3) **例题**：穿插在讲解中，选自 2026 年 2/3 月（F/M）与 5/6 月（M/J）9702 真题；
(4) **本章小结**：公式与要点一览表；
(5) **本章练习**：每章 6–10 道题。A Level 部分各主题真题数量有限，另配少量标注为“原创仿真题 Exam-style”的题目补足。

题目来源与评分标记 Sources and mark codes

题目来源标注格式为“9702/卷号/考季/年份题号”，例如 9702/42/F/M/26 Q1 表示 2026 年 2/3 月 Paper 42 第 1 题，9702/11/M/J/26 Q2 表示 2026 年 5/6 月 Paper 11 第 2 题。所有真题图均按原题重新绘制。

教师版的解答按 Cambridge 官方评分标准（mark scheme）给出评分点：

标记	含义
B1	独立得分点（independent mark），与其他步骤无关
M1	方法分（method mark），后续 A 分依赖于此
C1	过程分（compensatory mark），最终答案正确时自动给出
A1	答案分（accuracy mark），依赖于正确的方法

少数试卷（9702/21/M/J/26）未公布评分标准，相应解答为编者给出的参考解答与参考给分。教师版另含“教学提示”（课时建议、常见错误、教学与演示建议）。

致学生 To students

物理考试中，**定义要准确**（大量“per unit”“ratio”的表述）、**计算要写过程并带单位、图像要读准、解释要有因果链**。每章的“易错提醒”汇总了历年考官报告中反复出现的失分点，复习时请重点关注。

目录

前言与使用说明 Preface **i**

第一部分　AS Level **1**

第一章　物理量与单位 Physical quantities and units **2**

1.1 物理量 Physical quantities 2
1.2 国际单位制 SI units 3
1.3 误差与不确定度 Errors and uncertainties 5
1.4 标量与矢量 Scalars and vectors 8
1.5 本章小结 Summary 10
1.6 本章练习 Practice questions 10

第二章　运动学 Kinematics **14**

2.1 描述运动的物理量 14
2.2 运动图像 Graphs of motion 15
2.3 匀加速直线运动方程 Equations of uniformly accelerated motion 17
2.4 自由落体 Free fall 18
2.5 抛体运动 Projectile motion 19
2.6 本章小结 Summary 21
2.7 本章练习 Practice questions 21

第三章　动力学 Dynamics **25**

3.1 质量、力与牛顿运动定律 26
3.2 非匀变速运动：阻力与终端速度 28
3.3 动量守恒 Conservation of momentum 30
3.4 本章小结 Summary 33
3.5 本章练习 Practice questions 33

第四章　力、密度与压强 Forces, density and pressure **38**

4.1 力的转动效果 Turning effects of forces 38
4.2 平衡 Equilibrium 40
4.3 密度与压强 Density and pressure 42
4.4 本章小结 Summary 45
4.5 本章练习 Practice questions 45

第五章　功、能量与功率 Work, energy and power **49**

5.1 功 Work 49

5.2 重力势能与动能 . . . 50
5.3 能量守恒 Conservation of energy . . . 51
5.4 效率与功率 Efficiency and power . . . 52
5.5 本章小结 Summary . . . 54
5.6 本章练习 Practice questions . . . 54

第六章 固体的形变 Deformation of solids **57**
6.1 拉伸、压缩与胡克定律 . . . 57
6.2 应力、应变与杨氏模量 . . . 59
6.3 弹性形变与塑性形变 . . . 61
6.4 弹性势能 Elastic potential energy . . . 62
6.5 本章小结 Summary . . . 63
6.6 本章练习 Practice questions . . . 63

第七章 波 Waves **67**
7.1 行波 Progressive waves . . . 68
7.2 横波与纵波 Transverse and longitudinal waves . . . 72
7.3 声波的多普勒效应 Doppler effect for sound waves . . . 72
7.4 电磁波谱 Electromagnetic spectrum . . . 73
7.5 偏振 Polarisation . . . 74
7.6 本章小结 Summary . . . 75
7.7 本章练习 Practice questions . . . 76

第八章 叠加 Superposition **79**
8.1 叠加原理 Principle of superposition . . . 79
8.2 驻波 Stationary waves . . . 80
8.3 衍射 Diffraction . . . 83
8.4 干涉 Interference . . . 83
8.5 衍射光栅 The diffraction grating . . . 86
8.6 本章小结 Summary . . . 87
8.7 本章练习 Practice questions . . . 88

第九章 电学 Electricity **92**
9.1 电流 Electric current . . . 93
9.2 电势差与电功率 . . . 94
9.3 电阻与电阻率 . . . 95
9.4 本章小结 Summary . . . 98
9.5 本章练习 Practice questions . . . 99

第十章 直流电路 D.C. circuits **102**
10.1 电路符号 Circuit symbols . . . 103

10.2 电动势与内阻 . . . 103
10.3 基尔霍夫定律 Kirchhoff's laws . . . 105
10.4 分压电路 Potential dividers . . . 107
10.5 本章小结 Summary . . . 110
10.6 本章练习 Practice questions . . . 110

第十一章 粒子物理 Particle physics 114
11.1 原子、原子核与辐射 . . . 115
11.2 基本粒子 Fundamental particles . . . 118
11.3 本章小结 Summary . . . 120
11.4 本章练习 Practice questions . . . 121

第二部分 A Level 124

第十二章 圆周运动 Motion in a circle 125
12.1 圆周运动的运动学 . . . 125
12.2 向心加速度与向心力 . . . 126
12.3 本章小结 Summary . . . 129
12.4 本章练习 Practice questions . . . 130

第十三章 引力场 Gravitational fields 133
13.1 引力场与场强 . . . 133
13.2 万有引力定律 . . . 134
13.3 圆轨道与卫星 . . . 135
13.4 引力势与引力势能 . . . 136
13.5 本章小结 Summary . . . 139
13.6 本章练习 Practice questions . . . 139

第十四章 温度 Temperature 142
14.1 热平衡 Thermal equilibrium . . . 142
14.2 温标 Temperature scales . . . 143
14.3 比热容与比潜热 . . . 144
14.4 本章小结 Summary . . . 148
14.5 本章练习 Practice questions . . . 148

第十五章 理想气体 Ideal gases 151
15.1 物质的量 The mole . . . 151
15.2 理想气体状态方程 Equation of state . . . 152
15.3 气体动理论 Kinetic theory of gases . . . 152
15.4 本章小结 Summary . . . 156
15.5 本章练习 Practice questions . . . 156

第十六章 热力学 Thermodynamics **159**
16.1 内能 Internal energy . . . 159
16.2 气体做功 Work done by and on a gas . . . 160
16.3 热力学第一定律 First law of thermodynamics . . . 160
16.4 本章小结 Summary . . . 163
16.5 本章练习 Practice questions . . . 163

第十七章 振动 Oscillations **167**
17.1 描述振动的物理量 . . . 168
17.2 简谐运动 Simple harmonic motion . . . 168
17.3 简谐运动的能量 . . . 169
17.4 阻尼振动 Damped oscillations . . . 170
17.5 受迫振动与共振 Forced oscillations and resonance . . . 170
17.6 本章小结 Summary . . . 173
17.7 本章练习 Practice questions . . . 174

第十八章 电场 Electric fields **176**
18.1 电场与电场线 . . . 176
18.2 匀强电场 Uniform electric fields . . . 177
18.3 点电荷的电场 . . . 179
18.4 电势 Electric potential . . . 179
18.5 本章小结 Summary . . . 181
18.6 本章练习 Practice questions . . . 182

第十九章 电容 Capacitance **185**
19.1 电容 Capacitance . . . 185
19.2 电容器的串联与并联 . . . 186
19.3 电容器储存的能量 . . . 186
19.4 电容器的放电 Discharging a capacitor . . . 187
19.5 本章小结 Summary . . . 190
19.6 本章练习 Practice questions . . . 190

第二十章 磁场 Magnetic fields **193**
20.1 磁场与磁感线 . . . 194
20.2 磁场对电流的作用力 . . . 195
20.3 磁场对运动电荷的作用力 . . . 195
20.4 电磁感应 Electromagnetic induction . . . 200
20.5 本章小结 Summary . . . 202
20.6 本章练习 Practice questions . . . 202

第二十一章 交流电 Alternating currents **205**

21.1 交流电的特征 . . . 205
21.2 整流与平滑 Rectification and smoothing . . . 206
21.3 本章小结 Summary . . . 210
21.4 本章练习 Practice questions . . . 210

第二十二章 量子物理 Quantum physics **213**
22.1 光子 Photons . . . 214
22.2 光电效应 The photoelectric effect . . . 214
22.3 波粒二象性 Wave–particle duality . . . 216
22.4 能级与线状谱 . . . 217
22.5 本章小结 Summary . . . 218
22.6 本章练习 Practice questions . . . 219

第二十三章 核物理 Nuclear physics **222**
23.1 质能方程与结合能 . . . 223
23.2 放射性衰变 Radioactive decay . . . 225
23.3 本章小结 Summary . . . 227
23.4 本章练习 Practice questions . . . 228

第二十四章 医学物理 Medical physics **231**
24.1 超声波 Ultrasound . . . 232
24.2 X 射线 X-rays . . . 235
24.3 PET 扫描 Positron emission tomography . . . 237
24.4 本章小结 Summary . . . 238
24.5 本章练习 Practice questions . . . 238

第二十五章 天体物理与宇宙学 Astronomy and cosmology **241**
25.1 光度与标准烛光 . . . 241
25.2 恒星的温度与半径 . . . 243
25.3 红移、哈勃定律与大爆炸 . . . 244
25.4 本章小结 Summary . . . 246
25.5 本章练习 Practice questions . . . 246

第三部分 实验技能 Practical Skills **249**

第二十六章 实验技能 Practical skills (Papers 3 & 5) **250**
26.1 测量与读数 Measurement . . . 250
26.2 记录数据 Recording data . . . 251
26.3 不确定度 Uncertainties . . . 251
26.4 作图与图像分析 Graphs . . . 252
26.5 评价与改进（Paper 3 第 2 题） . . . 254

26.6 设计实验（Paper 5 第 1 题） 255
26.7 本章练习 Practice questions 257

附录 A 数据、公式与术语 Data, formulae and glossary **261**
A.1 考试数据表 Data 261
A.2 考试公式表 Formulae 261
A.3 SI 单位与词头 SI units and prefixes 263
A.4 命令词 Command words 263
A.5 中英术语对照表 Glossary 264

第一部分

AS Level

第一章 物理量与单位 Physical quantities and units

考纲要求 Syllabus learning outcomes

1.1.1 understand that all physical quantities consist of a numerical magnitude and a unit
理解所有物理量都由数值大小和单位两部分组成。

1.1.2 make reasonable estimates of physical quantities included within the syllabus
能够对考纲范围内的物理量作出合理的估算。

1.2.1 recall the following SI base quantities and their units: mass (kg), length (m), time (s), current (A), temperature (K)
记住 SI 基本量及其单位：质量 (kg)、长度 (m)、时间 (s)、电流 (A)、热力学温度 (K)。

1.2.2 express derived units as products or quotients of the SI base units and use the derived units for quantities listed in this syllabus as appropriate
能把导出单位表示为 SI 基本单位的乘积或商，并正确使用考纲中各物理量的导出单位。

1.2.3 use SI base units to check the homogeneity of physical equations
能用 SI 基本单位检验物理方程的齐次性（量纲一致性）。

1.2.4 recall and use the following prefixes and their symbols to indicate decimal submultiples or multiples of both base and derived units: pico (p), nano (n), micro (μ), milli (m), centi (c), deci (d), kilo (k), mega (M), giga (G), tera (T)
记住并使用词头 p, n, μ, m, c, d, k, M, G, T 表示单位的十进倍数和分数。

1.3.1 understand and explain the effects of systematic errors (including zero errors) and random errors in measurements
理解并解释系统误差（包括零点误差）和随机误差对测量的影响。

1.3.2 understand the distinction between precision and accuracy
理解精密度与准确度的区别。

1.3.3 assess the uncertainty in a derived quantity by simple addition of absolute or percentage uncertainties
通过绝对不确定度或百分比不确定度的简单相加，评估导出量的不确定度。

1.4.1 understand the difference between scalar and vector quantities and give examples of scalar and vector quantities included in the syllabus
理解标量与矢量的区别，并能举出考纲中的例子。

1.4.2 add and subtract coplanar vectors
能对共面矢量进行加减运算。

1.4.3 represent a vector as two perpendicular components
能把一个矢量分解为两个互相垂直的分量。

1.1 物理量 Physical quantities

物理学研究的是可以**测量**的量。一个**物理量** (*physical quantity*) 必须同时包含两部分：

$$\text{物理量} = \text{数值 (numerical magnitude)} \times \text{单位 (unit)}$$

例如 $m = 9.6\,\text{kg}$，其中 9.6 是数值，kg 是单位。只写“9.6”、只写“kg”、或只写“质量”都**不是**一个物理量的测量值。百分数（如 6.4%）是两个同类量的比值，本身没有单位，也不是对物理量的一次测量。

> **易错提醒 Common mistakes** 选择题常问“What represents a physical quantity?”——必须**同时**有数值和单位，如 7 N、9.6 J。单独的单位名称（newton, kilogram）或单独的物理量名称（power）都不是答案。

1.1.1 数量级估算 Estimation

考试中常要求对日常物理量作出合理估算（通常只需正确到数量级，即 10 的幂次）。下表列出应当熟记的典型数值：

物理量 Quantity	典型值 Typical value	备注
成年人质量	70 kg	体重约 700 N
汽车质量	1000 kg 至 1500 kg	大象约为汽车的 4 倍
一个苹果的重量	≈ 1 N	质量约 100 g
人步行速度	$1.5\,\text{m s}^{-1}$	跑步约 $5\,\text{m s}^{-1}$
高速公路车速	$30\,\text{m s}^{-1}$	约 $110\,\text{km h}^{-1}$
空气中的声速	$330\,\text{m s}^{-1}$	
光速	$3.0 \times 10^{8}\,\text{m s}^{-1}$	
一层楼高	3 m	
原子直径	10^{-10} m	
原子核直径	10^{-15} m 至 10^{-14} m	
可见光波长	400 nm 至 700 nm	
水的密度	$1000\,\text{kg m}^{-3}$	空气约 $1.2\,\text{kg m}^{-3}$
大气压	1.0×10^{5} Pa	
室温	293 K (20 °C)	
家用电压	230 V（英国）	灯泡功率约 60 W
足球体积	约 $6000\,\text{cm}^3$	半径约 11 cm

估算时先把已知量换成 SI 单位，然后保留一位有效数字进行计算。例如：大象最大动量 $p = mv \approx (4 \times 1200) \times (40/3.6) \approx 5 \times 10^{4}\,\text{kg m s}^{-1}$，最接近的选项是 $7 \times 10^{4}\,\text{kg m s}^{-1}$。

1.2 国际单位制 SI units

1.2.1 基本量与基本单位 Base quantities and base units

国际单位制 (Système International, SI) 选定少数几个**基本量** (*base quantity*)，其余所有物理量的单位都由它们导出。9702 考纲要求记住以下基本量：

基本量 Base quantity	符号	SI 基本单位 Base unit	单位符号
质量 mass	m	kilogram	kg
长度 length	l	metre	m
时间 time	t	second	s
电流 electric current	I	ampere	A
热力学温度 thermodynamic temperature	T	kelvin	K
物质的量 amount of substance	n	mole	mol

易错提醒 Common mistakes

- “基本量”是 **current（电流）**，“基本单位”是 **ampere**。题目问“SI base quantity”时，选 current 而不是 ampere；charge（电荷）是导出量，单位 coulomb (C = A s) 也是导出单位。
- 质量的基本单位是 **kg**，不是 g；温度的基本单位是 **K**，不是 °C。

1.2.2 导出单位 Derived units

导出量 (derived quantity) 由基本量通过定义式得到，其单位就是对应基本单位的乘积或商。推导时**一定要从定义式出发**：

导出量	定义式	导出单位	用 SI 基本单位表示
速度 velocity	$v = s/t$	$\mathrm{m\,s^{-1}}$	$\mathrm{m\,s^{-1}}$
加速度 acceleration	$a = \Delta v/t$	$\mathrm{m\,s^{-2}}$	$\mathrm{m\,s^{-2}}$
力 force	$F = ma$	newton (N)	$\mathrm{kg\,m\,s^{-2}}$
动量 momentum	$p = mv$	N s	$\mathrm{kg\,m\,s^{-1}}$
压强 pressure	$p = F/A$	pascal (Pa)	$\mathrm{kg\,m^{-1}\,s^{-2}}$
功/能 work, energy	$W = Fs$	joule (J)	$\mathrm{kg\,m^{2}\,s^{-2}}$
功率 power	$P = W/t$	watt (W)	$\mathrm{kg\,m^{2}\,s^{-3}}$
电荷 charge	$Q = It$	coulomb (C)	A s
电势差 p.d.	$V = W/Q$	volt (V)	$\mathrm{kg\,m^{2}\,s^{-3}\,A^{-1}}$
电阻 resistance	$R = V/I$	ohm (Ω)	$\mathrm{kg\,m^{2}\,s^{-3}\,A^{-2}}$
频率 frequency	$f = 1/T$	hertz (Hz)	$\mathrm{s^{-1}}$
密度 density	$\rho = m/V$	$\mathrm{kg\,m^{-3}}$	$\mathrm{kg\,m^{-3}}$
强度 intensity	$I = P/A$	$\mathrm{W\,m^{-2}}$	$\mathrm{kg\,s^{-3}}$

示例 1.1 电势差的基本单位 求伏特 (V) 用 SI 基本单位的表达式。

解：由 $V = W/Q$，而 W 的单位 $\mathrm{J = N\,m = kg\,m^{2}\,s^{-2}}$，$Q$ 的单位 $\mathrm{C = A\,s}$，所以

$$\mathrm{V} = \frac{\mathrm{kg\,m^{2}\,s^{-2}}}{\mathrm{A\,s}} = \mathrm{kg\,m^{2}\,s^{-3}\,A^{-1}}.$$

1.2.3 方程的齐次性 Homogeneity of equations

如果一个物理方程中**每一项**的 SI 基本单位都相同，则称该方程是**齐次的** (*homogeneous*)。

- 齐次是方程正确的**必要条件**，但不是充分条件：齐次方程仍可能因为数值系数（如 $\frac{1}{2}$）或缺项而错误。例如 $s = ut + at^2$ 是齐次的，但物理上是错误的（应为 $\frac{1}{2}at^2$）。
- 纯数（如 $\frac{1}{2}$、2π）和比值没有单位，不影响齐次性。

示例 1.2 检验齐次性 检验 $v^2 = u^2 + 2as$ 是否齐次。

解：左边单位 $(\mathrm{m\,s^{-1}})^2 = \mathrm{m^2\,s^{-2}}$；右边第一项 $\mathrm{m^2\,s^{-2}}$，第二项 $\mathrm{m\,s^{-2} \times m = m^2\,s^{-2}}$。三项单位相同，故方程齐次。

1.2.4 词头 Prefixes

词头	符号	倍数	词头	符号	倍数
pico	p	10^{-12}	deci	d	10^{-1}
nano	n	10^{-9}	kilo	k	10^{3}
micro	μ	10^{-6}	mega	M	10^{6}
milli	m	10^{-3}	giga	G	10^{9}
centi	c	10^{-2}	tera	T	10^{12}

易错提醒 Common mistakes 带词头的单位做乘方时，词头一起乘方：$1\,\mathrm{cm^3} = (10^{-2}\,\mathrm{m})^3 = 10^{-6}\,\mathrm{m^3}$；$1\,\mathrm{mm^2} = 10^{-6}\,\mathrm{m^2}$。这是计算密度、截面积、应力时最常见的错误来源。

1.3 误差与不确定度 Errors and uncertainties

1.3.1 系统误差与随机误差

定义 1.1 (系统误差 *Systematic error*)

An error that causes all readings to differ from the true value by the same amount (or the same proportion) in the same direction each time a measurement is taken.

使每次测量读数都朝**同一方向**偏离真实值相同大小（或相同比例）的误差。**不能**通过多次测量取平均来减小，只能通过校准仪器 (calibration)、修正零点误差或改进实验方法来消除。

零点误差 (*zero error*) 是一种典型的系统误差：当被测量为零时，仪器读数不为零。例如千分尺两砧合拢时读数为 +0.02 mm，则所有读数都偏大 0.02 mm，应从每次读数中减去。

定义 1.2 (随机误差 *Random error*)

An error that causes readings to be scattered about the true value, by different amounts and in different directions.

使读数在真实值附近**无规律地**分散（大小、方向都不同）的误差。可以通过**多次重复测量取平均值**或作图取最佳拟合线来减小其影响。

1.3.2 精密度与准确度 Precision and accuracy

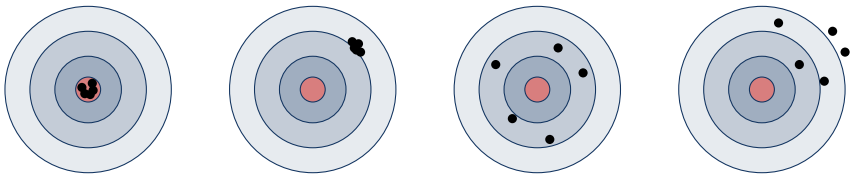
定义 1.3 (准确度与精密度 *Accuracy and precision*)

Accuracy: how close a measured value is to the true value.

Precision: how close repeated measured values are to each other (the size of the smallest division / the spread of readings).

准确度：测量值与真实值的接近程度——受**系统误差**影响。

精密度：多次测量值之间彼此接近的程度（也指读数的分辨程度、有效数字位数）——受**随机误差**影响。



精密且准确 precise & accurate　精密但不准确 precise, not accurate　准确但不精密 accurate, not precise　既不精密也不准确 neither

在读取仪器数值时，**精密度**取决于仪器最小分度：例如读数 62.65 cm 比 61.5 cm 更精密（读到 0.01 cm）；而哪一个更**准确**则要和真实值比较。

常用仪器	最小分度 (精密度)	典型不确定度
米尺 metre rule	1 mm	±1 mm（两端读数各 ±0.5 mm）
游标卡尺 vernier calipers	0.1 mm 或 0.02 mm	±0.1 mm
千分尺 micrometer screw gauge	0.01 mm	±0.01 mm
秒表 stopwatch	0.01 s	人反应时间约 ±0.2 s
数字电表 digital meter	最后一位	± 最后一位的 1

1.3.3 不确定度的表示与合成

测量结果写成 $x \pm \Delta x$，其中 Δx 为**绝对不确定度** (*absolute uncertainty*)。

分数不确定度 fractional uncertainty $= \dfrac{\Delta x}{x}$，　　百分比不确定度 percentage uncertainty $= \dfrac{\Delta x}{x} \times 100\%$

公式 1.1 (不确定度合成规则 *Combining uncertainties*)

(1) 加减：$y = a + b$ 或 $y = a - b$ $\Rightarrow$ $\Delta y = \Delta a + \Delta b$（绝对不确定度相加）。

(2) 乘除：$y = ab$ 或 $y = \dfrac{a}{b}$ $\Rightarrow$ $\dfrac{\Delta y}{y} = \dfrac{\Delta a}{a} + \dfrac{\Delta b}{b}$（百分比不确定度相加）。

(3) 乘方：$y = a^n$ $\Rightarrow$ $\dfrac{\Delta y}{y} = |n|\dfrac{\Delta a}{a}$（$n$ 可以是分数或负数）。

(4) 常数（如 $\pi, 2$）无不确定度。

一般地，$y = k\dfrac{a^m b^n}{c^p}$ 时，$\dfrac{\Delta y}{y} = m\dfrac{\Delta a}{a} + n\dfrac{\Delta b}{b} + p\dfrac{\Delta c}{c}$。

易错提醒 Common mistakes

- 两个量**相减**时，绝对不确定度仍然**相加**；当差值很小时，百分比不确定度会非常大。
- 最终结果的绝对不确定度一般保留 1 位有效数字，测量值的小数位与之对齐，例如 $g = (9.8 \pm 0.3)\ \mathrm{m\,s^{-2}}$。

示例 1.3 合成不确定度 单摆周期 $T = 2\pi\sqrt{l/g}$，测得 $l = (0.800 \pm 0.002)$ m，$T = (1.80 \pm 0.02)$ s。求 g 及其不确定度。

解：$g = \dfrac{4\pi^2 l}{T^2} = \dfrac{4\pi^2 \times 0.800}{1.80^2} = 9.75\ \mathrm{m\,s^{-2}}$。

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + 2\frac{\Delta T}{T} = \frac{0.002}{0.800} + 2 \times \frac{0.02}{1.80} = 0.0025 + 0.0222 = 0.0247$$

$\Delta g = 0.0247 \times 9.75 = 0.24\ \mathrm{m\,s^{-2}} \approx 0.2\ \mathrm{m\,s^{-2}}$，所以 $g = (9.8 \pm 0.2)\ \mathrm{m\,s^{-2}}$。

例题 1.1 9702/11/M/J/26 Q2

The surface area of one face of a cube is determined with an uncertainty of ±1.0%. This area is used to calculate the volume of the cube. What is the percentage uncertainty in the calculated volume of the cube?

A ±1.0% **B** ±1.5% **C** ±3.0% **D** ±6.0%

解答 Solution

答案 Answer：**B**

面积 $A = L^2$，体积 $V = L^3 = A^{3/2}$。所以 $\dfrac{\Delta V}{V} = \dfrac{3}{2}\dfrac{\Delta A}{A} = 1.5 \times 1.0\% = 1.5\%$。

注意不要先求 L 的不确定度再乘 3 而出错：$\Delta L/L = 0.5\%$，$\Delta V/V = 3 \times 0.5\% = 1.5\%$，结果一致。

例题 1.2 9702/14/M/J/26 Q4

At temperatures close to 0 K, the specific heat capacity c of a particular solid is given by $c = bT^3$, where T is the temperature and b is a constant, characteristic of the solid. The SI unit of specific heat capacity is $\mathrm{J\,kg^{-1}\,K^{-1}}$. What is the unit of constant b, expressed in SI base units?

A $\mathrm{m^2\,s^{-2}\,K^{-3}}$ **B** $\mathrm{m^2\,s^{-2}\,K^{-4}}$

C $\mathrm{kg\,m^2\,s^{-2}\,K^{-3}}$ **D** $\mathrm{kg\,m^2\,s^{-2}\,K^{-4}}$

解答 Solution

答案 Answer：**B**

$\mathrm{J\,kg^{-1}\,K^{-1}} = (\mathrm{kg\,m^2\,s^{-2}})\,\mathrm{kg^{-1}\,K^{-1}} = \mathrm{m^2\,s^{-2}\,K^{-1}}$。

$b = c/T^3$，单位为 $\mathrm{m^2\,s^{-2}\,K^{-1}/K^3 = m^2\,s^{-2}\,K^{-4}}$。

例题 1.3 9702/12/M/J/26 Q4

A metal rod is exactly 62.00 cm long. The length of the rod is measured using two different devices, X and Y. The reading on device X is 61.5 cm. The reading on device Y is 62.65 cm. Which device gives the more precise reading for the length, and which device gives the more accurate reading for the length?

	more precise reading	more accurate reading
A	X	X
B	X	Y
C	Y	X
D	Y	Y

解答 Solution

答案 Answer：C

精密度看读数分辨率：Y 读到 0.01 cm，X 只读到 0.1 cm，所以 Y 更精密。

准确度看与真实值的差距：$|61.5 - 62.00| = 0.50$ cm，$|62.65 - 62.00| = 0.65$ cm，X 更接近真实值，所以 X 更准确。

1.4 标量与矢量 Scalars and vectors

定义 1.4 (标量与矢量 *Scalar and vector*)

A **scalar** quantity has magnitude only. A **vector** quantity has both magnitude and direction.

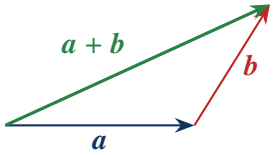
标量只有大小；**矢量**既有大小又有方向。

标量 Scalars	矢量 Vectors
distance, speed, mass, time, energy, work, power, pressure, density, temperature, charge, potential difference, resistance, volume, amplitude, frequency, wavelength	displacement, velocity, acceleration, force (including weight, tension, friction, upthrust, drag), momentum, impulse, electric/gravitational/magnetic field strength, torque (moment)

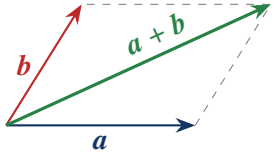
易错提醒 Common mistakes **功、能量、功率、压强**都是标量（虽然它们由矢量计算而来）；**重量 weight** 是力，是矢量，但质量 mass 是标量。振幅 amplitude 是标量。

1.4.1 矢量的加减 Adding and subtracting vectors

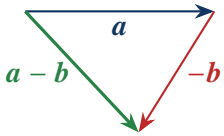
矢量相加用**首尾相接法**（三角形法则 triangle of vectors）或**平行四边形法则**。$\boldsymbol{a} - \boldsymbol{b} = \boldsymbol{a} + (-\boldsymbol{b})$，即把 $\boldsymbol{b}$ 反向后再相加。



三角形法则（首尾相接）



平行四边形法则

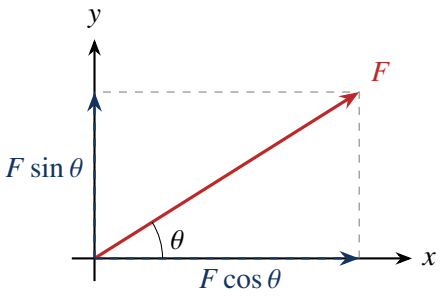


矢量相减

计算方法：

- 两矢量互相垂直：$R = \sqrt{a^2 + b^2}$，方向 $\tan\theta = b/a$。
- 两矢量夹角为 θ：用余弦定理 $R^2 = a^2 + b^2 + 2ab\cos\theta$（$\theta$ 为两矢量**尾对尾**时的夹角）；也可以用精确作图（scale drawing）求解。
- 多个力：先全部分解到两个互相垂直的方向上，分别求和，再合成。

1.4.2 矢量的分解 Resolving vectors



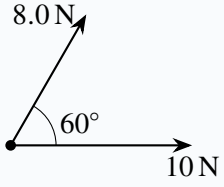
一个矢量 F 与 x 轴夹角 θ，可以分解为两个互相垂直的**分量** (*components*)：

$$F_x = F\cos\theta, \qquad F_y = F\sin\theta, \qquad F = \sqrt{F_x^2 + F_y^2}.$$

记忆方法：与角 θ“相邻”的分量用 cos，“对边”的分量用 sin。斜面上的物体，重力沿斜面分量为 $mg\sin\theta$，垂直斜面分量为 $mg\cos\theta$（θ 为斜面倾角）。

例题 1.4 9702/13/M/J/26 Q4

Two coplanar forces act on a small object. One force has magnitude 10 N. The other force has magnitude 8.0 N and acts at an angle of 60° to the first force.



What is the magnitude of the resultant of the two forces?

A 14 N **B** 16 N **C** 17 N **D** 18 N

解答 Solution

答案 Answer：**B**

方法一（分解）：$R_x = 10 + 8.0\cos 60° = 14.0$ N，$R_y = 8.0\sin 60° = 6.93$ N，$R = \sqrt{14.0^2 + 6.93^2} = 15.6 \approx 16$ N。

方法二（余弦定理）：$R^2 = 10^2 + 8^2 + 2(10)(8)\cos 60° = 244$，$R = 15.6$ N。

教学提示 Teacher's note 本章是全书基础，建议 3–4 课时。重点训练：(1) 从定义式推导基本单位；(2) 百分比不确定度的合成，尤其是乘方项；(3) 共点力的分解。Paper 1 每年前 3–4 题都出自本章，属于必拿分。学生最常犯的错误：把 current/ampere 混淆；cm^3 与 m^3 换算错误；把 accuracy 与 precision 说反。

1.5 本章小结 Summary

物理量/关系	公式	说明
SI 基本单位	kg, m, s, A, K, mol	记清基本**量**与基本**单位**
百分比不确定度	$\dfrac{\Delta x}{x} \times 100\%$	乘除：百分比相加
加减的不确定度	$\Delta y = \Delta a + \Delta b$	相减时也相加
乘方的不确定度	$\dfrac{\Delta y}{y} = n\dfrac{\Delta a}{a}$	$y = a^n$
矢量分解	$F_x = F\cos\theta$, $F_y = F\sin\theta$	θ 为与 x 轴夹角
两矢量合成	$R^2 = a^2 + b^2 + 2ab\cos\theta$	垂直时 $R = \sqrt{a^2 + b^2}$

1.6 本章练习 Practice questions

练习 1.1 9702/12/M/J/26 Q2

What are the base units for intensity?

A kg s^{-3} **B** kg s **C** $\text{kg m}^2\text{ s}^{-3}$ **D** $\text{kg m}^4\text{ s}^{-3}$

解答 Solution

答案 Answer：**A**

强度 $I = P/A$，单位 W m^{-2}。$\text{W} = \text{J s}^{-1} = \text{kg m}^2\text{ s}^{-3}$，所以 $\text{W m}^{-2} = \text{kg s}^{-3}$。

练习 1.2 9702/13/M/J/26 Q2

What are the advantages of taking repeat measurements and calculating a mean in an experiment?

	taking repeat measurements	calculating a mean
A	shows if the results are accurate	reduces the effects of random errors
B	shows if the results are accurate	reduces the effects of systematic errors
C	shows if the results are precise	reduces the effects of random errors
D	shows if the results are precise	reduces the effects of systematic errors

解答 Solution

答案 Answer：C

重复测量可以看出读数之间的分散程度，即是否**精密**；取平均只能减小**随机误差**的影响，对系统误差无效（系统误差使每次读数朝同一方向偏移，平均后依然存在）。

练习 1.3 9702/13/M/J/26 Q3

The value of a quantity X is calculated from the values of quantities Y and Z by using the equation

$$X = \frac{Y}{Z}.$$

The percentage uncertainty in the value of Y is twice the percentage uncertainty in the value of Z. What is the ratio $\dfrac{\text{percentage uncertainty in the value of } X}{\text{percentage uncertainty in the value of } Z}$?

A 1 **B** 3 **C** 4 **D** 5

解答 Solution

答案 Answer：B

除法：百分比不确定度相加。设 Z 的百分比不确定度为 z，则 Y 为 $2z$，X 为 $2z + z = 3z$，比值为 3。

练习 1.4 9702/14/M/J/26 Q3

What is a reasonable estimate for the volume of a fully-inflated spherical football?

A $6000\,\text{mm}^3$ **B** $60\,\text{cm}^3$ **C** $6000\,\text{cm}^3$ **D** $0.60\,\text{m}^3$

解答 Solution

答案 Answer：C

足球直径约 22 cm，半径 $r \approx 11\,\text{cm}$，$V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(11)^3 \approx 5600\,\text{cm}^3$，最接近 $6000\,\text{cm}^3$。

练习 1.5 9702/12/F/M/26 Q3

A student measures the current and the potential difference for a resistor in a circuit.

$$\text{current} = (50.00 \pm 0.01)\ \text{mA}, \qquad \text{potential difference} = (500.0 \pm 0.1)\ \text{mV}$$

The measurements are used to calculate the resistance of the resistor. What is the percentage uncertainty in the calculated resistance?

A 0.0002%　　**B** 0.0004%　　**C** 0.02%　　**D** 0.04%

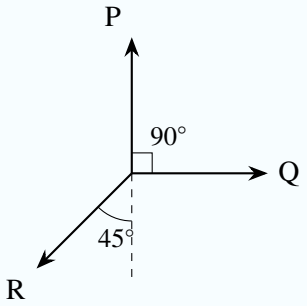
解答 Solution

答案 Answer：**D**

$R = V/I$，百分比不确定度相加：$\frac{0.01}{50.00} \times 100\% + \frac{0.1}{500.0} \times 100\% = 0.02\% + 0.02\% = 0.04\%$。

练习 1.6　9702/12/M/J/26 Q3

Three coplanar forces, P, Q and R, act at a point.



The three forces are in equilibrium. The magnitude of P is increased by 10%. The directions of the forces are unchanged. Which percentage increase, if any, is required to the magnitudes of Q and R for the forces to be once again in equilibrium?

	% increase in magnitude of Q	% increase in magnitude of R
A	10	10
B	0	10
C	$10\sqrt{2}$	$10\sqrt{2}$
D	10	$10\sqrt{2}$

解答 Solution

答案 Answer：**A**

竖直方向：$R\cos 45° = P$；水平方向：$R\sin 45° = Q$。方向不变时，三个力构成的矢量三角形形状不变（相似三角形），所以 P 增大 10% 时，Q 和 R 也都必须增大 10%。

练习 1.7　9702/22/M/J/26 Q1(a)

By placing one tick (✓) in each row, complete Table 1.1 to identify whether each of the listed quantities is a scalar or a vector.

quantity	scalar	vector
amplitude		
force		
momentum		
power		

[2]

解答 Solution

amplitude：scalar；force：vector；momentum：vector；power：scalar。

amplitude 与 power 为标量 **[B1]**

force 与 momentum 为矢量 **[B1]**

第二章 运动学 Kinematics

考纲要求 Syllabus learning outcomes

2.1.1 define and use distance, displacement, speed, velocity and acceleration
定义并使用路程、位移、速率、速度和加速度。

2.1.2 use graphical methods to represent distance, displacement, speed, velocity and acceleration
用图像表示路程、位移、速率、速度和加速度。

2.1.3 determine displacement from the area under a velocity–time graph
由速度–时间图像下的面积求位移。

2.1.4 determine velocity using the gradient of a displacement–time graph
由位移–时间图像的斜率求速度。

2.1.5 determine acceleration using the gradient of a velocity–time graph
由速度–时间图像的斜率求加速度。

2.1.6 derive, from the definitions of velocity and acceleration, equations that represent uniformly accelerated motion in a straight line
由速度和加速度的定义推导匀加速直线运动方程。

2.1.7 solve problems using equations that represent uniformly accelerated motion in a straight line, including the motion of bodies falling in a uniform gravitational field without air resistance
用匀加速直线运动方程解题，包括无空气阻力时物体在均匀重力场中的下落。

2.1.8 describe an experiment to determine the acceleration of free fall using a falling object
描述用下落物体测定自由落体加速度的实验。

2.1.9 describe and explain motion due to a uniform velocity in one direction and a uniform acceleration in a perpendicular direction
描述并解释一个方向匀速、垂直方向匀加速的运动（抛体运动）。

2.1 描述运动的物理量

定义 2.1 (路程与位移 *Distance and displacement*)

Distance: the total length of the path travelled (scalar).

Displacement: the distance in a straight line from a reference point in a specified direction (vector).

路程是运动轨迹的总长度，标量；**位移**是从参考点指向物体所在位置的有向直线距离，矢量。

定义 2.2 (速率与速度 *Speed and velocity*)

Speed: distance travelled per unit time.

Velocity: change in displacement per unit time (rate of change of displacement).

速率 = 路程 / 时间（标量）；**速度** = 位移的变化率 $v = \dfrac{\Delta s}{\Delta t}$（矢量）。

定义 2.3 (加速度 *Acceleration*)

Acceleration is the rate of change of velocity: $a = \dfrac{\Delta v}{\Delta t}$.

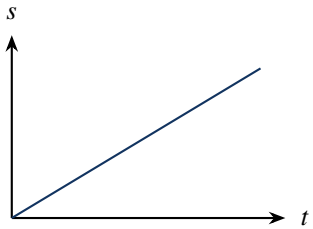
加速度是速度的变化率（矢量），单位 $\mathrm{m\,s^{-2}}$。速度大小改变或方向改变都意味着有加速度。

- **平均速度** = 总位移 / 总时间；**平均速率** = 总路程 / 总时间。沿圆周走 3/4 圈时，平均速率按弧长算，平均速度按弦长（起点到终点直线）算。
- **瞬时速度**是某一时刻的速度，等于 *s*–*t* 图在该点的**切线斜率**。
- “减速”(deceleration) 指加速度方向与速度方向相反，计算时 a 取负值。

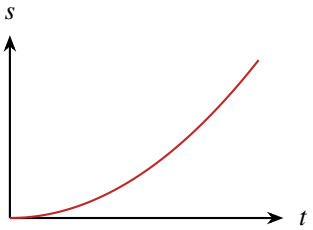
易错提醒 Common mistakes 定义要用“per unit time”或“rate of change”：velocity 的定义不能写成“speed in a given direction”（这是描述，CIE 不给定义分）。题目问 definition of velocity 时答 *rate of change of displacement*。

2.2 运动图像 Graphs of motion

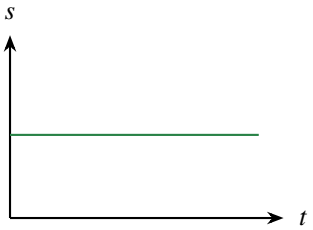
匀速：直线



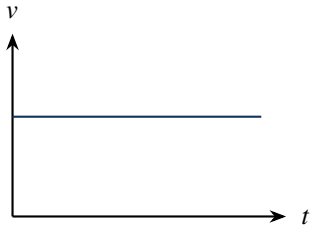
匀加速：抛物线



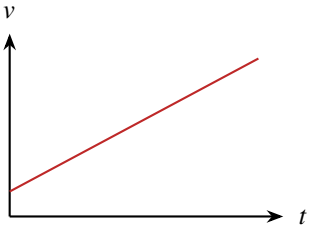
静止：水平线



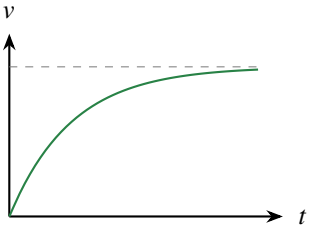
匀速



匀加速



加速度减小（趋于终端速度）

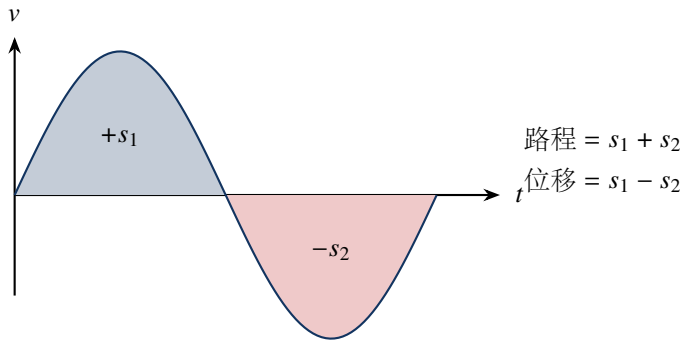


公式 2.1 (图像的斜率与面积 *Gradients and areas*)

图像	斜率 gradient	面积 area under graph
位移–时间 *s*–*t*	速度 v	无物理意义
速度–时间 *v*–*t*	加速度 a	位移 s
加速度–时间 *a*–*t*	（不要求）	速度的变化量 Δv

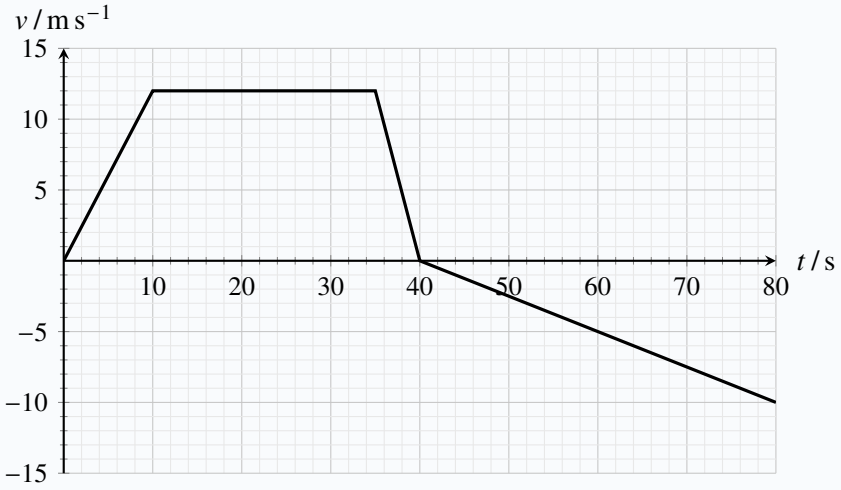
- *v*–*t* 图中，时间轴**下方**的面积表示反方向的位移。求**路程**时把各部分面积的绝对值相加；求**位移**时取代数和。
- 曲线 *s*–*t* 图求某时刻速度：在该点作**切线**，取切线上相距较远的两点计算斜率（取点间隔至少占图像一半以上，以减小读数误差）。

- 曲线 v–t 图下的面积可以用数方格估算：数出完整方格数（超过半格的计为一格），乘以每格代表的位移。



例题 2.1 9702/22/M/J/26 Q1(b)

A car moves in a straight line along a road. Figure 1.1 shows the variation with time t of the velocity v of the car.



(i) State the feature of the graph that represents the distance travelled by the car between $t = 0$ and $t = 40\,\text{s}$. [1]

(ii) Calculate the distance travelled by the car between $t = 0$ and $t = 40\,\text{s}$. [1]

(iii) Determine the maximum magnitude of the acceleration of the car. [2]

(iv) At $t = 25\,\text{s}$, the car is at a point P. At $t = 40\,\text{s}$, the car is at a distance of 150 m from P. After $t = 40\,\text{s}$, the car travels with a constant acceleration of $-0.25\,\text{m s}^{-2}$ until the car is again at point P. Calculate the speed of the car when it is at point P for the second time. [2]

(v) At $t = 0$, the displacement of the car is zero. Sketch the variation of the displacement of the car with t from $t = 0$ to $t = 80\,\text{s}$. Values for displacement are not required. [3]

解答 Solution

(i) 图线与时间轴之间的面积 (area under the graph)。 **[B1]**

(ii) 面积 $= \frac{1}{2}(10)(12) + (25)(12) + \frac{1}{2}(5)(12) = 60 + 300 + 30 = 390\,\text{m}$。 **[A1]**

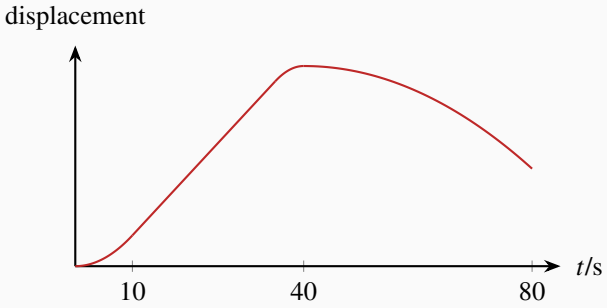
(iii) 最陡的一段是 $t = 35$ 到 $40\,\text{s}$：$a = \dfrac{0-12}{5} = -2.4\,\text{m s}^{-2}$。 **[C1]**

加速度大小最大值 $= 2.4\,\text{m s}^{-2}$。 **[A1]**

(iv) 从 $t = 40\,\text{s}$ 起初速 $u = 0$，$a = -0.25$，位移 $s = -150\,\text{m}$（返回 P）：$v^2 = u^2 + 2as = 0 + 2(-0.25)(-150) = 75$。 **[C1]**

$v = 8.7\,\text{m s}^{-1}$。 **[A1]**

(v) 草图要点：



- 从原点出发，先增大后减小，且不穿过 t 轴（$t = 80\,\text{s}$ 时位移 $390 - 200 = 190\,\text{m} > 0$）； **[B1]**
- 在 $t = 40\,\text{s}$ 处位移最大； **[B1]**
- 0–10 s 斜率增大；10–35 s 直线；35–40 s 斜率减小；40–80 s 斜率继续减小（变为负）。 **[B1]**

2.3 匀加速直线运动方程 Equations of uniformly accelerated motion

设物体初速度为 u，经过时间 t 后速度为 v，加速度 a 恒定，位移为 s。

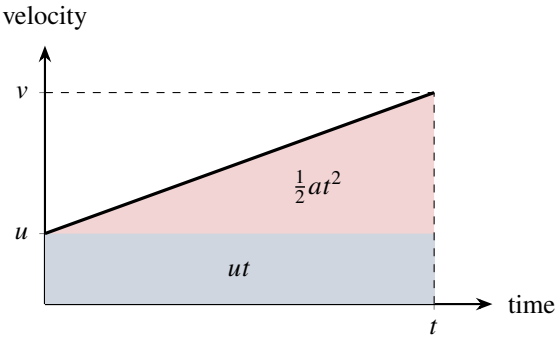
推导 (derivation)：

(1) 由加速度定义 $a = \dfrac{v-u}{t}$，得 $\boxed{v = u + at}$。

(2) 匀加速时平均速度为 $\dfrac{u+v}{2}$，位移 = 平均速度 × 时间，得 $\boxed{s = \dfrac{(u+v)}{2}t}$。这也是 v–t 图下梯形的面积。

(3) 把 (1) 代入 (2)：$s = \dfrac{u + (u+at)}{2}t$，得 $\boxed{s = ut + \frac{1}{2}at^2}$。

(4) 由 (1) 得 $t = \dfrac{v-u}{a}$，代入 (2)：$s = \dfrac{(u+v)(v-u)}{2a}$，得 $\boxed{v^2 = u^2 + 2as}$。



公式 2.2 (SUVAT 方程　*Equations of uniformly accelerated motion*)

$$v = u + at, \qquad s = \frac{u+v}{2}t, \qquad s = ut + \tfrac{1}{2}at^2, \qquad v^2 = u^2 + 2as.$$

适用条件：直线运动且加速度恒定。五个量 s, u, v, a, t 中已知任意三个即可求另外两个。（考试数据表给出 $s = ut + \frac{1}{2}at^2$ 与 $v^2 = u^2 + 2as$。）

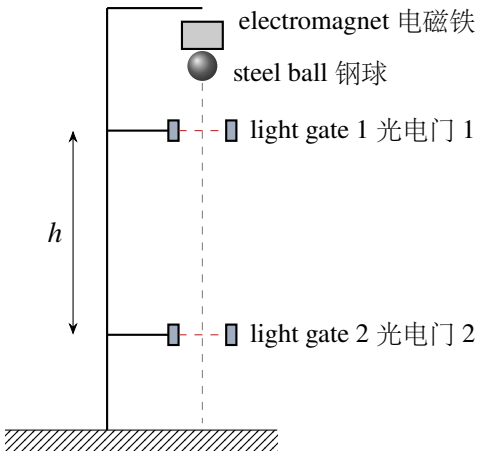
解题步骤：(1) 规定正方向；(2) 列出 s, u, v, a, t 中已知的三个和要求的一个；(3) 选不含“第五个量”的方程；(4) 带符号代入，答案注意正负号的物理意义。

考试技巧 Exam tip　对竖直上抛问题，通常取向上为正，则 $a = -g = -9.81\,\mathrm{m\,s^{-2}}$。从抛出到落回抛出点以下 h 处，位移 $s = -h$。整个过程可以用一个方程处理，**不需要**分成上升和下降两段。

2.4 自由落体 Free fall

在**无空气阻力**时，地球表面附近所有物体下落的加速度相同，称为**自由落体加速度** (*acceleration of free fall*) $g = 9.81\,\mathrm{m\,s^{-2}}$，方向竖直向下，与物体质量无关。

2.4.1 测定 g 的实验



方法一（电磁铁 + 光电门/计时器）：

(1) 切断电磁铁电流，钢球由静止释放，同时开始计时；钢球通过（或撞开）下方的触发板 (trapdoor) 时停止计时，得时间 t。

(2) 用米尺测量钢球底部到触发板的高度 h。

(3) 由 $h = \frac{1}{2}gt^2$（$u = 0$）得 $g = 2h/t^2$。

(4) 改变 h 重复多次，作 h–t^2 图，斜率 $= g/2$，所以 $g = 2\times$ 斜率。

方法二（两个光电门）：测出钢球通过光电门 1、2 的速度 $v_1 = d/\Delta t_1$，$v_2 = d/\Delta t_2$（d 为球直径），以及两门间距 h，则 $g = \dfrac{v_2^2 - v_1^2}{2h}$。

误差来源与改进：电磁铁剩磁使释放延迟（系统误差）——作 h–t^2 图可消除截距带来的影响；使用密度大、体积小的钢球以减小空气阻力；测 h 时视线与刻度垂直以避免视差。

例题 2.2 9702/14/M/J/26 Q6

A stone is thrown vertically upwards from a point that is 12 m above the sea. It then falls into the sea below after 3.4 s. Air resistance is negligible. At which speed is the stone released when it is thrown?

A $3.5\,\mathrm{m\,s^{-1}}$ **B** $6.6\,\mathrm{m\,s^{-1}}$ **C** $13\,\mathrm{m\,s^{-1}}$ **D** $20\,\mathrm{m\,s^{-1}}$

解答 Solution

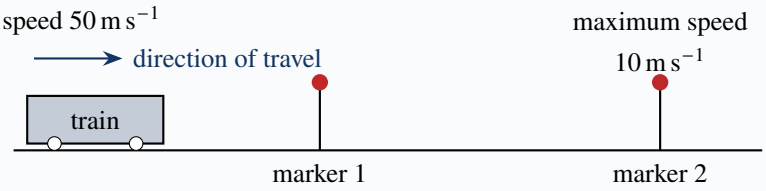
答案 Answer：C

取向上为正：$s=-12\,\mathrm{m}$，$a=-9.81\,\mathrm{m\,s^{-2}}$，$t=3.4\,\mathrm{s}$。

$$-12=u(3.4)-\tfrac{1}{2}(9.81)(3.4)^2=3.4u-56.7\ \Rightarrow\ u=\frac{44.7}{3.4}=13\,\mathrm{m\,s^{-1}}.$$

例题 2.3 9702/12/M/J/26 Q7

The diagram shows an arrangement to stop trains that are travelling too fast.



Trains coming from the left travel at a speed of $50\,\mathrm{m\,s^{-1}}$. At marker 1, the driver must apply the brakes so that the train decelerates uniformly in order to pass marker 2 at no more than $10\,\mathrm{m\,s^{-1}}$. The train carries a detector that notes the times when the train passes each marker and will apply an emergency brake if the time between passing marker 1 and marker 2 is less than 20 s. How far from marker 2 should marker 1 be placed?

A 200 m **B** 400 m **C** 500 m **D** 600 m

解答 Solution

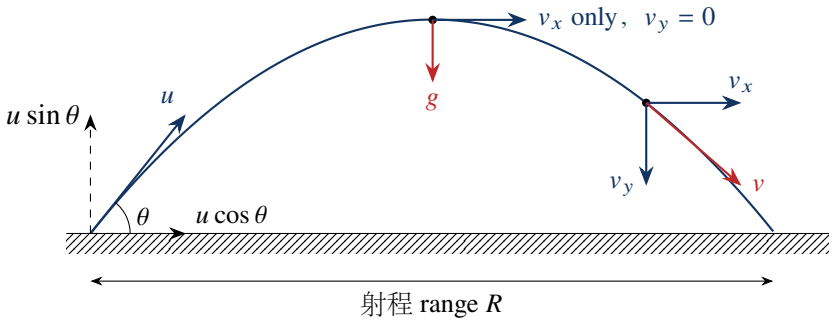
答案 Answer：D

临界情况：从 $50\,\mathrm{m\,s^{-1}}$ 匀减速到 $10\,\mathrm{m\,s^{-1}}$ 恰好用 20 s。$s=\dfrac{u+v}{2}t=\dfrac{50+10}{2}\times 20=600\,\mathrm{m}$。

2.5 抛体运动 Projectile motion

当物体在一个方向上**匀速**运动、而在垂直方向上**匀加速**运动时，两个方向的运动**相互独立**，只通过共同的时间 t 联系。忽略空气阻力的抛体运动就是这种情况：

- 水平方向：$a_x=0$，$v_x=u\cos\theta$ 恒定，$x=(u\cos\theta)\,t$；
- 竖直方向：$a_y=-g$，$v_y=u\sin\theta-gt$，$y=(u\sin\theta)\,t-\tfrac{1}{2}gt^2$。



- 最高点：竖直速度为零，但水平速度不为零，所以物体**不是静止**的；加速度仍为 g（向下）。
- 飞行时间由竖直方向决定：同一高度落回时 $t = \dfrac{2u\sin\theta}{g}$；最大高度 $H = \dfrac{(u\sin\theta)^2}{2g}$。
- 达到相同最大高度的两个抛体，竖直初速度相同，所以**飞行时间相同**，与水平速度无关。
- 水平抛出：$u_y = 0$，下落时间只由高度决定 $h = \frac{1}{2}gt^2$。
- 落地速度：$v = \sqrt{v_x^2 + v_y^2}$，方向 $\tan\alpha = v_y/v_x$。

有空气阻力时（定性）：水平速度逐渐减小，射程和最大高度都减小，轨迹不再对称——下降段比上升段更陡。

例题 2.4 9702/11/M/J/26 Q7

A ball is thrown horizontally with a speed of $10.0\,\mathrm{m\,s^{-1}}$ above horizontal ground. The ball hits the ground at a time of 3.0 s after being thrown. Air resistance is negligible. What is the speed of the ball just before it hits the ground?

A $10\,\mathrm{m\,s^{-1}}$ **B** $29\,\mathrm{m\,s^{-1}}$ **C** $31\,\mathrm{m\,s^{-1}}$ **D** $39\,\mathrm{m\,s^{-1}}$

解答 Solution

答案 Answer：C

$v_x = 10.0\,\mathrm{m\,s^{-1}}$ 不变；$v_y = gt = 9.81 \times 3.0 = 29.4\,\mathrm{m\,s^{-1}}$。$v = \sqrt{10.0^2 + 29.4^2} = 31\,\mathrm{m\,s^{-1}}$。（选 B 是忘了合成水平速度；选 D 是把两个分量直接相加。）

教学提示 Teacher's note 建议 4 课时。v–t 图的“面积—位移、斜率—加速度”必须反复练习，Paper 2 几乎每年都有图像题；草图题按 MS 要求：起点、最大值位置、各段斜率的变化趋势各占一分。竖直上抛题强调“规定正方向 + 一个方程到底”。实验题中常考 h–t^2 作图法求 g。

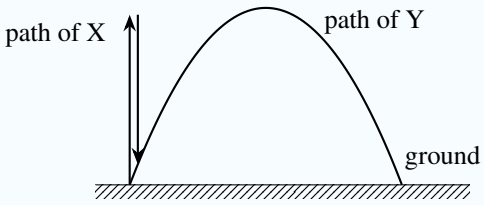
2.6 本章小结 Summary

物理量/关系	公式	说明
速度	$v = \Delta s/\Delta t$	s–t 图斜率
加速度	$a = \Delta v/\Delta t$	v–t 图斜率
位移	v–t 图下面积	轴下方为负
SUVAT	$v = u + at$；$s = \frac{u+v}{2}t$	恒定加速度直线运动
	$s = ut + \frac{1}{2}at^2$；$v^2 = u^2 + 2as$	
抛体（无阻力）	v_x 恒定，$a_y = -g$	两方向独立，时间相同

2.7 本章练习 Practice questions

练习 2.1 9702/13/M/J/26 Q6

Two projectiles, X and Y, are fired into the air from the same place on level ground and reach the same maximum height, as shown.



Projectile X is fired vertically upwards and projectile Y is fired at an angle to the horizontal. Air resistance is negligible. Which statement is correct?

A X and Y are at rest at their maximum heights.

B X and Y are fired with the same speed.

C X and Y take the same time to return to the ground.

D X and Y travel the same distance.

解答 Solution

答案 Answer：C

最大高度 $H = u_y^2/2g$ 相同 $\Rightarrow$ 竖直初速度 u_y 相同 $\Rightarrow$ 飞行时间 $2u_y/g$ 相同。Y 在最高点仍有水平速度，不是静止；Y 的初速度更大（还有水平分量）；Y 的路程更长。

练习 2.2 9702/13/M/J/26 Q7

A lead sphere is released from rest at point X, a long way above the surface of a planet. The sphere falls in a vacuum. After a time of 4.0 s, it has fallen through a vertical distance of 3.0 m. Assume the acceleration of free fall is constant. How far will the sphere have fallen from point X at a time of 20 s after its release?

A 15 m **B** 75 m **C** 80 m **D** 2000 m

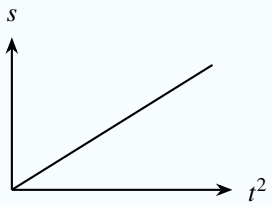
解答 Solution

答案 Answer：**B**

由静止开始 $s = \frac{1}{2}gt^2 \propto t^2$。时间变为 5 倍，距离变为 25 倍：$3.0 \times 25 = 75\,\text{m}$。

练习 2.3 9702/14/M/J/26 Q7

At time $t = 0$, an object moves from rest with constant acceleration in a straight line. At time t, the object is distance s from its rest position. A graph is drawn of s against t^2 as shown.



Which statement describes the acceleration of the object?

A It is equal to half the value of the gradient of the graph.

B It is equal to the value of the gradient of the graph.

C It is equal to twice the value of the gradient of the graph.

D It is equal to the reciprocal of the gradient of the graph.

解答 Solution

答案 Answer：**C**

$s = \frac{1}{2}at^2$，与 $y = mx$ 比较，斜率 $m = a/2$，所以 $a = 2\times$ 斜率。

练习 2.4 9702/12/F/M/26 Q5

The water surface in a deep well is 78.0 m below the top of the well. A person at the top of the well drops a heavy stone down the well. Air resistance is negligible. The speed of sound in the air is $330\,\text{m s}^{-1}$. What is the time interval between the person dropping the stone and hearing it hitting the water?

A 3.75 s **B** 3.99 s **C** 4.19 s **D** 4.22 s

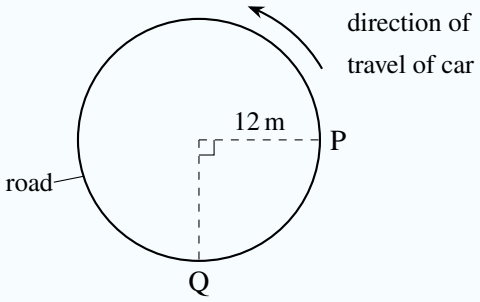
解答 Solution

答案 Answer：**D**

下落时间：$78.0 = \frac{1}{2}(9.81)t_1^2$，$t_1 = 3.99\,\text{s}$；声音上传时间 $t_2 = 78.0/330 = 0.236\,\text{s}$。总时间 = 3.99+0.24 = 4.22 s。

练习 2.5 9702/12/F/M/26 Q7

A car travels anticlockwise along a horizontal circular road of radius 12 m, as shown. The car takes a time of 4.0 s to move from position P to position Q.



What is the magnitude of the average velocity of the car for the journey from P to Q?

A $4.2\,\text{m s}^{-1}$ **B** $4.7\,\text{m s}^{-1}$ **C** $6.0\,\text{m s}^{-1}$ **D** $14\,\text{m s}^{-1}$

解答 Solution

答案 Answer：A

平均速度 = 位移 / 时间。逆时针从 P 到 Q 走了 3/4 圈，但位移只是 P、Q 间的弦长 = $12\sqrt{2} = 17.0\,\text{m}$。$\bar{v} = 17.0/4.0 = 4.2\,\text{m s}^{-1}$。（D 是平均速率：$\frac{3}{4} \times 2\pi \times 12/4.0 = 14$。）

练习 2.6 9702/24/M/J/26 Q1

(a) Underline all the vector quantities in the following list:

distance displacement speed velocity acceleration.

[1]

(b) A car of mass 1400 kg is initially travelling horizontally in a straight line at a constant speed of $15\,\text{m s}^{-1}$. The brakes are applied and the car decelerates uniformly to rest in a distance of 35 m. Calculate:

(i) the initial kinetic energy of the car [2]

(ii) the acceleration of the car [2]

(iii) the resultant force on the decelerating car. [2]

(c) The initial speed of the car in (b) is now doubled to $30\,\text{m s}^{-1}$. The brakes of the car are then applied and the car comes to rest. The resultant force on the car remains the same as in (b)(iii). Describe and explain the effect, if any, of doubling the initial speed on:

(i) the magnitude of the acceleration of the car [1]

(ii) the distance travelled by the car in coming to rest. [2]

解答 Solution

(a) 只划 displacement, velocity, acceleration。 **[A1]**

(b)(i) $E_k = \frac{1}{2}mv^2 = \frac{1}{2}(1400)(15)^2$ **[C1]**

$= 1.6 \times 10^5\,\text{J}$ **[A1]**

(ii) $0 = 15^2 + 2a(35)$ **[C1]**

$a = -3.2\,\text{m s}^{-2}$ **[A1]**

(iii) $F = ma = 1400 \times 3.2$ **[C1]**

$= 4500\,\text{N}$（也可用 $F = E_k/s = 1.6 \times 10^5/35 = 4600\,\text{N}$） **[A1]**

(c)(i) 不变：力和质量都不变，由 $F = ma$，加速度不变。 **[A1]**

(ii) 初动能变为原来的 4 倍（$E_k \propto v^2$），**[B1]**

力相同，所以制动距离也变为 4 倍，即 140 m。 **[B1]**

第三章 动力学 Dynamics

考纲要求 Syllabus learning outcomes

3.1.1 understand that mass is the property of an object that resists change in motion
理解质量是物体抵抗运动状态改变的性质（惯性）。

3.1.2 recall $F = ma$ and solve problems using it, understanding that acceleration and resultant force are always in the same direction
记住 $F = ma$ 并用其解题，理解加速度方向总与合力方向相同。

3.1.3 define and use linear momentum as the product of mass and velocity
定义并使用线动量（质量与速度之积）。

3.1.4 define and use force as rate of change of momentum
把力定义为动量的变化率并加以运用。

3.1.5 state and apply each of Newton's laws of motion
陈述并应用牛顿三大运动定律。

3.1.6 describe and use the concept of weight as the effect of a gravitational field on a mass and recall that the weight of an object is equal to the product of its mass and the acceleration of free fall
理解重量是引力场对质量的作用，记住 $W = mg$。

3.2.1 show a qualitative understanding of frictional forces and viscous/drag forces including air resistance (no treatment of the coefficients of friction and viscosity is required, and a simple model of drag force increasing as speed increases is sufficient)
定性理解摩擦力和黏滞力/阻力（含空气阻力），只需知道阻力随速度增大而增大。

3.2.2 describe and explain qualitatively the motion of objects in a uniform gravitational field with air resistance
定性描述并解释有空气阻力时物体在均匀重力场中的运动。

3.2.3 understand that objects moving against a resistive force may reach a terminal (constant) velocity
理解受阻力的物体可能达到终端（恒定）速度。

3.3.1 state the principle of conservation of momentum
陈述动量守恒定律。

3.3.2 apply the principle of conservation of momentum to solve simple problems including elastic and inelastic interactions between objects in both one and two dimensions (knowledge of the concept of coefficient of restitution is not required)
用动量守恒解决一维和二维的弹性、非弹性相互作用问题（不要求恢复系数）。

3.3.3 recall that, for a perfectly elastic collision, the relative speed of approach is equal to the relative speed of separation
记住完全弹性碰撞中，相对接近速率等于相对分离速率。

3.3.4 understand that, while the momentum of a system is always conserved in interactions between objects, some change in kinetic energy may take place
理解在相互作用中系统动量总是守恒的，但动能可能发生变化。

3.1 质量、力与牛顿运动定律

3.1.1 质量与惯性

质量 (*mass*) 是物体**抵抗运动状态改变**的性质（即惯性 inertia 的量度）。质量越大，同样的合力产生的加速度越小。质量是标量，单位 kg，与物体所在位置无关。

定义 3.1 (重量　*Weight*)

Weight is the effect of a gravitational field on a mass; it is the gravitational force acting on an object: $W = mg$.

重量是引力场对质量的作用力（矢量，竖直向下），$W = mg$，单位 N。同一物体在月球上质量不变，重量约为地球上的 1/6。

3.1.2 线动量

定义 3.2 (线动量　*Linear momentum*)

Linear momentum is the product of mass and velocity: $p = mv$.

动量是质量与速度的乘积，矢量，方向与速度相同，单位 $\mathrm{kg\,m\,s^{-1}}$（$= \mathrm{N\,s}$）。

3.1.3 牛顿三大定律

定律 3.1 (牛顿第一定律　*Newton's first law*)

A body will remain at rest or continue to move with constant velocity unless acted on by a resultant force.

物体将保持静止或匀速直线运动状态，除非受到**合力**作用。

定律 3.2 (牛顿第二定律　*Newton's second law*)

The resultant force acting on a body is proportional to the rate of change of its momentum (and in the same direction). With SI units: $F = \dfrac{\Delta p}{\Delta t}$.

物体所受合力等于其动量的变化率，方向与动量变化的方向相同。质量不变时 $F = \dfrac{\Delta(mv)}{\Delta t} = m\dfrac{\Delta v}{\Delta t} = ma$。

定义 3.3 (力　*Force*)

Force is the rate of change of momentum. （1 N 是使 1 kg 物体产生 $1\,\mathrm{m\,s^{-2}}$ 加速度所需的力。）

力是动量的变化率。注意：$F = ma$ 只在**质量不变**时成立；质量变化时（如火箭喷气、传送带上落沙）必须用 $F = \Delta p/\Delta t$。

要点：

- F 是**合力**（resultant force）。加速度方向**总是**与合力方向相同（但速度方向不一定与合力相同，例如减速运动、抛体运动）。
- 合力为零 ⇔ 加速度为零 ⇔ 静止或匀速直线运动，称为**平衡**（equilibrium）。在 s–t 图上，水平线（静止）和倾斜直线（匀速）都对应平衡。
- 冲量 (impulse) $F\Delta t = \Delta p$；F–t 图下的面积等于动量变化量。

定律 3.3 (牛顿第三定律　*Newton's third law*)

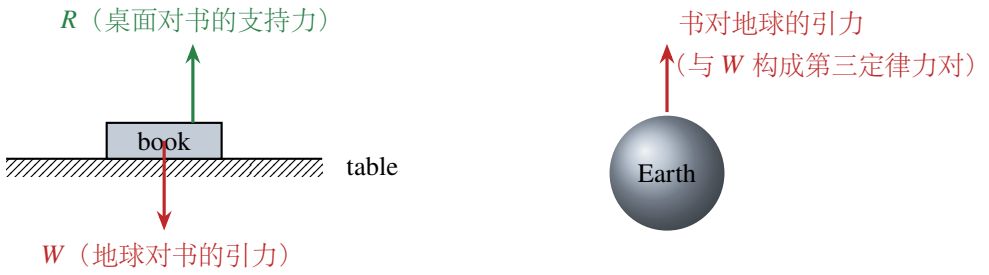
If body A exerts a force on body B, then body B exerts a force on body A that is equal in magnitude and opposite in direction.

若物体 A 对物体 B 施加一个力，则 B 对 A 施加一个大小相等、方向相反的力。

第三定律力对的特征（考试常考）：

(1) 大小相等、方向相反；
(2) 作用在**两个不同的物体**上（所以永远不会相互抵消）；
(3) 是**同种性质**的力（都是引力，或都是接触力/电磁力）；
(4) 沿同一直线，同时产生、同时消失。

两个物体**不必**接触（引力对就不接触）。



易错提醒 Common mistakes　书静止在桌面上时，重力 W 与支持力 R 大小相等方向相反，但它们**不是**第三定律力对——它们作用在同一物体（书）上，而且性质不同。W 的第三定律配对力是**书对地球的引力**；R 的配对力是**书对桌面的压力**。

例题 3.1　9702/11/M/J/26 Q9

A book rests on a table. The book has weight W. Newton's third law describes the relationship between pairs of forces. Which force pairs with W to make a Newton's third law pair of forces?

A the contact force acting on the book due to the table

B the contact force acting on the table due to the book

C the friction force between the book and the table

D the gravitational force acting on the Earth due to the book

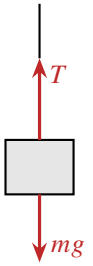
解答 Solution

答案 Answer：D

W 是地球对书的**引力**，它的配对力必须是同种力（引力）、作用在另一物体（地球）上：书对地球的引力。

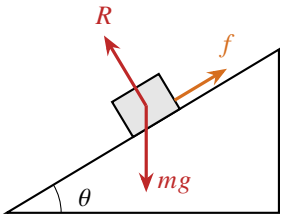
3.1.4 运用 $F = ma$ 解题

步骤：(1) 隔离物体，画**受力分析图** (*free-body diagram*)；(2) 规定正方向（通常取加速度方向）；(3) 合力 = 沿正方向的力之和 − 反方向的力之和；(4) 列 $F_{合} = ma$。



电梯/吊物加速上升

$T - mg = ma$



物体沿斜面下滑

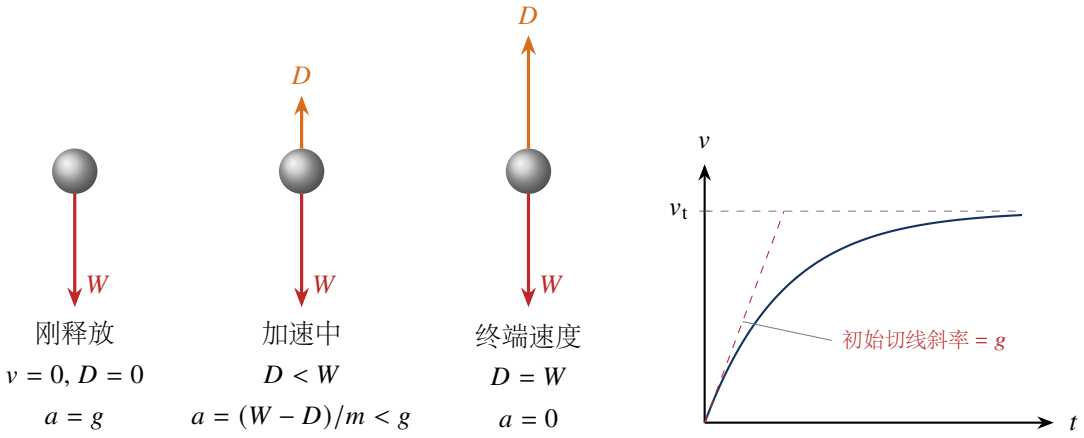
$mg\sin\theta - f = ma \quad R = mg\cos\theta$

3.2 非匀变速运动：阻力与终端速度

3.2.1 摩擦力与阻力

- **摩擦力** (*friction*)：两个接触表面相对运动（或有相对运动趋势）时产生，方向与相对运动（趋势）方向相反。
- **黏滞力/阻力** (*viscous force / drag*)：物体在流体（液体或气体）中运动时受到的阻力；空气中的阻力称为空气阻力 (air resistance)。阻力方向总与物体相对流体的运动方向相反，**大小随速度增大而增大**。
- 摩擦和阻力都把动能转化为内能（热），使系统机械能减少。

3.2.2 有空气阻力的下落与终端速度



物体从静止开始在空气中下落：

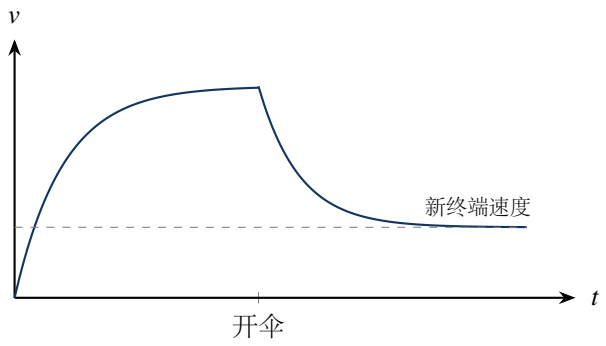
(1) 刚释放时速度为零，阻力为零，合力 = W，加速度 = g；

(2) 速度增大，阻力 D 增大，合力 $W - D$ 减小，加速度减小（v–t 图斜率减小）；

(3) 当 $D = W$ 时合力为零，加速度为零，速度达到最大并保持不变，称为**终端速度** (*terminal velocity*)。

（若物体在液体中下落，还要考虑浮力 U：终端速度时 $W = U + D$。）

跳伞运动员：以终端速度下落时打开降落伞，面积增大使阻力突然**大于**重量，合力向上，运动员**减速**；随速度减小阻力减小，直到再次 $D = W$，达到一个较小的新终端速度。



例题 3.2 9702/12/M/J/26 Q10

A hollow steel object is dropped from rest in air. The object reaches a terminal velocity v before hitting the ground. Which change results in an increase in the terminal velocity?

A Drop a solid steel object of the same dimensions from the same height.

B Drop the object from a greater height.

C Throw the object downwards from the same height with an initial velocity $0.5v$.

D Throw the object upwards from the same height with an initial velocity $0.5v$.

解答 Solution

答案 Answer：A

终端速度时 $D = W$。外形相同时阻力只取决于速度；实心物体重量更大，需要更大的阻力才能平衡，所以终端速度更大。改变高度或初速度不改变 $D = W$ 的条件，终端速度不变。

3.3 动量守恒 Conservation of momentum

定律 3.4 (动量守恒定律 *Principle of conservation of momentum*)

The total momentum of a system remains constant provided no resultant external force acts on the system.

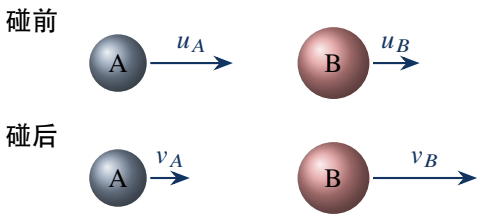
若系统不受外力（或所受合外力为零），则系统的总动量保持不变：作用前总动量 = 作用后总动量。

由牛顿定律推导：A、B 相互作用时间 Δt 内，由第三定律 $F_{AB} = -F_{BA}$，由第二定律 $\Delta p_A = F_{BA}\Delta t$，$\Delta p_B = F_{AB}\Delta t$，所以 $\Delta p_A + \Delta p_B = 0$，总动量不变。

3.3.1 弹性碰撞与非弹性碰撞

	完全弹性碰撞 perfectly elastic	非弹性碰撞 inelastic
总动量	守恒	守恒
总能量	守恒	守恒
总动能	**守恒**	**减少**（转化为内能、声能等）
相对速率	接近速率 = 分离速率	分离速率 < 接近速率

- **完全非弹性碰撞**：碰后两物体粘在一起，以共同速度运动，动能损失最大（但动量仍守恒）。
- **爆炸/反冲**：初总动量为零，炸开后各部分动量矢量和仍为零；总动能**增加**（化学能转化为动能）。
- 弹性碰撞的**相对速度关系**：$u_A - u_B = v_B - v_A$（一维，带符号）。这个式子与动量守恒联立，比解动能方程简单得多。



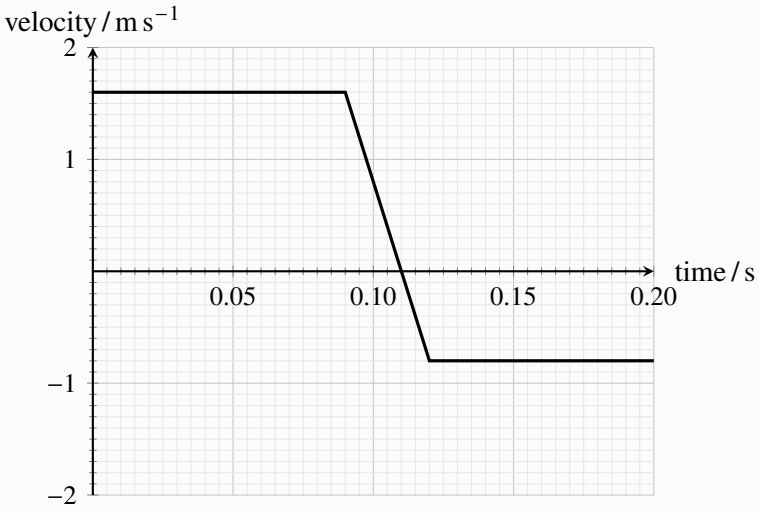
动量：$m_A u_A + m_B u_B = m_A v_A + m_B v_B$

弹性：$u_A - u_B = v_B - v_A$

例题 3.3 9702/23/M/J/26 Q1

(a) Define momentum. [1]

(b) A ball X of mass 0.15 kg moves along a horizontal frictionless surface. The ball collides elastically with ball Y that is initially stationary. Balls X and Y have equal diameters but different masses. Figure 1.1 shows the variation with time of the velocity of ball X before, during and after the collision.



(i) Determine the magnitude of the acceleration of ball X during the collision. [2]

(ii) Determine the magnitude of the change in momentum of ball X due to the collision. [2]

(iii) Determine the total distance travelled by ball X during the collision. [2]

(c) (i) By considering relative speeds, explain why the velocity of ball Y after the collision must be $+0.80\,\text{m s}^{-1}$. [2]

(ii) On Figure 1.1, sketch the variation of the velocity of ball Y with time between time = 0 and time = 0.20 s. [2]

(iii) Calculate the mass of ball Y. [2]

解答 Solution

(a) 质量与速度的乘积 (product of mass and velocity)。注意不能写 speed。 **[B1]**

(b)(i) $a = \dfrac{\Delta v}{\Delta t} = \dfrac{-0.8 - 1.6}{0.12 - 0.09}$ **[C1]**

$= -80$，大小 $80\,\text{m s}^{-2}$ **[A1]**

(ii) $\Delta p = m\Delta v = 0.15 \times (-0.8 - 1.6)$ **[C1]**

$= -0.36$，大小 $0.36\,\text{kg m s}^{-1}$ **[A1]**

(iii) 碰撞中 X 先减速到零（0.020 s内），再反向加速（0.010 s内）。路程是两段面积之和：

$s_1 = \frac{1}{2} \times 1.6 \times 0.020 = 0.016\,\text{m}$，$s_2 = \frac{1}{2} \times 0.80 \times 0.010 = 0.004\,\text{m}$ **[C1]**

总路程 $= 0.016 + 0.004 = 0.020\,\text{m}$ **[A1]**

(c)(i) 弹性碰撞：相对接近速率 = 相对分离速率。 **[B1]**

接近速率 $= 1.6 - 0 = 1.6$，所以 $v_Y - (-0.80) = 1.6$，$v_Y = +0.80\,\text{m s}^{-1}$。 **[B1]**

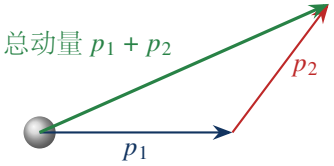
(ii) 从 $(0, 0)$ 到 $(0.09, 0)$ 的水平线；从 $(0.09, 0)$ 到 $(0.12, 0.80)$ 的直线；从 $(0.12, 0.80)$ 到 $(0.20, 0.80)$ 的水平线。 **[B1+B1]**

(iii) 动量守恒：$0.15 \times 1.6 = 0.15 \times (-0.80) + m_Y(0.80)$ **[C1]**

$0.36 = 0.80\,m_Y$，$m_Y = 0.45\,\text{kg}$ **[A1]**

3.3.2 二维动量守恒

二维问题中动量守恒对**每个方向分别成立**：把各动量分解到两个互相垂直的方向（x、y），分别列守恒方程；或直接画**动量矢量三角形**。

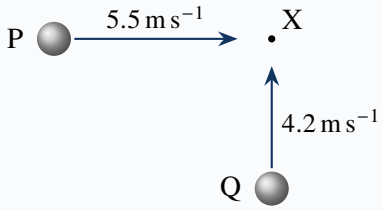


碰前总动量（矢量和）= 碰后总动量（矢量和）

x：$\sum p_x$ 守恒；　y：$\sum p_y$ 守恒

例题 3.4　9702/11/M/J/26 Q12

The diagram shows the view from above of two hard balls rolling along a horizontal table.



The balls each have a mass of 0.16 kg. The initial velocity of ball P is 5.5 m s^{-1} and the initial velocity of ball Q is 4.2 m s^{-1}. The initial velocity of ball P is perpendicular to the initial velocity of ball Q. The balls collide at point X where ball Q stops immediately. There are no resistive forces acting on the balls. What is the speed of ball P after the collision?

A 1.3 m s^{-1}　　**B** 5.5 m s^{-1}　　**C** 6.9 m s^{-1}　　**D** 9.7 m s^{-1}

解答 Solution

答案 Answer：C

Q 碰后静止，所以碰后 P 的动量等于碰前两球的总动量（矢量和）。两球质量相同，可直接合成速度：$v_P = \sqrt{5.5^2 + 4.2^2} = 6.9\,\text{m s}^{-1}$，方向介于原来两速度方向之间。

教学提示 Teacher's note　建议 5 课时。重点：(1) 第三定律力对的四个特征——选择题每年必考；(2) 终端速度的受力分析（开伞后为何减速）；(3) 弹性碰撞用“相对速度”关系代替动能方程，计算量小得多；(4) 力的定义“rate of change of momentum”是 Paper 2 高频定义题。注意区分“动量守恒”与“动能守恒”：所有碰撞动量都守恒，只有弹性碰撞动能才守恒。

3.4 本章小结 Summary

物理量/关系	公式	说明
动量	$p = mv$	矢量，$\mathrm{kg\,m\,s^{-1}}$
牛顿第二定律	$F = \dfrac{\Delta p}{\Delta t}$，$F = ma$	F 为合力
重量	$W = mg$	引力场对质量的作用
冲量	$F\Delta t = \Delta p$	F–t 图面积
动量守恒	$\sum p_{前} = \sum p_{后}$	无外力，矢量式
弹性碰撞	$u_A - u_B = v_B - v_A$	动能也守恒
终端速度	$D = W$（液体中 $D + U = W$）	合力为零

3.5 本章练习 Practice questions

练习 3.1 9702/11/M/J/26 Q11

A ball of mass 0.20 kg is falling vertically through a liquid and has not reached terminal velocity. The upthrust on the ball is 0.65 N. At one instant, the drag force on the ball is 0.50 N. What is the magnitude of the resultant force on the ball at this instant?

A 0.81 N **B** 1.5 N **C** 1.8 N **D** 2.0 N

解答 Solution

答案 Answer：A

向下：$W = 0.20 \times 9.81 = 1.96\,\mathrm{N}$；向上：浮力 0.65 N 与阻力 0.50 N（球向下运动，阻力向上）。合力 $= 1.96 - 0.65 - 0.50 = 0.81\,\mathrm{N}$（向下）。

练习 3.2 9702/12/M/J/26 Q11

A bowling ball of mass 4.50 kg falls a short distance vertically into a tray of sand. The bowling ball decelerates uniformly in the sand. The deceleration of the bowling ball when it is in contact with the sand is $8.00\times10^{2}\ \mathrm{m\,s^{-2}}$. What is the magnitude of the force exerted on the bowling ball by the sand as it decelerates?

A 44.1 N **B** 3560 N **C** 3600 N **D** 3640 N

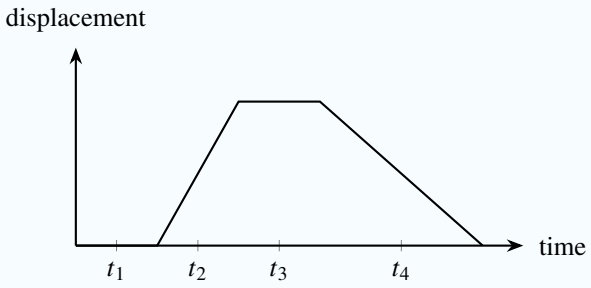
解答 Solution

答案 Answer：D

取向上为正，加速度向上 $a = +800$。$F - mg = ma$，$F = m(g + a) = 4.50 \times (9.81 + 800) = 3640\,\mathrm{N}$。（C 漏了重力。）

练习 3.3 9702/12/F/M/26 Q6

A toy car moves in a straight line. The graph shows the variation with time of the displacement of the car from its initial position.



A student investigates whether the forces acting on the car are in equilibrium at four different times, t_1, t_2, t_3 and t_4. At which times are the forces acting on the car in equilibrium?

A t_1, t_2, t_3 and t_4 **B** t_1 and t_3 only **C** t_1 only **D** t_3 only

解答 Solution

答案 Answer：A

四个时刻图线都是直线段：t_1、t_3 静止，t_2、t_4 匀速（斜率恒定）。加速度都为零，合力为零，都处于平衡。

练习 3.4 9702/14/M/J/26 Q8

A skydiver is falling at terminal velocity. The skydiver opens her parachute. Which quantity increases in magnitude immediately after the skydiver opens her parachute?

A the resultant force acting on the skydiver

B the velocity of the skydiver

C the distance between the skydiver and the ground

D the weight of the skydiver

解答 Solution

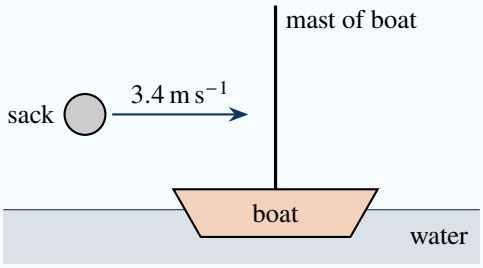
答案 Answer：A

开伞前合力为零；开伞瞬间阻力突然增大，超过重力，合力从零变为向上的较大值，所以合力的大小增大。速度随后减小，离地距离继续减小，重量不变。

练习 3.5 9702/22/F/M/26 Q2(a)(b)

(a) State the principle of conservation of momentum. [2]

(b) A stationary boat of mass 160 kg floats on water. The mast of the boat is vertical. A sack of mass 23 kg is thrown at the boat, as shown in Figure 2.1.



Immediately before colliding with the mast of the boat, the sack is travelling horizontally with a speed of $3.4\,\text{m s}^{-1}$. The sack collides with the mast and remains in contact with it, without rebounding. Assume that resistive forces acting on the boat during the collision are negligible.

(i) Calculate the horizontal speed v of the sack and boat immediately after the collision. [2]

(ii) The total kinetic energy of the sack and boat immediately before the collision is E_1. The total kinetic energy of the sack and boat immediately after the collision is E_2. Determine the ratio E_2/E_1. [3]

(iii) State and explain whether the collision between the sack and boat is elastic or inelastic. [1]

解答 Solution

(a) 系统的总动量保持不变（或：作用前总动量 = 作用后总动量），[M1]

条件：孤立系统 / 无合外力。[A1]

(b)(i) $23 \times 3.4 + 160 \times 0 = (23 + 160)v$ [C1]

$v = 0.43\,\text{m s}^{-1}$ [A1]

(ii) $E_\text{k} = \frac{1}{2}mv^2$ [C1]

$$\frac{E_2}{E_1} = \frac{\frac{1}{2}(183)(0.43)^2}{\frac{1}{2}(23)(3.4)^2} = \frac{16.9}{132.9}$$ [C1]

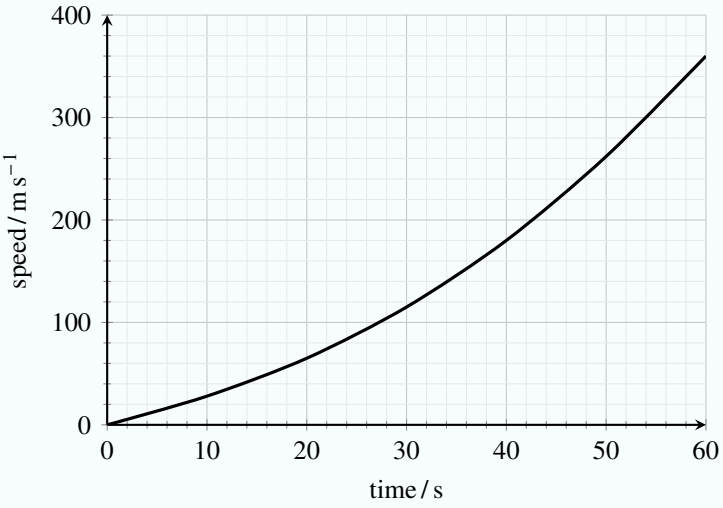
$= 0.13$ [A1]

(iii) 碰后总动能小于碰前总动能（$E_2 < E_1$），所以是非弹性碰撞。[B1]

练习 3.6 9702/21/M/J/26 Q1

(a) Define force. [1]

(b) A rocket is launched vertically upwards from a launch pad. The initial mass of the rocket, with all of its fuel, is $2.83 \times 10^6\,\text{kg}$. For the first 60 s after lift-off, the rocket engine burns fuel at a rate of $2.0 \times 10^4\,\text{kg s}^{-1}$ to produce a constant thrust force. Figure 1.1 shows the variation of the speed of the rocket with time for the first 60 s after lift-off.



Assume that the drag force acting on the rocket is negligible over this time.

(i) By considering the mass of unburnt fuel in the rocket, explain why the speed of the rocket does not increase linearly with time, despite the thrust force from the engine being constant. [2]

(ii) Describe how Figure 1.1 may be used to determine the acceleration of the rocket at lift-off. No calculations are required. [1]

(iii) The acceleration of the rocket at lift-off is 2.5 m s^{-2}. Determine a value for the thrust force T produced by the rocket engine. [2]

(iv) Determine the kinetic energy of the rocket at time 60 s after lift-off. [3]

(v) Use Figure 1.1 to estimate the height of the rocket above the launch pad at time 60 s after lift-off. Explain your reasoning. [2]

(c) The mass of the emitted exhaust gases is the same as the mass of fuel burnt by the engine. Use your answer in (b)(iii) to calculate the speed at which the exhaust gases are emitted from the rocket. [2]

(d) In practice, the drag force on the rocket is not negligible. State the effect of the drag force on the acceleration at lift-off and on the kinetic energy at time 60 s (less / same / greater), compared with the values that assumed negligible drag force. [2]

解答 Solution

（本卷未公布评分标准，以下为参考解答与参考给分。）

(a) 力是动量的变化率 (rate of change of momentum)。 **[B1]**

(b)(i) 燃料不断燃烧，火箭（含剩余燃料）的质量不断减小； **[B1]**

推力恒定、重力减小，合力增大而质量减小，由 $a = F/m$，加速度不断增大，所以速度–时间图的斜率增大，不是直线。 **[B1]**

(ii) 在 $t = 0$ 处作曲线的切线，切线的斜率即为起飞时的加速度。 **[B1]**

(iii) 起飞时 $T - mg = ma$：$T = m(g + a) = 2.83 \times 10^6 \times (9.81 + 2.5)$ **[C1]**

$T = 3.5 \times 10^7$ N **[A1]**

(iv) $t = 60$ s 时质量 $m = 2.83 \times 10^6 - 60 \times 2.0 \times 10^4 = 1.63 \times 10^6$ kg； **[C1]**

由图读出 $v = 360\,\mathrm{m\,s^{-1}}$，$E_\mathrm{k} = \frac{1}{2}(1.63\times10^{6})(360)^2$ **[C1]**

$= 1.1\times10^{11}\,\mathrm{J}$ **[A1]**

(v) 高度 = 速度–时间图下的面积（火箭竖直上升，速率 = 速度大小）。 **[B1]**

用梯形近似（每 10 s 一段）：$10\times(14+46.5+90+147.5+221+311)\approx 8.3\times10^{3}\,\mathrm{m}$，曲线下凸，实际略小，约 $8\times10^{3}\,\mathrm{m}$。 **[A1]**

(c) 推力等于排出气体的动量变化率：$T = \dfrac{\Delta m}{\Delta t}\,v$ **[C1]**

$v = \dfrac{3.48\times10^{7}}{2.0\times10^{4}} = 1.7\times10^{3}\,\mathrm{m\,s^{-1}}$ **[A1]**

(d) 起飞时速度为零，阻力为零，加速度**不变 (same)**； **[B1]**

飞行中阻力做负功、合力减小，60 s 时速度更小，动能**更小 (less)**。 **[B1]**

第四章 力、密度与压强 Forces, density and pressure

考纲要求 Syllabus learning outcomes

4.1.1 understand that the weight of an object may be taken as acting at a single point known as its centre of gravity
理解物体的重量可看作作用在一点上，该点称为重心。

4.1.2 define and apply the moment of a force
定义并应用力矩。

4.1.3 understand that a couple is a pair of forces that acts to produce rotation only
理解力偶是只产生转动效果的一对力。

4.1.4 define and apply the torque of a couple
定义并应用力偶矩。

4.2.1 state and apply the principle of moments
陈述并应用力矩原理。

4.2.2 understand that, for an object to be in equilibrium, there is no resultant force and no resultant torque
理解物体平衡的条件：合力为零且合力矩为零。

4.2.3 use a vector triangle to represent coplanar forces in equilibrium
用矢量三角形表示处于平衡的共面力。

4.3.1 define and use density
定义并使用密度。

4.3.2 define and use pressure
定义并使用压强。

4.3.3 derive, from the definitions of pressure and density, the equation $\Delta p = \rho g \Delta h$
由压强和密度的定义推导 $\Delta p = \rho g \Delta h$。

4.3.4 use the equation $\Delta p = \rho g \Delta h$
使用 $\Delta p = \rho g \Delta h$。

4.3.5 understand that the upthrust acting on an object in a fluid is due to a difference in hydrostatic pressure
理解流体中物体受到的浮力源于上下表面的静压强差。

4.3.6 calculate the upthrust acting on an object in a fluid using the equation $F = \rho g V$ (Archimedes' principle)
用 $F = \rho g V$（阿基米德原理）计算浮力。

4.1 力的转动效果 Turning effects of forces

4.1.1 重心

定义 4.1 (重心 *Centre of gravity*)

The centre of gravity of an object is the point at which the whole weight of the object may be considered to act.

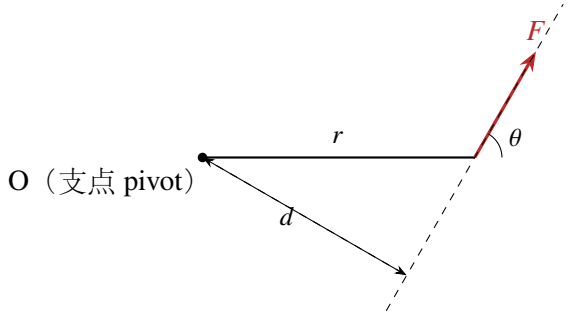
重心是物体全部重量可以看作集中作用的那一点。均匀规则物体的重心在其几何中心；悬挂物体静止时，重心一定在悬挂点的正下方（否则重力对悬挂点有力矩）。

4.1.2 力矩

定义 4.2 (力矩　*Moment of a force*)

The moment of a force (about a point) is the product of the force and the perpendicular distance from the point to the line of action of the force.

力矩 = 力 × 支点到力的**作用线**的**垂直距离**，单位 N m。



力矩 $= F \times d$

$d = r\sin\theta$（r 为 O 到作用点距离）

等价于 $(F\sin\theta) \times r$

易错提醒 Common mistakes　定义中必须出现“perpendicular distance”和“line of action”。常见错误：用支点到作用点的距离 r 直接乘 F（只有当力垂直于杆时才正确）。

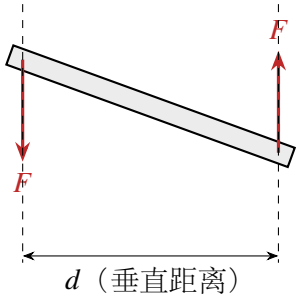
4.1.3 力偶与力偶矩

定义 4.3 (力偶与力偶矩　*Couple and torque of a couple*)

A **couple** is a pair of forces that are equal in magnitude, opposite in direction and act along parallel (but not the same) lines of action, so as to produce rotation only.

The **torque of a couple** is the product of one of the forces and the perpendicular distance between the lines of action of the forces.

力偶是一对大小相等、方向相反、作用线平行但不重合的力，合力为零，只产生转动。**力偶矩** = **其中一个力** × 两力作用线之间的垂直距离：$\tau = Fd$。力偶矩与转轴位置无关。

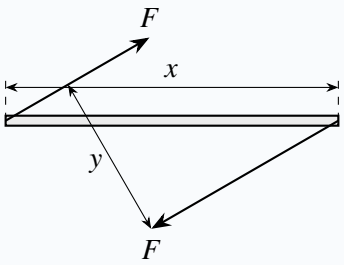


力偶矩 $\tau = F\,d$

合力 = 0（不平动）

例题 4.1 9702/11/M/J/26 Q13

Two forces, each of magnitude F, act on a beam of length x as shown.



The perpendicular distance between the forces is y and the two forces form a couple. What is the torque of the couple?

A Fx **B** $2Fx$ **C** Fy **D** $2Fy$

解答 Solution

答案 Answer：C

力偶矩 = 一个力 × 两力作用线间的垂直距离 = Fy。x 是梁长，不是垂直距离；也不要乘 2。

4.2 平衡 Equilibrium

定律 4.1 (力矩原理 *Principle of moments*)

For an object in equilibrium, the sum of the clockwise moments about any point is equal to the sum of the anticlockwise moments about the same point.

物体处于平衡时，对**任意一点**，顺时针力矩之和等于逆时针力矩之和。

公式 4.1 (平衡条件 *Conditions for equilibrium*)

(1) 合力为零 (resultant force is zero)：任意方向上的分力之和为零；

(2) 合力矩为零 (resultant torque/moment about any point is zero)。

解题技巧：

- 选取有未知力（尤其是支点反力）作用的点为力矩中心，可以消去该未知力。
- 均匀杆的重量作用在中点。
- 只有合力为零而合力矩不为零（如力偶）时物体会转动，不平衡；只有合力矩为零而合力不为零时物体会平动，也不平衡。

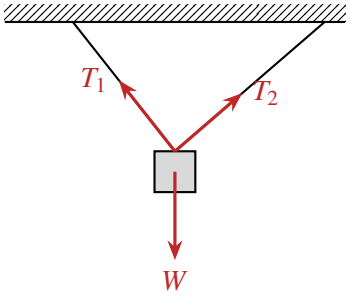
4.2.1 三力平衡与矢量三角形

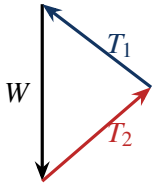
若物体在三个共面力作用下平衡，则：

- 三个力首尾相接组成一个**闭合三角形**（triangle of forces）；

- 三个力的作用线（若不平行）交于一点。

可用正弦定理、作图或分解法求解。



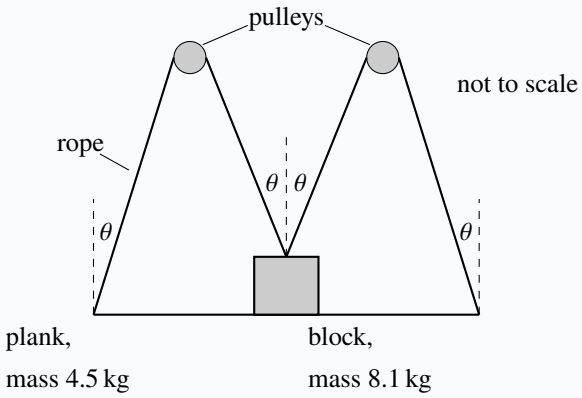


首尾相接成闭合三角形
⇒ 合力为零

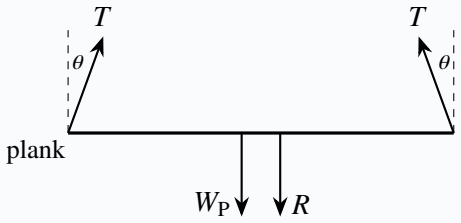
例题 4.2 9702/22/M/J/26 Q2

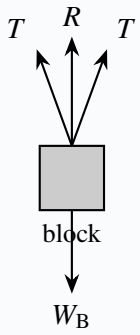
(a) State the conditions required for an object to be in equilibrium. [2]

(b) A block rests in equilibrium on the centre of a uniform horizontal plank. The block is attached to two ropes, each of which passes over a fixed pulley and then attaches to an end of the plank as shown in Figure 2.1.



Where the ropes attach to the block and to the plank, the angle between the ropes and the vertical is θ. There is negligible friction between the pulleys and the ropes. The mass of the block is 8.1 kg. The mass of the plank is 4.5 kg. Figure 2.2 shows the directions of the forces acting on the plank and the block. The weight of the plank is W_P, the weight of the block is W_B, the tension in each rope is T and the forces that the plank and the block exert on each other each have magnitude R.





(i) Use Figure 2.2 to show that $R + W_P = W_B - R$. [2]

(ii) Use the equation in (b)(i) to calculate the magnitude of R. [2]

解答 Solution

(a) 合力（任意方向）为零； [B1]

对任意点的合力矩为零。 [B1]

(b)(i) 木板竖直方向：$2T\cos\theta = R + W_P$ [M1]

木块竖直方向：$2T\cos\theta + R = W_B$，即 $2T\cos\theta = W_B - R$。两式联立得 $R + W_P = W_B - R$。 [A1]

（注意：T 与竖直方向夹角为 θ，竖直分量用 $\cos\theta$，误用 $\sin\theta$ 只得 1 分。）

(ii) $2R = W_B - W_P = (8.1 - 4.5) \times 9.81$ [C1]

$R = 18\,\mathrm{N}$ [A1]

4.3 密度与压强 Density and pressure

定义 4.4 (密度 *Density*)

Density is mass per unit volume: $\rho = \dfrac{m}{V}$.

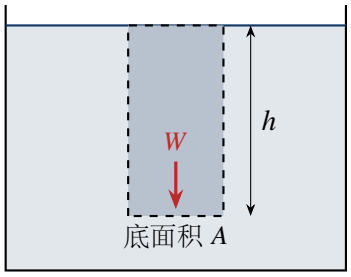
密度 = 质量 / 体积，单位 $\mathrm{kg\,m^{-3}}$。水的密度 $1000\,\mathrm{kg\,m^{-3}} = 1.00\,\mathrm{g\,cm^{-3}}$。

定义 4.5 (压强 *Pressure*)

Pressure is the (normal) force acting per unit area: $p = \dfrac{F}{A}$.

压强 = 垂直作用在单位面积上的力，单位 pascal，$1\,\mathrm{Pa} = 1\,\mathrm{N\,m^{-2}}$。压强是**标量**。

4.3.1 液体压强公式的推导



液柱体积 $V = Ah$

液柱质量 $m = \rho V = \rho Ah$

液柱重量 $W = mg = \rho Ahg$

压强 $p = \dfrac{W}{A} = \rho gh$

想象液面下深度 h 处有一个水平面积 A，其上方液柱的重量全部压在该面积上：

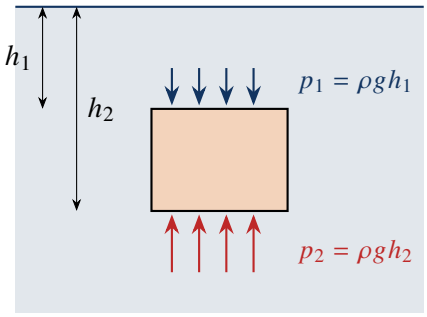
$$\Delta p = \frac{F}{A} = \frac{mg}{A} = \frac{\rho Vg}{A} = \frac{\rho(Ah)g}{A} = \rho gh.$$

公式 4.2 (液体压强 *Hydrostatic pressure*)

$$\Delta p = \rho g \Delta h$$

液体中两点的压强差只取决于液体密度和竖直高度差，与容器形状、截面积无关。液面下深度 h 处的总压强 $p = p_0 + \rho gh$（p_0 为液面上的大气压）。

4.3.2 浮力与阿基米德原理



侧面压力相互抵消
上表面向下 $F_1 = p_1 A$
下表面向上 $F_2 = p_2 A$
浮力 $U = F_2 - F_1 = \rho g(h_2 - h_1)A$
$= \rho g V$

浸在流体中的物体，下表面比上表面更深，受到的压强更大，因此向上的压力大于向下的压力，二者之差就是**浮力** (*upthrust*)。

定律 4.2 (阿基米德原理 *Archimedes' principle*)

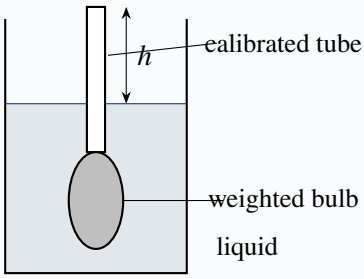
$$F = \rho g V$$

浮力等于物体排开的流体的重量。式中 ρ 是流体的密度，V 是物体浸入流体部分的体积（排开流体的体积）。

- 浮力与物体自身的密度、重量、浸没深度（完全浸没后）都无关。
- **漂浮**：浮力 = 重量，$\rho_{液} g V_{浸} = \rho_{物} g V_{物}$，所以浸入比例 $\dfrac{V_{浸}}{V_{物}} = \dfrac{\rho_{物}}{\rho_{液}}$。液体密度越大，物体浮得越高。

例题 4.3 9702/12/M/J/26 Q13

A hydrometer is a device for measuring the density of liquids. It consists of a weighted bulb at the bottom of a calibrated tube.



The hydrometer is floating in a liquid at room temperature. The length of the tube above the level of the liquid is h. Which change causes h to decrease?

A adding more liquid into the cylinder

B decreasing the density of the liquid

C increasing the air pressure above the liquid

D increasing the density of the liquid

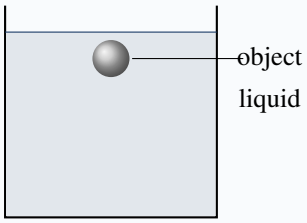
解答 Solution

答案 Answer：**B**

漂浮时浮力 = 重量（不变），$\rho g V_{浸} = W$。液体密度减小，需要排开更多体积，所以浸得更深，露出部分 h 减小。加液体不改变浸入体积；大气压同时作用于上下，不影响。

例题 4.4 9702/24/M/J/26 Q4

An object is held at rest just below the surface of a liquid of uniform density as shown in Figure 4.1.



The object is released and accelerates downwards until it reaches terminal velocity. The magnitude of the weight of the object is W. The magnitude of the upthrust acting on the object is U. The magnitude of the drag force acting on the object is D.

(a) By placing one tick in each row, complete Table 4.1 to show what happens to the magnitude of the three forces acting on the object as it accelerates.

force	increases	stays the same	decreases
W			
D			
U			

[3]

(b) State an expression, in terms of W, D and U, that shows the relationship between the forces acting on the object when the object is at its terminal velocity. [1]

(c) The object has a mass of 0.070 kg and a volume of $1.0 \times 10^{-5}\,\text{m}^3$. The density of the liquid is $1100\,\text{kg m}^{-3}$. Calculate D when the object is at terminal velocity. [3]

解答 Solution

(a) W： stays the same **[B1]**

D： increases（速度增大，阻力增大） **[B1]**

U： stays the same（完全浸没后排开体积不变，$U = \rho g V$ 不变） **[B1]**

(b) $W = D + U$ **[A1]**

(c) $W = 0.070 \times 9.81 = 0.69\,\text{N}$ **[C1]**

$U = \rho g V = 1100 \times 9.81 \times 1.0 \times 10^{-5} = 0.11\,\text{N}$ **[C1]**

$D = W - U = 0.69 - 0.11 = 0.58\,\text{N}$ **[A1]**

教学提示 Teacher's note 建议 4 课时。力矩题注意"垂直距离"的几何关系；三力平衡多用矢量三角形或正交分解。压强部分要求会**推导** $\Delta p = \rho g \Delta h$（Paper 2 曾考 3 分推导题），推导中每一步都要写出（$m = \rho V$，$V = Ah$，$W = mg$，$p = W/A$）。浮力的本质是上下表面的压强差，这一句解释也常考。

4.4 本章小结 Summary

物理量/关系	公式	说明
力矩	$M = Fd$	d 为到作用线的垂直距离
力偶矩	$\tau = Fd$	一个力 × 两作用线间距
平衡条件	$\sum F = 0$，$\sum M = 0$	力矩对任意点
密度	$\rho = m/V$	$\mathrm{kg\,m^{-3}}$
压强	$p = F/A$	$\mathrm{Pa = N\,m^{-2}}$
液体压强	$\Delta p = \rho g \Delta h$	只与竖直深度有关
浮力	$F = \rho g V$	ρ 为流体密度

4.5 本章练习 Practice questions

练习 4.1 9702/12/M/J/26 Q14

Torque is the result of a pair of forces acting to cause a turning effect. Torque is calculated as the product of force and distance. Which row gives the correct description of the force and the distance used to calculate the torque of a couple?

	force	distance
A	the magnitude of one force	the perpendicular distance between the two forces
B	the sum of the magnitudes of both forces	the perpendicular distance between the two forces
C	the magnitude of one force	the perpendicular distance from one force to the mid-point between the forces
D	the resultant of the two forces	the perpendicular distance from one force to the mid-point between the forces

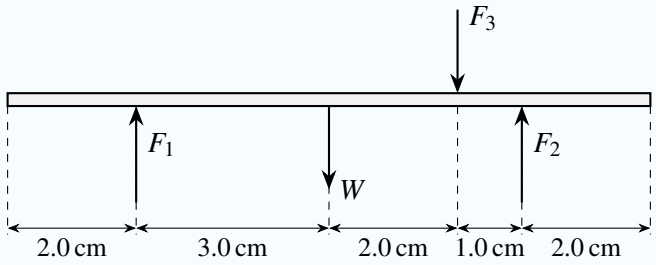
解答 Solution

答案 Answer：A

力偶矩 = 一个力的大小 × 两力作用线之间的垂直距离。

练习 4.2 9702/11/M/J/26 Q15

A horizontal uniform rod of length 10.0 cm is shown.



The rod is acted on by its weight W and by three forces F_1, F_2 and F_3. The rod is in equilibrium. All the forces are vertical. $F_2 = \frac{4}{3}F_3$. What is the ratio F_1/F_2?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 1 **D** 2

解答 Solution

答案 Answer：**B**

对重心（W 的作用点，$x = 5$ cm）取力矩，消去 W：

逆时针：$F_2 \times 3.0$；顺时针：$F_1 \times 3.0 + F_3 \times 2.0$。

$3F_2 = 3F_1 + 2F_3 = 3F_1 + 2 \times \frac{3}{4}F_2 = 3F_1 + 1.5F_2$，得 $F_1 = 0.5F_2$，比值 $= \frac{1}{2}$。

练习 4.3 9702/12/M/J/26 Q15

The density of paper is 800 kg m^{-3}. A typical sheet of paper has a width of 210 mm and a length of 300 mm. The thickness of a pack of 500 sheets of paper is 50 mm. What is the mass of a single sheet of paper?

A 0.5 g **B** 5 g **C** 50 g **D** 500 g

解答 Solution

答案 Answer：**B**

每张厚度 $= 50/500 = 0.10$ mm。体积 $V = 0.210 \times 0.300 \times 1.0 \times 10^{-4} = 6.3 \times 10^{-6}$ m^3。$m = \rho V = 800 \times 6.3 \times 10^{-6} = 5.0 \times 10^{-3}$ kg $= 5$ g。

练习 4.4 9702/12/M/J/26 Q16

Which statement about equilibrium is correct?

A If a system is in equilibrium then there must be no forces acting on it.

B If a system is in equilibrium then there must be no resultant torque on it.

C If there are no moments due to forces acting on a system then it must be in equilibrium.

D If there is no resultant force acting on a system then it must be in equilibrium.

解答 Solution

答案 Answer：**B**

平衡要求合力和合力矩**都**为零，所以“平衡 ⇒ 合力矩为零”正确。A：可以受力，只要合力为零；C、D 只满足一个条件，不一定平衡。

练习 4.5 9702/11/M/J/26 Q14

An object is submerged in a liquid. Which statement about the upthrust acting on the object is correct?

A It is proportional to the density of the liquid.

B It is proportional to the density of the object.

C It is proportional to the total depth of the liquid.

D It is proportional to the weight of the object.

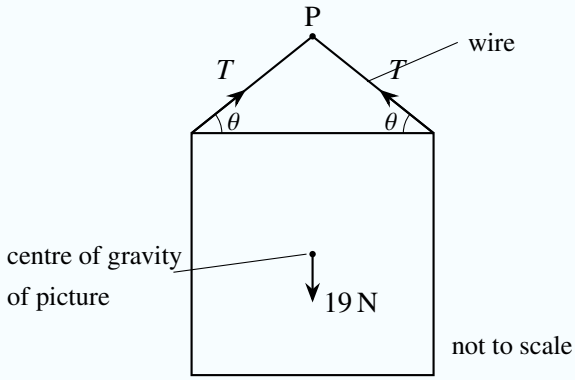
解答 Solution

答案 Answer：**A**

$F = \rho g V$，ρ 是液体密度，所以浮力与液体密度成正比，与物体密度、重量、液体深度无关。

练习 4.6 9702/22/F/M/26 Q4(a)

A uniform square picture is attached to a wire and hangs in equilibrium from a pin at point P, as shown in Figure 4.1.



The picture has weight 19 N and its centre of gravity is directly below P. The top side of the picture is horizontal. The wire forms an angle θ with the horizontal. The tension T in the wire is 13 N.

(i) Define the moment of a force about a point. [1]

(ii) Explain why the moment of the weight about P is zero. [1]

(iii) Calculate θ. [2]

解答 Solution

(i) 力与该点到力的作用线的垂直距离的乘积。 **[B1]**

(ii) 重力的作用线通过 P 点（P 到重力作用线的垂直距离为零），所以力矩为零。 **[B1]**

(iii) 竖直方向平衡：$2T \sin\theta = W$ **[C1]**

$$\sin\theta = \frac{19}{2\times 13} = 0.731,\quad \theta = 47^\circ$$ **[A1]**

第五章 功、能量与功率 Work, energy and power

考纲要求 Syllabus learning outcomes

5.1.1 understand the concept of work, and recall and use work done = force × displacement in the direction of the force
理解功的概念，记住并使用功 = 力 × 沿力方向的位移。

5.1.2 recall and apply the principle of conservation of energy
记住并应用能量守恒定律。

5.1.3 recall and understand that the efficiency of a system is the ratio of useful energy output from the system to the total energy input
记住并理解：系统效率是有用输出能量与总输入能量之比。

5.1.4 use the concept of efficiency to solve problems
用效率的概念解题。

5.1.5 define power as work done per unit time
把功率定义为单位时间内做的功。

5.1.6 solve problems using $P = W/t$
用 $P = W/t$ 解题。

5.1.7 derive $P = Fv$ and use it to solve problems
推导 $P = Fv$ 并用其解题。

5.2.1 derive, using $W = Fs$, the formula $\Delta E_\mathrm{P} = mg\Delta h$ for gravitational potential energy changes in a uniform gravitational field
用 $W = Fs$ 推导均匀重力场中重力势能变化 $\Delta E_\mathrm{P} = mg\Delta h$。

5.2.2 recall and use the formula $\Delta E_\mathrm{P} = mg\Delta h$ for gravitational potential energy changes in a uniform gravitational field
记住并使用 $\Delta E_\mathrm{P} = mg\Delta h$。

5.2.3 derive, using the equations of motion, the formula for kinetic energy $E_\mathrm{K} = \frac{1}{2}mv^2$
用运动学方程推导动能公式 $E_\mathrm{K} = \frac{1}{2}mv^2$。

5.2.4 recall and use $E_\mathrm{K} = \frac{1}{2}mv^2$
记住并使用 $E_\mathrm{K} = \frac{1}{2}mv^2$。

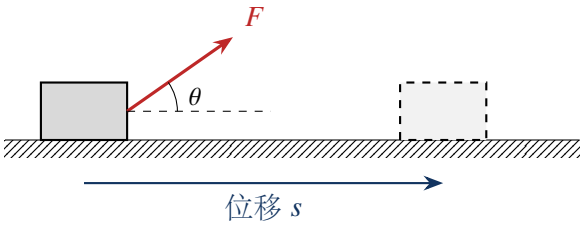
5.1 功 Work

定义 5.1 (功 *Work done*)

Work done is the product of the force and the displacement in the direction of the force: $W = Fs$.

功 = 力 × 沿力方向的位移。功是**标量**，单位 joule，1 J = 1 N m：1 N 的力使物体沿力方向移动 1 m 所做的功。做功的过程就是能量转移（转化）的过程：**做了多少功，就转移了多少能量**。

♣



$W = Fs\cos\theta$

沿位移方向的力分量 $F\cos\theta$

- 力与位移成角 θ 时，$W = Fs\cos\theta$。
- $\theta = 90°$ 时力不做功：如匀速圆周运动的向心力、水平移动物体时的重力和支持力。
- 力与位移方向相反（如摩擦力、阻力）时做负功，物体的能量减少（称为“克服阻力做功”work done against resistive forces）。
- F–s 图下的面积等于所做的功（适用于变力，如弹簧）。

5.2 重力势能与动能

5.2.1 重力势能 Gravitational potential energy

推导：在均匀重力场中把质量为 m 的物体**匀速**提高 Δh，所需外力等于重量 mg，外力做功

$$W = Fs = (mg)(\Delta h)$$

这些功全部储存为重力势能，所以

$$\boxed{\Delta E_{\mathrm{P}} = mg\Delta h}$$

只适用于 g 近似不变的地表附近（大距离的情形见第 13 章 $E_{\mathrm{P}} = -GMm/r$）。重力势能的变化只取决于**竖直**高度变化，与路径无关；水平移动物体时 E_{P} 不变。

5.2.2 动能 Kinetic energy

推导：质量为 m 的物体从静止开始，在恒定合力 F 作用下通过位移 s，速度达到 v。

$$v^2 = u^2 + 2as = 0 + 2as \implies as = \tfrac{1}{2}v^2$$

合力做功 $W = Fs = (ma)s = m(as) = \tfrac{1}{2}mv^2$。这些功全部转化为物体的动能，所以

$$\boxed{E_{\mathrm{K}} = \tfrac{1}{2}mv^2}$$

一般地，若初速度为 u：$Fs = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2$，即 $2Fs = mv^2 - mu^2$（合力做的功等于动能的变化——**动能定理**）。

易错提醒 Common mistakes

- $E_{\mathrm{K}} \propto v^2$：速度从 10 增加到 20 与从 70 增加到 80 时，动能增量**不同**，$\Delta E_{\mathrm{K}} \propto (v_2^2 - v_1^2)$。
- 动能与动量的关系：$E_{\mathrm{K}} = \dfrac{p^2}{2m}$。

5.3 能量守恒 Conservation of energy

定律 5.1 (能量守恒定律 *Principle of conservation of energy*)

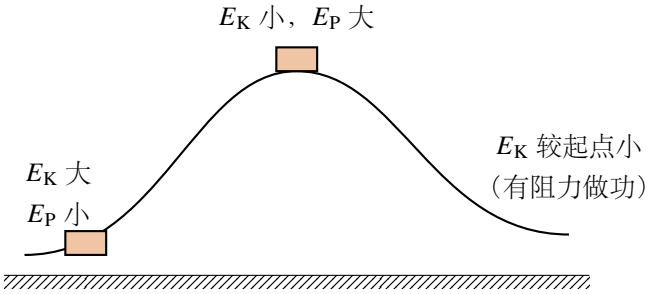
Energy cannot be created or destroyed; it can only be transferred from one form to another. The total energy of a closed system is constant.

能量既不能创生也不能消灭，只能从一种形式转化为另一种形式（或从一个物体转移到另一个物体），封闭系统的总能量保持不变。

有阻力时的能量方程（最常用）：

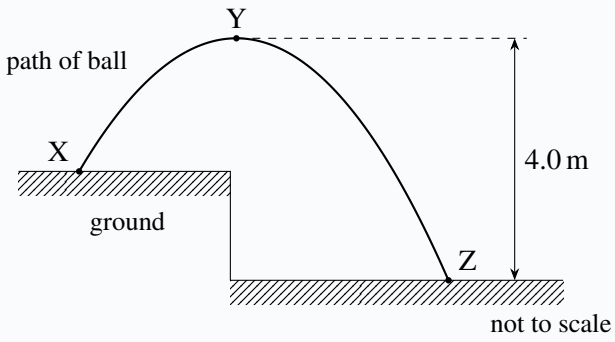
初始总能量 = 末总能量 + 克服阻力做的功

例如物体沿粗糙轨道运动：$E_{K1} + E_{P1} = E_{K2} + E_{P2} + F_{阻}s$。



例题 5.1 9702/13/M/J/26 Q16

A ball is projected from the ground at point X. The ball then reaches its maximum height at point Y before landing on lower ground at point Z. The diagram shows the path of the ball.



The ball leaves the ground at X with a kinetic energy of 5.2 J. The ball has a kinetic energy of 9.6 J when it hits the ground at Z. The weight of the ball is 2.0 N. The vertical height of Y above Z is 4.0 m. Air resistance is negligible. What is the vertical height of Y above X?

A 1.8 m **B** 2.0 m **C** 2.2 m **D** 2.6 m

解答 Solution

答案 Answer：A

X 到 Z：动能增加 9.6 − 5.2 = 4.4 J = 重力势能减少 = $W\Delta h$，所以 X 比 Z 高 4.4/2.0 = 2.2 m。Y 比

X 高 $4.0 - 2.2 = 1.8\,\text{m}$。

5.4 效率与功率 Efficiency and power

定义 5.2 (效率 *Efficiency*)

Efficiency is the ratio of the useful energy (or power) output from a system to the total energy (or power) input.

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100\% = \frac{\text{useful power output}}{\text{total power input}} \times 100\%$$

效率是有用输出能量与总输入能量之比，总小于 100%（总有一部分能量以热等形式浪费）。浪费的能量 = 总输入 − 有用输出。

定义 5.3 (功率 *Power*)

Power is the work done per unit time (rate of doing work): $P = \dfrac{W}{t}$.

功率是单位时间内所做的功（或能量转移的速率），单位 watt，$1\,\text{W} = 1\,\text{J}\,\text{s}^{-1}$。注意定义中写“work done per unit time”，不要写成“work done in unit time”。

5.4.1 $P = Fv$ 的推导

恒力 F 使物体以速度 v 沿力方向运动，在 Δt 时间内移动 Δs：

$$P = \frac{W}{\Delta t} = \frac{F\Delta s}{\Delta t} = F\frac{\Delta s}{\Delta t} = Fv.$$

公式 5.1 (功率与速度 *Power and velocity*)

$$P = Fv$$

F 是驱动力（牵引力），v 是沿力方向的速度。车辆匀速行驶时驱动力等于总阻力，所以 $P = F_{\text{阻}}v$；车速最大时 $v_{\max} = P/F_{\text{阻}}$。

例题 5.2 9702/11/M/J/26 Q19

Which equation is not required to derive the formula $P = Fv$?

A $W = Fs$　　**B** $v = \dfrac{s}{t}$　　**C** $F = ma$　　**D** $P = \dfrac{W}{t}$

解答 Solution

答案 Answer：C

推导只用到 $P = W/t$、$W = Fs$、$v = s/t$，不需要 $F = ma$。

例题 5.3 9702/12/M/J/26 Q17

An object of weight 3.0 N is moved vertically at a constant velocity v by an electric motor. The electric motor has an efficiency of 25%. The object travels a vertical distance of 1.2 m in 3.0 s. Assume that the useful output energy of the motor is equal to the increase in gravitational potential energy of the object. What is the electric power supplied to the motor?

A 0.3 W **B** 1.2 W **C** 4.8 W **D** 30 W

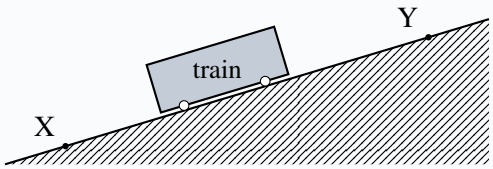
解答 Solution

答案 Answer：C

有用输出功率 $= \dfrac{mg\Delta h}{t} = \dfrac{3.0 \times 1.2}{3.0} = 1.2\,\text{W}$；输入功率 $= \dfrac{1.2}{0.25} = 4.8\,\text{W}$。

例题 5.4 9702/22/M/J/26 Q4

An electric train of mass 1.5×10^4 kg travels up a slope at a constant speed as shown in Figure 4.1.



At point X, the train has gravitational potential energy 7.4 MJ. At point Y, the train has gravitational potential energy 13.0 MJ. The train takes a time of 52 s to move from X to Y. The kinetic energy of the train is 1.3×10^5 J.

(a) Calculate the speed of the train. [3]

(b) The train is powered by an electric motor. The useful output power of the motor is 1.1×10^5 W.

(i) Define power. [1]

(ii) Determine the resistive force F acting on the train as it moves from point X to point Y. [3]

(iii) The current in the motor is 95 A. The potential difference (p.d.) across the motor is 1400 V. Calculate the percentage efficiency of the motor. [3]

解答 Solution

(a) $E_\text{K} = \frac{1}{2}mv^2$ **[C1]**

$v^2 = \dfrac{2 \times 1.3 \times 10^5}{1.5 \times 10^4}$ **[C1]**

$v = 4.2\,\text{m s}^{-1}$ **[A1]**

(b)(i) 单位时间内做的功 (work done per unit time)。 **[B1]**

(ii) 电机有用输出功 = 重力势能增加 + 克服阻力做功：$W = Pt$，$W_\text{阻} = Fs$ **[C1]**

$W_\text{阻} = 1.1 \times 10^5 \times 52 - (13.0 - 7.4) \times 10^6 = 1.2 \times 10^5\,\text{J}$ **[C1]**

$s = vt = 4.2 \times 52$，$F = \dfrac{1.2 \times 10^5}{4.2 \times 52} = 550\,\text{N}$ **[A1]**

(iii) 输入功率 $P = IV = 95 \times 1400 = 1.33 \times 10^5\,\text{W}$ **[C1]**

$$\text{效率} = \frac{\text{有用输出功率}}{\text{总输入功率}} \times 100\% = \frac{1.1 \times 10^5}{1.33 \times 10^5} \times 100\% = 83\% \quad \text{[C1] [A1]}$$

教学提示 Teacher's note 建议 3–4 课时。两个推导（$E_P = mg\Delta h$，$E_K = \frac{1}{2}mv^2$）和 $P = Fv$ 的推导都可能以"show that/derive"形式出现。综合题常把能量守恒、功率、效率与电学 $P = IV$ 联系在一起（如上题电动列车）。提醒学生：MJ、kJ 的单位换算是丢分重灾区；计算阻力时"有用功 = ΔE_P + 克服阻力做功"要写清楚。

5.5 本章小结 Summary

物理量/关系	公式	说明
功	$W = Fs\cos\theta$	标量，J
重力势能变化	$\Delta E_P = mg\Delta h$	均匀重力场
动能	$E_K = \frac{1}{2}mv^2$	合力做功 = ΔE_K
效率	$\frac{\text{有用输出}}{\text{总输入}} \times 100\%$	能量或功率
功率	$P = W/t = Fv$	$\mathrm{W = J\,s^{-1}}$
能量守恒	初能 = 末能 + 克服阻力做功	

5.6 本章练习 Practice questions

练习 5.1 9702/11/M/J/26 Q8

A parent and a child run from rest at the bottom of a stairway to the top of the stairway. Both take the same time. The mass of the parent is greater than the mass of the child. Which physical quantity for the parent and for the child has the same numerical value as a result of climbing the stairway?

A average kinetic energy

B average velocity

C gravitational potential energy gained

D power generated

解答 Solution

答案 Answer：**B**

位移相同、时间相同，所以平均速度相同。动能、势能、功率都与质量有关，质量不同则不同。

练习 5.2 9702/12/M/J/26 Q18

An alpha particle has 2.2×10^{-13} J of kinetic energy. What is the speed of the alpha particle?

A $4.1 \times 10^6\,\mathrm{m\,s^{-1}}$ **B** $5.8 \times 10^6\,\mathrm{m\,s^{-1}}$

C $8.1 \times 10^6\,\mathrm{m\,s^{-1}}$ **D** $1.2 \times 10^7\,\mathrm{m\,s^{-1}}$

解答 Solution

答案 Answer：C

α 粒子质量 $\approx 4\,\mathrm{u} = 4 \times 1.66 \times 10^{-27} = 6.64 \times 10^{-27}\,\mathrm{kg}$。$v = \sqrt{\dfrac{2E_\mathrm{K}}{m}} = \sqrt{\dfrac{2 \times 2.2 \times 10^{-13}}{6.64 \times 10^{-27}}} = 8.1 \times 10^6\,\mathrm{m\,s^{-1}}$。

练习 5.3 9702/13/M/J/26 Q19

A small diesel engine uses a volume of $1.5 \times 10^4\,\mathrm{cm^3}$ of fuel per hour to produce a useful power output of 40 kW. Assume that 34 kJ of energy is transferred to the engine when it uses $1.0\,\mathrm{cm^3}$ of fuel. What is the rate of transfer from the engine of energy that is wasted?

A 102 kW **B** 142 kW **C** 182 kW **D** 470 kW

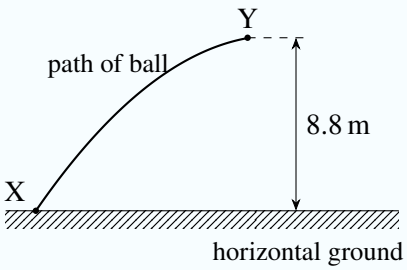
解答 Solution

答案 Answer：A

输入功率 $= \dfrac{1.5 \times 10^4 \times 34 \times 10^3}{3600} = 1.42 \times 10^5\,\mathrm{W} = 142\,\mathrm{kW}$。浪费功率 $= 142 - 40 = 102\,\mathrm{kW}$。

练习 5.4 9702/12/F/M/26 Q17

A small ball is projected into the air from horizontal ground. The diagram shows part of the path of the ball.



The ball leaves the ground at point X with a kinetic energy of 8.0 J. A short time later, the ball is at point Y where it has a kinetic energy of 3.5 J. Point Y is at a vertical height of 8.8 m above the ground. Air resistance is negligible. What is the mass of the ball?

A 0.041 kg **B** 0.052 kg **C** 0.093 kg **D** 0.13 kg

解答 Solution

答案 Answer：B

$\Delta E_\mathrm{P} = 8.0 - 3.5 = 4.5\,\mathrm{J} = mg\Delta h$，$m = \dfrac{4.5}{9.81 \times 8.8} = 0.052\,\mathrm{kg}$。

练习 5.5 9702/12/F/M/26 Q19

When the speed of an object is increased from 10 m s^{-1} to 20 m s^{-1}, its kinetic energy increases by E. What is the increase in the kinetic energy of the object when its speed increases from 70 m s^{-1} to 80 m s^{-1}?

A E **B** $4E$ **C** $5E$ **D** $7E$

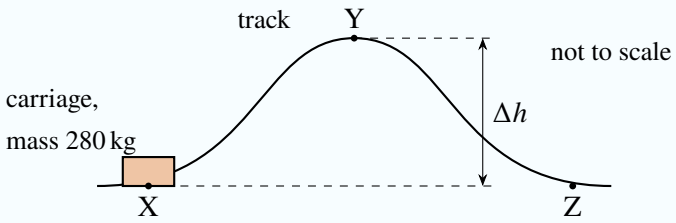
解答 Solution

答案 Answer: C

$\Delta E_K \propto v_2^2 - v_1^2$。$\dfrac{80^2 - 70^2}{20^2 - 10^2} = \dfrac{1500}{300} = 5$。

练习 5.6 9702/22/F/M/26 Q3

A rollercoaster carriage of mass 280 kg travels over a raised section of track, as illustrated in Figure 3.1.



At position X, the carriage has kinetic energy 16.1 kJ. At position Y, the carriage has kinetic energy 1.3 kJ. The length of track between X and Y is 23 m. In moving from X to Y the carriage does 5.1 kJ of work against resistive forces.

(a) (i) Calculate the average resistive force acting on the carriage between X and Y. [2]

(ii) Calculate the change in height Δh between X and Y. [3]

(b) The carriage travels further along the track to point Z. State and explain whether the speed of the carriage at Z is greater than, less than or the same as the speed of the carriage at X. [1]

解答 Solution

(a)(i) $F = \dfrac{W}{d} = \dfrac{5.1 \times 10^3}{23}$ **[C1]**

$= 220$ N **[A1]**

(ii) $\Delta E_P = 16.1 \times 10^3 - 1.3 \times 10^3 - 5.1 \times 10^3 = 9.7 \times 10^3$ J **[C1]**

$\Delta E_P = mg\Delta h$ **[C1]**

$\Delta h = \dfrac{9.7 \times 10^3}{280 \times 9.81} = 3.5$ m **[A1]**

(b) X、Z 高度相同，但从 X 到 Z 一直在克服阻力做功，Z 处动能小于 X 处，所以 Z 处速度**较小**。 **[B1]**

第六章 固体的形变 Deformation of solids

考纲要求 Syllabus learning outcomes

6.1.1 understand that deformation is caused by tensile or compressive forces (forces and deformations will be assumed to be in one dimension only)
理解形变由拉力或压力引起（只考虑一维情况）。

6.1.2 understand and use the terms load, extension, compression and limit of proportionality
理解并使用负载、伸长量、压缩量和比例极限等术语。

6.1.3 recall and use Hooke's law
记住并使用胡克定律。

6.1.4 recall and use the formula for the spring constant $k = F/x$
记住并使用弹簧劲度系数公式 $k = F/x$。

6.1.5 define and use the terms stress, strain and the Young modulus
定义并使用应力、应变和杨氏模量。

6.1.6 describe an experiment to determine the Young modulus of a metal in the form of a wire
描述测定金属丝杨氏模量的实验。

6.2.1 understand and use the terms elastic deformation, plastic deformation and elastic limit
理解并使用弹性形变、塑性形变和弹性极限。

6.2.2 understand that the area under the force–extension graph represents the work done
理解力–伸长量图下的面积表示所做的功。

6.2.3 determine the elastic potential energy of a material deformed within its limit of proportionality from the area under the force–extension graph
由力–伸长量图下的面积求比例极限内的弹性势能。

6.2.4 recall and use $E_P = \frac{1}{2}Fx = \frac{1}{2}kx^2$ for a material deformed within its limit of proportionality
记住并使用 $E_P = \frac{1}{2}Fx = \frac{1}{2}kx^2$（比例极限内）。

6.1 拉伸、压缩与胡克定律

6.1.1 基本术语

- **拉力** (*tensile force*)：使物体伸长的一对方向相反、向外的力；**压力** (*compressive force*)：使物体缩短的一对方向相反、向内的力。
- **负载** (*load*)：施加在物体上使其形变的力。
- **伸长量** (*extension*) x：拉伸后长度 − 原长；**压缩量** (*compression*)：原长 − 压缩后长度。（注意不是“拉伸后的长度”。）
- **比例极限** (*limit of proportionality*)：伸长量与负载成正比的最大负载点，超过此点 F–x 图不再是直线。

定律 6.1 (胡克定律 *Hooke's law*)

The extension of a spring (or wire) is proportional to the load (applied force), provided the limit of proportionality is not exceeded: $F = kx$.

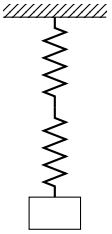
在比例极限内，弹簧（或金属丝）的伸长量与所受拉力成正比。

定义 6.1 (劲度系数 *Spring constant*)

The spring constant is the force per unit extension: $k = \dfrac{F}{x}$.

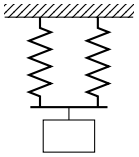
劲度系数 = 单位伸长量所需的力，单位 $\mathrm{N\,m^{-1}}$，等于 F–x 图直线部分的**斜率**。k 越大弹簧越"硬"；相同负载下 k 越小伸长越大。

6.1.2 弹簧的串联与并联



串联 series

$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$$



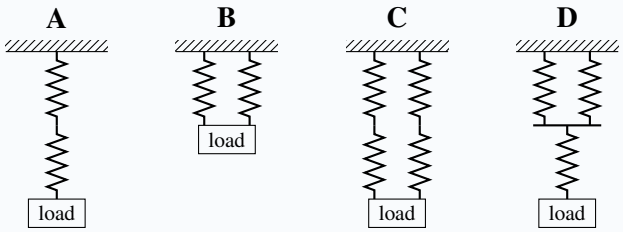
并联 parallel

$$k = k_1 + k_2$$

两根相同弹簧（各 k）：
串联 $\Rightarrow k/2$（每根受全部负载，伸长加倍）
并联 $\Rightarrow 2k$（每根只受一半负载）

例题 6.1 9702/11/M/J/26 Q22

Identical springs are joined in four arrangements. Which arrangement has the same spring constant as a single spring?



解答 Solution

答案 Answer：C

A：串联 $k/2$；B：并联 $2k$；C：每支串联为 $k/2$，两支并联 $= k/2 + k/2 = k$；D：$2k$ 与 k 串联，$\dfrac{1}{k_{总}} = \dfrac{1}{2k} + \dfrac{1}{k}$，$k_{总} = \dfrac{2}{3}k$。

6.2 应力、应变与杨氏模量

同种材料、不同粗细和长度的金属丝，劲度系数 k 不同。为描述**材料本身**的性质，引入应力和应变。

定义 6.2 (应力 *Stress*)

Stress is the force acting per unit cross-sectional area: $\sigma = \dfrac{F}{A}$.

应力 = 单位横截面积上的力，单位 Pa（$\mathrm{N\,m^{-2}}$）。圆截面 $A = \dfrac{\pi d^2}{4}$（d 为直径，注意不要误用直径当半径）。

定义 6.3 (应变 *Strain*)

Strain is the extension per unit original length: $\varepsilon = \dfrac{x}{L}$.

应变 = 伸长量 / 原长，是比值，**没有单位**。

定义 6.4 (杨氏模量 *Young modulus*)

The Young modulus is the ratio of stress to strain (stress per unit strain): $E = \dfrac{\sigma}{\varepsilon}$.

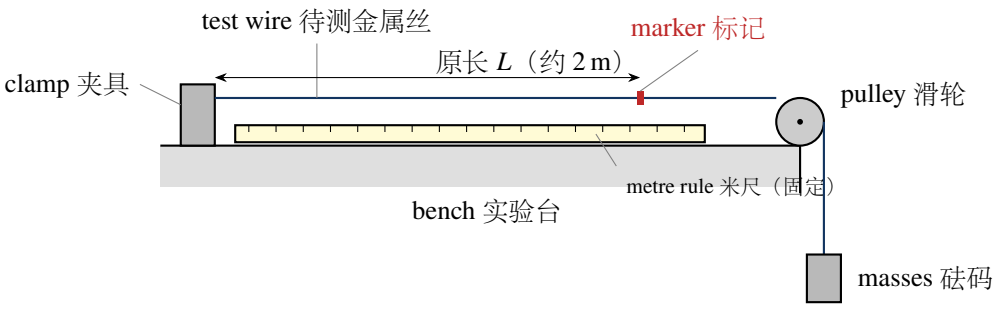
杨氏模量 = 应力 / 应变，单位 Pa。只在比例极限内有定义，等于应力–应变图直线部分的斜率。杨氏模量是**材料**的性质，与样品的长度、粗细无关。

把定义合并：

$$E = \frac{\sigma}{\varepsilon} = \frac{F/A}{x/L} = \frac{FL}{Ax} \quad \Rightarrow \quad k = \frac{F}{x} = \frac{EA}{L}.$$

同一材料，越粗越短的金属丝劲度系数越大。典型值：钢 2.0×10^{11} Pa，铜 1.3×10^{11} Pa，橡胶约 10^{7} Pa。

6.2.1 测定金属丝的杨氏模量



步骤：

(1) 用米尺测量夹具到标记的原长 L（≈ 2 m，长金属丝使伸长量足够大，减小百分比不确定度）。

(2) 用千分尺 (micrometer screw gauge) 在金属丝**不同位置、不同方向**测量直径 d 若干次取平均，$A = \pi d^2/4$。

(3) 先挂一个小砝码把金属丝拉直（消除弯曲），记下标记的初始读数。

(4) 逐次增加砝码，记录每次负载 $F = mg$ 和标记的位置，求伸长量 x。

(5) 逐次减载，检查伸长量能否恢复（确认未超过弹性极限）。

(6) 作 F–x 图，取直线部分斜率 $= F/x$，则 $E =$ 斜率 $\times \dfrac{L}{A}$。

注意事项：戴护目镜（金属丝可能断裂弹回）；伸长量很小，可用游标或行进显微镜 (travelling microscope) 读数；另一种方案是 Searle 装置（两根并排金属丝，一根作参照，消除温度变化和支架下垂的影响）。

例题 6.2 9702/23/M/J/26 Q2

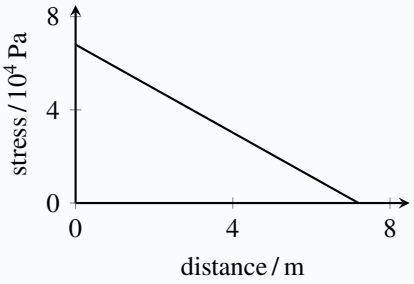
(a) Define the Young modulus of a material. [1]

(b) A long, thin thread of high-density polythene has a uniform cross-section. The thread is suspended from one end so that it hangs vertically. The thread has a mass of 0.018 kg and a cross-sectional area of $2.6 \times 10^{-6}\,\text{m}^2$. High-density polythene has a density of $960\,\text{kg m}^{-3}$ and a Young modulus of 8.0×10^{8} Pa.

(i) Show that the length of the thread is 7.2 m. [2]

(ii) Show that the stress at the top of the thread due to its own weight is 6.8×10^{4} Pa. [2]

(iii) Figure 2.1 shows the variation with distance from the top of the thread of the stress in the thread due to its own weight.



Suggest an explanation for the fact that the variation shown in Figure 2.1 is linear. [2]

(c) The thread in (b) is now laid horizontally on a frictionless bench. A tensile force is applied to the ends of the thread so that there is a uniform stress throughout the thread of 6.8×10^{4} Pa.

(i) Calculate the extension of the thread. [2]

(ii) Calculate the elastic potential energy of the thread. [2]

(d) Suggest, with a reason, whether the total extension of the vertically suspended thread in (b) is less than, equal to or greater than the answer in (c)(i). [1]

解答 Solution

(a) 应力与应变之比 (stress per unit strain)。 **[B1]**

(b)(i) $\rho = m/V$，$V = \dfrac{0.018}{960} = 1.875 \times 10^{-5}\,\text{m}^3$ **[C1]**

$L = \dfrac{V}{A} = \dfrac{0.018}{960 \times 2.6 \times 10^{-6}} = 7.2\,\text{m}$ **[A1]**

(ii) $\sigma = \dfrac{F}{A}$ [C1]

$= \dfrac{0.018 \times 9.81}{2.6 \times 10^{-6}} = 6.8 \times 10^4\ \text{Pa}$ [A1]

(iii) 线上某点所承受的重量（该点以下线段的重量）随距顶端距离的增大而线性减小； [B1]

截面均匀，所以该点的应力与其下方线段的重量成正比。 [B1]

(c)(i) $\varepsilon = \dfrac{\sigma}{E}$，$x = \dfrac{\sigma L}{E}$ [C1]

$x = \dfrac{6.8 \times 10^4 \times 7.2}{8.0 \times 10^8} = 6.1 \times 10^{-4}\ \text{m}$ [A1]

(ii) $E_\text{P} = \frac{1}{2}Fx = \frac{1}{2}(6.8 \times 10^4 \times 2.6 \times 10^{-6})(6.1 \times 10^{-4})$ [C1]

$= 5.4 \times 10^{-5}\ \text{J}$ [A1]

(d) 竖直悬挂时，除顶端外各处应力都小于 6.8×10^4 Pa，所以总伸长量**较小**（实际约为一半）。[B1]

6.3 弹性形变与塑性形变

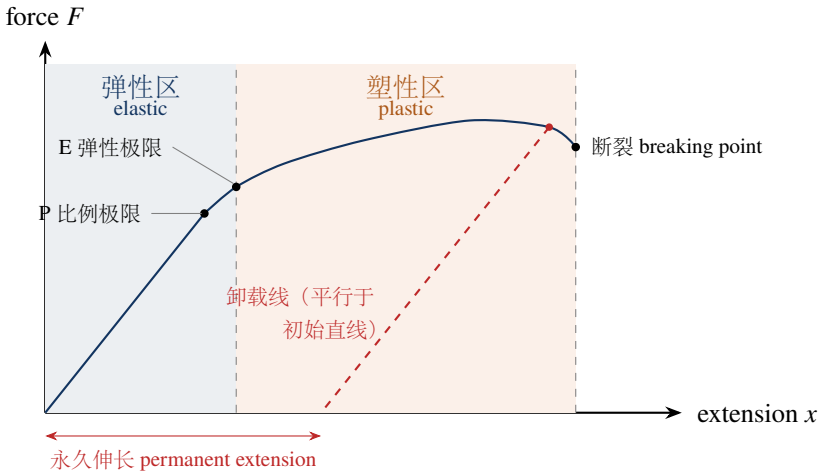
定义 6.5 (弹性形变与塑性形变 *Elastic and plastic deformation*)

Elastic deformation: the object returns to its original shape and size when the load is removed.

Plastic deformation: the object does not return to its original shape and size when the load is removed (permanent deformation).

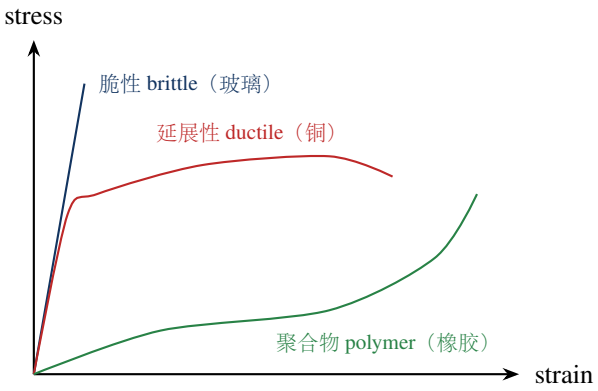
Elastic limit: the maximum load beyond which the deformation is no longer elastic (becomes plastic).

弹性形变：撤去负载后物体恢复原来的形状和大小（伸长量回到零）。**塑性形变**：撤去负载后不能恢复，留下永久形变。**弹性极限**：超过此负载后形变变为塑性的。弹性极限通常略高于（或等于）比例极限。



超过弹性极限后卸载，卸载线大致平行于原来的直线部分，但不回到原点，留下永久伸长。

典型材料的应力–应变图：



6.4 弹性势能 Elastic potential energy

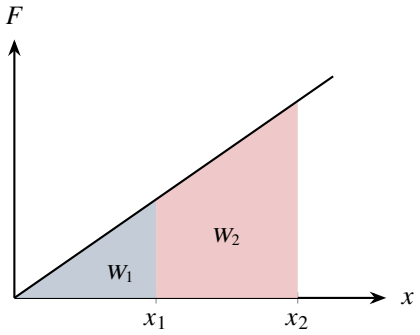
拉伸物体时外力做功。由于力随伸长量变化，功等于 F–x **图下的面积**。

在比例极限内（直线部分），面积是三角形：

公式 6.1 (弹性势能 *Elastic potential energy*)

$$E_{\mathrm{P}} = \frac{1}{2}Fx = \frac{1}{2}kx^2$$

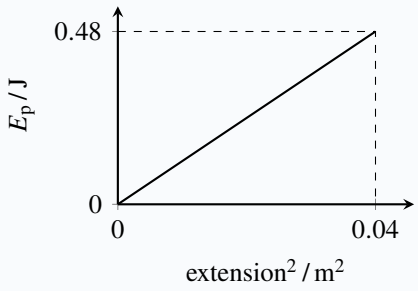
只适用于比例极限内。若在弹性极限内，所做的功全部以弹性势能储存，卸载时可全部释放；若发生塑性形变，部分功转化为内能，不能恢复。



从 x_1 拉到 x_2 所做的功是梯形面积 $= \frac{1}{2}k(x_2^2 - x_1^2)$，**不是** $\frac{1}{2}k(x_2 - x_1)^2$。例如从 0 到 x 与从 x 到 $2x$ 的功之比为 1 : 3。

例题 6.3 9702/11/M/J/26 Q21

The graph shows the variation of the elastic potential energy E_{p} stored in a spring with the square of the extension of the spring.



What is the spring constant of the spring?

A $0.0096\,\mathrm{N\,m^{-1}}$ **B** $6.0\,\mathrm{N\,m^{-1}}$ **C** $12\,\mathrm{N\,m^{-1}}$ **D** $24\,\mathrm{N\,m^{-1}}$

解答 Solution

答案 Answer：D

$E_p = \frac{1}{2}kx^2$，图像斜率 $= \frac{1}{2}k = \dfrac{0.48}{0.04} = 12$，所以 $k = 24\,\mathrm{N\,m^{-1}}$。（C 漏了因子 2。）

教学提示 Teacher's note 建议 4 课时。学生常见错误：(1) 截面积用 πd^2 而不是 $\pi d^2/4$；(2) 伸长量用最终长度；(3) 两点间弹性势能变化用 $\frac{1}{2}k(\Delta x)^2$。杨氏模量实验是 Paper 2/5 的经典考点，务必能写出完整步骤（测什么、用什么仪器、如何作图、如何从斜率求 E）以及安全措施。

6.5 本章小结 Summary

物理量/关系	公式	说明
胡克定律	$F = kx$	比例极限内
串/并联弹簧	$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$；$k = k_1 + k_2$	
应力	$\sigma = F/A$	Pa
应变	$\varepsilon = x/L$	无单位
杨氏模量	$E = \sigma/\varepsilon = \dfrac{FL}{Ax}$	材料性质
弹性势能	$E_P = \frac{1}{2}Fx = \frac{1}{2}kx^2$	F–x 图面积

6.6 本章练习 Practice questions

练习 6.1 9702/11/M/J/26 Q20

A metal wire is stretched by a tensile force. The tensile force is increased. The volume of the wire remains constant and the limit of proportionality is not exceeded. Which quantity remains constant?

A the cross-sectional area of the wire

B the strain of the wire

C the stress in the wire

D the Young modulus of the metal

解答 Solution

答案 Answer：D

杨氏模量是材料的性质，在比例极限内不变。拉力增大，应力、应变都增大；体积不变而长度增大，截面积减小。

练习 6.2 9702/12/M/J/26 Q19

An initially unstretched spring is stretched within its limit of proportionality to a length of 80 mm. The work done stretching the spring is 0.010 J. The spring constant of the spring is 22.5 N m^{-1}. What is the unstretched length of the spring?

A 30 mm **B** 50 mm **C** 60 mm **D** 110 mm

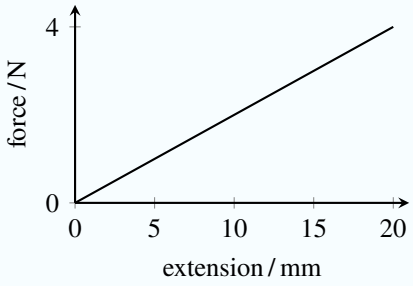
解答 Solution

答案 Answer：B

$\frac{1}{2}kx^2 = 0.010$，$x = \sqrt{\dfrac{2\times 0.010}{22.5}} = 0.030\,\text{m} = 30\,\text{mm}$。原长 $= 80 - 30 = 50\,\text{mm}$。

练习 6.3 9702/13/M/J/26 Q20

The graph shows the relationship between the applied force and the extension for a spring.



X is the work done by the force in extending the spring from 0 to 10 mm. Y is the work done by the force in extending the spring from 10 mm to 20 mm. What is the ratio $X : Y$?

A 1 : 1 **B** 1 : 2 **C** 1 : 3 **D** 1 : 4

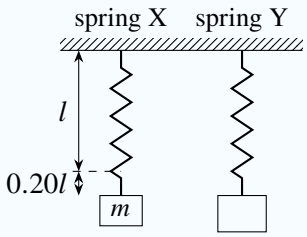
解答 Solution

答案 Answer：C

$X = \frac{1}{2}k(10)^2 = 50k$，$Y = \frac{1}{2}k(20^2 - 10^2) = 150k$，$X : Y = 1 : 3$。（或面积：三角形与梯形之比。）

练习 6.4 9702/13/M/J/26 Q21

Two springs X and Y are attached to a fixed surface. Blocks are attached to their free ends. The blocks hang vertically in equilibrium from the springs as shown.



The stretched lengths of X and Y are equal. Spring X has spring constant k, unstretched length l and extension $0.20l$. Spring Y has spring constant $3.0k$ and extension $0.10l$. The mass of the block attached to spring X is m. What is the mass of the block attached to spring Y?

A $0.67m$ **B** $1.0m$ **C** $1.5m$ **D** $3.0m$

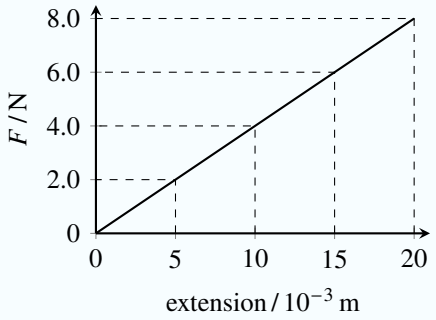
解答 Solution

答案 Answer：C

X：$mg = k(0.20l)$；Y：$Mg = 3.0k(0.10l) = 0.30kl$。$\dfrac{M}{m} = \dfrac{0.30}{0.20} = 1.5$。

练习 6.5 9702/14/M/J/26 Q20

The diagram shows a force–extension graph for a spring.



What is the value of the work done to increase the extension of the spring from 10×10^{-3} m to 20×10^{-3} m?

A 0.02 J **B** 0.06 J **C** 0.08 J **D** 0.12 J

解答 Solution

答案 Answer：B

梯形面积 $= \frac{1}{2}(4.0 + 8.0) \times (10 \times 10^{-3}) = 0.060\,\text{J}$。

练习 6.6 9702/24/M/J/26 Q2

(a) State Hooke's law. [1]

(b) In an experiment to determine the spring constant of a spring X, a student measures the original length

of the spring. The student then suspends an object in equilibrium from the spring and measures the final length of the spring. The spring does not extend beyond its limit of proportionality. The measurements taken by the student are shown in Table 2.1.

original length of spring X	(2.1 ± 0.1) cm
final length of spring X	(14.0 ± 0.1) cm
mass of suspended object	(0.300 ± 0.005) kg

Calculate:

(i) the extension, in cm, of spring X together with its absolute uncertainty [1]

(ii) the spring constant of spring X [3]

(iii) the percentage uncertainty in the value of the spring constant calculated in (b)(ii) due to the uncertainties in the measurements. [2]

(c) An identical object to the one in (b) is suspended in equilibrium from a spring Y that has a greater spring constant than spring X. Spring Y does not extend beyond its limit of proportionality. Describe and explain any difference in:

(i) the extension of the springs [1]

(ii) the elastic potential energy of the springs. [2]

解答 Solution

(a) 伸长量与所施加的力成正比（在比例极限内）。 **[B1]**

(b)(i) $x = 14.0 - 2.1 = 11.9\,\text{cm}$，不确定度相加 $0.1 + 0.1 = 0.2$：$(11.9 \pm 0.2)\,\text{cm}$ **[A1]**

(ii) $k = F/x$ **[C1]**

$= \dfrac{0.300 \times 9.81}{0.119}$ **[C1]**

$= 24.7\,\text{N m}^{-1}$ **[A1]**

(iii) $\dfrac{0.2}{11.9} + \dfrac{0.005}{0.300} = 1.68\% + 1.67\%$ **[C1]**

$= 3\%$ **[A1]**

(c)(i) Y 的伸长量较小：力相同，劲度系数较大（$x = F/k$）。 **[B1]**

(ii) Y 的弹性势能较小； **[B1]**

力相同而伸长量较小，$E_\text{P} = \frac{1}{2}Fx$（或 $E_\text{P} = \frac{F^2}{2k}$）。 **[B1]**

第七章 波 Waves

考纲要求 Syllabus learning outcomes

7.1.1 describe what is meant by wave motion as illustrated by vibration in ropes, springs and ripple tanks
描述绳、弹簧和水波槽中所展示的波动。

7.1.2 understand and use the terms displacement, amplitude, phase difference, period, frequency, wavelength and speed
理解并使用位移、振幅、相位差、周期、频率、波长和波速。

7.1.3 understand the use of the time-base and y-gain of a cathode-ray oscilloscope (CRO) to determine frequency and amplitude
理解用示波器的扫描时基和 y 增益测量频率和振幅。

7.1.4 derive, using the definitions of speed, frequency and wavelength, the wave equation $v = f\lambda$
由速度、频率、波长的定义推导波动方程 $v = f\lambda$。

7.1.5 recall and use $v = f\lambda$
记住并使用 $v = f\lambda$。

7.1.6 understand that energy is transferred by a progressive wave
理解行波传递能量。

7.1.7 recall and use intensity = power/area and intensity $\propto$ (amplitude)^2 for a progressive wave
记住并使用强度 = 功率 / 面积，以及强度 $\propto$ 振幅 ^{2}。

7.2.1 compare transverse and longitudinal waves
比较横波与纵波。

7.2.2 analyse and interpret graphical representations of transverse and longitudinal waves
分析和解读横波、纵波的图像。

7.3.1 understand that when a source of sound waves moves relative to a stationary observer, the observed frequency is different from the source frequency
理解声源相对静止观察者运动时，观察到的频率与声源频率不同（多普勒效应）。

7.3.2 use the expression $f_o = f_s v/(v \pm v_s)$ for the observed frequency when a source of sound waves moves relative to a stationary observer
使用 $f_o = f_s v/(v \pm v_s)$。

7.4.1 state that all electromagnetic waves are transverse waves that travel with the same speed c in free space
陈述所有电磁波都是横波，在真空中以相同速度 c 传播。

7.4.2 recall the approximate range of wavelengths in free space of the principal regions of the electromagnetic spectrum from radio waves to γ-rays
记住电磁波谱各主要区域（从无线电波到 γ 射线）在真空中的大致波长范围。

7.4.3 recall that wavelengths in the range 400–700 nm in free space are visible to the human eye
记住真空中波长 400–700 nm 的电磁波是人眼可见的。

7.5.1 understand that polarisation is a phenomenon associated with transverse waves
理解偏振是横波特有的现象。

7.5.2 recall and use Malus's law ($I = I_0 \cos^2\theta$) to calculate the intensity of a plane-polarised electromagnetic wave after transmission through a polarising filter or a series of polarising filters

记住并用马吕斯定律 $I = I_0 \cos^2 \theta$ 计算平面偏振电磁波通过一个或一系列偏振片后的强度（不要求计算非偏振光通过偏振片的情况）。

7.1 行波 Progressive waves

行波 (*progressive wave*) 是振动（扰动）在介质中的传播，它**传递能量**而**不传递物质**：介质中的质点只在各自的平衡位置附近振动，并不随波迁移。

定义 7.1 (行波 *Progressive wave*)

A progressive wave is a wave that transfers energy from one place to another (in the direction of travel of the wave) without transferring matter.

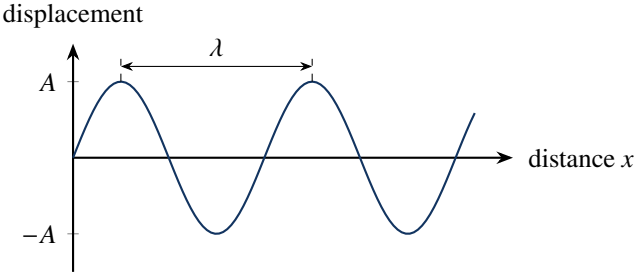
行波沿传播方向传递能量，而介质本身不发生迁移。

7.1.1 描述波的物理量

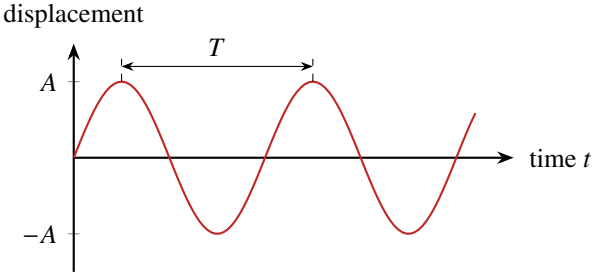
术语	英文定义	中文说明
位移 displacement x	distance (in a specified direction) of a point from its equilibrium position	质点偏离平衡位置的距离（矢量）
振幅 amplitude A	maximum displacement (from the equilibrium position)	最大位移
周期 period T	time for one complete oscillation (of a point on the wave)	完成一次全振动的时间
频率 frequency f	number of oscillations per unit time	$f = 1/T$，单位 Hz
波长 wavelength λ	distance between two adjacent points on a wave that are in phase (e.g. crest to crest)	相邻两个同相点之间的距离
波速 speed v	distance travelled by the wave(front) per unit time	波形（能量）传播的速度
相位差 phase difference	difference in the stages of oscillation of two points, measured in degrees or radians	一个周期 = 360° = 2π rad

7.1.2 波的两种图像

位移–距离图（某一时刻的波形，“快照”）



位移–时间图（某一质点的振动）



易错提醒 Common mistakes 两种图形状相同，但横轴含义不同：位移–**距离**图上两峰间距是**波长**；位移–**时间**图上两峰间隔是**周期**。读图前一定先看横轴。

相位差：波上相距 Δx 的两点，相位差 $\phi = \dfrac{\Delta x}{\lambda} \times 360° = \dfrac{2\pi \Delta x}{\lambda}$。相距整数个波长的点同相 (in phase)，相距半波长奇数倍的点反相 (antiphase，相位差 180°)。

判断质点运动方向：波向右传播时，某点下一时刻的位移等于它**左边**相邻点此刻的位移（“前一个点带动后一个点”）。

7.1.3 波动方程 $v = f\lambda$

推导：波在一个周期 T 内恰好传播一个波长 λ 的距离，所以

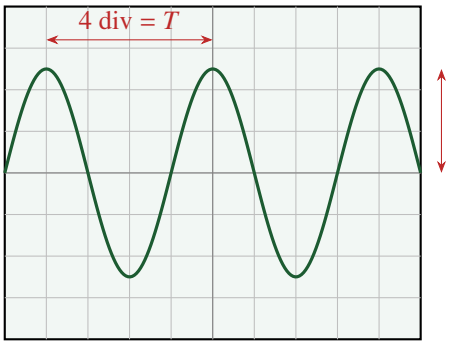
$$v = \frac{\text{distance}}{\text{time}} = \frac{\lambda}{T} = \lambda \times \frac{1}{T} = f\lambda.$$

公式 7.1 (波动方程 *Wave equation*)

$$v = f\lambda$$

对横波、纵波、电磁波都适用。波从一种介质进入另一种介质时，频率不变（由波源决定），波速和波长改变。

7.1.4 示波器 Cathode-ray oscilloscope



time-base 时基：$2\ \text{ms div}^{-1}$

y-gain 增益：$0.5\ \text{V div}^{-1}$

$T = 4 \times 2 = 8\ \text{ms}$，$f = 1/T = 125\ \text{Hz}$

$A = 2.5 \times 0.5 = 1.25\ \text{V}$

- **时基 time-base**：水平方向每格代表的时间（如 ms div^{-1}）。周期 T = 一个完整波形占的格数 × 时基。
- **y 增益 y-gain**：竖直方向每格代表的电压（如 V div^{-1}）。振幅 = 峰到中线的格数 × y 增益。
- 为减小读数误差，测多个周期的总格数再除以周期数；测峰–峰值再除以 2。

7.1.5 波的能量与强度

定义 7.2 (强度 *Intensity*)

Intensity is the power transmitted per unit area (normal to the direction of wave travel): $I = \dfrac{P}{A}$.

强度 = 垂直于传播方向的单位面积上传输的功率，单位 $\mathrm{W\,m^{-2}}$。

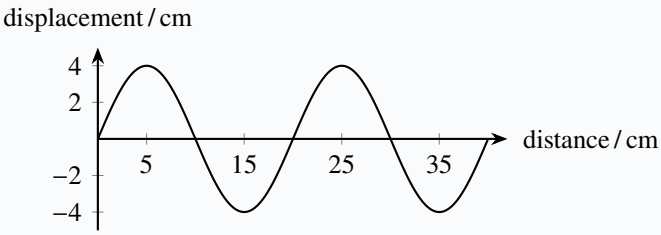
公式 7.2 (强度与振幅 *Intensity and amplitude*)

$$I \propto A^2$$

振幅加倍，强度变为 4 倍。点波源向四周均匀辐射时 $I = \dfrac{P}{4\pi r^2} \propto \dfrac{1}{r^2}$，所以振幅 $A \propto \dfrac{1}{r}$。

例题 7.1 9702/12/M/J/26 Q23

The graph shows the displacement of each particle in a transverse wave against distance at one instant in time.



The speed of the wave is $1.6\,\mathrm{m\,s^{-1}}$. What is the frequency of the wave?

A 0.050 Hz **B** 0.080 Hz **C** 5.0 Hz **D** 8.0 Hz

解答 Solution

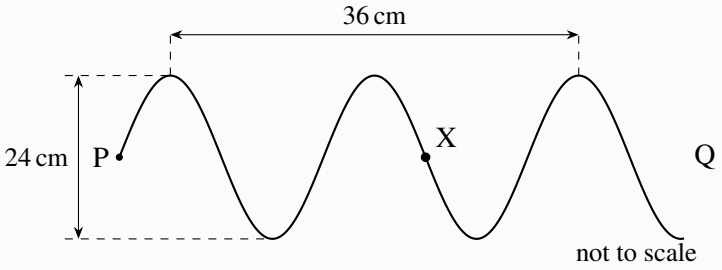
答案 Answer：D

从图读出波长 $\lambda = 25 - 5 = 20\,\mathrm{cm} = 0.20\,\mathrm{m}$（相邻两峰）。$f = v/\lambda = 1.6/0.20 = 8.0\,\mathrm{Hz}$。（B 是忘了把 cm 换成 m。）

例题 7.2 9702/21/M/J/26 Q3(a)(b)

(a) State what is meant by a progressive wave. [1]

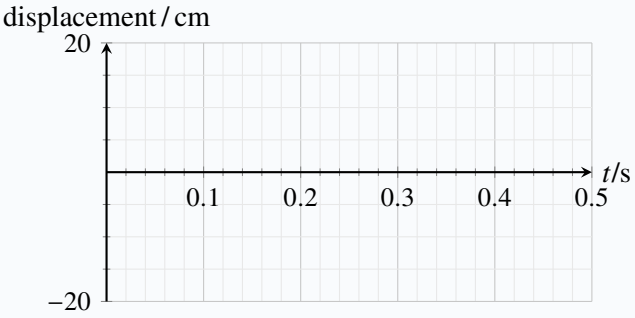
(b) A transverse progressive wave travels along a long string PQ from P to Q. Figure 3.1 shows the string at time $t = 0$.



A point on the string is labelled X. At $t = 0$, point X is at its equilibrium position. The speed of the wave on the string is $75\,\text{cm}\,\text{s}^{-1}$.

(i) Show that the period of the wave is 0.24 s. [2]

(ii) Sketch the variation of the displacement from the equilibrium position of point X with t between $t = 0$ and $t = 0.48$ s. [3]



解答 Solution

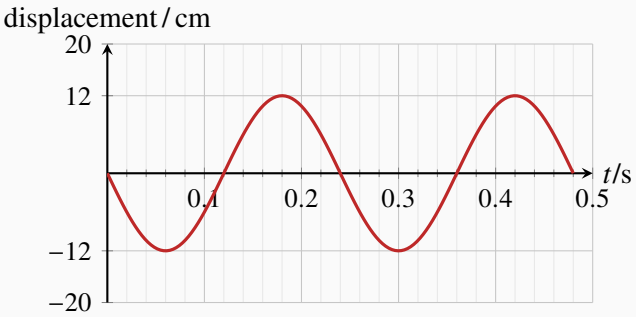
（本卷未公布评分标准，以下为参考解答。）

(a) 一种把能量从一处传到另一处（沿波的传播方向）而不传递物质的波。 **[B1]**

(b)(i) 36 cm 包含两个完整波长，$\lambda = 18$ cm。 **[C1]**

$T = \dfrac{\lambda}{v} = \dfrac{18}{75} = 0.24$ s **[A1]**

(ii) 振幅 = 24/2 = 12 cm。波向右（P→Q）传播，X 左侧是波谷，所以 X 下一时刻向下运动。

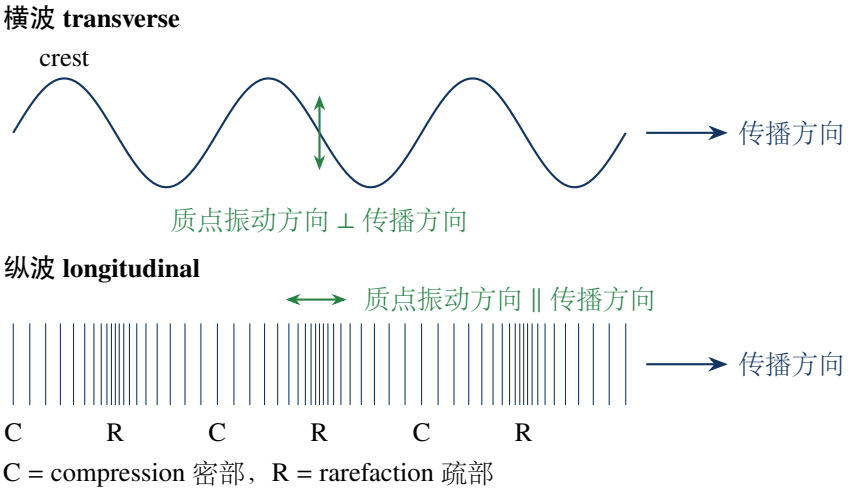


正弦曲线从原点开始先向负方向； **[B1]**

振幅 12 cm； **[B1]**

周期 0.24 s，共两个完整周期至 0.48 s。 **[B1]**

7.2 横波与纵波 Transverse and longitudinal waves



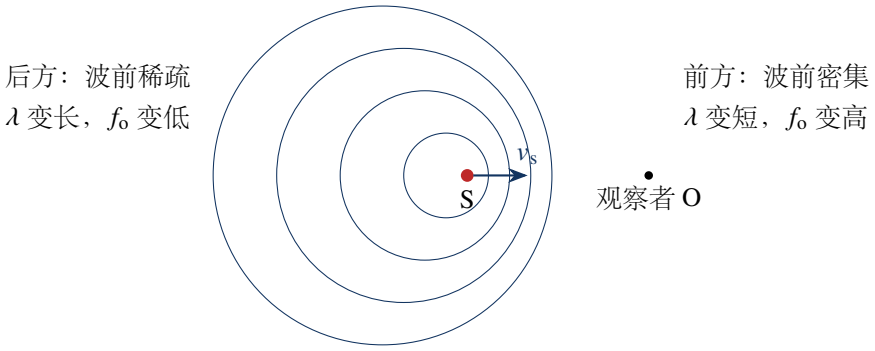
	横波 transverse	纵波 longitudinal
质点振动方向	垂直于能量传播方向	平行于能量传播方向
结构	波峰 crest、波谷 trough	密部 compression、疏部 rarefaction
例子	电磁波、绳波、水面波、地震 S 波	声波、弹簧纵波、地震 P 波、超声
能否偏振	能	不能

两种波都是行波，都满足 $v = f\lambda$，都能发生反射、折射、衍射、干涉。

纵波的图像表示：纵波也可以画成位移–距离的正弦图（把沿传播方向的位移画在纵轴上，规定向右为正）。在正弦图中，位移为零且两侧质点**向它靠拢**的点是密部中心，两侧质点**离开它**的点是疏部中心。

> **考试技巧 Exam tip** 声波在空气中是纵波，不能偏振；“sound and microwaves are longitudinal”是错误说法——微波是电磁波，是横波。

7.3 声波的多普勒效应 Doppler effect for sound waves



声源运动时，它在连续发出的两个波峰之间移动了一段距离，使前方的波前挤在一起（波长变短），

后方的波前拉开（波长变长）。声速 v 由介质决定，不变，所以观察者接收到的频率改变：

公式 7.3 (多普勒效应（声源运动，观察者静止） *Doppler effect*)

$$f_o = \frac{f_s v}{v \pm v_s}$$

声源远离观察者用“+”（$f_o < f_s$）；声源靠近观察者用“−”（$f_o > f_s$）。v 为声速，v_s 为声源速度。

推导要点：声源每个周期 T_s 内前进 $v_s T_s$，前方观察到的波长 $\lambda_o = vT_s - v_s T_s$，所以

$$f_o = \frac{v}{\lambda_o} = \frac{v}{(v - v_s)T_s} = \frac{f_s v}{v - v_s}.$$

匀速驶过观察者的火车鸣笛：靠近时观察到恒定的较高频率，经过时突然降为恒定的较低频率（f–t 图为两段水平线）。

例题 7.3 9702/12/M/J/26 Q24

The siren on an emergency vehicle emits a sound of constant frequency 1100 Hz. The driver in a stationary car hears the sound at a frequency of 1000 Hz. The speed of sound in air is 340 m s^{-1}. What is the velocity of the emergency vehicle?

A 31 m s^{-1} away from the stationary car

B 34 m s^{-1} away from the stationary car

C 31 m s^{-1} towards the stationary car

D 34 m s^{-1} towards the stationary car

解答 Solution

答案 Answer：B

观察频率降低，说明声源远离。$1000 = \dfrac{1100 \times 340}{340 + v_s}$，$340 + v_s = 374$，$v_s = 34\ \text{m s}^{-1}$。

7.4 电磁波谱 Electromagnetic spectrum

所有电磁波：

- 都是**横波**（振荡的电场和磁场互相垂直，且都垂直于传播方向）；
- 在真空中传播速度都相同，$c = 3.00 \times 10^8\ \text{m s}^{-1}$；
- 不需要介质，能在真空中传播；可以偏振。

γ-rays	X-rays	ultraviolet	visible	infrared	microwaves	radio waves
$< 10^{-11}$	10^{-11}–10^{-8}	10^{-8}–4×10^{-7}	400–700 nm	7×10^{-7}–10^{-3}	10^{-3}–10^{-1}	$> 10^{-1}$

波长 λ/m（真空中）：从左到右增大 ⟶；频率与光子能量从左到右减小

区域 region	波长范围（近似）	典型来源/应用
γ 射线	$< 10^{-11}$ m	放射性衰变；放疗、灭菌
X 射线	10^{-11} m – 10^{-8} m	X 光管；医学成像
紫外线	10^{-8} m – 4×10^{-7} m	太阳；消毒、荧光
可见光	4×10^{-7} m – 7×10^{-7} m	紫 400 nm → 红 700 nm
红外线	7×10^{-7} m – 10^{-3} m	热物体；遥控、热成像
微波	10^{-3} m – 10^{-1} m	微波炉、雷达、手机
无线电波	$> 10^{-1}$ m	广播、通信

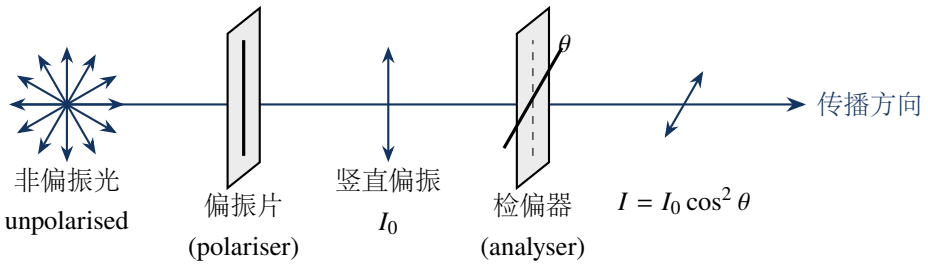
考试技巧 Exam tip　记住数量级即可：X 射线 ~ 10^{-10} m（约一个原子大小，0.1 nm），可见光 ~ 5×10^{-7} m，微波 ~ cm，无线电波 ≥ m。由频率判断区域时先用 $\lambda = c/f$ 换算。

7.5 偏振 Polarisation

定义 7.3 (偏振　*Polarisation*)

A transverse wave is (plane) polarised if its oscillations are in one direction only, in a plane normal to the direction of propagation (energy transfer).

横波的振动只限于（与传播方向垂直的平面内的）**一个方向**时，称为平面偏振波。偏振是**横波**特有的现象，纵波（振动方向与传播方向一致）不能偏振。偏振现象证明了光是横波。



定律 7.1 (马吕斯定律　*Malus's law*)

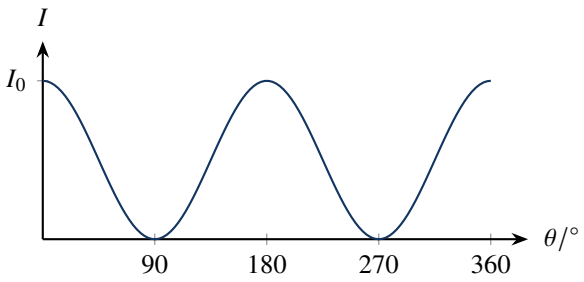
$$I = I_0\cos^2\theta$$

I_0 为入射平面偏振波的强度，θ 为入射波的偏振方向与偏振片透振轴 (transmission axis) 之间的夹角。

推导思路：把入射振幅 A_0 分解为平行于透振轴的分量 $A_0\cos\theta$（通过）和垂直分量（被吸收），而强度 ∝ 振幅 ^{2}，所以 $I = I_0\cos^2\theta$。

- $\theta = 0$：全部通过；$\theta = 90°$：完全不通过（"正交"crossed）。
- 旋转偏振片 360°，透射强度出现 2 次最大、2 次为零（周期 180° 的 $\cos^2$ 曲线）。

- 多片偏振片：逐片相乘，每次用**相邻两片**透振轴之间的夹角。两正交偏振片之间插入一片 45° 的偏振片，反而有光通过：$I = I_0 \cos^2 45° \cos^2 45° = I_0/4$。



例题 7.4 9702/11/M/J/26 Q23

A vertically polarised electromagnetic wave has an intensity of $3.2 \times 10^4\ \mathrm{W\,m^{-2}}$. The wave is incident on a polarising filter that has a transmission axis at an angle θ to the vertical. The transmitted wave has an intensity of $2.5 \times 10^4\ \mathrm{W\,m^{-2}}$. What is θ?

A 28° **B** 39° **C** 52° **D** 62°

解答 Solution

答案 Answer：A

$\cos^2\theta = \dfrac{2.5}{3.2} = 0.781$，$\cos\theta = 0.884$，$\theta = 28°$。（选 B 是忘了平方：$\cos\theta = 0.781$。）

教学提示 Teacher's note 建议 5 课时。易错点：(1) 区分两种波形图的横轴；(2) 多普勒公式符号（"靠近频率升高"用于检查）；(3) 马吕斯定律中 θ 是偏振方向与透振轴之间的夹角，多片时逐片计算；(4) 强度 $\propto$ 振幅平方，选择题常与面积变化结合（如振幅变 3 倍、面积变 3 倍 $\Rightarrow$ 功率变 27 倍）。可用 Slinky 弹簧演示横波与纵波，用两片偏振片和手机屏幕演示马吕斯定律。

7.6 本章小结 Summary

物理量/关系	公式	说明
波动方程	$v = f\lambda$，$f = 1/T$	频率由波源决定
相位差	$\phi = \dfrac{\Delta x}{\lambda} \times 360°$	
强度	$I = P/A$，$I \propto A^2$	点源 $I \propto 1/r^2$
多普勒效应	$f_o = \dfrac{f_s v}{v \pm v_s}$	远离 +，靠近 −
电磁波	$c = 3.00 \times 10^8\ \mathrm{m\,s^{-1}}$	横波；可见光 400–700 nm
马吕斯定律	$I = I_0 \cos^2\theta$	入射为平面偏振波

7.7 本章练习 Practice questions

练习 7.1 9702/11/M/J/26 Q26

A wave is incident on a surface of area S placed perpendicular to the direction of travel of the wave. The wave has amplitude R and the power incident on the surface is P. The amplitude of the wave is increased to $3R$ and the wave is now incident normally on a surface of area $3S$. What is the power incident on the surface of area $3S$?

A P **B** $3P$ **C** $9P$ **D** $27P$

解答 Solution

答案 Answer：D

强度 $I \propto$ 振幅^2，振幅变 3 倍，强度变 9 倍；功率 = $I\times$ 面积，面积变 3 倍，所以功率变为 $9\times3 = 27$ 倍。

练习 7.2 9702/13/M/J/26 Q25

Plane polarised light of intensity I is incident normally on a set of polarising filters. The transmission axis of the first filter is at 30° to the plane of polarisation of the incident light. The transmission axis of each successive filter is at 30° to the transmission axis of the previous filter. The intensity of the light emerging from the final filter is $0.24I$. How many polarising filters are in the set?

A 3 **B** 5 **C** 6 **D** 10

解答 Solution

答案 Answer：B

每片透过比例 $\cos^2 30° = 0.75$。$0.75^n = 0.24$，$n = \dfrac{\lg 0.24}{\lg 0.75} = 4.96 \approx 5$。

练习 7.3 9702/14/M/J/26 Q23

A police car has a two-tone siren emitting sound of frequencies of 700 Hz and 1000 Hz. The police car is travelling at a speed of 40.0 m s^{-1} towards a stationary observer. The speed of sound in the air is 340 m s^{-1}. What is the difference between the two frequencies of the sound that is heard by the observer?

A 268 Hz **B** 300 Hz **C** 335 Hz **D** 340 Hz

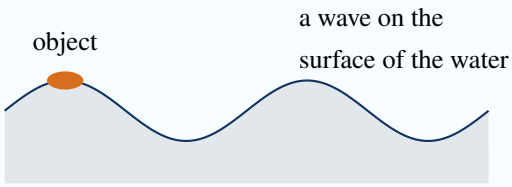
解答 Solution

答案 Answer：D

靠近：$f_o = f_s \times \dfrac{340}{340 - 40} = 1.133 f_s$。两频率差也乘以同一因子：$300 \times 1.133 = 340$ Hz。

练习 7.4 9702/14/M/J/26 Q27

The diagram shows an object floating in water through which a progressive wave is passing.



In time t, the object moves from maximum displacement at a peak of the wave to minimum displacement at a trough and then back to maximum displacement again. Through which vertical distance does the object move in time t?

A one wavelength

B twice the wavelength

C twice the amplitude

D four times the amplitude

解答 Solution

答案 Answer：D

物体只上下振动，不随波移动。从波峰到波谷移动 $2A$，再回到波峰又移动 $2A$，总路程 $4A$。

练习 7.5 9702/12/F/M/26 Q24

An electromagnetic wave has a frequency of 3.3×10^{14} Hz. Which principal region of the electromagnetic spectrum is this wave in?

A infrared

B visible light

C ultraviolet

D X-ray

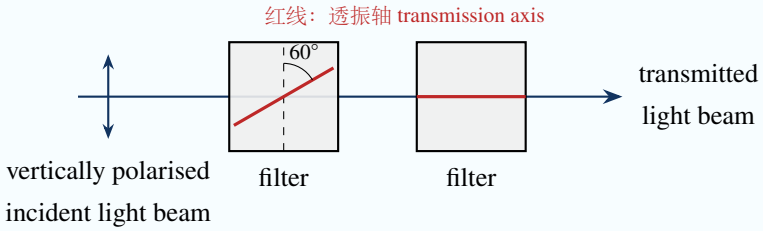
解答 Solution

答案 Answer：A

$\lambda = c/f = \dfrac{3.00 \times 10^8}{3.3 \times 10^{14}} = 9.1 \times 10^{-7}\,\text{m} = 910\,\text{nm} > 700\,\text{nm}$，属于红外线。

练习 7.6 9702/12/F/M/26 Q27

A beam of vertically polarised light of intensity I_0 is incident normally on a polarising filter. The transmission axis of the filter is at an angle of 60° to the vertical. The transmitted light beam is then incident normally on a second polarising filter. The transmission axis of the second filter is perpendicular to the vertical as shown.



The intensity of the light beam that emerges from the second filter is I_T. What is the ratio I_T/I_0?

A 0 **B** 0.19 **C** 0.25 **D** 0.43

解答 Solution

答案 Answer：**B**

第一片：与竖直成 $60°$，$I_1 = I_0 \cos^2 60° = 0.25I_0$。第二片透振轴水平，与第一片透振轴夹角 $90° - 60° = 30°$：$I_T = 0.25I_0 \cos^2 30° = 0.1875I_0 \approx 0.19I_0$。

第八章 叠加 Superposition

考纲要求 Syllabus learning outcomes

8.1.1 explain and use the principle of superposition
解释并使用叠加原理。

8.1.2 show an understanding of experiments that demonstrate stationary waves using microwaves, stretched strings and air columns (it will be assumed that end corrections are negligible)
理解用微波、拉紧的弦和空气柱演示驻波的实验（忽略端点修正）。

8.1.3 explain the formation of a stationary wave using a graphical method, and identify nodes and antinodes
用作图法解释驻波的形成，并识别波节和波腹。

8.1.4 understand how wavelength may be determined from the positions of nodes or antinodes of a stationary wave
理解如何由驻波波节或波腹的位置测定波长。

8.2.1 explain the meaning of the term diffraction
解释衍射的含义。

8.2.2 show an understanding of experiments that demonstrate diffraction including the qualitative effect of the gap width relative to the wavelength of the wave; for example diffraction of water waves in a ripple tank
理解演示衍射的实验，包括缝宽相对波长对衍射程度的定性影响（如水波槽）。

8.3.1 understand the terms interference and coherence
理解干涉和相干的含义。

8.3.2 show an understanding of experiments that demonstrate two-source interference using water waves in a ripple tank, sound, light and microwaves
理解用水波、声波、光和微波演示双源干涉的实验。

8.3.3 understand the conditions required if two-source interference fringes are to be observed
理解观察到双源干涉条纹所需的条件。

8.3.4 recall and use $\lambda = ax/D$ for double-slit interference using light
记住并使用双缝干涉公式 $\lambda = ax/D$。

8.4.1 recall and use $d \sin\theta = n\lambda$
记住并使用光栅方程 $d \sin\theta = n\lambda$。

8.4.2 describe the use of a diffraction grating to determine the wavelength of light (the structure and use of the spectrometer are not included)
描述用衍射光栅测定光波长的方法（不要求分光计）。

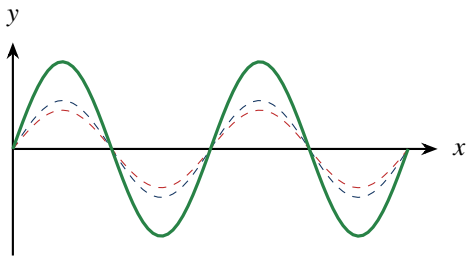
8.1 叠加原理 Principle of superposition

定律 8.1 (叠加原理 *Principle of superposition*)

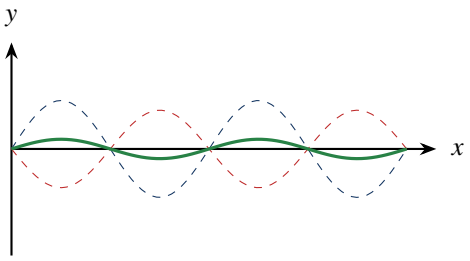
When two (or more) waves meet at a point, the resultant displacement is the sum of the displacements of the individual waves.

两列（或多列）波在某点相遇时，该点的合位移等于各列波单独引起的位移之**矢量和（代数和）**。注意是**位移**相加，不是振幅、强度或频率相加。

同相：相长干涉 constructive



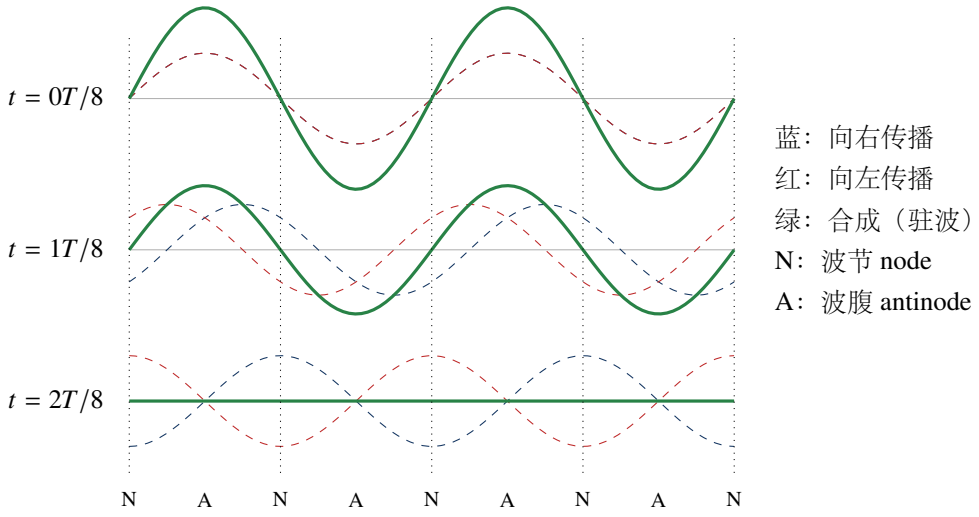
反相：相消干涉 destructive



8.2 驻波 Stationary waves

8.2.1 驻波的形成

当两列**频率（波长）相同、振幅相近、沿相反方向传播**的行波相遇时，叠加形成**驻波**(*stationary wave*)。实际中通常由一列入射波和它在固定端（或反射面）的反射波叠加而成。



作图法解释：在不同时刻画出两列反向行波，逐点把位移相加。可以看到某些点（N）的合位移**始终为零**——波节；相邻两波节中点（A）的振幅**最大**（$= 2A$）——波腹。合成波形只上下“呼吸”，不向任何方向移动，所以称为驻波。

定义 8.1 (波节与波腹　*Node and antinode*)

Node: a point on a stationary wave where the amplitude is zero (always).

Antinode: a point on a stationary wave where the amplitude is maximum.

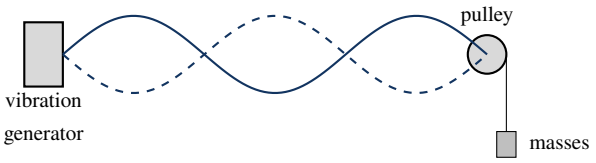
相邻两波节（或两波腹）间距 $= \frac{\lambda}{2}$；相邻波节与波腹间距 $= \frac{\lambda}{4}$。

8.2.2 驻波与行波的比较

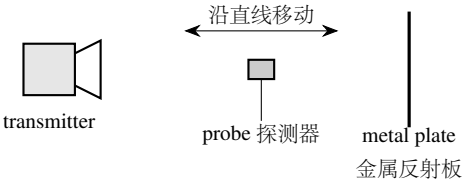
	行波 progressive	驻波 stationary
能量	沿传播方向传递	**不**传递能量（储存在波中）
振幅	所有点相同（无衰减时）	随位置变化：波节为零、波腹最大
相位	一个波长内各点相位都不同	相邻两波节之间各点**同相**；跨过一个波节相位差 180°
波长	相邻同相两点的距离	相邻两波节距离的 2 倍
频率	所有点相同	所有点（除波节）相同

8.2.3 演示驻波的实验

(a) 弦 stretched string

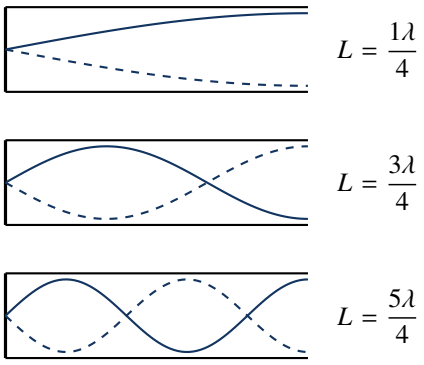


(b) 微波 microwaves

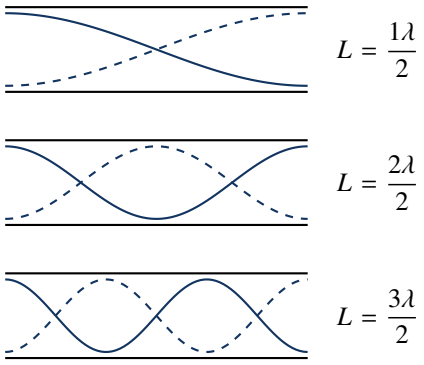


- **弦**：振动发生器使弦振动，波在滑轮处（固定端）反射。调节频率，当弦长为半波长整数倍时形成稳定驻波。两端都是**波节**：$L = n\dfrac{\lambda}{2}$，$f_n = \dfrac{nv}{2L}$（$n = 1, 2, 3, \ldots$）。
- **微波**：发射器正对金属板，探测器在两者之间移动，读数交替出现最大和最小。相邻两个最小读数（波节）的距离为 $\lambda/2$；测出多个波节间距取平均，$\lambda = 2\times$ 平均间距。金属板处为波节。
- **空气柱（声波）**：扬声器（或音叉）置于管口。**闭端是位移波节，开端是位移波腹**（忽略端点修正）。

一端封闭 closed at one end



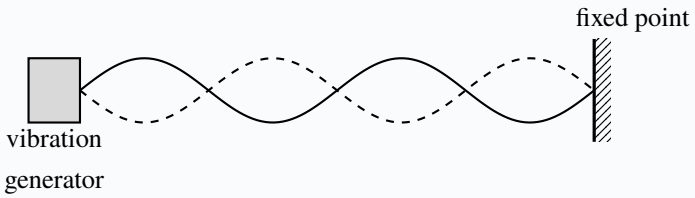
两端开口 open at both ends



一端封闭的管只能形成 $L = \lambda/4,\ 3\lambda/4,\ 5\lambda/4, \ldots$（奇数倍 $\lambda/4$）。用共鸣管测声速：改变水面高度找到相邻两次共鸣的空气柱长度 L_1、L_2，则 $L_2 - L_1 = \lambda/2$，$v = f\lambda$（此法可消去端点修正）。

例题 8.1 9702/11/M/J/26 Q27

A string is stretched between a vibration generator and a fixed point. When the vibration generator is vibrating at a frequency f, a stationary wave with five nodes is created on the string, as shown. There is a node at the end of the string that is attached to the vibration generator.



The frequency of vibration of the vibration generator is slowly increased. What is the next frequency that produces a stationary wave on the string?

A $1.25f$ **B** $1.50f$ **C** $1.75f$ **D** $2.00f$

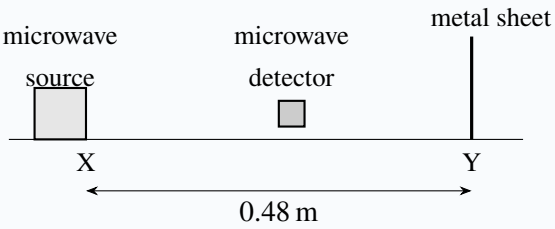
解答 Solution

答案 Answer：A

5 个波节 $\Rightarrow$ 4 个“环”（loop），$L = 4 \times \frac{\lambda}{2}$。下一个驻波有 5 个环。波速不变，$f \propto$ 环数，所以 $f' = \frac{5}{4}f = 1.25f$。

例题 8.2 9702/13/M/J/26 Q30

A microwave source is placed on a horizontal surface at position X. A shiny metal sheet is placed at position Y that is a distance of 0.48 m from X.



The microwaves are reflected back from the metal sheet towards the source. A microwave detector is moved from X to Y and all positions of minimum microwave intensity are recorded. Intensity minima were detected at X and Y and also at 15 positions in between X and Y. The speed of the microwaves is $3.0 \times 10^8\,\text{m s}^{-1}$. What is the frequency of the microwaves emitted by the source?

A $4.7 \times 10^9\,\text{Hz}$ **B** $5.0 \times 10^9\,\text{Hz}$

C $9.4 \times 10^9\,\text{Hz}$ **D** $1.0 \times 10^{10}\,\text{Hz}$

解答 Solution

答案 Answer：B

共 17 个波节，相邻波节间隔 16 个，每个间隔 $\lambda/2$：$16 \times \frac{\lambda}{2} = 0.48$，$\lambda = 0.060\,\text{m}$。$f = \frac{c}{\lambda} = \frac{3.0 \times 10^8}{0.060} =$

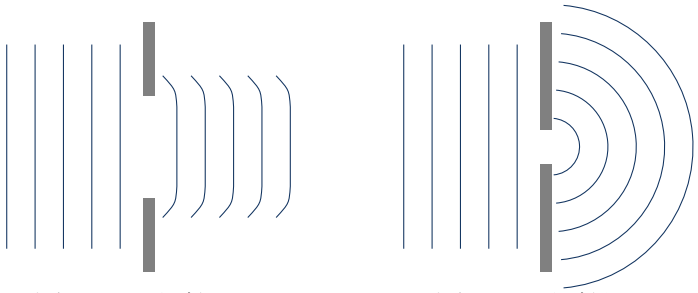
5.0×10^9 Hz。（A 是误用 17 个间隔。）

8.3 衍射 Diffraction

定义 8.2 (衍射 *Diffraction*)

Diffraction is the spreading of a wave as it passes through a gap or around the edge of an obstacle.

衍射：波通过狭缝或绕过障碍物边缘时发生的**展开（偏离直线传播）**现象。衍射前后波长、频率、波速都不变。



缝宽 $\gg \lambda$：衍射不明显　　缝宽 $\approx \lambda$：衍射最明显

- 缝宽 (gap width) 与波长**相近**（或更小）时衍射最显著；缝宽远大于波长时，波几乎直线通过，只在边缘略有弯曲。
- 衍射程度取决于 λ/缝宽 的比值：缝变宽或波长变短 $\Rightarrow$ 衍射减弱。
- 水波槽实验：用直的振动条产生平面波，通过可调宽度的缝，用频闪灯观察波前形状。
- 生活实例：声波（$\lambda \sim$ m）能绕过门口，所以能听到门外的声音；光波（$\lambda \sim 5 \times 10^{-7}$ m）通过门口几乎没有衍射。

8.4 干涉 Interference

定义 8.3 (干涉与相干 *Interference and coherence*)

Interference: the superposition of two (or more) coherent waves to give a resultant wave whose intensity (amplitude) varies with position.

Coherence: two waves (sources) are coherent if they have a constant phase difference (and therefore the same frequency).

干涉：两列（或多列）**相干**波叠加，产生强度随位置变化的稳定图样。**相干**：两列波有**恒定的相位差**（因此频率必然相同）。

观察到稳定干涉条纹的条件：

(1) 两波源相干（恒定相位差，同频率）；

(2) 两列波为同种波，若为横波则偏振方向相同（或有平行分量）；

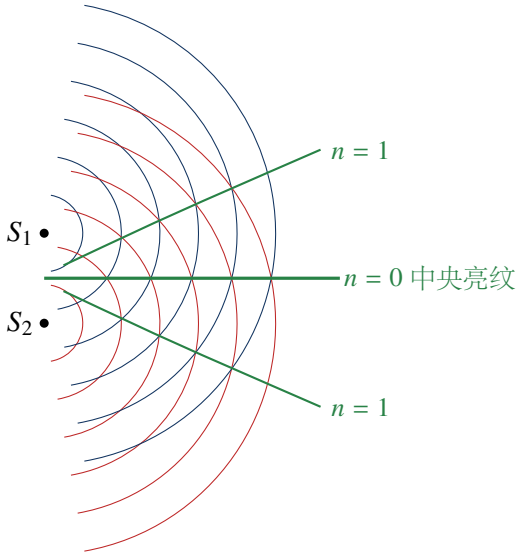
(3) 振幅相近（使暗纹足够暗，条纹清晰）；

(4) 两列波在相遇处有重叠（需要衍射使光从两缝展开并重叠）。

光程差与相位差：设两波源**同相**，到某点的路程差（path difference）为 Δ：

	路程差	相位差
相长干涉（亮/强）	$\Delta = n\lambda$	$0, 360°, \ldots$
相消干涉（暗/弱）	$\Delta = (n + \frac{1}{2})\lambda$	$180°, 540°, \ldots$

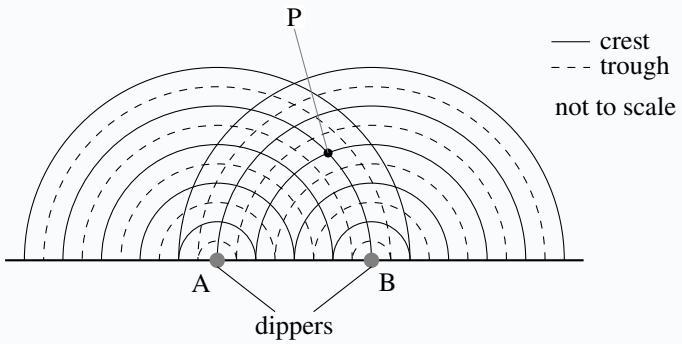
若两波源本身反相，则条件互换。



水波槽中两个同相振子：
实线（波峰）与实线交点：相长
波峰与波谷交点：相消
相长点连成的线称为“antinodal lines”

例题 8.3 9702/22/M/J/26 Q3

Two oscillating dippers are used to create progressive transverse water waves on the surface of a lake. The waves are shown from above in Figure 3.1.



Wave crests are represented by a solid line. Wave troughs are represented by a dashed line. The waves superpose and an interference pattern is formed on the surface of the water. The dippers oscillate in phase with each other and with a frequency of 3.0 Hz.

(a) State what is meant by a progressive wave. [1]

(b) (i) Calculate the period of the waves. [1]

(ii) Points A and B are at the centre of the dippers. Point P is shown on Figure 3.1. The distance from A to P is 3.9 m. The distance from B to P is 2.4 m. Determine the speed of the waves. [3]

(c) A ball of mass 0.44 kg is placed at P, where it floats on the surface of the lake. The amplitude of the vertical oscillation of the ball is 15 cm.

(i) Calculate the difference in the gravitational potential energy ΔE of the ball between its highest and lowest points. [3]

(ii) The frequency of oscillation of the dippers is increased to 4.0 Hz without changing their amplitude of oscillation. The ball remains at the same position on the surface of the lake. State and explain whether the amplitude of the oscillation of the ball increases, decreases or remains the same. [2]

解答 Solution

(a) 沿传播方向传递能量（而不传递物质）的波。 **[B1]**

(b)(i) $T = 1/f = 1/3.0 = 0.33\,\text{s}$ **[A1]**

(ii) 由图，P 处两列波都是波峰，且 P 位于 A 的第 4 个波峰、B 的第 3 个波峰上，路程差为一个波长：$\lambda = 3.9 - 2.4 = 1.5\,\text{m}$ **[C1]**

$v = f\lambda = 3.0 \times 1.5$ **[C1]**

$= 4.5\,\text{m s}^{-1}$ **[A1]**

(iii) $\Delta E = mg\Delta h$ **[C1]**

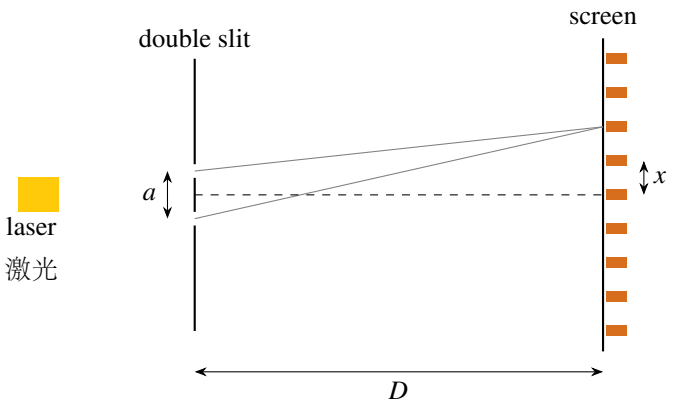
$= 0.44 \times 9.81 \times (2 \times 0.15)$（最高点与最低点相差 2 倍振幅） **[C1]**

$= 1.3\,\text{J}$ **[A1]**

(iv) 频率变为 4.0 Hz 后 $\lambda = 4.5/4.0 = 1.125\,\text{m}$，路程差 1.5 m 不再是波长的整数倍，P 处两列波不再同相； **[B1]**

所以振幅**减小**。 **[B1]**

8.4.1 杨氏双缝干涉 Young's double-slit experiment



公式 8.1 (双缝干涉条纹间距 *Double-slit interference*)

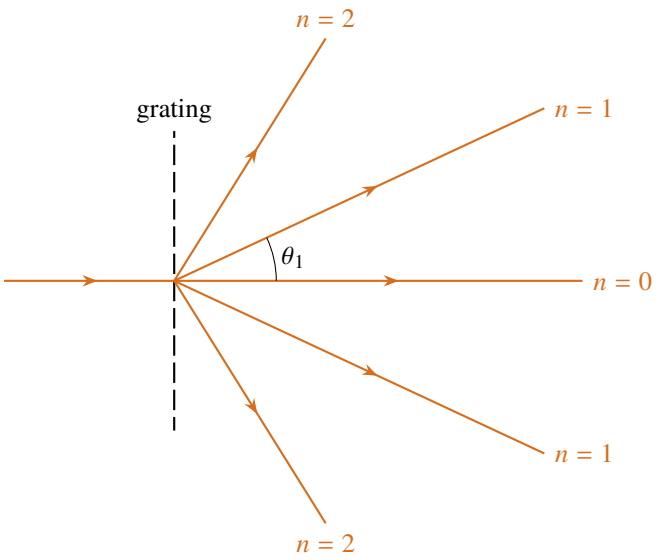
$$\lambda = \frac{ax}{D} \quad \text{即} \quad x = \frac{\lambda D}{a}$$

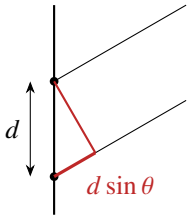
a：双缝间距 (slit separation)；x：相邻亮纹（或暗纹）的间距 (fringe separation)；D：双缝到屏的距离。适用条件 $D \gg a$。

- 增大 D 或减小 a 或用长波长（红光）⇒ 条纹变宽。
- 测 x 时测量**多条**条纹的总宽度再除以间隔数（n 条亮纹之间有 $n-1$ 个间隔），以减小百分比不确定度。
- 白光照射：中央为白色亮纹，两侧为彩色条纹（紫在内、红在外）。
- 也可用声波（两个由同一信号发生器驱动的扬声器）、微波（发射器 + 双缝金属板）演示两源干涉。

8.5 衍射光栅 The diffraction grating

衍射光栅是刻有大量等间距平行狭缝的板。光正入射时，只有在满足相邻缝光程差为波长整数倍的方向上，所有缝出射的光同相，形成非常**明亮**、**尖锐**的极大。





相邻缝光程差 $= d\sin\theta = n\lambda$

公式 8.2 (光栅方程 *Diffraction grating equation*)

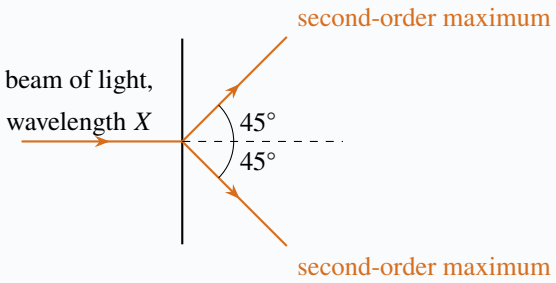
$$d\sin\theta = n\lambda$$

d：相邻缝间距（$d = 1/N$，N 为每米刻线数，注意每 mm 刻线数要换算）；θ：第 n 级极大与入射方向（中央极大）的夹角；n：级数 (order)，$n = 0, 1, 2, \ldots$。

- **最高级数**：$\sin\theta \le 1$，$n_{\max}$ = 不超过 d/λ 的最大整数（$\sin\theta = 1$ 即 90° 时观察不到，若 d/λ 恰为整数，最高级数要减 1）。可观察到的总条数 $= 2n_{\max} + 1$。
- 光栅比双缝更好：极大更亮（缝多，透光多）、更窄更尖锐，角度可以更大且测量更精确。
- **测定波长**：激光正入射光栅，屏上测出第 n 级亮点到中央亮点的距离 y 和光栅到屏距离 L，$\tan\theta = y/L$，再由 $\lambda = d\sin\theta/n$ 计算。测两侧对称亮点之间的距离再取一半，可减小误差。
- 白光：中央为白色，各级为光谱，紫光偏折最小、红光偏折最大（λ 越大 θ 越大）；高级次光谱可能互相重叠。

例题 8.4 9702/12/M/J/26 Q27

The diagram shows a beam of light of wavelength X that is incident normally on a diffraction grating.



The diagram shows only the second-order intensity maxima that are produced at an angle of 45° to the direction of the incident beam of light. A replacement beam of light of a different wavelength Y is now incident normally on the diffraction grating. This new wavelength also produces intensity maxima that are at an angle of 45° to the direction of the incident beam of light. What is a possible value of the ratio X/Y?

A 0.25 **B** 0.75 **C** 1.25 **D** 1.50

解答 Solution

答案 Answer：D

同一角度：$d\sin 45° = 2X = nY$，$\dfrac{X}{Y} = \dfrac{n}{2}$，$n$ 为正整数（$n \neq 2$，因为波长不同）。可能值 $0.5, 1.5, 2, \ldots$，选项中只有 1.50（$n = 3$）。

教学提示 Teacher's note 建议 6 课时，是 AS 最“概念化”的一章。Paper 2 常考：(1) 驻波形成的文字解释——必须提到“反射”、“两列波沿相反方向传播”、“相同频率/波长”、“叠加”；(2) 相干的定义“constant phase difference”；(3) $d\sin\theta = n\lambda$ 求最高级数。实验演示推荐：Melde 弦驻波、微波驻波与双缝、激光双缝与光栅。学生常错：把“5 个波节”当作 5 个半波长；光栅刻线数单位换算。

8.6 本章小结 Summary

物理量/关系	公式	说明
叠加原理	$y = y_1 + y_2$	位移相加
驻波	节–节 $= \lambda/2$；节–腹 $= \lambda/4$	不传递能量
两端固定弦	$L = n\lambda/2$	两端为波节
一端封闭管	$L = (2n-1)\lambda/4$	闭端节、开端腹
相长/相消	$\Delta = n\lambda$ / $(n+\frac{1}{2})\lambda$	两源同相时
双缝干涉	$\lambda = ax/D$	相邻亮纹间距 x
光栅	$d\sin\theta = n\lambda$	$d = 1/N$

8.7 本章练习 Practice questions

练习 8.1 9702/11/M/J/26 Q30

Green light of wavelength 550 nm passes through a double slit. This produces an interference pattern with a fringe spacing of 5.0 cm on a screen. Which changes to the experimental arrangement result in a fringe spacing of 2.5 cm?

	slit separation	distance between slits and screen
A	doubled	unchanged
B	unchanged	doubled
C	halved	doubled
D	doubled	halved

解答 Solution

答案 Answer：A

$x = \lambda D/a$。要 x 减半：a 加倍、D 不变。B：x 加倍；C：x 变 4 倍；D：x 变为 1/4。

练习 8.2 9702/12/M/J/26 Q26

Water waves diffract as they pass through a gap. Which combination produces the greatest amount of diffraction?

	wavelength / m	gap width / m
A	10^{-1}	10^{0}
B	10^{-1}	10^{-2}
C	10^{0}	10^{0}
D	10^{0}	10^{2}

解答 Solution

答案 Answer：C

缝宽与波长**相近**时衍射最显著。（B 的缝宽远小于波长，几乎没有波能通过；A、D 缝宽远大于波长，衍射很弱。）

练习 8.3 9702/14/M/J/26 Q29

Light of wavelength 500 nm is incident normally on a double slit. A flat screen is placed parallel to the plane of the double slit. The screen is at a distance of 3.5 m from the double slit. A series of bright and dark fringes is formed on the screen. The distance from the centre of the first bright fringe to the centre of the tenth bright fringe is 22 mm. What is the separation of the two slits?

A 0.080 mm **B** 0.35 mm **C** 0.72 mm **D** 0.80 mm

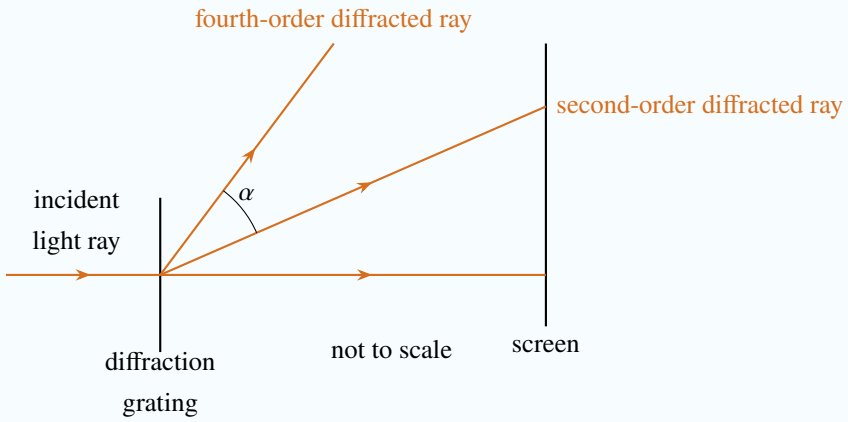
解答 Solution

答案 Answer：C

第 1 到第 10 条亮纹之间有 9 个间隔：$x = 22/9 = 2.44\,\text{mm}$。$a = \dfrac{\lambda D}{x} = \dfrac{500\times10^{-9}\times3.5}{2.44\times10^{-3}} = 7.2\times10^{-4}\,\text{m} = 0.72\,\text{mm}$。（D 是误用 10 个间隔。）

练习 8.4 9702/13/M/J/26 Q29

Light of wavelength 400 nm passes normally through a diffraction grating with a line spacing of 2.0×10^{-6} m and forms a diffraction pattern on a screen.



What is the angle α between the second-order and the fourth-order diffracted rays?

A 24° **B** 30° **C** 53° **D** 77°

解答 Solution

答案 Answer：B

$\sin\theta_2 = \dfrac{2\times400\times10^{-9}}{2.0\times10^{-6}} = 0.40$, $\theta_2 = 23.6°$; $\sin\theta_4 = 0.80$, $\theta_4 = 53.1°$。$\alpha = 53.1-23.6 = 29.5 \approx 30°$。（注意不能用 $\theta \propto n$。）

练习 8.5 9702/21/M/J/26 Q3(c)

A string is fixed to walls at ends P and Q. An oscillator causes transverse oscillations of the string of period 0.24 s. The speed of waves on the string is 75 cm s^{-1}. The distance between P and Q is 45 cm.

(i) Explain how a stationary wave is formed on the string. [2]

(ii) Figure 3.3 shows the string at one instant in time.

P●- - - -

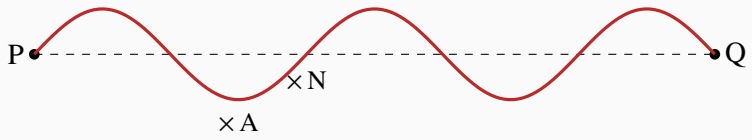
解答 Solution

（本卷未公布评分标准，以下为参考解答。）

(i) 波沿弦传播并在固定端（墙）反射； **[B1]**

入射波与反射波频率（波长）相同、沿相反方向传播，二者叠加形成驻波。 **[B1]**

(ii) $\lambda = vT = 75 \times 0.24 = 18\,\text{cm}$，$PQ = 45\,\text{cm} = 2.5\lambda$，即 5 个半波长（5 个环）。图中时刻弦是直的（各点位移为零），经过 $0.060\,\text{s} = T/4$ 后各点到达最大位移：



正弦形状、两端为零； **[B1]**

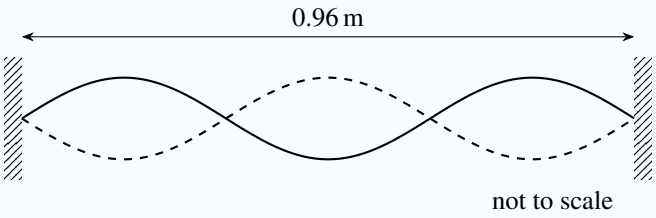
共 5 个半波长（4 个内部波节，等间距 9 cm）。 **[B1]**

(iii) N 标在两端或每隔 9 cm 的位置（如距 P 9 cm 处）；A 标在两相邻波节中点（如距 P 4.5 cm 处）。 **[B1+B1]**

练习 8.6 9702/22/F/M/26 Q5

(a) For a progressive wave, state what is meant by: (i) period; (ii) amplitude. [2]

(b) A stretched string is fixed at both ends and made to vibrate with a frequency of 25 Hz. A stationary wave is formed on the string. The distance between the fixed ends of the string is 0.96 m. Figure 5.1 shows the string at one instant.



(i) Draw a cross (×) at the position of an antinode. [1]

(ii) Show that the speed of a progressive wave on the string is $16\,\text{m}\,\text{s}^{-1}$. [3]

(iii) The frequency of the vibrations is increased to 30 Hz. The speed of the wave on the string does not change. State and explain whether a stationary wave is formed. [2]

(c) A beam of light of wavelength 4.5×10^{-7} m is incident normally on a diffraction grating with 6.7×10^{5} lines per metre. The light passes through the grating and forms a pattern of bright spots on a screen that is parallel to the grating. The angle formed at the diffraction grating between the central bright spot and the bright spot at position X is 37°.

(i) Calculate the order n of the bright spot formed at X. [3]

(ii) The diffraction grating is replaced by a grating with fewer lines per metre. State whether the distance between adjacent bright spots on the screen increases, decreases or remains the same. [1]

解答 Solution

(a)(i) 完成一次全振动的时间（波上某质点）。 **[B1]**

(ii) （质点的）最大位移。 **[B1]**

(b)(i) 标在任一波峰或波谷正上方（如距左端 0.16 m 处）。 **[B1]**

(ii) 图中有 3 个环：$\lambda = 0.96 \times \dfrac{2}{3} = 0.64\,\text{m}$ **[M1]**

$v = f\lambda$ **[C1]**

$= 25 \times 0.64 = 16\,\text{m s}^{-1}$ **[A1]**

(iii) $\lambda = 16/30 = 0.53\,\text{m}$; **[B1]**

0.96 m 不是 $\lambda/2 = 0.267$ m 的整数倍（$0.96/0.267 = 3.6$），所以**不能**形成驻波。 **[B1]**

(c)(i) $n = \dfrac{d \sin\theta}{\lambda}$ **[C1]**

$= \dfrac{(1/6.7 \times 10^5) \sin 37°}{4.5 \times 10^{-7}}$ **[C1]**

$= 2$ **[A1]**

(ii) 减小 (decreases)：刻线数减少 $\Rightarrow$ d 增大 $\Rightarrow$ $\sin\theta = n\lambda/d$ 减小，相邻亮点更靠近。 **[B1]**

第九章 电学 Electricity

考纲要求 Syllabus learning outcomes

9.1.1 understand that an electric current is a flow of charge carriers
理解电流是电荷载流子的流动。

9.1.2 understand that the charge on charge carriers is quantised
理解载流子所带电荷是量子化的。

9.1.3 recall and use $Q = It$
记住并使用 $Q = It$。

9.1.4 use, for a current-carrying conductor, the expression $I = Anvq$, where n is the number density of charge carriers
对载流导体使用 $I = Anvq$，其中 n 是载流子数密度。

9.2.1 define the potential difference across a component as the energy transferred per unit charge
把元件两端的电势差定义为单位电荷转移的能量。

9.2.2 recall and use $V = W/Q$
记住并使用 $V = W/Q$。

9.2.3 recall and use $P = VI$, $P = I^2R$ and $P = V^2/R$
记住并使用 $P = VI$，$P = I^2R$，$P = V^2/R$。

9.3.1 define resistance
定义电阻。

9.3.2 recall and use $V = IR$
记住并使用 $V = IR$。

9.3.3 sketch the I–V characteristics of a metallic conductor at constant temperature, a semiconductor diode and a filament lamp
画出恒温金属导体、半导体二极管和白炽灯的 I–V 特性曲线。

9.3.4 explain that the resistance of a filament lamp increases as current increases because its temperature increases
解释白炽灯电阻随电流增大而增大，是因为温度升高。

9.3.5 state Ohm's law
陈述欧姆定律。

9.3.6 recall and use $R = \rho L/A$
记住并使用 $R = \rho L/A$。

9.3.7 understand that the resistance of a light-dependent resistor (LDR) decreases as the light intensity increases
理解光敏电阻的阻值随光强增大而减小。

9.3.8 understand that the resistance of a thermistor decreases as the temperature increases (it will be assumed that thermistors have a negative temperature coefficient)
理解热敏电阻的阻值随温度升高而减小（假定为负温度系数）。

9.1 电流 Electric current

定义 9.1 (电流 *Electric current*)

An electric current is a flow of charge carriers. Current is the rate of flow of charge: $I = \dfrac{\Delta Q}{\Delta t}$.

电流是电荷载流子的定向流动，大小等于电荷的流动速率。电流是 SI **基本量**，单位 ampere (A)。♣

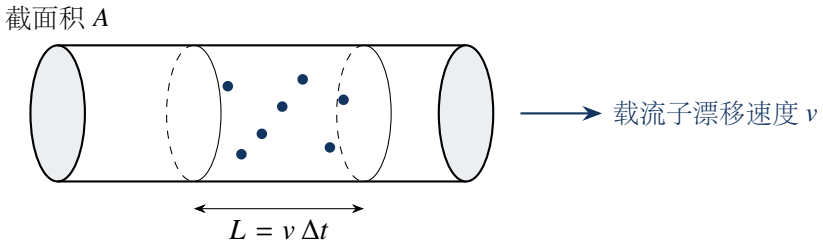
- **载流子** (*charge carriers*)：金属中是自由电子；电解液中是正负离子；半导体中是电子和空穴。
- 电荷的单位 coulomb：1 C = 1 A s，即 1 A 电流在 1 s 内通过某截面的电荷量。
- **电荷量子化** (quantisation of charge)：任何带电体的电荷都是元电荷 $e = 1.60 \times 10^{-19}\,\mathrm{C}$ 的整数倍，$Q = \pm Ne$。所以 $2.4 \times 10^{-19}\,\mathrm{C}$ 这样的电荷不可能出现。
- **约定电流方向** (conventional current)：从电源正极经外电路流向负极，与电子实际流动方向相反。

公式 9.1 (电荷与电流 *Charge and current*)

$$Q = It$$

通过的电子数 $N = \dfrac{Q}{e} = \dfrac{It}{e}$。♠

9.1.1 微观表达式 $I = Anvq$



设导体截面积 A，单位体积内载流子数（数密度）n，每个载流子电荷 q，平均漂移速度 v。在 Δt 内载流子移动 $L = v\Delta t$，通过截面的载流子都在体积 AL 内：

$$\Delta Q = (n\,A\,v\Delta t)\,q \quad \Rightarrow \quad I = \frac{\Delta Q}{\Delta t} = Anvq.$$

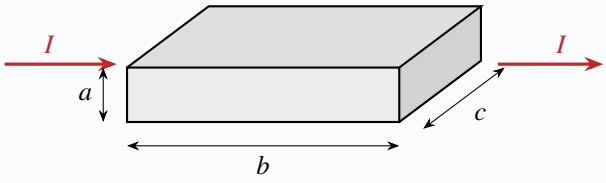
公式 9.2 (电流的微观表达式 *Current and drift velocity*)

$$I = Anvq$$

n 的单位是 m^{-3}。金属 $n \sim 10^{28}\,\mathrm{m}^{-3}$，漂移速度极小（$\sim \mathrm{mm\,s}^{-1}$）。同一导线中电流相同时，截面积越小的部分漂移速度越大；半导体 n 小得多，同样电流下漂移速度大得多。♠

例题 9.1 9702/12/M/J/26 Q31

The diagram shows a metal block.



The block has sides of length a, b and c as shown, and its volume is V. Each charge carrier has a charge $-q$ and the number density of the charge carriers in the metal is n. It takes each charge carrier an average time of t to pass through the block. What is an expression for the current I?

A $I = nqabc$ **B** $I = \dfrac{nqV}{t}$ **C** $I = \dfrac{nqbc}{t}$ **D** $I = nqaV$

解答 Solution

答案 Answer：**B**

电流沿 b 方向，截面积 $A = ac$，漂移速度 $v = b/t$。$I = Anvq = (ac)\,n\,\dfrac{b}{t}\,q = \dfrac{nqV}{t}$。（也可理解为：块中全部载流子 nV 个、电荷 nVq，在时间 t 内全部通过。）

9.2 电势差与电功率

定义 9.2 (电势差 *Potential difference*)

The potential difference (p.d.) across a component is the energy transferred (from electrical to other forms) per unit charge passing through it: $V = \dfrac{W}{Q}$.

电势差（电压）：单位电荷通过元件时转化为其他形式的能量。单位 volt，$1\,\mathrm{V} = 1\,\mathrm{J\,C^{-1}}$。

公式 9.3 (电功率 *Electrical power*)

$$P = VI = I^2R = \frac{V^2}{R}, \qquad W = VIt = VQ$$

推导：$P = \dfrac{W}{t} = \dfrac{VQ}{t} = VI$；再代入 $V = IR$ 得另两式。

考试技巧 Exam tip 选择公式的技巧：串联元件电流相同，比较功率用 $P = I^2R$（电阻大的功率大）；并联元件电压相同，用 $P = V^2/R$（电阻小的功率大）。

9.3 电阻与电阻率

定义 9.3 (电阻 *Resistance*)

Resistance is the ratio of the potential difference across a component to the current in it: $R = \dfrac{V}{I}$.

电阻 = 两端电势差与通过电流之比，单位 ohm，$1\,\Omega = 1\,\mathrm{V\,A^{-1}}$。注意：$R = V/I$ 是电阻的**定义**，对任何元件在任何工作点都成立；而“$V \propto I$”才是欧姆定律。

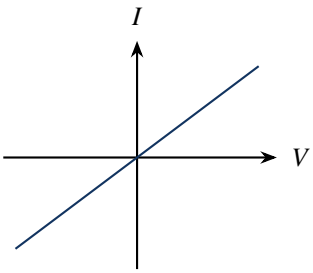
定律 9.1 (欧姆定律 *Ohm's law*)

For a metallic conductor at constant temperature, the current in it is proportional to the potential difference across it.

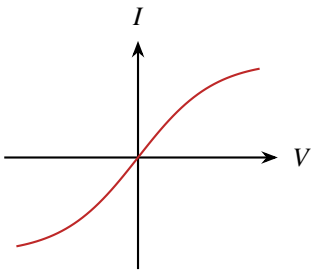
恒温下金属导体中的电流与其两端电势差成正比。遵守欧姆定律的元件称为 ohmic conductor，其 I–V 图是过原点的直线。

9.3.1 I–V 特性曲线

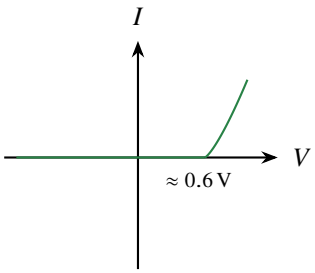
恒温金属导体



白炽灯 filament lamp



半导体二极管 diode



- **金属导体（恒温）**：过原点直线，电阻不变。
- **白炽灯**：曲线逐渐变平（斜率减小）——电流增大，灯丝温度升高，金属离子晶格振动加剧，与自由电子碰撞更频繁，阻碍电子定向移动，**电阻增大**。正反向对称。（若画 V–I 图，曲线则逐渐变陡。）
- **半导体二极管**：反向几乎无电流（电阻极大）；正向电压低于约 0.6 V 时电流很小，超过后电流急剧增大（电阻迅速减小）。只允许电流单向通过。

易错提醒 Common mistakes　从 I–V 图求某点电阻：$R = V/I$ 用**该点的坐标**之比，**不是**该点切线斜率的倒数（除非是过原点直线）。

9.3.2 电阻率

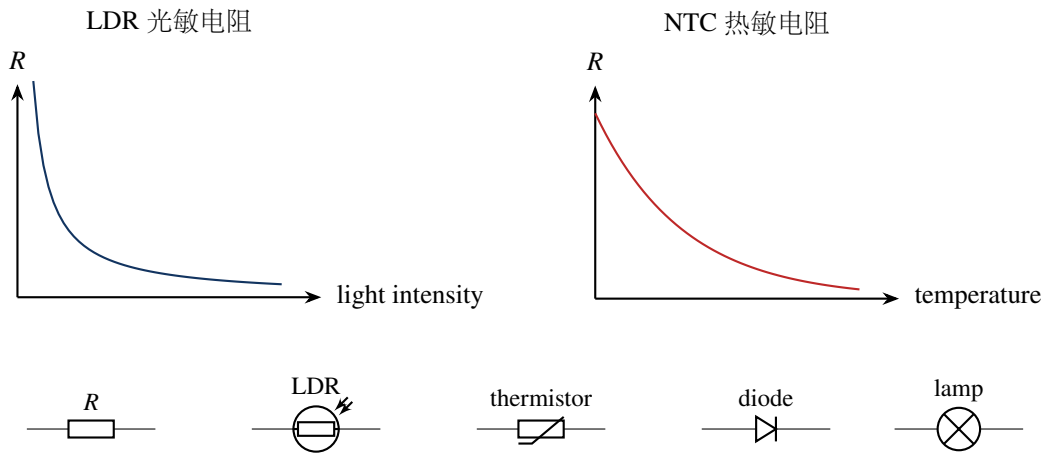
公式 9.4 (电阻率 *Resistivity*)

$$R = \frac{\rho L}{A}$$

ρ 为电阻率 (resistivity)，单位 Ω m，是材料的性质（随温度变化）；L 为长度；A 为横截面积。

- 金属拉长（体积不变）：长度变为 k 倍，截面积变为 $1/k$，电阻变为 k^2 倍。直径变为 1/2（体积不变）：A 变为 1/4，L 变为 4 倍，R 变为 16 倍。
- 典型值：铜 $1.7 \times 10^{-8}\,\Omega\,\text{m}$，镍铬合金 $1.1 \times 10^{-6}\,\Omega\,\text{m}$。金属的电阻率随温度升高而增大。

9.3.3 光敏电阻与热敏电阻



- **LDR**：光强增大，半导体中被激发的载流子增多，电阻**减小**。
- **NTC 热敏电阻**：温度升高，载流子数目增多，电阻**减小**（考纲假定热敏电阻均为负温度系数）。

例题 9.2 9702/24/M/J/26 Q6

A nichrome wire of cross-sectional area $5.7 \times 10^{-8}\,\text{m}^2$ is connected to a power supply. The potential difference (p.d.) across the wire is 230 V. The power dissipated in the wire is 1.0 kW.

(a) Show that the resistance of the wire is 53 Ω. [2]

(b) The resistivity of nichrome is $1.3 \times 10^{-6}\,\Omega\,\text{m}$. Calculate the length of the wire. [2]

(c) The average drift speed of the charge carriers (free electrons) in the nichrome wire is $5.4 \times 10^{-3}\,\text{m s}^{-1}$. Calculate the number density of charge carriers of nichrome. [3]

解答 Solution

(a) $P = V^2/R$ [C1]

$R = \dfrac{V^2}{P} = \dfrac{230^2}{1000} = 53\,\Omega$ [A1]

(b) $R = \rho L/A$, $L = \dfrac{RA}{\rho} = \dfrac{53 \times 5.7 \times 10^{-8}}{1.3 \times 10^{-6}}$ [C1]

$= 2.3\,\text{m}$ [A1]

(c) $n = \dfrac{I}{Aqv}$, $I = P/V = 1000/230 = 4.35\,\text{A}$ [C1]

$n = \dfrac{4.35}{5.7\times10^{-8}\times1.6\times10^{-19}\times5.4\times10^{-3}}$ [C1]

$= 8.8\times10^{28}\,\text{m}^{-3}$ [A1]

例题 9.3 9702/12/M/J/26 Q33

A metal wire with circular cross-section has length L and diameter d. The resistance of the wire is R. The wire is now melted and reformed into a new wire of diameter $0.50d$. The volume of the wire is unchanged. What is the resistance of the new wire?

A R **B** $2R$ **C** $4R$ **D** $16R$

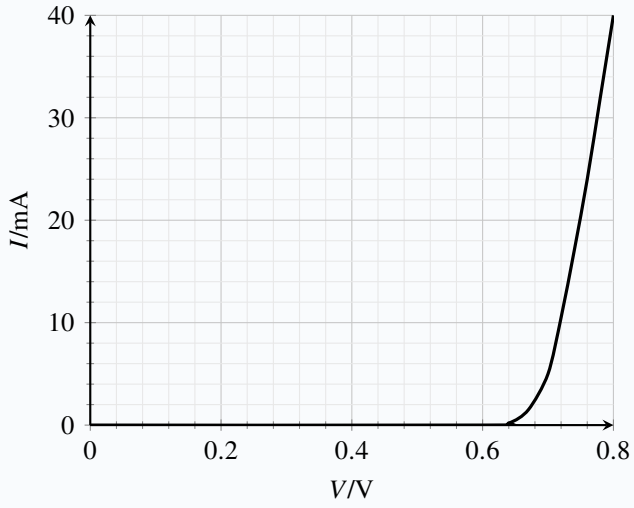
解答 Solution

答案 Answer: D

直径减半 $\Rightarrow$ 截面积 $A' = A/4$；体积不变 $\Rightarrow L' = 4L$。$R' = \dfrac{\rho(4L)}{A/4} = 16R$。

例题 9.4 9702/24/M/J/26 Q5

The I–V characteristic of an electrical component X is shown in Figure 5.1.



(a) State the name of electrical component X. [1]

(b) A metallic conductor has a constant resistance of 40 Ω. On Figure 5.1, sketch the I–V characteristic of this conductor between $V = 0$ and $V = 0.80$ V. [2]

(c) Component X is connected in series with a resistor of resistance 20 Ω, an ammeter and a cell. The current in the ammeter is 5.0 mA and the cell has negligible internal resistance. Show that the electromotive force (e.m.f.) of the cell is 0.80 V. [3]

(d) The terminals of the cell in (c) are now reversed without making any other changes to the circuit. State the reading on the ammeter. [1]

(e) The resistor of resistance 20 Ω and component X are now connected in parallel to the 0.80 V cell, with

an ammeter measuring the total current. The current in component X is not negligible. Use Figure 5.1 to determine the current in the ammeter. [3]

解答 Solution

(a) （半导体）二极管 diode。 **[A1]**

(b) 过原点的直线； **[B1]**

经过 (0.80 V, 20 mA)（$I = 0.80/40 = 0.020$ A）。 **[B1]**

(c) 由图，$I = 5.0$ mA 时 X 两端电压 = 0.70 V； **[C1]**

电阻两端电压 $= 0.005 \times 20 = 0.10$ V； **[C1]**

e.m.f. $= 0.70 + 0.10 = 0.80$ V。 **[A1]**

(d) 0（二极管反接，几乎无电流）。 **[A1]**

(e) 电阻电流 $= 0.80/20 = 0.040$ A； **[C1]**

X 两端 0.80 V，由图 X 的电流 = 40 mA = 0.040 A； **[C1]**

总电流 = 0.080 A。 **[A1]**

教学提示 Teacher's note 建议 5 课时。三个高频定义：电流（flow of charge carriers / rate of flow of charge）、电势差（energy transferred per unit charge）、电阻（ratio of p.d. to current）。$I = Anvq$ 的计算常结合 $P = VI$ 求电流；电阻率题注意直径换半径、mm 换 m。I–V 图的读图与草图是 Paper 2 常考 3–4 分题；解释白炽灯电阻变化时必须写“温度升高”。

9.4 本章小结 Summary

物理量/关系	公式	说明
电荷	$Q = It$	量子化：$Q = Ne$
微观电流	$I = Anvq$	n：数密度
电势差	$V = W/Q$	J/C
电功率	$P = VI = I^2R = V^2/R$	
电阻	$R = V/I$	定义，任何点适用
电阻率	$R = \rho L/A$	Ω m
LDR / NTC	光强↑ / 温度↑ $\Rightarrow R\downarrow$	

9.5 本章练习 Practice questions

练习 9.1 9702/11/M/J/26 Q31

What is the definition of the potential difference (p.d.) across a component?

A the energy transferred per unit charge

B the energy transferred per unit current

C the power transferred per unit charge

D the power transferred per unit current

解答 Solution

答案 Answer：A

练习 9.2 9702/13/M/J/26 Q32

A wire carries a current of 2.0 A for 1.0 hour. How many electrons pass a point in the wire in this time?

A 1.2×10^{-15} **B** 7.2×10^{3} **C** 1.3×10^{19} **D** 4.5×10^{22}

解答 Solution

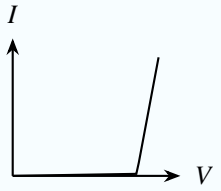
答案 Answer：D

$Q = It = 2.0 \times 3600 = 7200\,\text{C}$；$N = Q/e = 7200/(1.60 \times 10^{-19}) = 4.5 \times 10^{22}$。

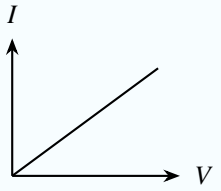
练习 9.3 9702/13/M/J/26 Q33

Which graph shows the I–V characteristic of a filament lamp?

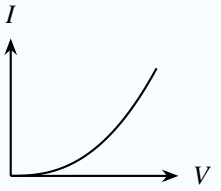
A



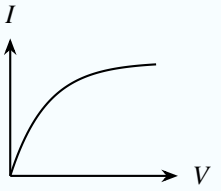
B



C



D



解答 Solution

答案 Answer：D

白炽灯电流增大，温度升高，电阻增大，I–V 图斜率逐渐减小。A 为二极管（正向），B 为恒温金属，C 为电阻随电压减小的元件（如热敏电阻）。

练习 9.4 9702/13/M/J/26 Q34

A straight copper wire, of diameter 4.2×10^{-4} m, has a number density of free electrons of $8.5 \times 10^{28}\,\text{m}^{-3}$. In a given time interval, a charge of 0.15 C moves through the wire. What is the average displacement of the free electrons along the wire in this time interval?

A 3.3×10^{-8} m　　**B** 2.0×10^{-5} m

C 8.0×10^{-5} m　　**D** 2.5×10^{-4} m

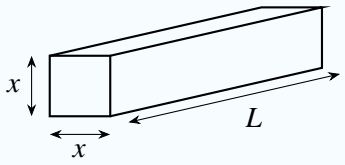
解答 Solution

答案 Answer：C

$Q = nAxq$（体积 Ax 内的电荷全部通过），$A = \pi(2.1 \times 10^{-4})^2 = 1.39 \times 10^{-7}\,\text{m}^2$。$x = \dfrac{Q}{nAq} = \dfrac{0.15}{8.5 \times 10^{28} \times 1.39 \times 10^{-7} \times 1.6 \times 10^{-19}} = 8.0 \times 10^{-5}$ m。

练习 9.5 9702/12/F/M/26 Q34

Conducting putty is a substance that conducts electricity and can be moulded into various shapes. A cuboid made from this putty has length L and width x. The resistance between the square faces of the cuboid is 15 Ω.



Some more of the putty is moulded into a cylinder that has length L. The resistance between the circular faces of the cylinder is also 15 Ω. What is the radius of the cylinder?

A $\dfrac{x}{2}$　　**B** $\dfrac{x}{\sqrt{\pi}}$　　**C** $\dfrac{x}{\pi}$　　**D** $\dfrac{2x}{\sqrt{\pi}}$

解答 Solution

答案 Answer：B

同材料、同长度、同电阻 ⇒ 截面积相同：$\pi r^2 = x^2$，$r = x/\sqrt{\pi}$。

练习 9.6 9702/12/M/J/26 Q32

An electrical component is connected into a circuit. Which component and which change causes an increase in the resistance of the component?

	component	change
A	filament lamp	increase in electric current
B	light-dependent resistor (LDR)	increase in light intensity
C	semiconductor diode	increase in electric current
D	thermistor	increase in temperature

解答 Solution

答案 Answer：**A**

白炽灯电流增大，温度升高，电阻增大。B、D 电阻减小；二极管正向电流增大时电阻减小。

第十章 直流电路 D.C. circuits

考纲要求 Syllabus learning outcomes

10.1.1 recall and use the circuit symbols shown in section 6 of the syllabus
记住并使用考纲第 6 节所列的电路符号。

10.1.2 draw and interpret circuit diagrams containing the circuit symbols shown in section 6 of the syllabus
绘制并解读包含这些符号的电路图。

10.1.3 define and use the electromotive force (e.m.f.) of a source as energy transferred per unit charge in driving charge around a complete circuit
定义并使用电动势：电源驱动单位电荷绕完整电路一周所转移的能量。

10.1.4 distinguish between e.m.f. and potential difference (p.d.) in terms of energy considerations
从能量角度区分电动势与电势差。

10.1.5 understand the effects of the internal resistance of a source of e.m.f. on the terminal potential difference
理解电源内阻对端电压的影响。

10.2.1 recall Kirchhoff's first law and understand that it is a consequence of conservation of charge
记住基尔霍夫第一定律，理解它是电荷守恒的结果。

10.2.2 recall Kirchhoff's second law and understand that it is a consequence of conservation of energy
记住基尔霍夫第二定律，理解它是能量守恒的结果。

10.2.3 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in series
用基尔霍夫定律推导串联电阻公式。

10.2.4 use the formula for the combined resistance of two or more resistors in series
使用串联电阻公式。

10.2.5 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in parallel
用基尔霍夫定律推导并联电阻公式。

10.2.6 use the formula for the combined resistance of two or more resistors in parallel
使用并联电阻公式。

10.2.7 use Kirchhoff's laws to solve simple circuit problems
用基尔霍夫定律解决简单电路问题。

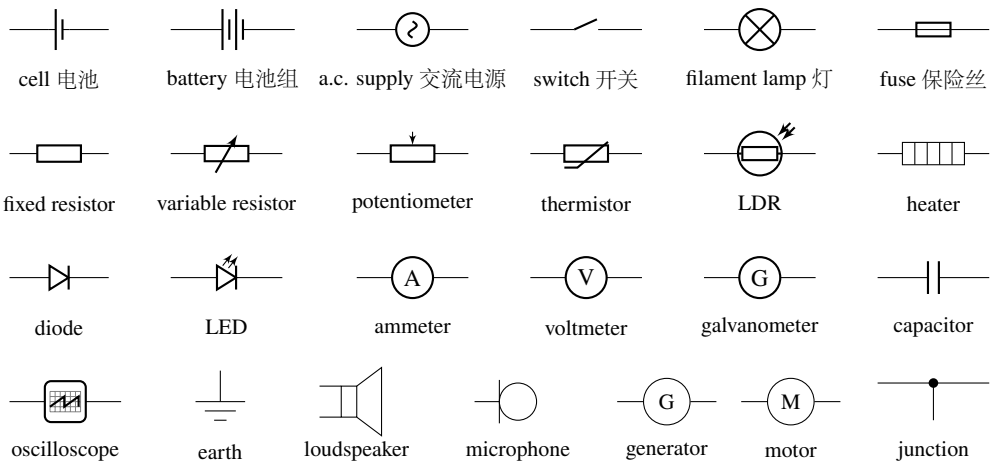
10.3.1 understand the principle of a potential divider circuit
理解分压电路的原理。

10.3.2 recall and use the principle of the potentiometer as a means of comparing potential differences
记住并使用电位差计比较电势差的原理。

10.3.3 understand the use of a galvanometer in null methods
理解检流计在零示法中的应用。

10.3.4 explain the use of thermistors and light-dependent resistors in potential dividers to provide a potential difference that is dependent on temperature and light intensity
解释在分压电路中用热敏电阻和光敏电阻获得随温度、光强变化的电势差。

10.1 电路符号 Circuit symbols



10.2 电动势与内阻

定义 10.1 (电动势 *Electromotive force (e.m.f.)*)

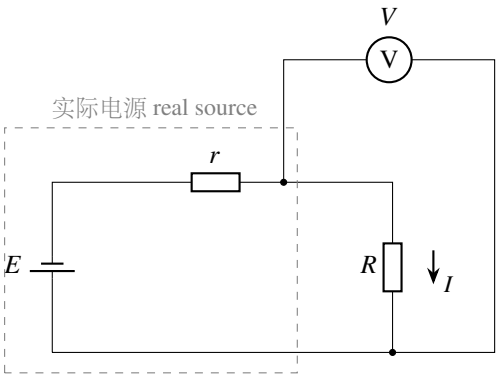
The e.m.f. of a source is the energy transferred (from other forms to electrical energy) per unit charge in driving charge around a complete circuit: $E = \dfrac{W}{Q}$.

电动势：电源驱动单位电荷绕**完整电路**一周时，由其他形式能量（化学能、机械能等）转化为电能的量。单位 V。

e.m.f. 与 p.d. 的区别（从能量转化的方向看）：

	电动势 e.m.f.	电势差 p.d.
能量转化	其他形式能 → 电能	电能 → 其他形式能（热、光等）
发生位置	电源内部	电路元件（负载）上
定义	驱动单位电荷绕**整个电路**一周所做的功	单位电荷通过**元件**时转化的电能

10.2.1 内阻与端电压

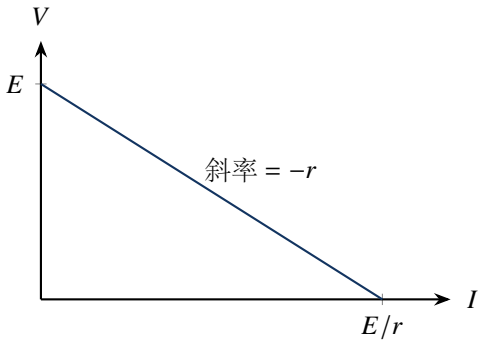


实际电源可以看作理想电动势 E 与**内阻** (*internal resistance*) r 串联。电流 I 流过内阻时有"lost volts" Ir，所以电源两端的**端电压** (*terminal p.d.*) V 小于 E：

公式 10.1 (全电路方程　*E.m.f. and internal resistance*)

$$E = V + Ir = I(R + r) \qquad \Rightarrow \qquad V = E - Ir$$

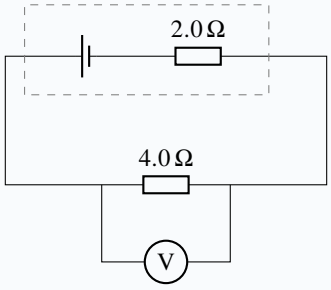
- 外电阻 R 减小（如并联更多元件）$\Rightarrow$ 电流 I 增大 $\Rightarrow$ 内阻上电压 Ir 增大 $\Rightarrow$ 端电压 V 减小。
- 电路断开（$I = 0$）时，$V = E$：用理想（高阻）电压表直接测电源两端，读数等于电动势。
- 内阻消耗功率 I^2r，使电源发热、效率降低；短路时 $I = E/r$ 很大，危险。



测定 E 和 r：改变外电阻，记录多组 (I, V)，作 V–I 图：纵截距 $= E$，斜率 $= -r$。

例题 10.1　9702/12/M/J/26 Q35

The diagram shows a circuit with a battery connected to a resistor with a resistance of 4.0 Ω.



The battery has an internal resistance of 2.0 Ω. The voltmeter reads 12 V. A second resistor of resistance

4.0 Ω is connected in parallel to the first resistor. What is the new reading on the voltmeter?

A 6.0 V **B** 9.0 V **C** 12 V **D** 24 V

解答 Solution

答案 Answer：B

原来：$I = 12/4.0 = 3.0\,\text{A}$，$E = I(R+r) = 3.0\times6.0 = 18\,\text{V}$。并联后外电阻 2.0 Ω，$I = 18/(2.0+2.0) = 4.5\,\text{A}$，读数 $V = 4.5\times2.0 = 9.0\,\text{V}$。

10.3 基尔霍夫定律 Kirchhoff's laws

定律 10.1 (基尔霍夫第一定律 *Kirchhoff's first law*)

The sum of the currents entering a junction is equal to the sum of the currents leaving the junction: $\sum I_{\text{in}} = \sum I_{\text{out}}$.

流入任一节点的电流之和等于流出该节点的电流之和。它是**电荷守恒**的结果：电荷不会在节点处积累或消失。

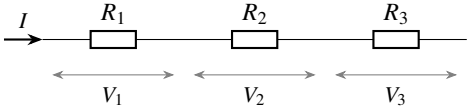
定律 10.2 (基尔霍夫第二定律 *Kirchhoff's second law*)

Around any closed loop in a circuit, the sum of the e.m.f.s is equal to the sum of the p.d.s: $\sum E = \sum IR$.

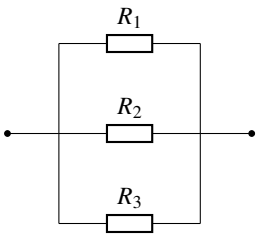
沿电路中任一闭合回路，电动势之和等于电势差之和。它是**能量守恒**的结果：单位电荷绕回路一周，获得的电能等于转化掉的电能。

使用第二定律的符号规则：选定回路绕行方向；电动势方向（从负极到正极）与绕行方向一致取正；电流方向与绕行方向一致时 IR 取正。只有电阻没有电源的回路，各 IR 的代数和为零。

10.3.1 串联与并联电阻公式的推导



串联 series：电流相同



并联 parallel：电压相同

串联：同一电流 I 流过各电阻。由第二定律，总电压 $V = V_1 + V_2 + V_3$，即 $IR = IR_1 + IR_2 + IR_3$，消去 I：

$$R = R_1 + R_2 + R_3$$

并联：各电阻两端电压都为 V。由第一定律，$I = I_1 + I_2 + I_3$，即 $\dfrac{V}{R} = \dfrac{V}{R_1} + \dfrac{V}{R_2} + \dfrac{V}{R_3}$，消去 V：

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

考试技巧 Exam tip 两个电阻并联：$R = \dfrac{R_1 R_2}{R_1 + R_2}$。$n$ 个相同电阻 R_0 并联：$R = R_0/n$。并联总电阻**小于**其中最小的一个。

例题 10.2 9702/22/M/J/26 Q5

(a) Three resistors of resistances R_1, R_2 and R_3 are connected in a network in parallel. The currents in the resistors are I_1, I_2 and I_3 respectively. The total current entering the network is I_T. The potential difference (p.d.) across the resistors is V. Derive the equation

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

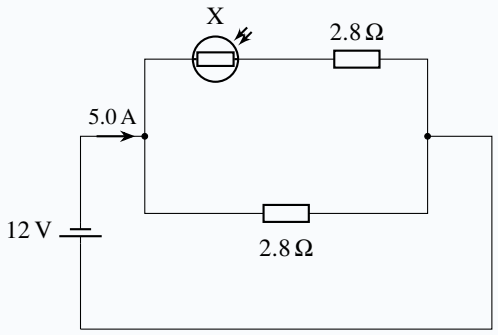
where R_T is the total resistance of the network. [2]

(b) Two identical resistance wires, each of resistance 2.8 Ω, are connected in parallel to a battery with negligible internal resistance and electromotive force (e.m.f.) 12 V.

(i) Calculate the current in the battery. [2]

(ii) Each resistance wire has a cross-sectional area of 6.1×10^{-6} m^2. The wires are made from a metal with a number density of charge carriers (free electrons) of 8.5×10^{28} m^{-3}. Calculate the average drift speed of the charge carriers in the wires. [2]

(c) A component X is added to the circuit in (b) as shown in Figure 5.3.



两根电阻丝 resistance wires（各 2.8 Ω）

The current in the battery is now 5.0 A. The resistance of each resistance wire remains 2.8 Ω.

(i) State the name of component X. [1]

(ii) Determine the resistance of component X. [3]

(iii) A change in the environment of component X causes the current in the battery to decrease. The resistance of the resistance wires remains constant. State and explain how the environment of X has changed. [2]

解答 Solution

(a) 由基尔霍夫第一定律 $I_T = I_1 + I_2 + I_3$，各电阻电压都为 V：$\dfrac{V}{R_T} = \dfrac{V}{R_1} + \dfrac{V}{R_2} + \dfrac{V}{R_3}$ **[B1]**

两边除以 V 得 $\dfrac{1}{R_T} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}$ **[B1]**

(b)(i) $R_T = \left(\dfrac{1}{2.8} + \dfrac{1}{2.8}\right)^{-1} = 1.4\,\Omega$ **[C1]**

$I = 12/1.4 = 8.6\,\text{A}$ **[A1]**

(ii) 每根导线电流 $8.6/2 = 4.3\,\text{A}$：$v = \dfrac{I}{Anq} = \dfrac{4.3}{6.1\times10^{-6}\times8.5\times10^{28}\times1.6\times10^{-19}}$ **[C1]**

$= 5.2\times10^{-5}\,\text{m s}^{-1}$ **[A1]**

(c)(i) 光敏电阻 LDR。 **[B1]**

(ii) 下支路电流 $= 12/2.8 = 4.29\,\text{A}$，上支路电流 $I_X = 5.0 - 4.29 = 0.71\,\text{A}$ **[C1]**

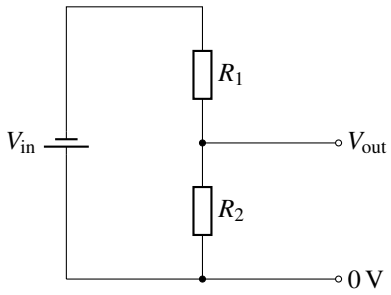
上支路总电阻 $= 12/0.71 = 16.8\,\Omega$ **[C1]**

$R_X = 16.8 - 2.8 = 14\,\Omega$ **[A1]**

(iii) 电池电流减小 $\Rightarrow$ 总电阻增大 $\Rightarrow$ X 的电阻增大； **[M1]**

所以照射到 X 上的光强**减弱**（环境变暗）。 **[A1]**

10.4 分压电路 Potential dividers



$$V_{out} = \frac{R_2}{R_1 + R_2} V_{in}$$

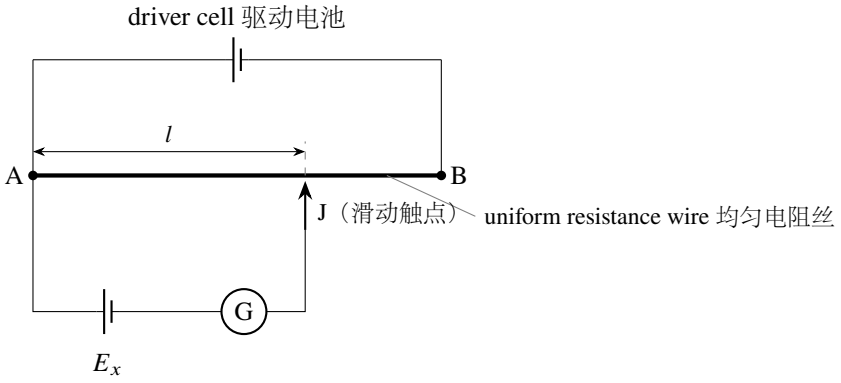
$\dfrac{V_1}{V_2} = \dfrac{R_1}{R_2}$（串联电压按电阻比分配）

两个电阻串联在电源两端，电流相同，电压按电阻成正比分配。输出电压可从其中一个电阻两端引出。

传感器电路：把 R_2 换成热敏电阻或 LDR：

- 热敏电阻在下（R_2 位置）：温度升高 $\Rightarrow$ R_2 减小 $\Rightarrow$ V_{out} **减小**。若要温度升高时输出增大，把热敏电阻与定值电阻交换位置。
- LDR 在下：光变暗 $\Rightarrow$ R_{LDR} 增大 $\Rightarrow$ V_{out} **增大**（可用于路灯自动开启电路）。
- 用可变电阻代替定值电阻，可以调节触发的温度或光强。

10.4.1 电位差计与零示法 Potentiometer and null methods

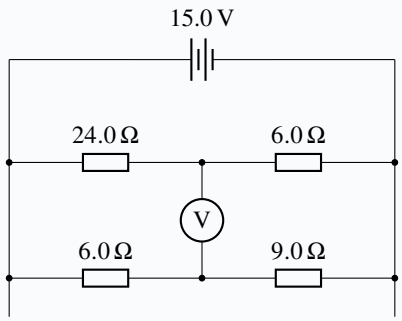


原理：均匀电阻丝 AB 两端加恒定电压，丝上电势沿长度均匀下降，任意一段的电压与长度成正比（$V \propto l$）。把待测电池（或电压）与检流计 G 接到 A 和滑动触点 J 之间，移动 J 直到**检流计读数为零**（平衡点 balance point）：此时待测电压等于 AJ 段的电压，且待测支路中没有电流。

- 比较两个电动势：$\dfrac{E_1}{E_2} = \dfrac{l_1}{l_2}$。
- **零示法 null method** 的优点：平衡时待测电池中无电流，没有内阻上的“lost volts”，测得的是真正的电动势；检流计只需判断有无电流，不需要校准。
- 待测电池的正极必须与驱动电池的正极接在同一端（A），否则找不到平衡点；待测电压必须小于 AB 两端电压。

例题 10.3 9702/11/M/J/26 Q36

A circuit consists of a battery, a high-resistance voltmeter and four fixed resistors as shown. The battery has an electromotive force (e.m.f.) of 15.0 V and negligible internal resistance.



What is the reading on the voltmeter?

A 3.0 V **B** 6.0 V **C** 9.0 V **D** 12.0 V

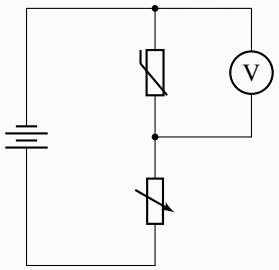
解答 Solution

答案 Answer：B

上支路：24.0 Ω 两端电压 $= 15.0 \times \dfrac{24}{30} = 12.0$ V；下支路：6.0 Ω 两端电压 $= 15.0 \times \dfrac{6}{15} = 6.0$ V。两中点相对左端的电势降落分别为 12.0 V 和 6.0 V，电压表读数 $= 12.0 - 6.0 = 6.0$ V。

例题 10.4 9702/13/M/J/26 Q35

The diagram shows a thermistor and a variable resistor connected to a battery.



The temperature of the thermistor decreases. What happens to the reading on the voltmeter, and which change to the resistance of the variable resistor is needed to cause the voltmeter reading to return to its initial value?

	voltmeter reading	change needed to resistance of variable resistor
A	decreases	decrease
B	decreases	increase
C	increases	decrease
D	increases	increase

解答 Solution

答案 Answer：D

温度降低，热敏电阻阻值增大，分得电压增大，电压表读数**增大**。要使读数恢复，需让可变电阻也按比例增大（两者电阻之比恢复原值），即**增大**可变电阻。

教学提示 Teacher's note 建议 6 课时。重点难点：(1) e.m.f. 与 p.d. 的定义及区别（能量转化方向）；(2) $E = V + Ir$ 的综合题，如并联后端电压变化；(3) 基尔霍夫定律分别对应电荷守恒与能量守恒（选择题高频）；(4) 串并联公式的推导（2 分"derive"题）；(5) 传感器分压电路的定性分析——要求学生写出完整因果链："温度↓ → R_{th}↑ → 其分得电压↑"。零示法注意讲清"平衡时检流计无电流"的意义。

10.5 本章小结 Summary

物理量/关系	公式	说明
电动势	$E = W/Q$	绕整个电路
内阻	$E = V + Ir = I(R + r)$	V–I 图斜率 $-r$
基尔霍夫第一定律	$\sum I_{\text{in}} = \sum I_{\text{out}}$	电荷守恒
基尔霍夫第二定律	$\sum E = \sum IR$	能量守恒
串联	$R = R_1 + R_2 + \cdots$	
并联	$1/R = 1/R_1 + 1/R_2 + \cdots$	
分压	$V_{\text{out}} = \dfrac{R_2}{R_1 + R_2} V_{\text{in}}$	
电位差计	$E_1/E_2 = l_1/l_2$	零示法

10.6 本章练习 Practice questions

练习 10.1 9702/12/M/J/26 Q34

A battery with internal resistance is connected by wires to an external resistor. Which statement describes the electromotive force (e.m.f.) of the battery?

A It is the energy transferred per unit charge in driving charge through a connecting wire.

B It is the energy transferred per unit charge in driving charge through the complete circuit.

C It is the energy transferred per unit charge in driving charge through the external resistor.

D It is the energy transferred per unit charge in driving charge through the internal resistance.

解答 Solution

答案 Answer：B

练习 10.2 9702/12/M/J/26 Q36

Kirchhoff's first and second laws are a consequence of the conservation of different quantities. What are those quantities?

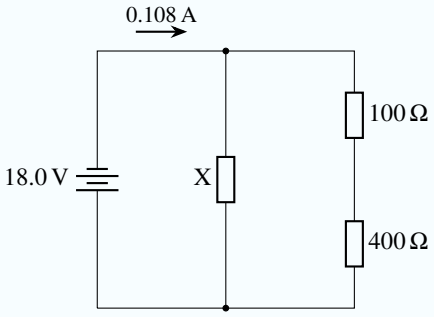
	Kirchhoff's first law	Kirchhoff's second law
A	charge	energy
B	energy	current
C	current	charge
D	energy	charge

解答 Solution

答案 Answer：**A**

练习 10.3 9702/12/M/J/26 Q37

A battery of electromotive force (e.m.f.) 18.0 V and negligible internal resistance is connected to a network of three fixed resistors as shown.



The current in the battery is 0.108 A. What is the resistance of resistor X?

A 125 Ω **B** 167 Ω **C** 250 Ω **D** 500 Ω

解答 Solution

答案 Answer：**C**

右支路 500 Ω 的电流 = 18.0/500 = 0.036 A；X 的电流 = 0.108 − 0.036 = 0.072 A；R_X = 18.0/0.072 = 250 Ω。

练习 10.4 9702/12/F/M/26 Q37

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected in series with a resistor of resistance 6.0 Ω and a variable resistor of resistance from zero to 4.0 Ω. A voltmeter is connected across the variable resistor. The resistance of the variable resistor is changed. What is the range of the voltmeter reading?

A 0 V – 2.4 V **B** 0 V – 3.6 V **C** 2.4 V – 6.0 V **D** 3.6 V – 6.0 V

解答 Solution

答案 Answer：**A**

可变电阻为 0 时读数 0；为 4.0 Ω 时读数 = $6.0 \times \dfrac{4.0}{10.0}$ = 2.4 V。

练习 10.5 9702/23/M/J/26 Q4

A light-dependent resistor (LDR) is connected across the terminals of a cell that has internal resistance. A voltmeter is connected across the LDR.

(a) Draw a circuit diagram to show this arrangement. [3]

(b) The cell has an electromotive force (e.m.f.) of 4.90 V. At a particular light intensity, the LDR has a

resistance of 230 Ω and a potential difference (p.d.) across it of 4.78 V.

(i) Identify one possible source of systematic error in measuring the p.d. across the LDR using a digital voltmeter. [1]

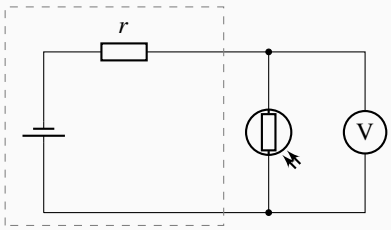
(ii) Calculate the power dissipated in the LDR. [2]

(iii) Determine the internal resistance of the cell. [2]

(iv) The intensity of the light illuminating the LDR is gradually decreased. State what happens (decreases / unchanged / increases) to: the resistance of the LDR; the current in the cell; the p.d. across the LDR. [3]

解答 Solution

(a)



电池符号（可含内阻）; **[B1]**

LDR 符号正确并接在电池两端; **[B1]**

电压表与 LDR 并联。 **[B1]**

(b)(i) 零点误差 / 校准误差（或接线接触电阻）。 **[B1]**

(ii) $P = V^2/R = 4.78^2/230$ **[C1]**

$= 0.099\,\text{W}$ **[A1]**

(iii) $E = V + Ir$，$I = 4.78/230 = 0.0208\,\text{A}$：$r = \dfrac{4.90 - 4.78}{0.0208}$ **[C1]**

$= 5.8\,\Omega$ **[A1]**

(iv) LDR 电阻：increases; **[B1]**

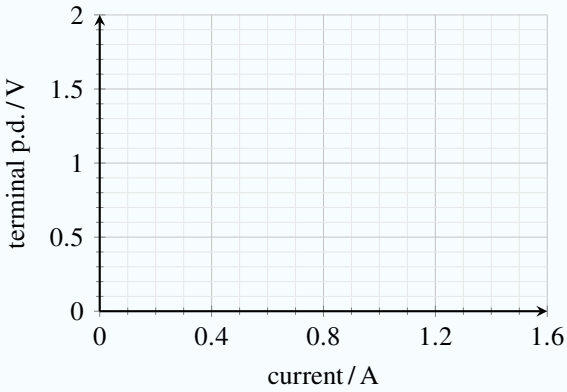
电池电流：decreases; **[B1]**

LDR 两端电压：increases（电流减小，内阻上的 lost volts 减小）。 **[B1]**

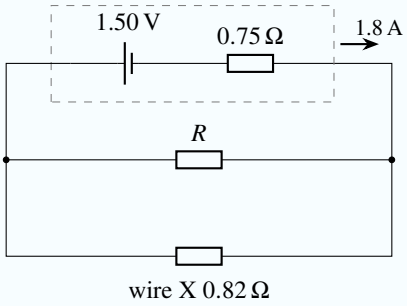
练习 10.6 9702/22/F/M/26 Q6

(a) State what is meant by an electric current. [1]

(b) A cell has electromotive force (e.m.f.) 1.50 V and internal resistance 0.75 Ω. Draw the variation of the terminal potential difference (p.d.) with current in the cell for current between 0 A and 1.6 A. [2]



(c) A metal wire X has length 6.8 m and cross-sectional area 2.0×10^{-7} m^2. The resistance of the wire is 0.82 Ω. The wire is connected in parallel with a fixed resistor of resistance R, and this combination is connected to the cell in (b).



The current in the cell is 1.8 A.

(i) Calculate the resistivity ρ of the metal from which wire X is made. [2]

(ii) Calculate R. [3]

(iii) Calculate the power dissipated in the internal resistance of the cell. [2]

解答 Solution

(a) 电荷载流子的流动。 **[B1]**

(b) 从 (0, 1.50 V) 出发、斜率为负的直线; **[M1]**

经过 (1.60 A, 0.30 V)（$V = 1.50 - 1.6 \times 0.75 = 0.30$）。 **[A1]**

(c)(i) $\rho = \dfrac{RA}{L} = \dfrac{0.82 \times 2.0 \times 10^{-7}}{6.8}$ **[C1]**

$= 2.4 \times 10^{-8}\ \Omega\,\text{m}$ **[A1]**

(ii) 端电压 $= 1.50 - 1.8 \times 0.75 = 0.15$ V **[C1]**

定值电阻的电流 $= 1.8 - 0.15/0.82 = 1.6$ A **[C1]**

$R = 0.15/1.6 = 0.093\ \Omega$ **[A1]**

(iii) $P = I^2 r = 1.8^2 \times 0.75$ **[C1]**

$= 2.4$ W **[A1]**

第十一章 粒子物理 Particle physics

考纲要求 Syllabus learning outcomes

11.1.1 infer from the results of the α-particle scattering experiment the existence and small size of the nucleus
由 α 粒子散射实验结果推断原子核的存在及其很小的尺寸。

11.1.2 describe a simple model for the nuclear atom to include protons, neutrons and orbital electrons
描述包含质子、中子和核外电子的简单核式原子模型。

11.1.3 understand that an element can exist in various isotopic forms, each with a different number of neutrons in the nucleus
理解元素可以有多种同位素，各同位素的中子数不同。

11.1.4 use the notation ${}^{A}_{Z}X$ for the representation of nuclides
使用 ${}^{A}_{Z}X$ 表示核素。

11.1.5 understand that nucleon number and charge are conserved in nuclear processes
理解核反应中核子数和电荷守恒。

11.1.6 describe the composition, mass and charge of α-, β- and γ-radiations (both β^- (electrons) and β^+ (positrons) are included)
描述 α、β（含 β^- 和 β^+）、γ 辐射的组成、质量和电荷。

11.1.7 understand that an antiparticle has the same mass but opposite charge to the corresponding particle, and that a positron is the antiparticle of an electron
理解反粒子与对应粒子质量相同、电荷相反；正电子是电子的反粒子。

11.1.8 state that (electron) antineutrinos are produced during β^- decay and (electron) neutrinos are produced during β^+ decay
陈述 β^- 衰变产生（电子）反中微子，β^+ 衰变产生（电子）中微子。

11.1.9 understand that α-particles have discrete energies but that β-particles have a continuous range of energies because (anti)neutrinos are emitted in β-decay
理解 α 粒子能量是分立的，而 β 粒子能量连续分布，因为 β 衰变同时放出（反）中微子。

11.1.10 represent α- and β-decay by a radioactive decay equation
用衰变方程表示 α 衰变和 β 衰变。

11.1.11 use the unified atomic mass unit (u) as a unit of mass
使用原子质量单位 u 作为质量单位。

11.2.1 understand that a quark is a fundamental particle and that there are six flavours (types) of quark: up, down, strange, charm, top and bottom
理解夸克是基本粒子，有六种味：上、下、奇、粲、顶、底。

11.2.2 recall and use the charge of each flavour of quark and understand that its respective antiquark has the opposite charge
记住并使用各种夸克的电荷，理解反夸克电荷相反。

11.2.3 recall that protons and neutrons are not fundamental particles and describe protons and neutrons in terms of their quark composition
记住质子和中子不是基本粒子，并用夸克组成描述它们。

11.2.4 understand that a hadron may be either a baryon (consisting of three quarks) or a meson (consisting of one

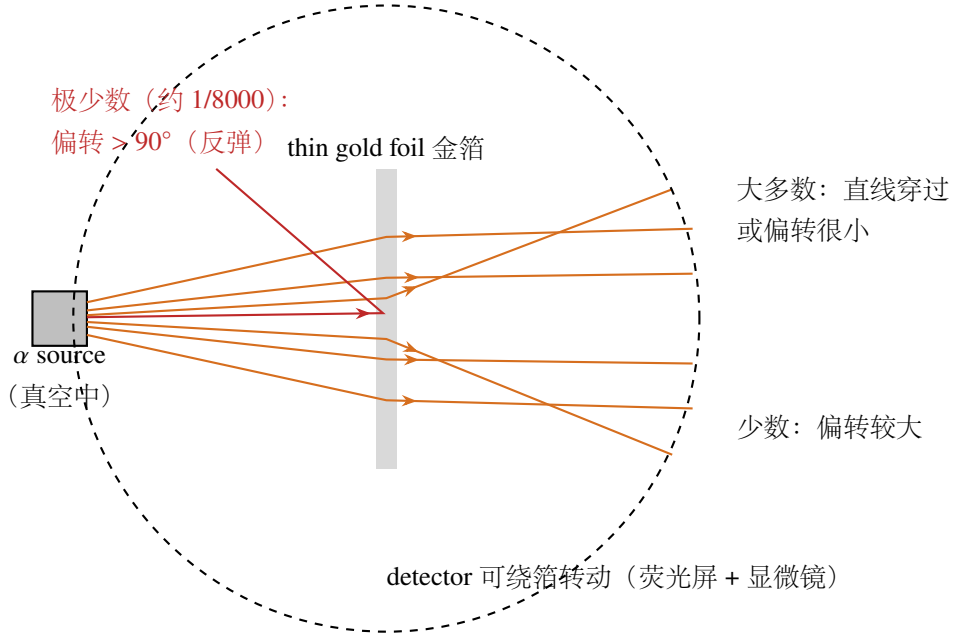
quark and one antiquark)
理解强子分为重子（三个夸克）和介子（一个夸克和一个反夸克）。

11.2.5 describe the changes to quark composition that take place during β^- and β^+ decay
描述 β^- 和 β^+ 衰变中夸克组成的变化。

11.2.6 recall that electrons and neutrinos are fundamental particles called leptons
记住电子和中微子是称为轻子的基本粒子。

11.1 原子、原子核与辐射

11.1.1 α 粒子散射实验 α-particle scattering



观察 observation	结论 conclusion
绝大多数 α 粒子直线穿过或偏转极小	原子内部绝大部分是**空的** (mostly empty space)
少数 α 粒子发生较大角度偏转	原子中心有一个**带正电**的区域（与 α 粒子相互排斥）
极少数 α 粒子偏转超过 90°（几乎反弹）	原子的正电荷和**绝大部分质量**集中在一个**极小的**核内

易错提醒 Common mistakes 散射实验**不能**说明原子核由质子和中子组成——中子是 1932 年 Chadwick 才发现的。选择题常以此作为干扰项（"which conclusion cannot be drawn"）。

原子直径约 10^{-10} m，原子核直径约 10^{-15} m 至 10^{-14} m，二者相差约 10^4–10^5 倍。

11.1.2 核式原子模型与核素符号

原子由带正电的**原子核**（含质子和中子，统称**核子** (*nucleons*)）和核外绕核运动的**电子**组成。

粒子	电荷	质量 / u	位置
质子 proton	$+e$	1.007	核内
中子 neutron	0	1.009	核内
电子 electron	$-e$	0.00055	核外

公式 11.1 (核素符号 *Nuclide notation*)

$$^{A}_{Z}\mathrm{X}$$

A：核子数（质量数）nucleon number = 质子数 + 中子数；Z：质子数 proton number（原子序数）；中子数 $N = A - Z$。

定义 11.1 (同位素 *Isotopes*)

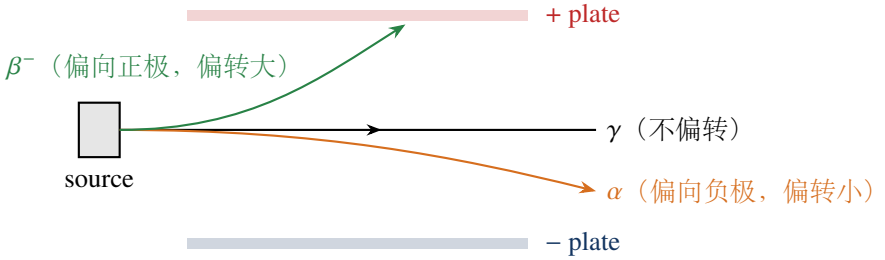
Isotopes are nuclei (atoms) of the same element with the same number of protons but different numbers of neutrons.

同位素：质子数相同（同一元素）而中子数不同的原子核。化学性质相同，物理性质（质量、稳定性）不同。例：$^{12}_{6}\mathrm{C}$、$^{13}_{6}\mathrm{C}$、$^{14}_{6}\mathrm{C}$。

原子质量单位 (unified atomic mass unit)：$1\,\mathrm{u} = 1.66 \times 10^{-27}\,\mathrm{kg}$，定义为一个 $^{12}_{6}\mathrm{C}$ 原子质量的 1/12。估算时质子、中子质量都可取 1 u。

11.1.3 三种辐射

	α	β^-	β^+	γ
本质	氦核 $^{4}_{2}\mathrm{He}$	电子 $^{0}_{-1}\mathrm{e}$	正电子 $^{0}_{+1}\mathrm{e}$	电磁波（光子）
电荷	$+2e$	$-e$	$+e$	0
质量	4 u	$\approx 1/1840$ u	$\approx 1/1840$ u	0
速度	约 $0.05c$	可达 $0.99c$	可达 $0.99c$	c
电离能力	强	中	中	弱
穿透（阻挡物）	纸、几 cm 空气	几 mm 铝	（遇电子湮灭）	几 cm 铅
能量分布	**分立**	**连续**	**连续**	分立



11.1.4 衰变方程与守恒定律

核反应中**核子数守恒、电荷（质子数）守恒**（以及质能、动量守恒）。

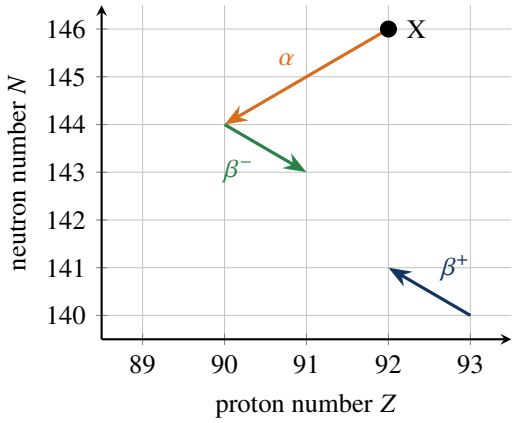
公式 11.2 (三种衰变 *Decay equations*)

$$\alpha：\ {}^{A}_{Z}\mathrm{X} \to {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\alpha$$

$$\beta^-：\ {}^{A}_{Z}\mathrm{X} \to {}^{A}_{Z+1}\mathrm{Y} + {}^{0}_{-1}\beta + \bar{\nu}$$

$$\beta^+：\ {}^{A}_{Z}\mathrm{X} \to {}^{A}_{Z-1}\mathrm{Y} + {}^{0}_{+1}\beta + \nu$$

γ 衰变：原子核从激发态跃迁到较低能态，放出光子，A、Z 都不变。



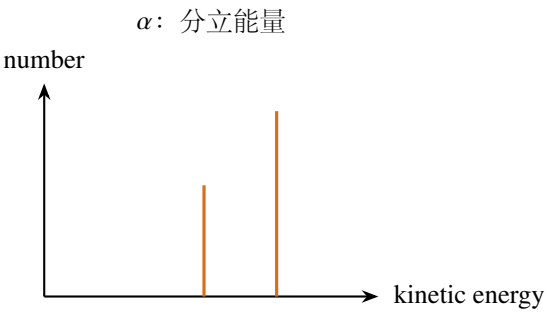
在 N–Z 图上：α 衰变向左下移动 $(-2,-2)$；β^- 衰变向右下 $(+1,-1)$；β^+ 衰变向左上 $(-1,+1)$。

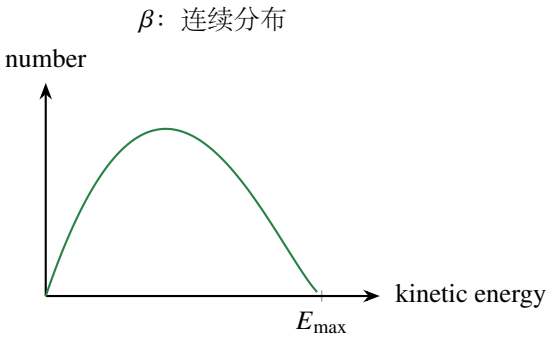
同位素判断：一次 α 衰变使 Z 减 2，两次 β^- 衰变使 Z 加 2，所以“1 个 α + 2 个 β^-”后回到同一元素的另一种同位素（A 减 4）。

11.1.5 反粒子与中微子

- **反粒子** (*antiparticle*)：与对应粒子**质量相同、电荷相反**。正电子 e^+ 是电子的反粒子；反质子带 $-e$。
- 粒子与其反粒子相遇会**湮灭**（annihilation），质量全部转化为能量（γ 光子）——详见第 23 章。
- **中微子** ν：电中性、质量极小（近似为零）、几乎不与物质相互作用。β^- 衰变放出**反中微子** $\bar{\nu}$，β^+ 衰变放出**中微子** ν。

能量分布：





α 衰变只产生两个粒子，由动量守恒和能量守恒，α 粒子的能量是确定的（分立值）。β 衰变每次释放的**总能量一定**，但这部分能量在 β 粒子和（反）中微子之间**分配方式每次不同**，所以 β 粒子的动能从零到最大值 $E_{\max}$ 连续分布。这正是 Pauli 预言中微子存在的依据。

例题 11.1 9702/12/M/J/26 Q40

A nucleus decays in a series of stages. At each stage, an α-particle or β^- particle is emitted. Which sequence of emitted particles forms a nucleus that is an isotope of the original nucleus?

A $\alpha, \alpha, \beta^-, \beta^-$　　**B** β^-, β^-, α　　**C** $\alpha, \alpha, \alpha, \alpha$　　**D** $\beta^-, \beta^-, \beta^-$

解答 Solution

答案 Answer：B

同位素要求 Z 不变。B：$+1+1-2=0$（正确）。A：$-2-2+1+1=-2$；C：-8；D：$+3$。

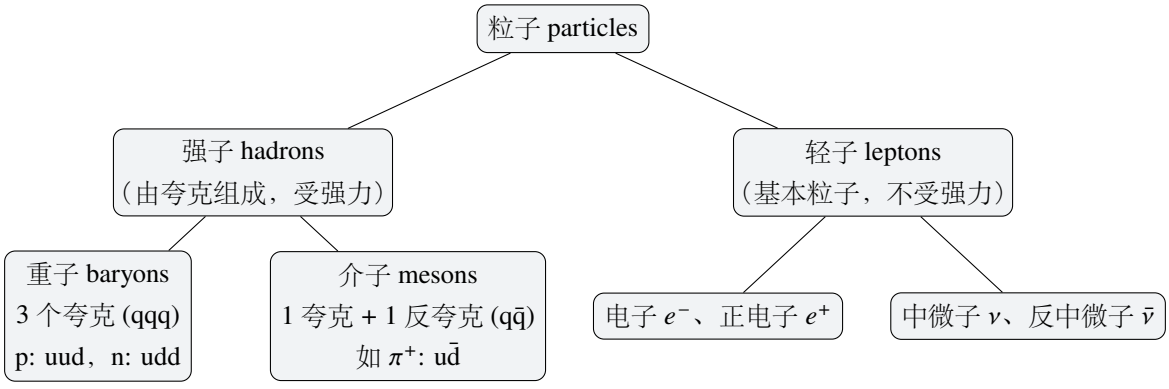
11.2 基本粒子 Fundamental particles

11.2.1 夸克

夸克 quark	符号	电荷	反夸克	符号	电荷
up 上	u	$+\frac{2}{3}e$	anti-up	$\bar{\text{u}}$	$-\frac{2}{3}e$
down 下	d	$-\frac{1}{3}e$	anti-down	$\bar{\text{d}}$	$+\frac{1}{3}e$
charm 粲	c	$+\frac{2}{3}e$	anti-charm	$\bar{\text{c}}$	$-\frac{2}{3}e$
strange 奇	s	$-\frac{1}{3}e$	anti-strange	$\bar{\text{s}}$	$+\frac{1}{3}e$
top 顶	t	$+\frac{2}{3}e$	anti-top	$\bar{\text{t}}$	$-\frac{2}{3}e$
bottom 底	b	$-\frac{1}{3}e$	anti-bottom	$\bar{\text{b}}$	$+\frac{1}{3}e$

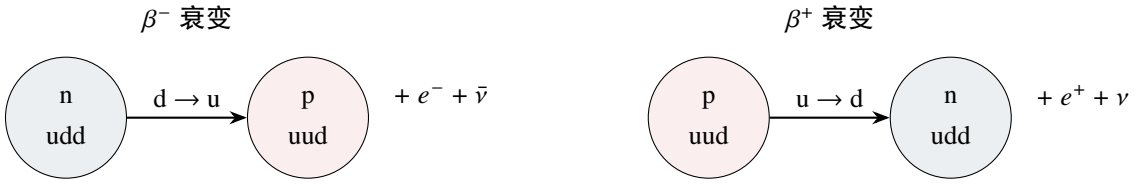
记忆：u、c、t 电荷 $+\frac{2}{3}e$；d、s、b 电荷 $-\frac{1}{3}e$。

11.2.2 粒子的分类



- **基本粒子** (fundamental particles)：没有内部结构，不能再分。考纲涉及的基本粒子：六种夸克（及反夸克）、轻子（电子、中微子等）。
- 质子、中子**不是**基本粒子：质子 uud（电荷 $\frac{2}{3}+\frac{2}{3}-\frac{1}{3}=+1$），中子 udd（电荷 $\frac{2}{3}-\frac{1}{3}-\frac{1}{3}=0$）。
- 重子的电荷只能是整数：如 $\Delta^{++}=\text{uuu}$（$+2e$）、$\Delta^{-}=\text{ddd}$（$-e$）。
- 介子由一个夸克和一个反夸克组成，电荷为 0 或 $\pm e$。

11.2.3 β 衰变中的夸克变化



- β^-：一个中子变为质子，$\text{n}\to\text{p}+e^-+\bar{\nu}$，夸克层面 $\text{d}\to\text{u}+e^-+\bar{\nu}$（电荷 $-\frac{1}{3}=+\frac{2}{3}-1$）。
- β^+：一个质子变为中子，$\text{p}\to\text{n}+e^++\nu$，夸克层面 $\text{u}\to\text{d}+e^++\nu$（只发生在核内，因为需要能量）。

例题 11.2 9702/11/M/J/26 Q40

A positively charged meson consists of a quark and an antiquark. What could be the quark and antiquark?

A charm and antiup

B down and antitop

C strange and antibottom

D up and antistrange

解答 Solution

答案 Answer：D

A：$+\frac{2}{3}-\frac{2}{3}=0$；B：$-\frac{1}{3}-\frac{2}{3}=-1$；C：$-\frac{1}{3}+\frac{1}{3}=0$；D：$+\frac{2}{3}+\frac{1}{3}=+1$（正确）。

例题 11.3 9702/22/M/J/26 Q6

(a) One of the conclusions from the α-particle scattering experiment was that the atom is mostly empty space. State the observation from the experiment that provides evidence for this conclusion. [1]

(b) An α-particle travelling with a speed of $5.60\times10^{6}\,\text{m s}^{-1}$ collides with a stationary gold nucleus of mass 197 u. After the collision, the α-particle is travelling with a speed of $5.38\times10^{6}\,\text{m s}^{-1}$ in the opposite direction to its initial motion. No external forces act on this system.

(i) Show that the momentum of the α-particle before the collision is $3.72\times10^{-20}\,\text{kg m s}^{-1}$. [2]

(ii) Use conservation of momentum to calculate the speed of the gold nucleus after the collision. [2]

(iii) The absolute uncertainty in each of the measured speeds of the α-particle is $\pm5\times10^{4}\,\text{m s}^{-1}$. Calculate the percentage uncertainty in the change in velocity of the α-particle. [2]

(c) Determine the number of up quarks in a nucleus of $^{2}_{1}\mathrm{H}$. [1]

(d) Describe the general quark composition of a meson. [1]

解答 Solution

(a) 绝大多数 α 粒子直接穿过（或偏转极小）。 **[B1]**

(b)(i) $p = mv$ **[C1]**

$= 4 \times 1.66 \times 10^{-27} \times 5.60 \times 10^{6} = 3.72 \times 10^{-20}\ \mathrm{kg\,m\,s^{-1}}$ **[A1]**

(ii) 取 α 初速度方向为正：$4\mathrm{u} \times 5.60 \times 10^{6} = 197\mathrm{u} \times v - 4\mathrm{u} \times 5.38 \times 10^{6}$ **[C1]**

$v = 2.2 \times 10^{5}\ \mathrm{m\,s^{-1}}$ **[A1]**

(iii) 速度变化量 $= 5.60\times10^{6} - (-5.38\times10^{6}) = 1.098 \times 10^{7}\ \mathrm{m\,s^{-1}}$，绝对不确定度 $= 5\times10^{4} + 5\times10^{4} = 1 \times 10^{5}\ \mathrm{m\,s^{-1}}$ **[C1]**

百分比不确定度 $= \dfrac{1 \times 10^{5}}{1.098 \times 10^{7}} \times 100\% = 0.9\%$ **[A1]**

(c) $^{2}_{1}\mathrm{H}$ 含 1 个质子 (uud) 和 1 个中子 (udd)，up 夸克共 3 个。 **[A1]**

(d) 一个夸克和一个反夸克。 **[B1]**

教学提示 Teacher's note 建议 4 课时。本章概念多、计算少。务必让学生熟记：夸克电荷表；质子 uud、中子 udd；$\beta^{\pm}$ 衰变的夸克变化与（反）中微子归属；α 散射的“观察—结论”对应。Paper 2 常见 3 分题：“解释 β 粒子能量连续”——关键词：each decay releases the same total energy; shared between β particle and (anti)neutrino in different proportions。

11.3 本章小结 Summary

物理量/关系	公式	说明
核素	$^{A}_{Z}\mathrm{X}$，$N = A - Z$	核子数、电荷守恒
α 衰变	$A-4$，$Z-2$	能量分立
β^{-} 衰变	$Z+1$；$\mathrm{d} \to \mathrm{u}$；放 $\bar{\nu}$	能量连续
β^{+} 衰变	$Z-1$；$\mathrm{u} \to \mathrm{d}$；放 ν	能量连续
夸克电荷	u, c, t: $+\frac{2}{3}e$；d, s, b: $-\frac{1}{3}e$	反夸克相反
重子 / 介子	qqq / $\mathrm{q\bar{q}}$	统称强子
轻子	$e^{\pm}$，ν，$\bar{\nu}$	基本粒子
原子质量单位	$1\,\mathrm{u} = 1.66 \times 10^{-27}\ \mathrm{kg}$	

11.4 本章练习 Practice questions

练习 11.1 9702/11/M/J/26 Q38

A stream of identical particles flows between two points. The stream of particles does not form an electric current. What could the particles be?

A α-particles

B β^- particles

C neutrons

D protons

解答 Solution

答案 Answer：C

电流是电荷的流动；中子不带电，所以不形成电流。

练习 11.2 9702/12/M/J/26 Q38

A nucleus P undergoes β^+ decay, becoming nucleus Q and emitting two particles X and Y as shown.

$$\mathrm{P} \rightarrow \mathrm{Q} + \mathrm{X} + \mathrm{Y}$$

Which two particles are X and Y?

A electron and antineutrino

B electron and neutrino

C positron and antineutrino

D positron and neutrino

解答 Solution

答案 Answer：D

练习 11.3 9702/14/M/J/26 Q39

An unstable nucleus ${}^{11}_{6}\mathrm{C}$ decays in a single step to form a stable nucleus. During the decay, a nucleon of the unstable nucleus is changed when an up quark in the nucleon becomes a down quark. What is the stable nucleus that is formed?

A ${}^{11}_{5}\mathrm{B}$ **B** ${}^{11}_{7}\mathrm{N}$ **C** ${}^{12}_{6}\mathrm{C}$ **D** ${}^{10}_{6}\mathrm{C}$

解答 Solution

答案 Answer：A

u → d：质子变为中子（β^+ 衰变），A 不变，Z 减 1：${}^{11}_{5}\mathrm{B}$。

练习 11.4 9702/12/F/M/26 Q40

Uranium, $^{238}_{92}$U, decays into radium, $^{226}_{88}$Ra, after a sequence of alpha and beta-minus decays. Which row gives the correct number of alpha decays and beta-minus decays?

	α	β^-
A	2	3
B	3	2
C	2	0
D	3	0

解答 Solution

答案 Answer：**B**

A 减少 12 ⇒ 3 次 α（Z 减 6 至 86）；Z 实际减少 4，所以还需 2 次 β^-（Z 加 2）。

练习 11.5 9702/23/M/J/26 Q5

(a) State what is meant by a meson. [1]

(b) The elementary charge is e. Explain, in terms of quark composition, how a baryon may have an overall charge of: (i) $-1e$; (ii) $+2e$. [3]

(c) The nuclide $^{18}_{8}$X undergoes β^+ decay to produce nuclide $^{P}_{Q}$Y and one other particle Z, as shown in the equation

$$^{18}_{8}\text{X} \rightarrow {}^{P}_{Q}\text{Y} + {}^{R}_{S}\beta + {}^{0}_{0}\text{Z}.$$

(i) State the values of P, Q, R and S. [2]

(ii) State the name of particle Z. [1]

(iii) State the net change in quark composition as nuclide X turns into nuclide Y. [1]

(iv) Explain why the emission of particle Z results in the β^+ particles from the decays of different nuclei of $^{18}_{8}$X having a continuous range of kinetic energies. [2]

解答 Solution

(a) 由一个夸克和一个反夸克组成的强子。 **[B1]**

(b) 夸克电荷的大小只有 $\frac{2}{3}e$ 或 $\frac{1}{3}e$；重子由三个夸克组成。 **[C1]**

$-1e$：$(-\frac{1}{3}) + (-\frac{1}{3}) + (-\frac{1}{3})$，如 ddd。 **[A1]**

$+2e$：$(+\frac{2}{3}) + (+\frac{2}{3}) + (+\frac{2}{3})$，如 uuu。 **[A1]**

(c)(i) $P = 18$，$R = 0$； **[A1]**

$Q = 7$，$S = +1$。 **[A1]**

(ii) （电子）中微子 neutrino。 **[B1]**

(iii) 一个 up 夸克变为 down 夸克（u → d）。 **[B1]**

(iv) 每次衰变释放的总能量相同； **[B1]**

这些能量在 β^+ 粒子与中微子之间分配，每次分配比例不同，所以 β^+ 粒子动能连续分布。 [B1]

练习 11.6 9702/24/M/J/26 Q7

(a) (i) State what is meant by isotopes. [2]

(ii) A thorium-232 nucleus ($^{232}_{90}$Th) undergoes radioactive decay to form another nucleus. A sequence of further decays occurs until a nucleus is formed that is another isotope of thorium. In each decay, either a single α-particle or a single β^- particle is emitted. Determine the minimum numbers of α-particles and β^- particles that could be emitted during the sequence of decays to form the other isotope of thorium from thorium-232. [2]

(iii) State the nucleon number and proton number of the thorium isotope formed by the sequence of decays. [2]

(b) (i) State the quark composition of a proton. [1]

(ii) Describe the change to the quark composition of the proton when it undergoes β^+ decay. [1]

解答 Solution

(a)(i) 同一元素（质子数相同）的原子核， [B1]

中子数（核子数）不同。 [B1]

(ii) 最少 1 个 α 粒子， [A1]

2 个 β^- 粒子。 [A1]

(iii) $^{228}_{90}$Th：核子数 228， [A1]

质子数 90。 [A1]

(b)(i) up, up, down (uud)。 [B1]

(ii) 一个 up 夸克变为 down 夸克。 [B1]

第二部分

A Level

第十二章 圆周运动 Motion in a circle

考纲要求 Syllabus learning outcomes

12.1.1 define the radian and express angular displacement in radians
定义弧度，并用弧度表示角位移。

12.1.2 understand and use the concept of angular speed
理解并使用角速度的概念。

12.1.3 recall and use $\omega = 2\pi/T$ and $v = r\omega$
记住并使用 $\omega = 2\pi/T$ 和 $v = r\omega$。

12.2.1 understand that a resultant force of constant magnitude that is always perpendicular to the velocity of an object causes the object to move in a circle (uniform circular motion)
理解大小不变、始终垂直于速度的合力使物体做（匀速）圆周运动。

12.2.2 recall and use centripetal acceleration $a = r\omega^2$ and $a = v^2/r$
记住并使用向心加速度 $a = r\omega^2$，$a = v^2/r$。

12.2.3 recall and use centripetal force $F = mr\omega^2$ and $F = mv^2/r$
记住并使用向心力 $F = mr\omega^2$，$F = mv^2/r$。

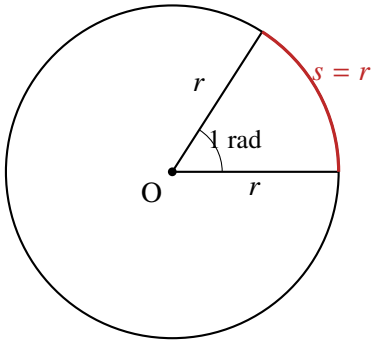
12.1 圆周运动的运动学

12.1.1 弧度与角位移

定义 12.1 (弧度 *Radian*)

One radian is the angle subtended at the centre of a circle by an arc of length equal to the radius of the circle.

1 弧度是圆心角所对的弧长等于半径时的角。一般地 $\theta = \dfrac{s}{r}$（s 为弧长），所以一整圈 $= 2\pi$ rad $= 360°$，1 rad $\approx 57.3°$。



角位移 angular displacement：
物体绕圆心转过的角度 θ（rad）
$\theta = s/r$

12.1.2 角速度

定义 12.2 (角速度　*Angular speed*)

Angular speed is the rate of change of angular displacement (angle swept out per unit time): $\omega = \dfrac{\Delta\theta}{\Delta t}$.

角速度 = 单位时间内转过的角度，单位 $\mathrm{rad\,s^{-1}}$。匀速圆周运动中 ω 恒定。

公式 12.1 (角速度、周期、线速度　*Angular speed, period and speed*)

$$\omega = \frac{2\pi}{T} = 2\pi f, \qquad v = r\omega$$

推导：一周弧长 $2\pi r$，用时 T，$v = \dfrac{2\pi r}{T} = r\omega$。

- 同一转动刚体（如转盘、地球自转）上各点的**角速度相同**，线速度与半径成正比（$v = r\omega$）。
- 线速度方向沿**切线**，时刻在变；转盘上关于圆心对称的两点线速度方向相反。

12.2 向心加速度与向心力

12.2.1 为什么需要加速度

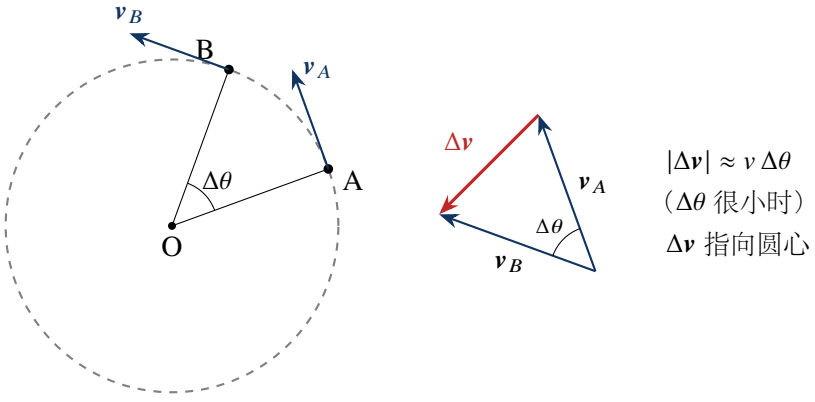
做匀速圆周运动的物体速率不变，但**速度方向不断改变**，所以速度在变化——一定有加速度，由牛顿第二定律一定受到合力作用。

定义 12.3 (匀速圆周运动　*Uniform circular motion*)

Motion in a circle at constant speed: the velocity has constant magnitude but continuously changing direction (tangential to the circle); the acceleration has constant magnitude and is always perpendicular to the velocity, directed towards the centre of the circle.

匀速圆周运动：速度大小不变、方向沿切线不断改变；加速度大小不变、始终垂直于速度并指向圆心。产生这种运动的条件：受到**大小恒定且始终垂直于速度**的合力。

12.2.2 向心加速度的推导



物体在 Δt 内从 A 运动到 B，转过 $\Delta\theta$。速度矢量三角形中两边长都为 v，夹角 $\Delta\theta$，当 $\Delta\theta$ 很小时 $\Delta v \approx v\Delta\theta$，方向指向圆心：

$$a = \frac{\Delta v}{\Delta t} = v\frac{\Delta\theta}{\Delta t} = v\omega = \frac{v^2}{r} = r\omega^2.$$

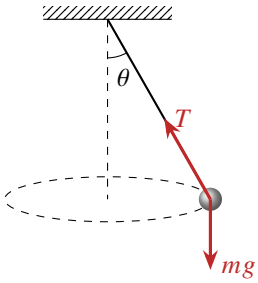
公式 12.2 (向心加速度与向心力　*Centripetal acceleration and force*)

$$a = r\omega^2 = \frac{v^2}{r}, \qquad F = mr\omega^2 = \frac{mv^2}{r}$$

方向都指向圆心。

易错提醒 Common mistakes　**向心力不是一种新的力**，而是物体所受**合力**（或合力沿半径方向的分量）的名称。受力分析图中**不要**画“centripetal force”，应画出真实的力（重力、拉力、摩擦力、支持力、电场力、磁场力、引力等），再说明它们的合力提供向心力。向心力与速度垂直，**不做功**，动能不变。

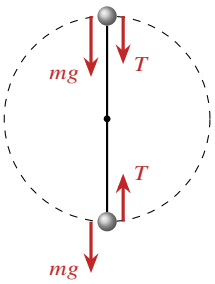
12.2.3 常见模型



圆锥摆 conical pendulum

$T\cos\theta = mg$

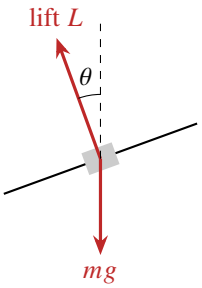
$T\sin\theta = mv^2/r$



竖直圆周

顶点：$T + mg = mv^2/r$

底点：$T - mg = mv^2/r$



倾斜转弯

$L\cos\theta = mg$

$L\sin\theta = mv^2/r$

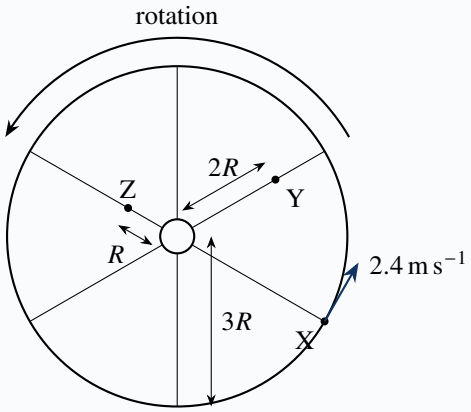
$\tan\theta = \dfrac{v^2}{rg}$

- **水平转弯的汽车**：地面对轮胎的侧向摩擦力提供向心力，$f = mv^2/r$；最大摩擦力决定最大安全速度。
- **竖直平面内的圆周运动**（绳拉小球）：在最高点，最小速度满足 $mg = mv^2/r$，即 $v_{\min} = \sqrt{gr}$（此时绳拉力为零）；最低点绳的拉力最大。竖直圆周中速率一般不恒定（动能与势能转换），但每一点仍满足“沿半径方向的合力 $= mv^2/r$”。
- **卫星、行星**：万有引力提供向心力（第 13 章）；**原子中电子**：库仑力提供向心力（第 18 章）；**磁场中带电粒子**：洛伦兹力提供向心力（第 20 章）。

例题 12.1　9702/42/F/M/26 Q1

(a) Define the radian. [1]

(b) Three children are playing on a roundabout (rotatable horizontal platform) at a playground. Child X is standing on the edge of the roundabout. This child is pushing so that the edge of the roundabout moves in a circle at a constant speed of 2.4 m s^{-1}, as shown in the view from above in Figure 1.1.



The radius of the roundabout is $3R$. Child Y is sitting on the roundabout at a distance $2R$ from the centre. Child Z is sitting at a distance R from the centre. Child Y is moving in a circle with an angular speed of $0.75\,\text{rad}\,\text{s}^{-1}$. Calculate:

(i) the radius of the roundabout [2]

(ii) the frequency of rotation of the roundabout [2]

(iii) the centripetal acceleration of child Y. [2]

(c) For the roundabout in (b), compare the linear and angular velocities for child X and child Z at any instant in time: for each quantity state whether the magnitude for X is less than / equal to / greater than that for Z, and whether the directions are the same or opposite. [3]

解答 Solution

(a) 圆心角所对弧长等于半径时的角。 **[B1]**

(b)(i) 同一转盘角速度相同，$\omega = 0.75\,\text{rad}\,\text{s}^{-1}$。$v = r\omega$ **[C1]**

$3R = 2.4/0.75 = 3.2\,\text{m}$ **[A1]**

(ii) $\omega = 2\pi f$ **[C1]**

$f = 0.75/2\pi = 0.12\,\text{Hz}$ **[A1]**

(iii) $a = r\omega^2$，$r_Y = 2R = \frac{2}{3} \times 3.2 = 2.13\,\text{m}$ **[C1]**

$a = 2.13 \times 0.75^2 = 1.2\,\text{m}\,\text{s}^{-2}$ **[A1]**

(c) 线速度：大小 X 大于 Z（$v \propto r$）； **[B1]**

线速度方向：相反（X、Z 位于圆心两侧的同一直径上）； **[B1]**

角速度：大小相等、方向相同。 **[B1]**

例题 12.2 原创仿真题 Exam-style

A small object moves in a horizontal circle of radius 0.50 m. It completes 120 revolutions per minute at constant speed. What is the magnitude of its centripetal acceleration?

A $3.9\,\text{m}\,\text{s}^{-2}$ **B** $7.9\,\text{m}\,\text{s}^{-2}$ **C** $39\,\text{m}\,\text{s}^{-2}$ **D** $79\,\text{m}\,\text{s}^{-2}$

解答 Solution

答案 Answer：D

$f = 120/60 = 2.0\,\text{Hz}$，$\omega = 2\pi f = 12.6\,\text{rad s}^{-1}$。$a = r\omega^2 = 0.50 \times 12.6^2 = 79\,\text{m s}^{-2}$。（注意转速要先换算为 Hz，再乘 2π。）

例题 12.3 原创仿真题 Exam-style

A small ball of mass 0.20 kg is attached to a light string of length 0.80 m. The ball moves in a horizontal circle at constant speed so that the string makes an angle of 30° with the vertical (a conical pendulum).

(a) Draw and label the forces acting on the ball. [1]

(b) Calculate the tension in the string. [2]

(c) Calculate the speed of the ball and the period of its motion. [4]

解答 Solution

(a) 只有两个力：沿绳指向悬点的拉力 T，竖直向下的重力 mg（不画“向心力”）。 **[B1]**

(b) 竖直方向平衡：$T\cos 30^\circ = mg$ **[C1]**

$$T = \frac{0.20 \times 9.81}{\cos 30^\circ} = 2.3\,\text{N}$$ **[A1]**

(c) 半径 $r = 0.80 \sin 30^\circ = 0.40\,\text{m}$ **[C1]**

水平方向：$T\sin 30^\circ = \dfrac{mv^2}{r}$，即 $mg\tan 30^\circ = \dfrac{mv^2}{r}$ **[C1]**

$$v = \sqrt{gr\tan 30^\circ} = \sqrt{9.81 \times 0.40 \times 0.577} = 1.5\,\text{m s}^{-1}$$ **[A1]**

$$\text{周期} = \frac{2\pi r}{v} = \frac{2\pi \times 0.40}{1.505} = 1.7\,\text{s}$$ **[A1]**

教学提示 Teacher's note 建议 3 课时。核心是让学生建立“合力指向圆心 = 向心力”的观念，而不是把向心力当作额外的力。可以用细绳甩橡皮塞（测量 F 与 v^2、r 的关系）做演示实验。定义题：弧度、角速度、匀速圆周运动的描述（Paper 4 2026 年两套卷都考了）。答“describe uniform circular motion”时需同时提到速度（大小不变、方向沿切线）与加速度（大小不变、垂直于速度/指向圆心）。

12.3 本章小结 Summary

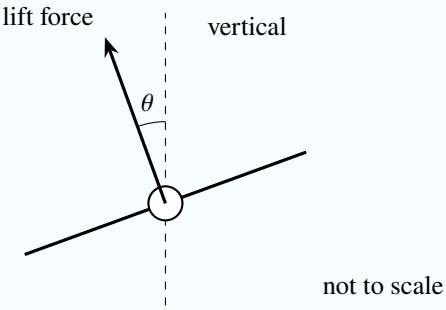
物理量/关系	公式	说明
弧度	$\theta = s/r$	2π rad = 360°
角速度	$\omega = \Delta\theta/\Delta t = 2\pi/T = 2\pi f$	rad s^{-1}
线速度	$v = r\omega$	沿切线
向心加速度	$a = r\omega^2 = v^2/r$	指向圆心
向心力	$F = mr\omega^2 = mv^2/r$	合力，不做功

12.4 本章练习 Practice questions

练习 12.1 9702/42/M/J/26 Q1

(a) With reference to velocity and acceleration, describe uniform circular motion. [2]

(b) An aircraft is travelling in a horizontal circle of radius 7200 m at a speed of 180 m s^{-1}. In order to move in a circle, the aircraft has to tilt its wings so that the lift force from the wings is not vertical, as shown in the view from behind in Figure 1.2.



(i) With reference to the forces acting on the aircraft, explain why the circular motion of the aircraft requires the lift force to be in the direction shown rather than vertically upwards. [2]

(ii) On Figure 1.2, draw an arrow from the aircraft to represent the direction of the resultant force on the aircraft. [1]

(iii) Calculate the centripetal acceleration of the aircraft. [2]

(iv) Determine the angle θ at which the aircraft is tilted. Explain your reasoning. [3]

解答 Solution

(a) 加速度始终垂直于速度（指向圆心）; **[B1]**

速度与加速度大小都不变（方向不断变化）。 **[B1]**

(b)(i) 飞机受的另一个力只有重力，它是竖直的（没有水平分量）; **[B1]**

升力必须有水平分量来提供向心加速度（向心力）。 **[B1]**

(ii) 从飞机出发、水平指向左（指向圆心）的箭头。 **[B1]**

(iii) $a = v^2/r$ **[C1]**

$= 180^2/7200 = 4.5\ \text{m s}^{-2}$ **[A1]**

(iv) 升力竖直分量平衡重力：$L\cos\theta = mg$；水平分量提供向心加速度：$L\sin\theta = ma$ **[B1]**

$\tan\theta = a/g$ **[C1]**

$\tan\theta = 4.5/9.81$，$\theta = 25°$ **[A1]**

练习 12.2 原创仿真题 Exam-style

An object moves in a horizontal circle at constant speed. Which quantity is constant?

A velocity

B acceleration

C kinetic energy

D momentum

解答 Solution

答案 Answer：C

速率不变，动能 $\frac{1}{2}mv^2$ 不变。速度、加速度、动量都是矢量，方向不断变化。

练习 12.3 原创仿真题 Exam-style

A car travels around a bend on a horizontal road. The bend is an arc of a circle of radius 40 m. The maximum sideways frictional force between the tyres and the road is 0.60 times the weight of the car. What is the maximum speed at which the car can travel around the bend without skidding?

A $4.9\,\mathrm{m\,s^{-1}}$ **B** $15\,\mathrm{m\,s^{-1}}$ **C** $24\,\mathrm{m\,s^{-1}}$ **D** $235\,\mathrm{m\,s^{-1}}$

解答 Solution

答案 Answer：B

摩擦力提供向心力：$0.60mg = \dfrac{mv^2}{r}$，$v = \sqrt{0.60 \times 9.81 \times 40} = 15\,\mathrm{m\,s^{-1}}$。与质量无关。

练习 12.4 原创仿真题 Exam-style

What is the angular speed of the second hand of a clock?

A $0.0017\,\mathrm{rad\,s^{-1}}$ **B** $0.10\,\mathrm{rad\,s^{-1}}$ **C** $1.0\,\mathrm{rad\,s^{-1}}$ **D** $6.3\,\mathrm{rad\,s^{-1}}$

解答 Solution

答案 Answer：B

$T = 60\,\mathrm{s}$，$\omega = 2\pi/60 = 0.105\,\mathrm{rad\,s^{-1}}$。（A 是分针的角速度。）

练习 12.5 原创仿真题 Exam-style

A ball of mass 0.15 kg is attached to a light inextensible string and is whirled in a vertical circle of radius 0.60 m. Air resistance is negligible.

(a) Show that the minimum speed of the ball at the top of the circle for the string to remain taut is $2.4\,\mathrm{m\,s^{-1}}$. [2]

(b) The ball has this minimum speed at the top. Calculate the speed of the ball at the bottom of the circle. [2]

(c) Calculate the tension in the string when the ball is at the bottom of the circle. [3]

解答 Solution

(a) 最高点绳刚好不松弛时 $T = 0$，重力提供向心力：$mg = \dfrac{mv^2}{r}$ **[C1]**

$v = \sqrt{gr} = \sqrt{9.81 \times 0.60} = 2.4\,\mathrm{m\,s^{-1}}$ **[A1]**

(b) 能量守恒：$\frac{1}{2}mv_\mathrm{b}^2 = \frac{1}{2}mv_\mathrm{t}^2 + mg(2r)$ **[C1]**

$v_\mathrm{b}^2 = 5.89 + 4 \times 9.81 \times 0.60 = 29.4$，$v_\mathrm{b} = 5.4\,\mathrm{m\,s^{-1}}$ **[A1]**

(c) 最低点：$T - mg = \dfrac{mv_\mathrm{b}^2}{r}$ **[C1]**

$T = 0.15 \times 9.81 + \dfrac{0.15 \times 29.4}{0.60}$ **[C1]**

$= 1.47 + 7.36 = 8.8\,\mathrm{N}$ **[A1]**

第十三章 引力场 Gravitational fields

考纲要求 Syllabus learning outcomes

13.1.1 understand that a gravitational field is an example of a field of force and define gravitational field as force per unit mass
理解引力场是一种力场，并把引力场强度定义为单位质量所受的力。

13.1.2 represent gravitational fields by means of field lines
用场线表示引力场。

13.2.1 understand that, for a point outside a uniform sphere, the mass of the sphere may be considered to be a point mass at its centre
理解对于均匀球体外部的点，球体质量可视为集中在球心的质点。

13.2.2 recall and use Newton's law of gravitation $F = Gm_1m_2/r^2$ for the force between two point masses
记住并使用牛顿万有引力定律 $F = Gm_1m_2/r^2$。

13.2.3 analyse circular orbits in gravitational fields by relating the gravitational force to the centripetal acceleration it causes
把引力与其产生的向心加速度联系起来分析引力场中的圆轨道。

13.2.4 understand that a satellite in a geostationary orbit remains at the same point above the Earth's surface, with an orbital period of 24 hours, orbiting from west to east, directly above the Equator
理解地球同步静止卫星位于赤道正上方，自西向东运行，周期 24 小时，始终在地面同一点上方。

13.3.1 derive, from Newton's law of gravitation and the definition of gravitational field, the equation $g = GM/r^2$ for the gravitational field strength due to a point mass
由万有引力定律和场强定义推导点质量的场强 $g = GM/r^2$。

13.3.2 recall and use $g = GM/r^2$
记住并使用 $g = GM/r^2$。

13.3.3 understand why g is approximately constant for small changes in height near the Earth's surface
理解地表附近高度变化不大时 g 近似不变的原因。

13.4.1 define gravitational potential at a point as the work done per unit mass in bringing a small test mass from infinity to the point
把某点引力势定义为把单位质量的小试探质量从无穷远移到该点所做的功。

13.4.2 use $\phi = -GM/r$ for the gravitational potential in the field due to a point mass
使用点质量引力势 $\phi = -GM/r$。

13.4.3 understand how the concept of gravitational potential leads to the gravitational potential energy of two point masses and use $E_P = -GMm/r$
理解由引力势得到两质点的引力势能，并使用 $E_P = -GMm/r$。

13.1 引力场与场强

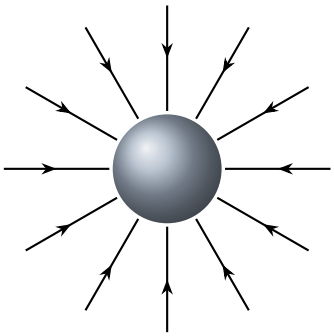
场 (*field*) 是空间中某种力作用的区域。**引力场** (*gravitational field*) 是质量在其周围产生的、对其他质量施加引力的区域。

定义 13.1 (引力场强度 *Gravitational field strength*)

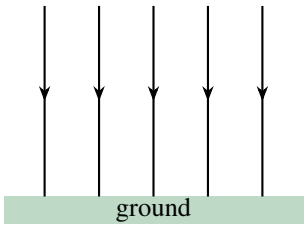
The gravitational field strength at a point is the gravitational force per unit mass acting on a small mass placed at that point: $g = \dfrac{F}{m}$.

引力场强度 = 放在该点的小质量所受的引力 / 其质量，单位 $\mathrm{N\,kg^{-1}}$（$= \mathrm{m\,s^{-2}}$），矢量，方向与引力相同。在地表附近数值上等于自由落体加速度。

13.1.1 场线



径向场 radial field
（孤立球体或质点外部）



均匀场 uniform field
（地表附近小范围）

- 场线的方向表示小质量受力方向（引力总是**吸引**，所以场线指向产生场的质量）。
- 场线越密，场强越大。场线互不相交。
- 地表附近的小区域里，场线可近似为**平行、等间距、竖直向下**的直线——均匀场。

13.2 万有引力定律

定律 13.1 (牛顿万有引力定律 *Newton's law of gravitation*)

Any two point masses attract each other with a force that is proportional to the product of their masses and inversely proportional to the square of their separation:

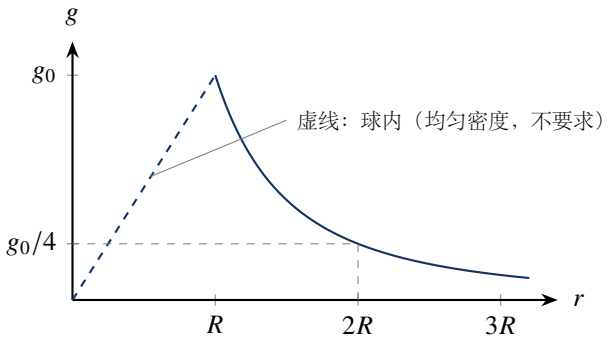
$$F = \frac{Gm_1m_2}{r^2}$$

$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ 为万有引力常量。r 为两**质点**间距离；对于均匀球体，在其外部可以把全部质量看作集中在**球心**，r 取球心之间的距离。

13.2.1 点质量的场强 $g = GM/r^2$

推导：在距质量 M 为 r 处放一个小质量 m，所受引力 $F = \dfrac{GMm}{r^2}$。由场强定义：

$$g = \frac{F}{m} = \frac{GM}{r^2}.$$



地表附近 g 近似不变：地球半径 $R \approx 6400\,\mathrm{km}$，在高度 $h \ll R$ 的范围内（如几 km），$\dfrac{g(R+h)}{g(R)} = \left(\dfrac{R}{R+h}\right)^2 \approx 1$，场线近似平行，可以视为均匀场，所以 $\Delta E_\mathrm{P} = mg\Delta h$ 适用。

13.3 圆轨道与卫星

卫星（或行星）绕中心天体做匀速圆周运动时，**万有引力提供向心力**：

$$\frac{GMm}{r^2} = \frac{mv^2}{r} = mr\omega^2.$$

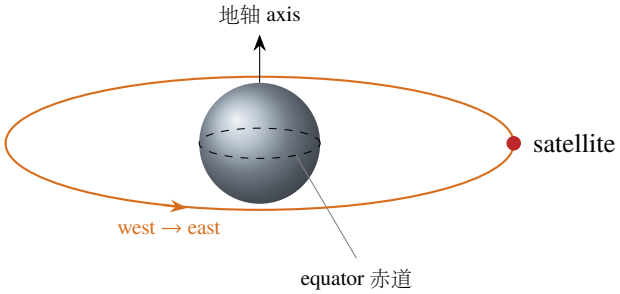
公式 13.1 (轨道关系　*Circular orbits*)

$$v = \sqrt{\frac{GM}{r}}, \qquad \omega = \sqrt{\frac{GM}{r^3}}, \qquad T^2 = \frac{4\pi^2}{GM}r^3 \ (T^2 \propto r^3)$$

卫星的轨道速度、周期与卫星自身质量无关；轨道越高，速度越小、周期越长。由已知的 r 和 T 可求中心天体质量 $M = \dfrac{4\pi^2 r^3}{GT^2}$。

推导 $T^2 \propto r^3$：$\dfrac{GMm}{r^2} = mr\left(\dfrac{2\pi}{T}\right)^2 \Rightarrow T^2 = \dfrac{4\pi^2 r^3}{GM}$。

13.3.1 地球同步静止轨道 Geostationary orbit



周期 $T = 24\,\mathrm{h}$；位于赤道正上方；
自西向东（与地球自转同向）；
轨道半径约 $4.2 \times 10^7\,\mathrm{m}$（高度约 $3.6 \times 10^7\,\mathrm{m}$）

定义 13.2 (地球同步静止轨道　*Geostationary orbit*)

An orbit in which a satellite remains above the same point on the Earth's surface: it has a period of 24 hours, is directly above the Equator and moves from west to east (same direction as the Earth's rotation).

卫星始终位于地面同一点上方的轨道：周期 24 小时、在赤道平面内、自西向东运行。地面天线无需转动即可对准卫星，适用于通信、电视广播。

13.4 引力势与引力势能

定义 13.3 (引力势 *Gravitational potential*)

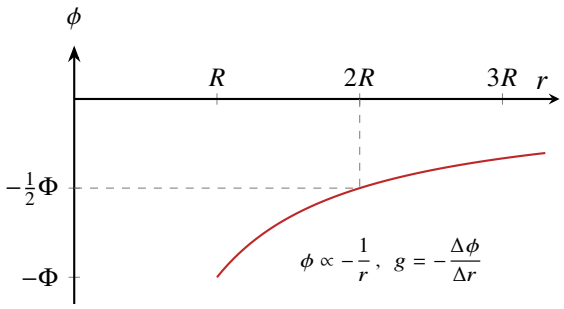
The gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point.

引力势：把单位质量的小试探质量从**无穷远**移到该点所做的功。单位 $\mathrm{J\,kg^{-1}}$，标量。

公式 13.2 (点质量的引力势 *Potential due to a point mass*)

$$\phi = -\frac{GM}{r}$$

为什么引力势总是负的：规定无穷远处引力势为零。引力是吸引力，把质量从无穷远移近时，引力对它做正功（外界做负功），系统能量减少；反过来要把质量移到无穷远，外界必须做功。所以除无穷远外各点的势都小于零。



- 场强等于引力势梯度的负值：$g = -\dfrac{\Delta\phi}{\Delta r}$（$\phi$–$r$ 图切线斜率的大小即为场强大小）。
- 两点间移动质量 m，外力做功 $W = m\Delta\phi$。

公式 13.3 (引力势能 *Gravitational potential energy*)

$$E_{\mathrm{P}} = m\phi = -\frac{GMm}{r}$$

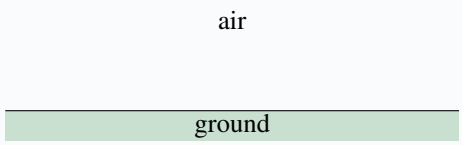
两质点系统的引力势能。$\Delta E_{\mathrm{P}} = GMm\left(\dfrac{1}{r_1} - \dfrac{1}{r_2}\right)$（从 r_1 移到 $r_2 > r_1$ 时势能增加）。高度变化与 R 可比时必须用此式，不能用 $mg\Delta h$。

例题 13.1 9702/42/M/J/26 Q2

(a) (i) Define gravitational field at a point. [1]

(ii) A horizontal section through a small piece of ground at the Earth's surface and the air at a small distance above it is shown. Draw four field lines to represent the gravitational field of the Earth

in the air above the ground. [2]



(b) The Earth may be considered to be an isolated uniform sphere of radius 6.37×10^6 m.

(i) By considering the gravitational field strength at the Earth's surface, calculate a value for the mass of the Earth. Give your answer to three significant figures. [2]

(ii) Determine, to three significant figures, the gravitational potential at the Earth's surface. Give a unit with your answer. [2]

(iii) A satellite of mass 96.0 kg is in circular orbit at a height of 1.93×10^6 m above the Earth's surface. Determine the gravitational potential energy of the satellite. [2]

解答 Solution

(a)(i) 单位质量所受的力。 **[B1]**

(ii) 四条平行、等间距、垂直于地面的直线； **[B1]**

箭头指向地面。 **[B1]**

(b)(i) $g = GM/R^2$ **[C1]**

$$M = \frac{9.81 \times (6.37 \times 10^6)^2}{6.67 \times 10^{-11}} = 5.97 \times 10^{24}\,\text{kg}$$ **[A1]**

(ii) $\phi = -\dfrac{GM}{R} = -9.81 \times 6.37 \times 10^6$ **[C1]**

$= -6.25 \times 10^7\,\text{J kg}^{-1}$（必须带负号和单位） **[A1]**

(iii) $E_\text{P} = -\dfrac{GMm}{r} = -\dfrac{6.67 \times 10^{-11} \times 5.97 \times 10^{24} \times 96.0}{(6.37 + 1.93) \times 10^6}$ **[C1]**

$= -4.61 \times 10^9\,\text{J}$ **[A1]**

例题 13.2 9702/44/M/J/26 Q2

A satellite is in geostationary orbit around the Earth. The orbit is circular.

(a) (i) State what is meant by a geostationary orbit. [1]

(ii) State two properties of the orbit. [2]

(b) The orbit has radius R and period T.

(i) Determine an expression, in terms of R and T, for the speed v of the satellite in its orbit. [1]

(ii) Use Newton's law of gravitation to show that the mass M of the Earth is given by $M = \dfrac{4\pi^2 R^3}{GT^2}$ where G is the gravitational constant. [2]

(c) The Earth may be considered to be a sphere of radius 6370 km. Geostationary satellites orbit at a height of 35 900 km above the surface of the Earth. Determine a value, to three significant figures, for the mean density of the Earth. [4]

解答 Solution

(a)(i) 卫星始终位于地球表面同一点的上方。 **[B1]**

(ii) 任两点：在赤道平面内；自西向东运行；周期 24 小时。 **[B2]**

(b)(i) $v = \dfrac{2\pi R}{T}$ **[A1]**

(ii) $\dfrac{GMm}{R^2} = mR\omega^2$ **[M1]**

代入 $\omega = 2\pi/T$：$\dfrac{GM}{R^2} = \dfrac{4\pi^2 R}{T^2}$，整理得 $M = \dfrac{4\pi^2 R^3}{GT^2}$。 **[A1]**

(c) 密度 = 质量 / 体积 **[C1]**

$$M = \frac{4\pi^2[(6370 + 35\,900)\times 10^3]^3}{6.67\times 10^{-11}\times(24\times 3600)^2} = 5.99\times 10^{24}\,\text{kg}$$ **[C1]**

$$V = \tfrac{4}{3}\pi(6370\times 10^3)^3 = 1.083\times 10^{21}\,\text{m}^3$$ **[C1]**

$$\rho = 5530\,\text{kg m}^{-3}$$ **[A1]**

例题 13.3 原创仿真题 Exam-style

A spacecraft of mass 500 kg is moved from the surface of the Earth to a height equal to three times the Earth's radius above the surface. The mass of the Earth is 5.97×10^{24} kg and its radius is 6.37×10^6 m.

(a) Calculate the change in gravitational potential energy of the spacecraft. [3]

(b) Explain why the expression $\Delta E_P = mg\Delta h$ cannot be used for this calculation. [1]

解答 Solution

(a) 初始 $r_1 = R$，终了 $r_2 = R + 3R = 4R$。 **[C1]**

$$\Delta E_P = GMm\left(\frac{1}{R} - \frac{1}{4R}\right) = \frac{3}{4}\cdot\frac{GMm}{R}$$ **[C1]**

$$= 0.75\times\frac{6.67\times 10^{-11}\times 5.97\times 10^{24}\times 500}{6.37\times 10^6} = 2.3\times 10^{10}\,\text{J}\ \text{（增加）}$$ **[A1]**

(b) 高度变化与地球半径相当，g 随高度显著减小（在 $4R$ 处只有地表的 1/16），不能视为常量。 **[B1]**

教学提示 Teacher's note 建议 5 课时。三个定义（field strength、potential、geostationary orbit）必须精确：potential 的定义要写"work done per unit mass"与"from infinity"。卫星题统一套路"$\dfrac{GMm}{r^2} = \dfrac{mv^2}{r}$"，注意 r 是到地心的距离（高度 + 半径），这是最常见失分点。画 g–x 图时注意 g 为矢量，在球两侧符号相反（指向球心）。

13.5 本章小结 Summary

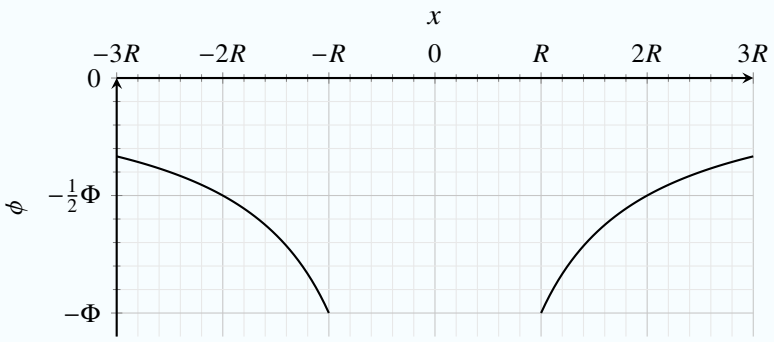
物理量/关系	公式	说明
场强定义	$g = F/m$	$\mathrm{N\,kg^{-1}}$
万有引力	$F = Gm_1m_2/r^2$	球体视为球心质点
点质量场强	$g = GM/r^2$	$\propto 1/r^2$
圆轨道	$GM/r^2 = v^2/r = r\omega^2$	$T^2 \propto r^3$
引力势	$\phi = -GM/r$	无穷远为零
引力势能	$E_\mathrm{P} = -GMm/r$	$g = -\Delta\phi/\Delta r$
同步卫星	$T = 24\,\mathrm{h}$，赤道上方，西向东	

13.6 本章练习 Practice questions

练习 13.1 9702/41/M/J/26 Q1

(a) Define gravitational potential at a point. [2]

(b) A uniform spherical planet has radius R. The gravitational potential at its surface is $-\Phi$. P is a point on a line that passes through the centre of the planet, at variable displacement x from the centre. Figure 1.2 shows the variation with x of the gravitational potential ϕ due to the planet in the region outside the planet close to its surface.



(i) Explain why ϕ is negative for all values of x. [2]

(ii) State an expression, in terms of Φ and R, for the mass M of the planet. Identify any other symbols that you use. [2]

(iii) Determine an expression, in terms of Φ and R, for the gravitational field strength g_0 at the surface of the planet. [1]

(iv) Sketch the variation of the gravitational field g with x between $x = -3R$ and $x = 3R$. Do not include the region inside the sphere between $x = -R$ and $x = R$. [3]

解答 Solution

(a) 单位质量所做的功， [B1]

把（小）质量从无穷远移到该点。 [B1]

(b)(i) 规定无穷远处势为零； [B1]

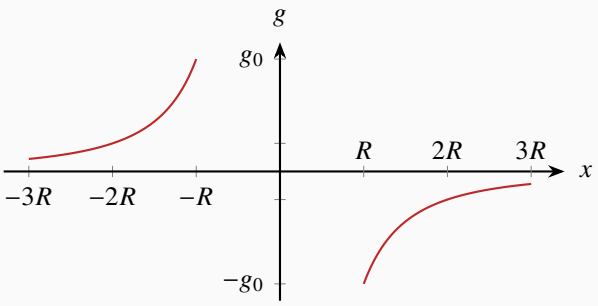
两质量相互吸引，相互靠近时引力做功（或：把它们分开需要外界做功），所以势小于零。 [B1]

(ii) $M = \dfrac{\Phi R}{G}$ [M1]

G 为万有引力常量。 [A1]

(iii) $g_0 = \dfrac{GM}{R^2} = \dfrac{\Phi}{R}$ [A1]

(iv) 取向右为 x 正方向，g 指向球心：



曲线经过 $(\pm R, \pm g_0)$、$(\pm 2R, \pm \frac{1}{4} g_0)$； [B1]

$|g|$ 随距离增大而减小，曲线斜率逐渐变平； [B1]

左侧在第二象限（$g > 0$），右侧在第四象限（$g < 0$）。 [B1]

练习 13.2 原创仿真题 Exam-style

The acceleration of free fall at the surface of the Earth is $9.81\ \mathrm{m\,s^{-2}}$. What is the gravitational field strength at a height above the Earth's surface equal to the radius of the Earth?

A $2.45\ \mathrm{N\,kg^{-1}}$ **B** $4.91\ \mathrm{N\,kg^{-1}}$ **C** $9.81\ \mathrm{N\,kg^{-1}}$ **D** $19.6\ \mathrm{N\,kg^{-1}}$

解答 Solution

答案 Answer：A

距地心 $2R$，$g \propto 1/r^2$，变为 $1/4$：$9.81/4 = 2.45\ \mathrm{N\,kg^{-1}}$。（B 是误用 $1/r$。）

练习 13.3 原创仿真题 Exam-style

Two satellites X and Y are in circular orbits around the same planet. The radius of the orbit of Y is four times the radius of the orbit of X. What is the ratio $\dfrac{\text{orbital speed of X}}{\text{orbital speed of Y}}$?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** 2 **D** 4

解答 Solution

答案 Answer：C

$v = \sqrt{GM/r} \propto r^{-1/2}$，$\dfrac{v_X}{v_Y} = \sqrt{\dfrac{r_Y}{r_X}} = \sqrt{4} = 2$。

练习 13.4 原创仿真题 Exam-style

A satellite in a circular orbit of radius r has period T. Another satellite orbits the same planet in a circular orbit of radius $2r$. What is its period?

A $\sqrt{2}T$ **B** $2T$ **C** $2\sqrt{2}T$ **D** $4T$

解答 Solution

答案 Answer：C

$T^2 \propto r^3$，$T' = T \times 2^{3/2} = 2\sqrt{2}T \approx 2.8T$。

练习 13.5 原创仿真题 Exam-style

Planet A has mass M and radius R. Planet B has mass $2M$ and radius $4R$. What is the ratio $\dfrac{\text{gravitational potential at the surface of B}}{\text{gravitational potential at the surface of A}}$?

A $\frac{1}{8}$ **B** $\frac{1}{2}$ **C** 2 **D** 8

解答 Solution

答案 Answer：B

$\phi = -GM/r \propto M/r$：$\dfrac{2M/4R}{M/R} = \dfrac{1}{2}$（两者都为负值，比值为正）。

第十四章 温度 Temperature

考纲要求 Syllabus learning outcomes

14.1.1 understand that (thermal) energy is transferred from a region of higher temperature to a region of lower temperature
理解热能从高温区域传向低温区域。

14.1.2 understand that regions of equal temperature are in thermal equilibrium
理解温度相等的区域处于热平衡。

14.2.1 understand that a physical property that varies with temperature may be used for the measurement of temperature and state examples of such properties, including the density of a liquid, volume of a gas at constant pressure, resistance of a metal, e.m.f. of a thermocouple
理解随温度变化的物理性质可用于测温，并举例：液体密度、定压气体体积、金属电阻、热电偶电动势。

14.2.2 understand that the scale of thermodynamic temperature does not depend on the property of any particular substance
理解热力学温标不依赖于任何特定物质的性质。

14.2.3 convert temperatures between kelvin and degrees Celsius and recall that $T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$
在开尔文与摄氏度之间换算，记住 $T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$。

14.2.4 understand that the lowest possible temperature is zero kelvin on the thermodynamic temperature scale and that this is known as absolute zero
理解热力学温标上最低可能温度为 0 K，即绝对零度。

14.3.1 define and use specific heat capacity
定义并使用比热容。

14.3.2 describe an experiment to measure the specific heat capacity of a solid or a liquid
描述测定固体或液体比热容的实验。

14.3.3 define and use specific latent heat and distinguish between specific latent heat of fusion and specific latent heat of vaporisation
定义并使用比潜热，区分熔化比潜热和汽化比潜热。

14.3.4 describe an experiment to measure specific latent heat of vaporisation
描述测定汽化比潜热的实验。

14.1 热平衡 Thermal equilibrium

- 两个物体接触时，热能（thermal energy，热量 heat）总是**从高温物体传向低温物体**，直到两者温度相等。
- 温度是决定热能传递**方向**的物理量；内能（第 16 章）是物体所含能量的**多少**。热的物体不一定内能大。

定义 14.1 (热平衡 *Thermal equilibrium*)

Two objects are in thermal equilibrium when they are at the same temperature, so that there is no net transfer of thermal energy between them (when in thermal contact).

两物体温度相同、相互接触时**没有净热能传递**，称为处于热平衡。温度计测温正是利用了“温度计与被测物达到热平衡”。

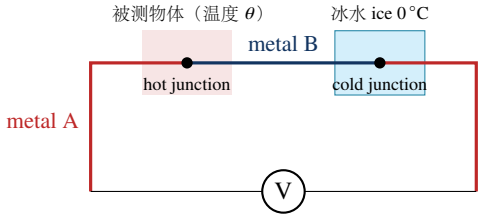
♣

14.2 温标 Temperature scales

14.2.1 测温性质

任何随温度**单调、可重复**变化的物理性质都可以用来测温，称为**测温性质** (*thermometric property*)：

温度计	测温性质	特点
液体玻璃温度计	液体体积（密度）	简单、直接读数；范围有限，热容较大、反应慢
定容/定压气体温度计	气体压强 / 定压下体积	精确，范围很宽，用于校准；体积大、慢、不方便
电阻温度计（铂丝）	金属电阻	精确、范围宽，可远程读数；反应较慢
热敏电阻	半导体电阻	灵敏（小范围内变化大）、体积小；非线性、范围窄
热电偶	两种金属接点处的 e.m.f.	体积极小、热容小、反应快、范围宽，可远程读数；非线性，需校准

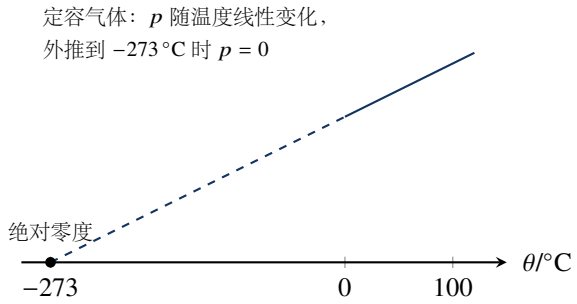


热电偶：e.m.f. 取决于两接点的温度差
（冷端保持 0 °C，由 e.m.f. 求 θ）

易错提醒 Common mistakes 不同测温性质随温度的变化规律不同（多数不是严格线性），所以按不同性质定标的经验温标，只在定标点（如 0 °C 和 100 °C）上读数一致，在其他温度可能不同。

14.2.2 热力学温标

- **热力学温标** (*thermodynamic (absolute) temperature scale*)：理论温标，**不依赖于任何特定物质**的性质。单位 kelvin (K)，是 SI 基本单位。
- **绝对零度** (*absolute zero*)：0 K，可能达到的最低温度。此时物质的内能最小（分子动能最小）。在理想气体模型中，绝对零度时气体压强（或定压时体积）外推为零。
- 换算：$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$。温度**差**在两种温标下数值相同：$\Delta T = 1\,\mathrm{K} \Leftrightarrow \Delta\theta = 1\,^\circ\mathrm{C}$。



14.3 比热容与比潜热

14.3.1 比热容 Specific heat capacity

定义 14.2 (比热容 *Specific heat capacity*)

The specific heat capacity of a substance is the thermal energy per unit mass required to raise the temperature of the substance by one degree (one kelvin): $c = \dfrac{Q}{m\Delta\theta}$.

比热容：使单位质量的物质温度升高 1 K（1 °C）所需的热能，单位 $\mathrm{J\,kg^{-1}\,K^{-1}}$。水的比热容约 $4200\,\mathrm{J\,kg^{-1}\,K^{-1}}$。

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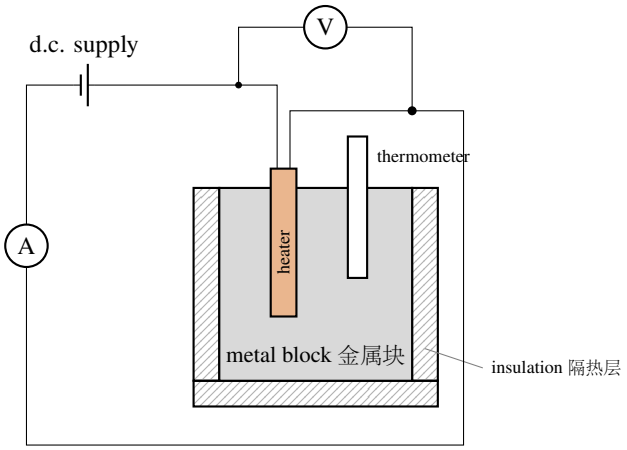
公式 14.1 (热量公式 *Heating without change of state*)

$$Q = mc\Delta\theta$$

混合问题：（无热量损失时）高温物体放出的热量 = 低温物体吸收的热量。

♠

14.3.2 测定比热容的实验（电加热法）



步骤：

(1) 称出金属块质量 m；把加热器和温度计插入块上的孔中（孔内滴少量油改善热接触）；块外包隔热材料。

(2) 记录初温 θ_1；接通电源，同时启动秒表，记录电压 V、电流 I（保持恒定）。

(3) 加热一段时间 t 后断电，继续观察，记录**最高**温度 θ_2。

(4) $VIt = mc(\theta_2 - \theta_1)$，得 $c = \dfrac{VIt}{m(\theta_2 - \theta_1)}$。

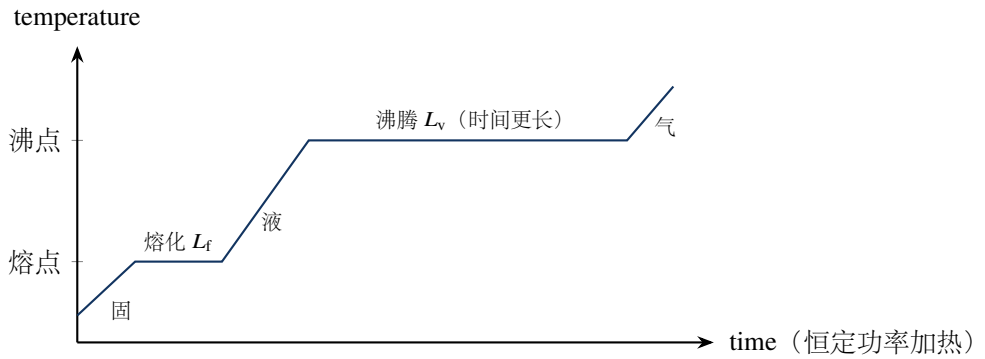
误差：热量散失使测得的 $\Delta\theta$ 偏小，c 偏大。改进：加隔热层；初温比室温低几度、末温比室温高同样几度，使吸热与散热相互抵消；作 θ–t 图用斜率 $\dfrac{\Delta\theta}{\Delta t} = \dfrac{P}{mc}$ 求 c。液体的比热容可用同样方法（把液体装在隔热容器中并搅拌），或用连续流动法。

14.3.3 比潜热 Specific latent heat

定义 14.3 (比潜热 *Specific latent heat*)

The specific latent heat of fusion (vaporisation) is the thermal energy per unit mass required to change a substance from solid to liquid (liquid to gas) without any change in temperature: $L = \dfrac{Q}{m}$.

熔化比潜热 L_f：单位质量的物质在**温度不变**的条件下由固态变为液态所需的热能。**汽化比潜热** L_v：单位质量的物质在温度不变的条件下由液态变为气态所需的热能。单位 $\mathrm{J\,kg^{-1}}$。$Q = mL$。

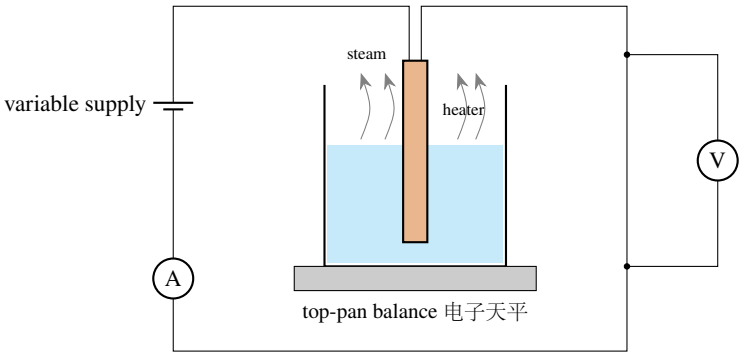


分子解释：状态变化时温度不变，分子平均动能不变；吸收的能量用于**增大分子势能**（克服分子间作用力、增大分子间距）。

为什么 $L_\mathrm{v} > L_\mathrm{f}$（同一物质）：

- 熔化时分子间距只略微增大；汽化时分子要完全分离，间距增大很多，需要克服的分子间作用力做功多得多；
- 汽化时体积大大增加，还要**对外界大气做功**。

14.3.4 测定汽化比潜热的实验（双功率法）



(1) 把装有沸水的容器放在电子天平上，加热器保持水持续沸腾；

(2) 功率 $P_1 = V_1 I_1$ 时，测时间 t 内蒸发的质量 m_1；

(3) 改变功率为 P_2，测相同时间 t 内蒸发的质量 m_2；

(4) 设单位时间散热为 h（两次温度相同，均为沸点，散热速率相同）：$P_1 t = m_1 L_\mathrm{v} + ht$，$P_2 t = m_2 L_\mathrm{v} + ht$，相减消去 h：

$$L_\mathrm{v} = \frac{(P_2 - P_1)t}{m_2 - m_1}.$$

例题 14.1 9702/44/M/J/26 Q3

(a) State what is meant by two objects being in thermal equilibrium. [2]

(b) An insulated container with negligible specific heat capacity contains a mass of 63 g of water at an initial temperature of 78.0 °C. A thermistor with a mass of 8.5 g is at an initial temperature of 25.0 °C. The thermistor is connected into a circuit containing an ammeter that is accurately calibrated to give readings of temperature in °C. The thermistor is immersed in the water. When thermal equilibrium is reached, the ammeter indicates a temperature of 76.8 °C.

(i) Explain why the thermistor does not measure the temperature as 78.0 °C. [2]

(ii) The specific heat capacity of water is 4.2 J g^{-1} K^{-1}. Calculate a value for the specific heat capacity, in J g^{-1} K^{-1}, of the material of the thermistor. [3]

(c) Suggest, with a reason, how the initial temperature of the water in (b) could be measured more accurately than using the thermistor in (b). [2]

解答 Solution

(a) 两物体温度相同； **[B1]**

（接触时）两者之间没有净热能传递。 **[B1]**

(b)(i) 热敏电阻自身升温需要吸收热能； **[B1]**

这些热能来自水，使水降温。 **[B1]**

(ii) $Q = mc\Delta\theta$ **[C1]**

水放热 = 热敏电阻吸热：$63 \times 4.2 \times (78.0 - 76.8) = 8.5 \times c \times (76.8 - 25.0)$ **[C1]**

$c = 0.72\,\mathrm{J\,g^{-1}\,K^{-1}}$ **[A1]**

(c) 用热电偶（或更小的热敏电阻）： **[B1]**

质量（热容）更小，从水中吸收的热能更少。（或：先把热敏电阻预热到约 78 °C 再放入，温度变化小，吸热少。） [B1]

例题 14.2 9702/42/F/M/26 Q3

(a) Define specific latent heat of fusion. [2]

(b) Explain why substances usually have a higher specific latent heat for vaporisation than for fusion. [2]

(c) An ice cube of mass 15.6 g at a temperature of −18 °C is removed from a freezer. The ice cube is immediately placed in an insulated beaker that contains a mass of 132.4 g of water at a temperature of 34 °C. The specific heat capacity of water is 4.19 kJ kg^{-1} °C^{-1}. The specific heat capacity of ice is 2.11 kJ kg^{-1} °C^{-1}. The specific latent heat of fusion of water is 336 kJ kg^{-1}. The beaker has negligible specific heat capacity. The contents of the beaker are left to reach thermal equilibrium. Calculate the temperature, in °C, of the water in the beaker when thermal equilibrium is reached. [4]

解答 Solution

(a) 单位质量的热能， [B1]

使物质在温度不变的情况下由固态变为液态所需。 [B1]

(b) 汽化时体积变化（分子间距增大）大得多； [B1]

体积增大需要对大气做功（所以需要提供更多热能）。 [B1]

(c) $Q = mc\Delta\theta$ [C1]

能量守恒：水降温放出 = 冰升温到 0 °C 吸收 + 冰熔化吸收 + 冰化成的水升温吸收 [C1]

$132.4 \times 4.19 \times (34 - \theta) = 15.6 \times 2.11 \times 18 + 15.6 \times 336 + 15.6 \times 4.19 \times \theta$ [C1]

$18\,862 - 554.8\theta = 592 + 5242 + 65.4\theta$，$\theta = 21$ °C [A1]

例题 14.3 原创仿真题 Exam-style

A block of aluminium of mass 1.00 kg is heated by an electric heater of power 50.0 W for 300 s. The temperature of the block rises by 15.8 K.

(a) Calculate a value for the specific heat capacity of aluminium. [2]

(b) The accepted value is 900 J kg^{-1} K^{-1}. Explain why the experimental value is larger than the accepted value and suggest one improvement to the experiment. [2]

解答 Solution

(a) $Pt = mc\Delta\theta$ [C1]

$$c = \frac{50.0 \times 300}{1.00 \times 15.8} = 949\ \text{J kg}^{-1}\ \text{K}^{-1}$$ [A1]

(b) 部分热量散失到周围，测得的温升偏小，所以计算出的 c 偏大； [B1]

改进：给铝块包上隔热材料（或在孔中加油改善热接触、使初温低于室温与末温高于室温的差相等）。 [B1]

教学提示 Teacher's note 建议 4 课时。两个实验（比热容电加热法、汽化比潜热）必须能完整描述：测量量、仪器、计算式、误差来源与改进。双功率法消去散热的思想也出现在 Paper 5 规划题中。定义题注意"per unit mass"与"without change in temperature"。比较 L_v 与 L_f 的解释是 2026 F/M 真题原题。

14.4 本章小结 Summary

物理量/关系	公式	说明
热力学温度	$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$	不依赖物质性质
比热容	$Q = mc\Delta\theta$	$\mathrm{J\,kg^{-1}\,K^{-1}}$
比潜热	$Q = mL$	温度不变
电加热法	$c = \dfrac{VIt}{m\Delta\theta}$	散热使 c 偏大
双功率法	$L_v = \dfrac{(P_2 - P_1)t}{m_2 - m_1}$	消去散热

14.5 本章练习 Practice questions

练习 14.1 9702/41/M/J/26 Q2(b)

Ideal gases may be used to measure the temperature of the environment in a laboratory experiment. In some situations, however, it is preferable to use a thermocouple thermometer.

(i) Explain how a thermometer that uses an ideal gas can be advantageous compared with a thermometer that uses a liquid in glass. [1]

(ii) Describe a laboratory situation in which a thermocouple thermometer is preferable to a thermometer that uses an ideal gas. [1]

解答 Solution

(i) 理想气体温度计直接测量热力学（绝对）温度，读数不依赖于具体物质的性质（或：测温范围比液体温度计宽得多）。 **[B1]**

(ii) 测量很小质量（很小体积）物体的温度，或温度快速变化的场合（热电偶热容小、反应快，也可以远程读数）。 **[B1]**

练习 14.2 原创仿真题 Exam-style

Which physical property is **not** suitable for use as the basis of a thermometer?

A the resistance of a metal wire

B the e.m.f. of a thermocouple

C the mass of a fixed quantity of liquid

D the volume of a gas at constant pressure

解答 Solution

答案 Answer：C

质量不随温度变化，不能作为测温性质。

练习 14.3 原创仿真题 Exam-style

The temperature of an object increases by 25 °C. What is the increase in its thermodynamic temperature?

A 25 K **B** 248 K **C** 298 K **D** 298.15 K

解答 Solution

答案 Answer：A

温度差在两种温标中数值相等。

练习 14.4 原创仿真题 Exam-style

A solid substance is heated at a constant rate. It takes 2.0 minutes to melt completely and, later, 15 minutes to boil away completely. There are no energy losses. What is the ratio $\dfrac{\text{specific latent heat of vaporisation}}{\text{specific latent heat of fusion}}$ of the substance?

A 0.13 **B** 2.0 **C** 7.5 **D** 15

解答 Solution

答案 Answer：C

功率与质量相同，$L \propto$ 时间：$15/2.0 = 7.5$。

练习 14.5 原创仿真题 Exam-style

Water in an electric kettle of power 2.0 kW is boiling. The kettle continues to supply energy for a further 60 s. The specific latent heat of vaporisation of water is $2.26 \times 10^6\,\text{J kg}^{-1}$. Energy losses are negligible. What mass of water is converted to steam in this time?

A 0.019 kg **B** 0.053 kg **C** 0.53 kg **D** 1.9 kg

解答 Solution

答案 Answer：B

$m = \dfrac{Pt}{L_\text{v}} = \dfrac{2000 \times 60}{2.26 \times 10^6} = 0.053\,\text{kg}$。

练习 14.6 原创仿真题 Exam-style

In an experiment to determine the specific latent heat of vaporisation L of water, water in a beaker is kept boiling by an electric heater. The beaker stands on a balance. When the input power is 40.0 W, the mass of water evaporated in 300 s is 4.2 g. When the input power is 60.0 W, the mass of water evaporated in 300 s is 6.9 g.

(a) Explain why the experiment is repeated with two different powers. [2]

(b) Calculate L. [2]

(c) Determine the rate of loss of thermal energy to the surroundings. [2]

解答 Solution

(a) 实验中有热量散失到周围；两次水都在沸点，散热速率相同， **[B1]**

两式相减可以消去散热项。 **[B1]**

(b) $(P_2 - P_1)t = (m_2 - m_1)L$ **[C1]**

$$L = \frac{(60.0 - 40.0) \times 300}{(6.9 - 4.2) \times 10^{-3}} = 2.2 \times 10^6 \,\mathrm{J\,kg^{-1}}$$ **[A1]**

(c) $P_1 t = m_1 L + ht$ **[C1]**

$$h = 40.0 - \frac{4.2 \times 10^{-3} \times 2.22 \times 10^6}{300} = 40.0 - 31.1 = 8.9\,\mathrm{W}$$ **[A1]**

第十五章 理想气体 Ideal gases

考纲要求 Syllabus learning outcomes

15.1.1 understand that amount of substance is an SI base quantity with the base unit mol
理解物质的量是 SI 基本量，基本单位为 mol。

15.1.2 use molar quantities where one mole of any substance is the amount containing a number of particles of that substance equal to the Avogadro constant N_A
使用摩尔量：1 mol 任何物质所含的粒子数等于阿伏加德罗常数 N_A。

15.2.1 understand that a gas obeying $pV \propto T$, where T is the thermodynamic temperature, is known as an ideal gas
理解满足 $pV \propto T$（T 为热力学温度）的气体称为理想气体。

15.2.2 recall and use the equation of state for an ideal gas expressed as $pV = nRT$, where n = amount of substance (number of moles) and as $pV = NkT$, where N = number of molecules
记住并使用理想气体状态方程 $pV = nRT$ 与 $pV = NkT$。

15.2.3 recall that the Boltzmann constant k is given by $k = R/N_A$
记住玻尔兹曼常量 $k = R/N_A$。

15.3.1 state the basic assumptions of the kinetic theory of gases
陈述气体动理论的基本假设。

15.3.2 explain how molecular movement causes the pressure exerted by a gas and derive and use the relationship $pV = \frac{1}{3}Nm\langle c^2\rangle$, where $\langle c^2\rangle$ is the mean-square speed
解释分子运动如何产生气体压强，推导并使用 $pV = \frac{1}{3}Nm\langle c^2\rangle$。

15.3.3 understand that the root-mean-square speed $c_{\text{r.m.s.}}$ is given by $\sqrt{\langle c^2\rangle}$
理解方均根速率 $c_{\text{r.m.s.}} = \sqrt{\langle c^2\rangle}$。

15.3.4 compare $pV = \frac{1}{3}Nm\langle c^2\rangle$ with $pV = NkT$ to deduce that the average translational kinetic energy of a molecule is $\frac{3}{2}kT$, and recall and use this expression
比较两式得出分子平均平动动能为 $\frac{3}{2}kT$，并记住和使用该式。

15.1 物质的量 The mole

定义 15.1 (摩尔与阿伏加德罗常数 *Mole and Avogadro constant*)

The **mole** is the amount of substance containing a number of particles equal to the Avogadro constant, $N_A = 6.02 \times 10^{23}\ \text{mol}^{-1}$. The **Avogadro constant** is the number of particles per unit amount of substance (per mole).

物质的量是 SI 基本量，单位 mol。1 mol 任何物质含有 N_A 个粒子。粒子数 $N = nN_A$；质量 = 物质的量 × 摩尔质量。

♣

- 摩尔质量（g/mol）在数值上等于相对分子质量：如 O_2 约 32 g/mol。
- 一个分子的质量 $m = \dfrac{M_{\text{molar}}}{N_A}$，或 $m = A \times 1.66 \times 10^{-27}$ kg（A 为核子数/相对分子质量）。

15.2 理想气体状态方程 Equation of state

实验发现，一定量的气体在压强不太大、温度不太低时满足：

- Boyle 定律：T 不变，$p \propto 1/V$；
- Charles 定律：p 不变，$V \propto T$；
- 压强定律：V 不变，$p \propto T$。

合起来 $pV \propto T$（T 必须用热力学温度）。

定义 15.2 (理想气体　*Ideal gas*)

An ideal gas is a gas that obeys $pV \propto T$ (at all pressures, volumes and temperatures), where T is the thermodynamic temperature.

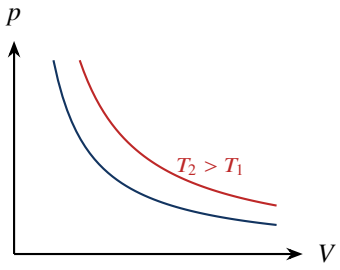
理想气体：在任何压强、体积、温度下都严格满足 $pV \propto T$ 的气体。真实气体在低压、高温下近似为理想气体。

公式 15.1 (理想气体状态方程　*Equation of state for an ideal gas*)

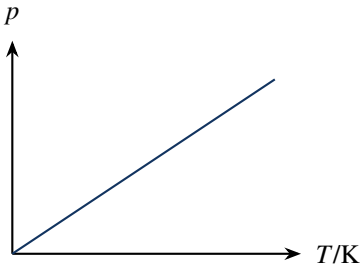
$$pV = nRT \quad \text{或} \quad pV = NkT$$

n：物质的量（mol）；$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$；$N$：分子数；$k = \dfrac{R}{N_\mathrm{A}} = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$（玻尔兹曼常量）。对一定量气体 $\dfrac{p_1V_1}{T_1} = \dfrac{p_2V_2}{T_2}$。

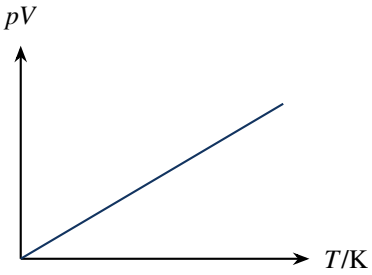
等温线 pV = 常量



定容：$p \propto T$



pV–T：斜率 nR



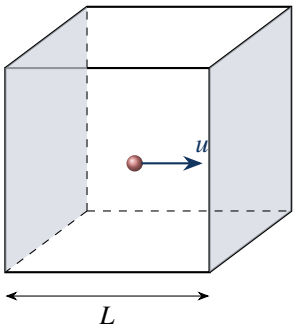
易错提醒 Common mistakes　状态方程中 T 必须用 K；V 用 $\mathrm{m^3}$（$1\,\mathrm{L} = 10^{-3}\,\mathrm{m^3}$，$1\,\mathrm{cm^3} = 10^{-6}\,\mathrm{m^3}$）。

15.3 气体动理论 Kinetic theory of gases

15.3.1 基本假设

(1) 气体由大量分子组成，分子做**永不停息的无规则运动**（random motion）；
(2) 分子本身的体积与气体所占体积相比**可以忽略**；
(3) 除碰撞外，分子之间**没有相互作用力**（所以分子势能为零）；
(4) 分子间及分子与器壁的碰撞都是**完全弹性**的；
(5) 碰撞时间与两次碰撞之间的时间相比**可以忽略**（碰撞是瞬时的）。

15.3.2 压强的微观解释与推导



一次碰撞动量变化：$\Delta p = 2mu$

往返时间：$t = 2L/u$

平均力：$F = \dfrac{2mu}{2L/u} = \dfrac{mu^2}{L}$

压强：$p = \dfrac{F}{L^2} = \dfrac{mu^2}{L^3} = \dfrac{mu^2}{V}$

推导 $pV = \frac{1}{3}Nm\langle c^2\rangle$：

(1) 边长为 L 的立方体容器中，一个质量为 m 的分子以速度分量 u 垂直撞击某一面（阴影面），弹性碰撞后以 $-u$ 反弹，动量变化大小为 $2mu$。

(2) 分子到达对面再返回需要时间 $\dfrac{2L}{u}$，所以每秒撞击该面的次数为 $\dfrac{u}{2L}$。

(3) 器壁对分子的平均力 $= \dfrac{\Delta p}{\Delta t} = \dfrac{2mu}{2L/u} = \dfrac{mu^2}{L}$；由牛顿第三定律，分子对器壁的平均力大小相同。

(4) 该面所受压强 $p = \dfrac{F}{L^2} = \dfrac{mu^2}{L^3} = \dfrac{mu^2}{V}$。

(5) N 个分子速度不同，用 $\langle u^2\rangle$ 代替 u^2，总压强 $p = \dfrac{Nm\langle u^2\rangle}{V}$。

(6) 分子沿三个方向随机运动，$\langle c^2\rangle = \langle u^2\rangle + \langle v^2\rangle + \langle w^2\rangle = 3\langle u^2\rangle$，所以 $\langle u^2\rangle = \frac{1}{3}\langle c^2\rangle$。

公式 15.2 (气体动理论压强公式　*Kinetic theory equation*)

$$pV = \tfrac{1}{3}Nm\langle c^2\rangle \qquad 或 \qquad p = \tfrac{1}{3}\rho\langle c^2\rangle$$

$\langle c^2\rangle$ 为方均速率 (*mean-square speed*)；$c_{\text{r.m.s.}} = \sqrt{\langle c^2\rangle}$ 为方均根速率 (*root-mean-square speed*)。

压强的微观解释：分子撞击器壁时动量发生变化，器壁对分子施力；由第三定律分子对器壁施加大小相等的反作用力。大量分子频繁撞击产生的平均力除以面积即为压强。

考试技巧 Exam tip　计算方均根速率：先平方、再求平均、最后开方。例如五个速率 357, 424, 509, 286, 435：$\langle c^2\rangle = \dfrac{357^2 + 424^2 + 509^2 + 286^2 + 435^2}{5} = 167\,465$，$c_{\text{r.m.s.}} = 409\ \text{m s}^{-1}$（不等于平均速率 402）。

15.3.3 温度的微观意义

比较 $pV = \frac{1}{3}Nm\langle c^2\rangle$ 与 $pV = NkT$：

$$\tfrac{1}{3}Nm\langle c^2\rangle = NkT \quad \Rightarrow \quad \tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT.$$

公式 15.3 (分子平均平动动能　*Mean translational kinetic energy*)

$$\langle E_{\text{K}}\rangle = \tfrac{1}{2}m\langle c^2\rangle = \tfrac{3}{2}kT$$

分子平均平动动能只取决于热力学温度，与气体种类无关。$c_{\text{r.m.s.}} = \sqrt{\dfrac{3kT}{m}} \propto \sqrt{T}$。气体总动能 $= \frac{3}{2}NkT = \frac{3}{2}nRT$——对单原子理想气体，这就是它的内能。

- 同温度下，轻分子的方均根速率更大（$c_{\text{r.m.s.}} \propto 1/\sqrt{m}$），但平均动能相同。
- T 加倍，$\langle E_{\text{K}} \rangle$ 加倍，$c_{\text{r.m.s.}}$ 变为 $\sqrt{2}$ 倍。

例题 15.1 9702/42/M/J/26 Q4

(a) State the two assumptions of the kinetic theory of gases that relate to collisions of particles with each other and with the walls of the container of the gas. [2]

(b) A single particle of an ideal gas has mass m. The particle is moving at speed u in a cubic box of side length L. The particle is moving in a direction that is perpendicular to two (shaded) faces, continually rebounding from one face and then the other. Determine an expression, in terms of some or all of m, u and L, for:

(i) the magnitude of the change in momentum of the particle during each collision with a shaded face [1]

(ii) the time taken between consecutive collisions of the particle with one of the shaded faces [1]

(iii) the mean force exerted on the particle by one of the shaded faces [1]

(iv) the mean pressure p exerted over the area of each of the shaded faces by the particle. Explain your reasoning. [2]

(c) An ideal gas consists of N particles moving in many different directions with a mean-square speed of $\langle c^2 \rangle$. Explain how the equation in (b)(iv) is modified to arrive at the more general equation $pV = \frac{1}{3}Nm\langle c^2 \rangle$ where V is the volume of the gas. [3]

解答 Solution

(a) 碰撞是瞬时的（碰撞时间远小于碰撞间隔）； [B1]

碰撞是完全弹性的。 [B1]

(b)(i) $2mu$ [A1]

(ii) $\dfrac{2L}{u}$ [A1]

(iii) $F = \dfrac{\Delta p}{\Delta t} = \dfrac{2mu}{2L/u} = \dfrac{mu^2}{L}$ [A1]

(iv) 由牛顿第三定律，粒子对面的力与面对粒子的力大小相等； [B1]

$p = \dfrac{F}{A} = \dfrac{mu^2/L}{L^2} = \dfrac{mu^2}{L^3}$ [A1]

(c) 立方体体积 $V = L^3$，所以 $pV = mu^2$； [B1]

粒子向三个方向运动，u^2 换为 $\frac{1}{3}\langle c^2 \rangle$； [B1]

共有 N 个粒子，乘以 N。 [B1]

例题 15.2 9702/41/M/J/26 Q2(a)(c)(d)

(a) State what is meant by an ideal gas. [2]

(b) A very small sample of an ideal gas consists of five molecules, each of mass 3.34×10^{-27} kg. At a particular instant, the speeds of the molecules are 357 m s^{-1}, 424 m s^{-1}, 509 m s^{-1}, 286 m s^{-1} and 435 m s^{-1}. Determine, to three significant figures, the root-mean-square (r.m.s.) speed c_0 of the molecules. [2]

(c) A larger sample of the gas in (b) consists of 5.0×10^{23} molecules. These molecules have the same r.m.s. speed as the molecules of the gas in (b).

(i) Calculate the amount of gas. [2]

(ii) Determine the thermodynamic temperature T_0 of the gas. [2]

(iii) The temperature of the gas is now gradually increased. Sketch the variation of the r.m.s. speed of the molecules of the gas with thermodynamic temperature, starting from the point (T_0, c_0). [3]

解答 Solution

(a) 满足 $pV\propto T$ 的气体， [M1]

T 为热力学温度。 [A1]

(b) $\langle c^2\rangle=\dfrac{357^2+424^2+509^2+286^2+435^2}{5}=\dfrac{837\,327}{5}=167\,465$ [C1]

$c_0=\sqrt{167\,465}=409\,\text{m s}^{-1}$ [A1]

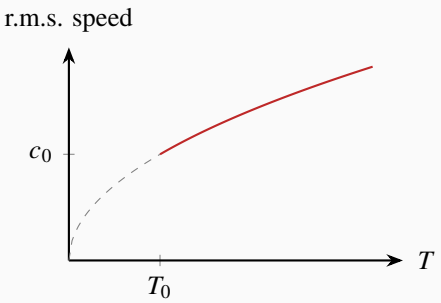
(c)(i) $N=nN_\text{A}$ [C1]

$n=\dfrac{5.0\times10^{23}}{6.02\times10^{23}}=0.83\,\text{mol}$ [A1]

(ii) $\frac{1}{2}m\langle c^2\rangle=\frac{3}{2}kT$ [C1]

$T_0=\dfrac{3.34\times10^{-27}\times409^2}{3\times1.38\times10^{-23}}=13.5\,\text{K}$ [A1]

(iii) $c_\text{r.m.s.}\propto\sqrt{T}$:



从 (T_0, c_0) 开始、斜率为正； [B1]

斜率逐渐减小； [B1]

反向延长通过原点。 [B1]

教学提示 Teacher's note 建议 4 课时。动理论推导是 Paper 4 的经典"长推导题"，2026 年 M/J 42 卷把它拆成小问考查：要强调每一步的物理依据（弹性碰撞 ⇒ 动量变化 $2mu$；第三定律；三维 ⇒ $\frac{1}{3}$；N 个分子）。五条假设至少要能写出四条。注意区分 $\langle c^2\rangle$、$\langle c\rangle^2$ 与 $c_\text{r.m.s.}$。

15.4 本章小结 Summary

物理量/关系	公式	说明
摩尔	$N = nN_A$	$N_A = 6.02 \times 10^{23}\ \text{mol}^{-1}$
状态方程	$pV = nRT = NkT$	T 用 K
玻尔兹曼常量	$k = R/N_A$	$1.38 \times 10^{-23}\ \text{J K}^{-1}$
动理论	$pV = \frac{1}{3}Nm\langle c^2\rangle$	$p = \frac{1}{3}\rho\langle c^2\rangle$
平均动能	$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$	只与 T 有关
方均根速率	$c_{\text{r.m.s.}} = \sqrt{3kT/m}$	$\propto \sqrt{T}$

15.5 本章练习 Practice questions

练习 15.1 9702/42/F/M/26 Q5

(a) State what is represented by the Avogadro constant. [1]

(b) A gas is in a sealed container that has a fixed volume. The thermodynamic temperature T of the gas is varied. Figure 5.1 shows how the pressure p of the gas varies with NT, where N is the number of molecules of the gas. Figure 5.2 shows how the mean-square speed $\langle c^2\rangle$ of the molecules of the gas varies with T. Figure 5.3 shows how the total kinetic energy E_K of all the molecules of the gas varies with T.

Figure 5.1

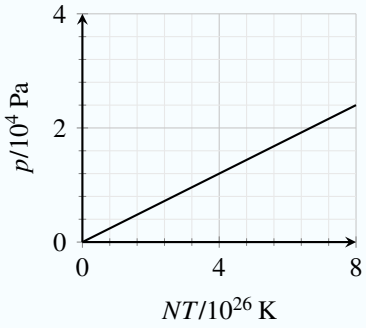


Figure 5.2

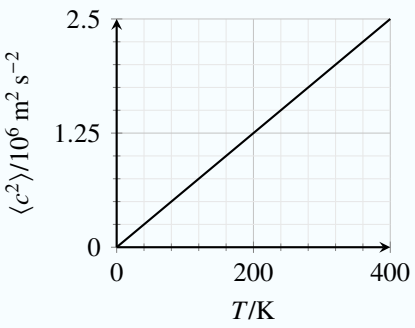
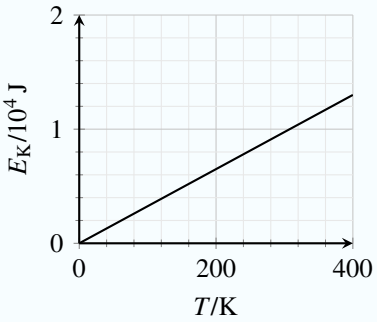


Figure 5.3



Determine three conclusions about the gas that can be drawn from Figures 5.1, 5.2 and 5.3. The

conclusions may be qualitative or quantitative. [3]

(c) One assumption of the kinetic theory of gases is that the volume of the molecules is negligible compared with the volume of the gas.

(i) State, with a reason, whether this assumption applies to the gas in (b). [1]

(ii) State two other assumptions of the kinetic theory of gases. [2]

解答 Solution

(a) 单位物质的量所含的粒子数。 **[B1]**

(b) 任三点（每点 1 分）： **[B3]**

- 图线都是过原点的直线 $\Rightarrow$ 气体是理想气体；
- 图 5.1 斜率 $= k/V = \dfrac{2.4\times10^{4}}{8\times10^{26}} = 3.0\times10^{-23}$，所以 $V = 0.46\,\mathrm{m}^3$；
- 图 5.2：$\langle c^2\rangle = 3kT/m$，斜率 6250，分子质量 $m = 3k/6250 = 6.6\times10^{-27}\,\mathrm{kg}$（摩尔质量 $4.0\times10^{-3}\,\mathrm{kg\,mol^{-1}}$，可能是氦）；
- 图 5.3：$E_\mathrm{K} = \frac{3}{2}NkT$，$N = \dfrac{1.3\times10^{4}}{1.5\times1.38\times10^{-23}\times400} = 1.6\times10^{24}$（即 2.6 mol）；
- 气体质量 $= Nm = 0.010\,\mathrm{kg}$，密度 $= 0.022\,\mathrm{kg\,m^{-3}}$。

(c)(i) 适用：气体是理想气体（满足 $pV \propto T$）。 **[B1]**

(ii) 任两点：分子做永不停息的无规则运动；碰撞是完全弹性的；除碰撞外分子间无作用力；碰撞是瞬时的。 **[B2]**

练习 15.2 原创仿真题 Exam-style

A container of volume $1.0\,\mathrm{m}^3$ holds an ideal gas at a pressure of $1.0\times10^{5}\,\mathrm{Pa}$ and a temperature of 300 K. Approximately how many molecules of gas are in the container?

A 2.4×10^{22} **B** 2.4×10^{25}

C 4.0×10^{1} **D** 6.0×10^{23}

解答 Solution

答案 Answer：**B**

$N = \dfrac{pV}{kT} = \dfrac{1.0\times10^{5}\times1.0}{1.38\times10^{-23}\times300} = 2.4\times10^{25}$。（C 是 $n = pV/RT$ 的摩尔数。）

练习 15.3 原创仿真题 Exam-style

The thermodynamic temperature of an ideal gas is doubled. By what factor does the root-mean-square speed of its molecules increase?

A $\sqrt{2}$ **B** 2 **C** $2\sqrt{2}$ **D** 4

解答 Solution

答案 Answer：**A**

$c_\mathrm{r.m.s.} \propto \sqrt{T}$。

练习 15.4 原创仿真题 Exam-style

Which statement is **not** an assumption of the kinetic theory of an ideal gas?

A The molecules are in continuous random motion.

B All the molecules move with the same speed.

C The collisions of the molecules are perfectly elastic.

D There are no intermolecular forces except during collisions.

解答 Solution

答案 Answer：**B**

分子速率各不相同（有分布），所以才需要用方均速率 $\langle c^2\rangle$。

练习 15.5 原创仿真题 Exam-style

Oxygen may be treated as an ideal gas. A molecule of oxygen has a mass of 32 u.

(a) Calculate the mean translational kinetic energy of an oxygen molecule at 300 K. [2]

(b) Calculate the root-mean-square speed of oxygen molecules at 300 K. [2]

(c) State, with a reason, how the r.m.s. speed of helium atoms (mass 4 u) at 300 K compares with that of oxygen molecules. [2]

解答 Solution

(a) $\langle E_K\rangle = \frac{3}{2}kT = 1.5 \times 1.38 \times 10^{-23} \times 300$ **[C1]**

$= 6.2 \times 10^{-21}$ J **[A1]**

(b) $m = 32 \times 1.66 \times 10^{-27} = 5.31 \times 10^{-26}$ kg，$c_{\text{r.m.s.}} = \sqrt{\dfrac{2\langle E_K\rangle}{m}} = \sqrt{\dfrac{2 \times 6.21 \times 10^{-21}}{5.31 \times 10^{-26}}}$ **[C1]**

$= 480\ \text{m s}^{-1}$ **[A1]**

(c) 同温度下平均动能相同，$c_{\text{r.m.s.}} \propto 1/\sqrt{m}$； **[B1]**

氦的方均根速率为氧的 $\sqrt{32/4} = 2.8$ 倍（约 $1.4 \times 10^3\ \text{m s}^{-1}$）。 **[B1]**

第十六章 热力学 Thermodynamics

考纲要求 Syllabus learning outcomes

16.1.1 understand that internal energy is determined by the state of the system and that it can be expressed as the sum of a random distribution of kinetic and potential energies associated with the molecules of a system
理解内能由系统状态决定，可表示为系统中分子随机分布的动能与势能之和。

16.1.2 relate a rise in temperature of an object to an increase in its internal energy
把物体温度升高与其内能增加联系起来。

16.2.1 recall and use $W = p\Delta V$ for the work done when the volume of a gas changes at constant pressure and understand the difference between the work done by the gas and the work done on the gas
记住并使用定压下气体体积变化做功 $W = p\Delta V$，区分气体对外做功与外界对气体做功。

16.2.2 recall and use the first law of thermodynamics $\Delta U = q + W$ expressed in terms of the increase in internal energy, the heating of the system (energy transferred to the system by heating) and the work done on the system
记住并使用热力学第一定律 $\Delta U = q + W$（内能增量 = 对系统加热传入的能量 + 外界对系统做的功）。

16.1 内能 Internal energy

定义 16.1 (内能 *Internal energy*)

The internal energy of a system is the sum of a random distribution of the kinetic and potential energies of the molecules (particles) in the system.

内能：系统内所有分子**无规则运动**的动能与分子间势能之**和**。内能是**状态函数**，只由系统当前状态（p、V、T 等）决定，与系统如何到达这一状态无关。

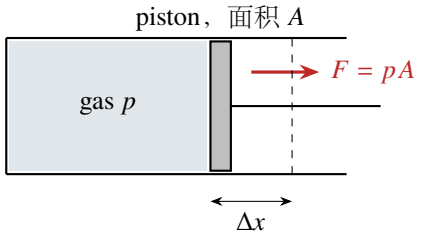
- **分子动能**：与温度有关。温度升高 $\Rightarrow$ 分子平均动能增大 $\Rightarrow$ 内能增大。
- **分子势能**：与分子间距（物态）有关。熔化、汽化时温度不变，动能不变，但势能增大，所以内能增大。
- **理想气体**：分子间无作用力，势能为零，内能 = 分子总动能：

$$U = \frac{3}{2}NkT = \frac{3}{2}nRT = \frac{3}{2}pV \quad \text{(单原子)}$$

所以理想气体的内能**只与温度有关**：$U \propto T$，$\Delta U \propto \Delta T$。

易错提醒 Common mistakes 内能中的“动能”指分子**无规则**运动的动能，不包括物体整体运动的动能；势能指分子间的势能，不包括物体整体的重力势能。把一杯水举高或使其高速运动，其内能不变。

16.2 气体做功 Work done by and on a gas



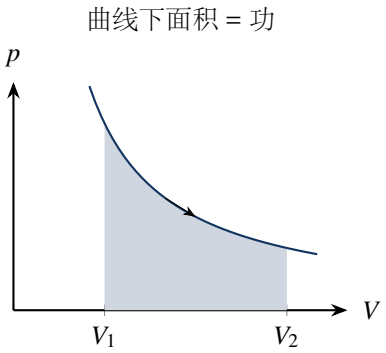
$$W = F\Delta x = pA\Delta x = p\Delta V$$

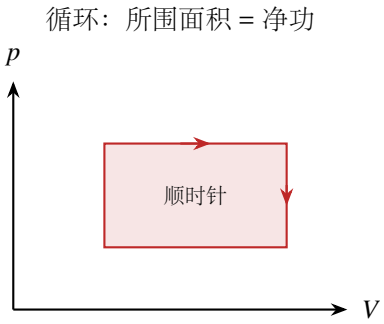
公式 16.1 (定压下气体做功 *Work done at constant pressure*)

$$W = p\Delta V$$

气体膨胀（$\Delta V > 0$）：气体对外做功，外界对气体做负功；气体被压缩（$\Delta V < 0$）：外界对气体做功（正功）。

p–V **图**：曲线下的面积等于气体做的功（无论压强是否恒定）。循环过程中，闭合曲线所围面积等于一个循环中的净功：顺时针循环气体对外做净功，逆时针循环外界对气体做净功。





16.3 热力学第一定律 First law of thermodynamics

定律 16.1 (热力学第一定律 *First law of thermodynamics*)

$$\Delta U = q + W$$

The increase in internal energy of a system is equal to the sum of the energy transferred to the system by heating and the work done on the system.

系统内能的**增加量** = **对系统加热**传入的热能 + **外界对系统**做的功。它是能量守恒在热学中的表述。

符号	ΔU	q	W
正 +	内能增加（T 升高）	系统吸热	外界对系统做功（压缩）
负 −	内能减少（T 降低）	系统放热	系统对外做功（膨胀）

易错提醒 Common mistakes CIE 的约定是 W 为**外界对系统做的功**（work done **on** the system）。气体膨胀时 W 取负值。许多教材写 $\Delta U = Q - W$（W 为系统对外做功），考试时一定按题目给出的形式使用。

16.3.1 特殊过程（理想气体）

过程	特征	第一定律	说明
等容 constant volume	$\Delta V = 0$，$W = 0$	$\Delta U = q$	吸收的热全部使内能增加
等压 constant pressure	$W = -p\Delta V$	$\Delta U = q - p\Delta V$	吸热一部分用于对外做功
等温 isothermal	$\Delta T = 0$，$\Delta U = 0$	$q = -W$	吸热全部用于对外做功
绝热 adiabatic（快速）	$q = 0$	$\Delta U = W$	快速压缩升温，快速膨胀降温
循环 cycle	回到初态	$\Delta U = 0$	净吸热 = 对外净功

考试技巧 Exam tip 由于理想气体 $U \propto T$，而 $pV \propto T$，所以 $U \propto pV$。在 p–V 图上任意两点的内能之比等于 pV 乘积之比——这是快速求 ΔU 的捷径（如定压下体积变为 80%，内能也变为 80%）。

例题 16.1 9702/42/M/J/26 Q3

(a) Define specific heat capacity. [2]

(b) A sample of an ideal gas contains 6.00×10^{23} molecules, each of mass 6.64×10^{-27} kg. The gas is at a pressure of 1.01×10^5 Pa. It has an initial volume of 0.0243 m^3, an initial internal energy of 3676 J and an initial temperature of 296 K. The gas is now cooled at constant pressure so that its volume becomes 80% of the initial volume.

(i) Show that the magnitude of the work done on the gas is 491 J. [2]

(ii) Explain whether the work done on the gas is positive or negative. [1]

(iii) Explain why the increase in internal energy of the gas is −735 J. [2]

(iv) Determine the thermal energy Q transferred to the gas. [2]

(v) Calculate a value for the specific heat capacity c of the gas for this change. [2]

(c) The gas in (b) is now heated at constant volume so that its temperature returns to 296 K. Use the first law of thermodynamics to explain, without calculation, whether the specific heat capacity for this change is less than, the same as, or greater than the answer in (b)(v). [3]

解答 Solution

(a) 单位质量的热能， [B1]

使温度改变 1 K 所需（per unit change in temperature）。 [B1]

(b)(i) $W = p\Delta V$ [C1]

$= 1.01 \times 10^5 \times (1.00 - 0.80) \times 0.0243 = 491\,\text{J}$ [A1]

(ii) 体积减小（气体被压缩），所以外界对气体做的功为**正**。 [B1]

(iii) 定压下 $T \propto V$，而内能 $\propto T$； [B1]

所以内能减为初值的 80%，增量 $= -0.20 \times 3676 = -735\,\text{J}$。 [B1]

(iv) $\Delta U = Q + W$：$-735 = Q + 491$ [C1]

$Q = -1226\,\text{J}$（放热 1230 J） [A1]

(v) $\Delta T = 0.20 \times 296 = 59\,\text{K}$ [C1]

$$c = \frac{1226}{6.00 \times 10^{23} \times 6.64 \times 10^{-27} \times 59} = 5200\,\text{J kg}^{-1}\,\text{K}^{-1}$$ [A1]

(c) 温度变化相同，内能变化相同； [B1]

等容过程不做功； [B1]

所需热能更少（只需 ΔU），所以比热容**更小**。 [B1]

例题 16.2 9702/42/F/M/26 Q4

The first law of thermodynamics may be written as $\Delta U = Q + W$ where W is the work done on a system.

(a) State the meaning, in this equation, of the symbols ΔU and Q. [2]

(b) An ideal gas at a pressure of 2.0×10^5 Pa has an initial volume of $0.048\,\text{m}^3$, an initial thermodynamic temperature of 560 K, and an initial internal energy of 14.4 kJ. The volume of the gas is reduced at constant pressure to a final volume of $0.012\,\text{m}^3$.

(i) Show that the magnitude of the work done on the gas is 7200 J. [2]

(ii) Explain whether the work done on the gas is positive or negative. [1]

(iii) Explain why the final internal energy of the gas is 3.6 kJ. [2]

(c) For the reduction of the volume of the gas in (b), calculate Q. Give a unit with your answer. [2]

解答 Solution

(a) ΔU：系统内能的增加量； [B1]

Q：传给系统的热能。 [B1]

(b)(i) $W = p\Delta V$ [C1]

$= 2.0 \times 10^5 \times (0.048 - 0.012) = 7200\,\text{J}$ [A1]

(ii) 气体被压缩，外界对气体做正功。 [B1]

(iii) 定压下 $V \propto T$，而 $U \propto T$； [B1]

体积变为 1/4，所以内能变为 14.4/4 = 3.6 kJ。 [B1]

(c) $\Delta U = 3.6 - 14.4 = -10.8\,\text{kJ}$ [C1]

$Q = \Delta U - W = -10.8 - 7.2 = -18.0\,\text{kJ}$ [A1]

教学提示 Teacher's note 建议 4 课时。最大难点是符号：反复强调 CIE 中 W 是"work done ON the system"。三步法：(1) 判断 W 正负（压缩为正）；(2) 由温度变化判断 ΔU（理想气体 $U \propto T \propto pV$）；(3) 由第一定律求 q。"三个结论"开放题（2026 M/J 41 卷）要求从图中得出关于 W、q、ΔU 的结论并量化，建议训练学生按"每个过程一条 + 整个循环一条"的模式作答。

16.4 本章小结 Summary

物理量/关系	公式	说明
内能	分子随机动能 + 势能之和	状态函数
理想气体内能	$U = \frac{3}{2}NkT$	只与 T 有关
气体做功	$W = p\Delta V$	p–V 图面积
第一定律	$\Delta U = q + W$	W：外界对系统做功
等容 / 等温 / 绝热	$W = 0$ / $\Delta U = 0$ / $q = 0$	
循环	$\Delta U = 0$	

16.5 本章练习 Practice questions

练习 16.1 9702/41/M/J/26 Q3

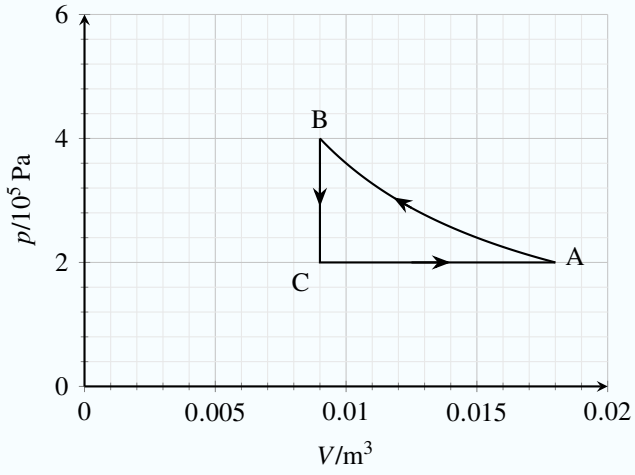
(a) State what is meant by the internal energy of a system. [2]

(b) An ideal gas containing 8.7×10^{23} molecules is at a temperature of 300 K and a pressure of 2.0×10^{5} Pa.

(i) Show that the volume of the gas is $0.018\ m^3$. [2]

(ii) Determine the internal energy of the gas. Explain your reasoning. [3]

(c) The gas in (b) undergoes a sequence of changes AB, BC and CA in its pressure p and volume V, as shown in Figure 3.1.



Determine three conclusions that can be drawn from Figure 3.1 about the individual and overall effects

of these changes on the energy of the gas. The conclusions may be qualitative or quantitative, and may refer to any of work done, transfer of thermal energy and change in internal energy. [3]

解答 Solution

(a) 分子势能与动能之和; **[B1]**

（分子）无规则运动的能量。 **[B1]**

(b)(i) $pV = NkT$ **[C1]**

$$V = \frac{8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300}{2.0 \times 10^{5}} = 0.018\,\text{m}^3$$ **[A1]**

(ii) 理想气体分子势能为零（无分子间力）; **[B1]**

$U = \frac{3}{2}NkT$ **[C1]**

$= 1.5 \times 8.7 \times 10^{23} \times 1.38 \times 10^{-23} \times 300 = 5400\,\text{J}$ **[A1]**

(c) 任三点（需指明对应哪个过程）: **[B3]**

- AB: pV 不变（等温），内能不变；外界对气体做的功 = 气体放出的热;
- BC: 等容，不做功; pV 减半，内能减半（减少 2.7 kJ），放出的热 = 2.7 kJ;
- CA: 等压膨胀，气体对外做功 $p\Delta V = 2.0 \times 10^5 \times 0.009 = 1.8$ kJ；内能加倍（增加 2.7 kJ）；吸热 = 4.5 kJ;
- 整个循环: 内能无净变化；逆时针循环，外界对气体做净功，气体净放热，二者相等。

练习 16.2 9702/44/M/J/26 Q4

(a) State the first law of thermodynamics. [2]

(b) A sample of an ideal gas contains 3.58×10^{24} molecules. The sample is at a temperature of 298 K and a pressure of 1.84×10^5 Pa.

(i) Show that the volume of the gas is $0.0800\,\text{m}^3$. [2]

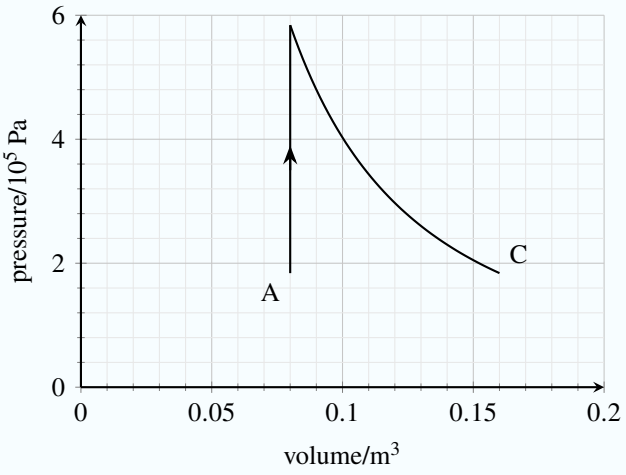
(ii) Calculate the amount of gas n in the sample. [1]

(iii) Determine the average translational kinetic energy E_K of one molecule of the gas. [2]

(iv) Explain why the average potential energy of one molecule of the gas is zero. [1]

(v) Calculate the internal energy U_A of the gas. [1]

(c) The gas in (b) undergoes two changes AB and BC, as shown in Figure 4.1.



In change AB, the gas is heated at constant volume from its initial pressure of 1.84×10^5 Pa to a pressure of 5.84×10^5 Pa. In change BC, the gas rapidly expands and returns to its original pressure with no transfer of thermal energy. The volume of the gas after this expansion is $0.160\,\text{m}^3$.

(i) Determine an expression, in terms of U_A, for the thermal energy Q supplied to the gas during change AB. Explain your reasoning. [2]

(ii) Determine an expression, in terms of U_A, for the work W done on the gas during change BC. Explain your reasoning. [2]

解答 Solution

(a) 内能增加量 = 外界对系统做的功 + 对系统加热传入的能量。 [C1+A1]

(b)(i) $pV = NkT$ [C1]

$$V = \frac{3.58\times10^{24}\times1.38\times10^{-23}\times298}{1.84\times10^5} = 0.0800\,\text{m}^3$$ [A1]

(ii) $n = \dfrac{3.58\times10^{24}}{6.02\times10^{23}} = 5.95\,\text{mol}$ [A1]

(iii) $E_\text{K} = \frac{3}{2}kT$ [C1]

$= 1.5\times1.38\times10^{-23}\times298 = 6.17\times10^{-21}\,\text{J}$ [A1]

(iv) 理想气体分子间没有作用力。 [B1]

(v) $U_\text{A} = 3.58\times10^{24}\times6.17\times10^{-21} = 2.21\times10^4\,\text{J}$ [A1]

(c)(i) 等容，不做功，$Q = \Delta U$；$U \propto pV$，$U_\text{B} = \dfrac{5.84}{1.84}U_\text{A}$： [B1]

$$Q = \frac{5.84-1.84}{1.84}U_\text{A} = 2.2U_\text{A}$$ [A1]

(ii) 无热传递，$W = \Delta U$；C 点 pV 是 A 点的 2 倍，$U_\text{C} = 2U_\text{A}$： [B1]

$W = U_\text{C} - U_\text{B} = 2U_\text{A} - 3.17U_\text{A} = -1.2U_\text{A}$ [A1]

练习 16.3 原创仿真题 Exam-style

A gas is compressed rapidly so that no thermal energy is transferred to or from the gas. Which row describes the work done on the gas W and the change in internal energy ΔU?

	W	ΔU
A	negative	decrease
B	negative	increase
C	positive	decrease
D	positive	increase

解答 Solution

答案 Answer：D

压缩：外界对气体做正功；$q = 0$，$\Delta U = W > 0$，内能增加（温度升高）。

练习 16.4 原创仿真题 Exam-style

A block of ice at 0 °C melts to form water at 0 °C. Which statement about the molecules is correct?

A Their mean kinetic energy increases and their potential energy is unchanged.

B Their mean kinetic energy is unchanged and their potential energy increases.

C Both their mean kinetic energy and their potential energy increase.

D Neither their mean kinetic energy nor their potential energy changes.

解答 Solution

答案 Answer：B

温度不变 ⇒ 平均动能不变；吸收的热能用于增大分子势能，内能增加。

练习 16.5 原创仿真题 Exam-style

An ideal gas expands at a constant pressure of 1.5×10^{5} Pa from a volume of 2.0×10^{-3} m^3 to 5.0×10^{-3} m^3. During the expansion 1200 J of thermal energy is supplied to the gas. What is the increase in internal energy of the gas?

A 450 J **B** 750 J **C** 1200 J **D** 1650 J

解答 Solution

答案 Answer：B

外界对气体做功 $W = -p\Delta V = -1.5\times10^{5}\times3.0\times10^{-3} = -450$ J。$\Delta U = q + W = 1200 - 450 = 750$ J。

第十七章 振动 Oscillations

考纲要求 Syllabus learning outcomes

17.1.1 understand and use the terms displacement, amplitude, period, frequency, angular frequency and phase difference in the context of oscillations, and express the period in terms of both frequency and angular frequency
在振动中理解并使用位移、振幅、周期、频率、角频率和相位差，并用频率和角频率表示周期。

17.1.2 understand that simple harmonic motion occurs when acceleration is proportional to displacement from a fixed point and in the opposite direction
理解当加速度与偏离某固定点的位移成正比且方向相反时发生简谐运动。

17.1.3 use $a = -\omega^2 x$ and recall and use, as a solution to this equation, $x = x_0 \sin \omega t$
使用 $a = -\omega^2 x$，并记住和使用其解 $x = x_0 \sin \omega t$。

17.1.4 use the equations $v = v_0 \cos \omega t$ and $v = \pm\omega\sqrt{x_0^2 - x^2}$
使用 $v = v_0 \cos \omega t$ 和 $v = \pm\omega\sqrt{x_0^2 - x^2}$。

17.1.5 analyse and interpret graphical representations of the variations of displacement, velocity and acceleration for simple harmonic motion
分析并解读简谐运动中位移、速度、加速度变化的图像。

17.2.1 describe the interchange between kinetic and potential energy during simple harmonic motion
描述简谐运动中动能与势能的相互转化。

17.2.2 recall and use $E = \frac{1}{2}m\omega^2 x_0^2$ for the total energy of a system undergoing simple harmonic motion
记住并使用简谐运动总能量 $E = \frac{1}{2}m\omega^2 x_0^2$。

17.3.1 understand that a resistive force acting on an oscillating system causes damping
理解阻力作用于振动系统导致阻尼。

17.3.2 understand and use the terms light, critical and heavy damping and sketch displacement–time graphs illustrating these types of damping
理解并使用轻阻尼、临界阻尼、重阻尼，并画出相应的位移–时间图。

17.3.3 understand that resonance involves a maximum amplitude of oscillations and that this occurs when an oscillating system is forced to oscillate at its natural frequency
理解共振指振幅达到最大，发生在系统被迫以其固有频率振动时。

17.1 描述振动的物理量

术语	含义
位移 x displacement	偏离平衡位置的距离（有方向）
振幅 x_0 amplitude	偏离平衡位置的最大位移
周期 T period	完成一次全振动所需的时间
频率 f frequency	单位时间内完成全振动的次数，$f = 1/T$
角频率 ω angular frequency	$\omega = 2\pi f = \dfrac{2\pi}{T}$，单位 $\mathrm{rad\,s^{-1}}$
相位差 phase difference	两个振动在同一时刻所处阶段之差，$\Delta\phi = \dfrac{\Delta t}{T} \times 2\pi$

17.2 简谐运动 Simple harmonic motion

定义 17.1 (简谐运动　*Simple harmonic motion*)

Simple harmonic motion is motion in which the acceleration is proportional to the displacement from a fixed point (the equilibrium position) and is always directed towards that point (opposite in direction to the displacement).

简谐运动：加速度与偏离某固定点（平衡位置）的位移**成正比**，且方向**始终与位移相反**（指向平衡位置）的运动。

公式 17.1 (简谐运动的定义式　*Defining equation of s.h.m.*)

$$a = -\omega^2 x$$

负号表示加速度与位移方向相反。a–x 图是过原点、斜率为 $-\omega^2$ 的直线。

17.2.1 简谐运动的方程与图像

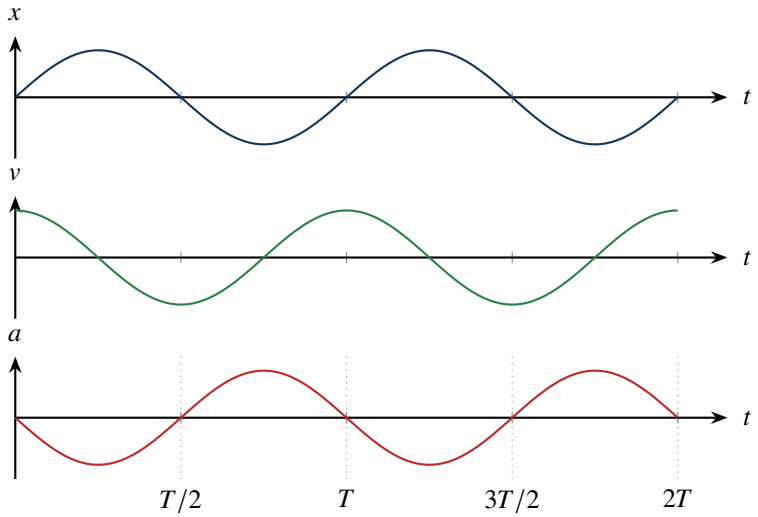
若 $t = 0$ 时物体在平衡位置并向正方向运动：

公式 17.2 (位移、速度、加速度　*Displacement, velocity, acceleration*)

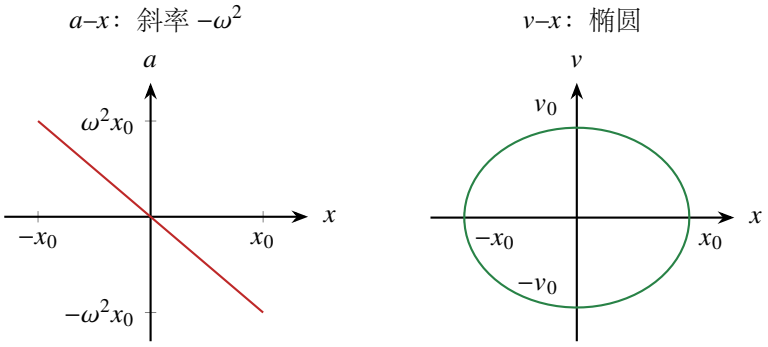
$$x = x_0 \sin \omega t, \qquad v = v_0 \cos \omega t \ (v_0 = \omega x_0), \qquad a = -\omega^2 x_0 \sin \omega t$$

$$v = \pm\omega\sqrt{x_0^2 - x^2}$$

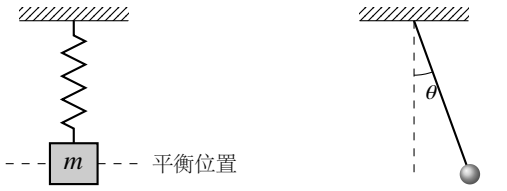
（若 $t = 0$ 时在最大位移处释放，则 $x = x_0 \cos \omega t$，$v = -v_0 \sin \omega t$。）



- v 比 x 超前 $\dfrac{\pi}{2}$（90°）；a 与 x 反相（相差 π）。
- 平衡位置（$x=0$）：速度最大 $v_0=\omega x_0$，加速度为零。
- 最大位移处（$x=\pm x_0$）：速度为零，加速度最大 $a_0=\omega^2 x_0$。
- 周期**与振幅无关**（等时性）。
- v–x 图是椭圆（或圆），与 x 轴交于 $\pm x_0$，与 v 轴交于 $\pm\omega x_0$。



17.2.2 常见简谐振子



弹簧振子　　单摆（小角度）

$T=2\pi\sqrt{m/k}$　　$T=2\pi\sqrt{l/g}$

（以上周期公式不在考纲“必须记住”之列，但常出现在题目信息中；关键是能证明 $a\propto -x$。）

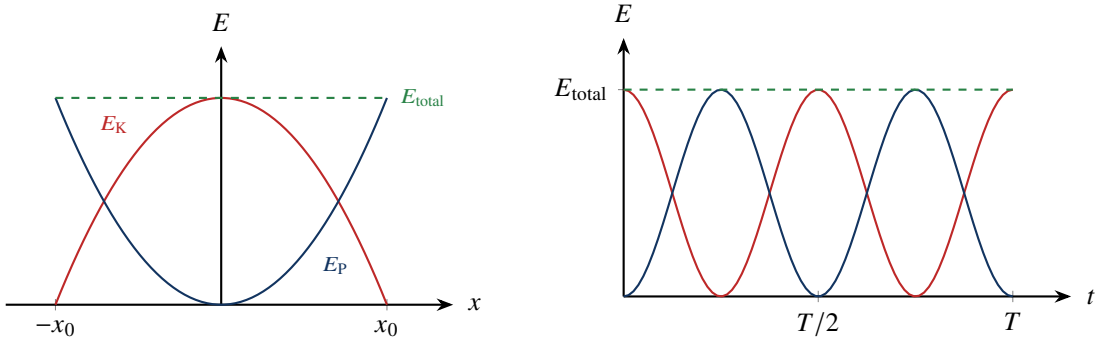
17.3 简谐运动的能量

振动过程中动能与势能不断相互转化，无阻力时总能量守恒：

- 平衡位置：动能最大，势能最小（为零）；
- 最大位移处：动能为零，势能最大。

公式 17.3 (简谐运动的能量 *Energy in s.h.m.*)

$$E_{\text{total}} = \tfrac{1}{2}m\omega^2 x_0^2, \qquad E_{\text{K}} = \tfrac{1}{2}m\omega^2(x_0^2 - x^2), \qquad E_{\text{P}} = \tfrac{1}{2}m\omega^2 x^2$$



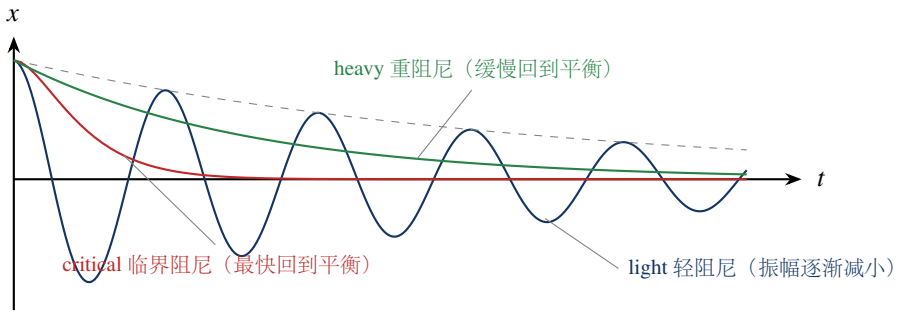
注意：动能、势能随时间变化的频率是振动频率的 **2 倍**（一个周期内各达到两次最大值）。

17.4 阻尼振动 Damped oscillations

定义 17.2 (阻尼 *Damping*)

Damping is the loss of energy (and hence decrease in amplitude) of an oscillating system caused by a resistive force (such as friction or air resistance) acting on it.

阻尼：振动系统受到阻力（摩擦、空气阻力等）作用，能量不断损失、振幅逐渐减小的现象。



- **轻阻尼** (light)：仍然振动，振幅随时间逐渐（近似指数）减小，周期基本不变。
- **临界阻尼** (critical)：不振动，在**最短时间内**回到平衡位置。应用：汽车减震器、电表指针。
- **重阻尼** (heavy)：不振动，缓慢回到平衡位置，所需时间比临界阻尼长。

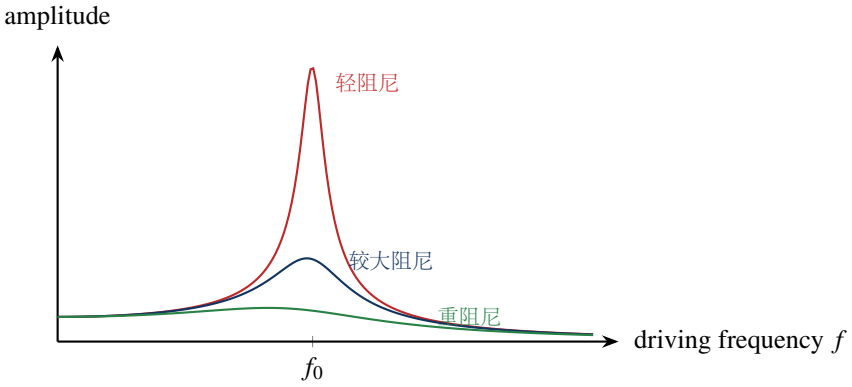
17.5 受迫振动与共振 Forced oscillations and resonance

- **固有频率** (*natural frequency*) f_0：系统自由振动（无外力驱动、无阻尼）时的频率。
- **受迫振动** (*forced oscillation*)：系统在周期性驱动力作用下的振动，稳定后以**驱动频率**振动。

定义 17.3 (共振 *Resonance*)

Resonance is the (large) maximum amplitude of oscillation of a system that occurs when the driving (forcing) frequency is equal to the natural frequency of the system.

共振：当驱动频率等于系统的固有频率时，系统振幅达到最大的现象。此时驱动力对系统的能量传递效率最高。



阻尼增大的影响：(1) 各频率下振幅都减小，峰值降低；(2) 共振峰变宽（变平）；(3) 峰值对应频率略低于 f_0。**有益的共振**：微波炉加热食物中的水分子、乐器的共鸣、MRI；**有害的共振**：桥梁、建筑在风或地震中振动，机器零件的共振——可通过增加阻尼或改变固有频率来避免。

例题 17.1 9702/42/M/J/26 Q5

A block of mass 27 g is suspended from a spring so that it rests in equilibrium. The block is displaced downwards through a small distance. The block is released and then oscillates vertically with simple harmonic motion. The oscillations have a period of 1.7 s and an energy of 13 μJ.

(a) (i) Show that the angular frequency of the oscillations is 3.7 rad s^{-1}. [1]

(ii) State what is meant by the amplitude of the oscillations. [1]

(iii) Calculate the amplitude of the oscillations. [3]

(b) Sketch the variation of the kinetic energy E_K of the block with displacement x from the equilibrium position for one complete oscillation of the block. Add a suitable scale to the x-axis. [3]

解答 Solution

(a)(i) $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{1.7} = 3.7\,\text{rad s}^{-1}$ **[A1]**

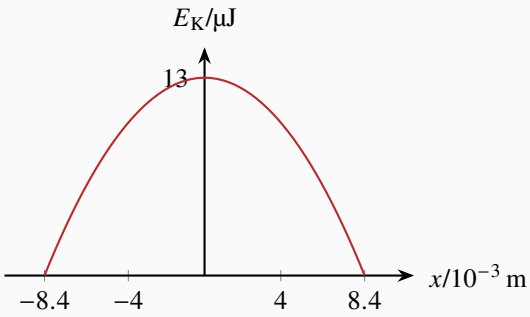
(ii) 偏离平衡位置的最大位移（的大小）。 **[B1]**

(iii) $E = \frac{1}{2}m\omega^2 x_0^2$ **[C1]**

$13 \times 10^{-6} = \frac{1}{2} \times 27 \times 10^{-3} \times 3.7^2 \times x_0^2$ **[C1]**

$x_0 = 8.4 \times 10^{-3}\,\text{m}$ **[A1]**

(b)



倒抛物线（圆顶形），峰值 13 μJ 位于 $x = 0$； [B1]

关于 E_K 轴对称，两端在 $E_K = 0$； [B1]

x 轴标度使端点在 $\pm 8.4 \times 10^{-3}$ m。 [B1]

例题 17.2 9702/44/M/J/26 Q5

A metal ball is attached to the lower end of a spring. The upper end of the spring is attached to a platform that is initially stationary.

(a) The ball is pulled down by a small distance and released so that it oscillates with simple harmonic motion. The oscillations have an amplitude of 2.5 cm and a maximum speed of 4.4 cm s^{-1}.

(i) Sketch the variation of the velocity v of the ball with the displacement x from its equilibrium position as the ball undergoes one full oscillation. [3]

(ii) Show that the frequency of the oscillations is 0.28 Hz. [2]

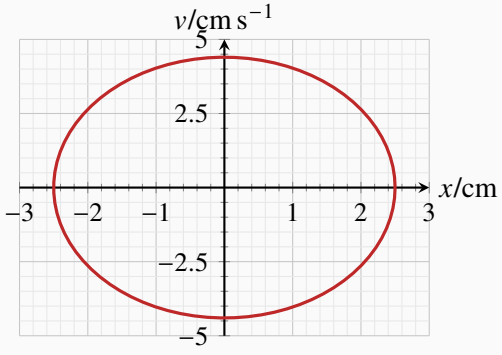
(b) The ball is returned to rest at its equilibrium position. The platform is now moved up and down by a motor so that it oscillates vertically with simple harmonic motion of constant amplitude but variable frequency. The oscillating platform causes the ball to oscillate.

(i) State what is meant by resonance. [2]

(ii) The frequency f of the oscillations of the platform slowly increases from 0 to 0.40 Hz. Sketch the variation with f of the amplitude of the oscillations of the ball. [2]

解答 Solution

(a)(i)



闭合曲线（椭圆）包围原点； [B1]

与 x 轴交于 $\pm 2.5\,\text{cm}$； [B1]

与 v 轴交于 $\pm 4.4\,\text{cm s}^{-1}$。 [B1]

(ii) $v_0 = \omega x_0$，$\omega = 2\pi f$ [C1]

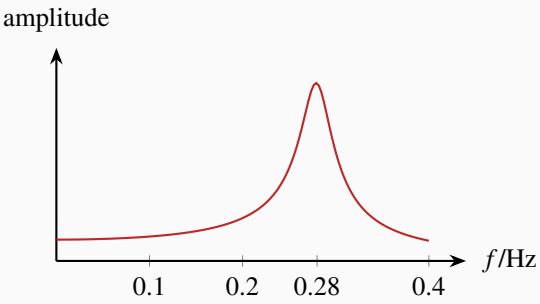
$$f = \frac{4.4}{2\pi \times 2.5} = 0.28\,\text{Hz}$$ [A1]

(b)(i) 振幅达到最大， [B1]

当驱动频率等于系统的固有频率时。 [B1]

(ii) 从 $f = 0$ 开始振幅不为零，连续上升到一个单峰后连续下降； [B1]

峰值在 $f = 0.28\,\text{Hz}$。 [B1]



教学提示 Teacher's note 建议 5 课时。重点：(1) 简谐运动的定义（"proportional"与"opposite direction/towards a fixed point"两个要点缺一不可）；(2) 四类图像（x–t、v–t、a–x、v–x）和能量图；(3) 共振的定义与阻尼对共振曲线的影响。数据表给出了 $x = x_0 \sin \omega t$、$v = v_0 \cos \omega t$、$v = \pm\omega\sqrt{x_0^2 - x^2}$、$a = -\omega^2 x$，但 $E = \frac{1}{2}m\omega^2 x_0^2$ 需要记忆。

17.6 本章小结 Summary

物理量/关系	公式	说明
角频率	$\omega = 2\pi f = 2\pi/T$	
定义式	$a = -\omega^2 x$	$a \propto -x$
位移	$x = x_0 \sin \omega t$	或 $x_0 \cos \omega t$
速度	$v = v_0 \cos \omega t$；$v = \pm\omega\sqrt{x_0^2 - x^2}$	$v_0 = \omega x_0$
最大加速度	$a_0 = \omega^2 x_0$	在最大位移处
总能量	$E = \frac{1}{2}m\omega^2 x_0^2$	$\propto$ 振幅 ^{2}
共振	驱动频率 = 固有频率	振幅最大

17.7 本章练习 Practice questions

练习 17.1 9702/42/F/M/26 Q6(a)(b)

(a) State what is meant by simple harmonic motion. [2]

(b) A rectangular wire frame is suspended by a thin thread and oscillates about its vertical axis. The oscillations are simple harmonic. The displacement s of the edge of the frame from its equilibrium position varies with time t according to

$$s = 3.2 \sin 6.9t$$

where s is in mm and t is in s.

(i) Calculate the period T of the oscillations. [2]

(ii) Show that the maximum speed of the edges of the frame is $22\,\text{mm}\,\text{s}^{-1}$. [2]

解答 Solution

(a) 加速度与位移成正比， [B1]

且方向始终与位移相反（指向一个固定点）。 [B1]

(b)(i) $\omega = 6.9\,\text{rad}\,\text{s}^{-1}$，$T = \dfrac{2\pi}{\omega}$ [C1]

$= \dfrac{2\pi}{6.9} = 0.91\,\text{s}$ [A1]

(ii) $v_0 = \omega x_0$ [C1]

$= 6.9 \times 3.2 = 22\,\text{mm}\,\text{s}^{-1}$ [A1]

练习 17.2 原创仿真题 Exam-style

A particle undergoes simple harmonic motion. Its maximum displacement is 4.0 cm and its maximum acceleration is $1.6\,\text{m}\,\text{s}^{-2}$. What is the frequency of the oscillations?

A 0.32 Hz **B** 1.0 Hz **C** 2.0 Hz **D** 6.3 Hz

解答 Solution

答案 Answer：**B**

$a_0 = \omega^2 x_0$，$\omega = \sqrt{1.6/0.040} = 6.32\,\text{rad}\,\text{s}^{-1}$，$f = \omega/2\pi = 1.0\,\text{Hz}$。

练习 17.3 原创仿真题 Exam-style

An object oscillates with simple harmonic motion. At one instant its displacement is half of the amplitude. What fraction of the total energy is kinetic energy at this instant?

A $\frac{1}{4}$ **B** $\frac{1}{2}$ **C** $\frac{3}{4}$ **D** $\frac{\sqrt{3}}{2}$

解答 Solution

答案 Answer：**C**

$E_\text{P} \propto x^2$，$x = x_0/2$ 时 $E_\text{P} = \frac{1}{4}E$，所以 $E_\text{K} = \frac{3}{4}E$。

练习 17.4 原创仿真题 Exam-style

The suspension system of a car is designed so that, after the car passes over a bump, the car body returns to its equilibrium position as quickly as possible without oscillating. What type of damping is this?

A no damping

B light damping

C critical damping

D heavy damping

解答 Solution

答案 **Answer**：**C**

练习 17.5 原创仿真题 Exam-style

A mass on a spring oscillates with simple harmonic motion of amplitude 0.050 m and period 0.80 s. At time $t = 0$ the mass is at its maximum positive displacement.

(a) Write down an equation for the displacement x of the mass in terms of t. [2]

(b) Calculate the speed of the mass when its displacement is 0.030 m. [2]

(c) On the same axes, sketch graphs showing the variation with time of the displacement and of the acceleration of the mass for one period. Label your graphs. [2]

解答 Solution

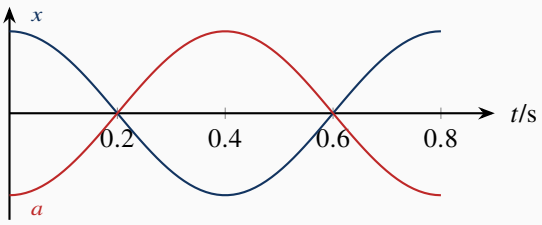
(a) $\omega = 2\pi/0.80 = 7.85\,\text{rad s}^{-1}$；$t = 0$ 时在正最大位移，用余弦： **[C1]**

$x = 0.050\cos(7.85t)$（x 单位 m，t 单位 s） **[A1]**

(b) $v = \omega\sqrt{x_0^2 - x^2} = 7.85\sqrt{0.050^2 - 0.030^2}$ **[C1]**

$= 7.85 \times 0.040 = 0.31\,\text{m s}^{-1}$ **[A1]**

(c) x：余弦曲线从 +0.050 开始；a：与 x 反相的余弦曲线，从 $-\omega^2 x_0 = -3.1\,\text{m s}^{-2}$ 开始，二者在同一时刻过零。 **[B1+B1]**



第十八章 电场 Electric fields

考纲要求 Syllabus learning outcomes

18.1.1 understand that an electric field is an example of a field of force and define electric field as force per unit positive charge
理解电场是一种力场，把电场强度定义为单位正电荷所受的力。

18.1.2 recall and use $F = qE$ for the force on a charge in an electric field
记住并使用 $F = qE$。

18.1.3 represent electric fields by means of field lines
用电场线表示电场。

18.2.1 recall and use $E = \Delta V/\Delta d$ to calculate the field strength of the uniform field between charged parallel plates
记住并使用 $E = \Delta V/\Delta d$ 计算平行板间匀强电场的场强。

18.2.2 describe the effect of a uniform electric field on the motion of charged particles
描述匀强电场对带电粒子运动的影响。

18.3.1 understand that, for any point outside a spherical conductor, the charge on the sphere may be considered to act as a point charge at its centre
理解对球形导体外的任一点，球上的电荷可视为位于球心的点电荷。

18.3.2 recall and use Coulomb's law $F = Q_1Q_2/(4\pi\varepsilon_0 r^2)$ for the force between two point charges in free space
记住并使用真空中两点电荷间的库仑定律。

18.4.1 recall and use $E = Q/(4\pi\varepsilon_0 r^2)$ for the electric field strength due to a point charge in free space
记住并使用真空中点电荷的场强公式。

18.5.1 define electric potential at a point as the work done per unit positive charge in bringing a small test charge from infinity to the point
把某点电势定义为把单位正的小试探电荷从无穷远移到该点所做的功。

18.5.2 recall and use the fact that the electric field at a point is equal to the negative of potential gradient at that point
记住并使用：某点电场强度等于该点电势梯度的负值。

18.5.3 use $V = Q/(4\pi\varepsilon_0 r)$ for the electric potential in the field due to a point charge
使用点电荷电势 $V = Q/(4\pi\varepsilon_0 r)$。

18.5.4 understand how the concept of electric potential leads to the electric potential energy of two point charges and use $E_P = Qq/(4\pi\varepsilon_0 r)$
理解由电势得到两点电荷的电势能，并使用 $E_P = Qq/(4\pi\varepsilon_0 r)$。

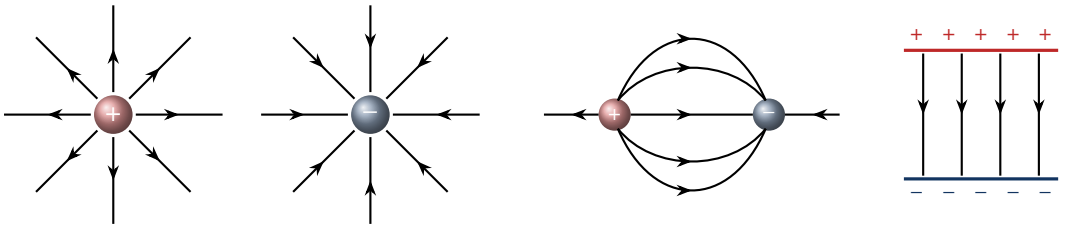
18.1 电场与电场线

定义 18.1 (电场强度 *Electric field strength*)

The electric field strength at a point is the force per unit positive charge acting on a small (test) charge placed at that point: $E = \dfrac{F}{q}$.

电场强度 = 单位**正**电荷所受的电场力，单位 $\mathrm{N\,C^{-1}}$（$=\mathrm{V\,m^{-1}}$），矢量，方向与正电荷受力方向相同。电荷 q 在电场中受力 $F = qE$；负电荷受力方向与 E 相反。

18.1.1 电场线



正点电荷　负点电荷　等量异号电荷　平行板：匀强电场

- 电场线从正电荷出发，终止于负电荷（或无穷远）；方向表示正电荷受力方向；疏密表示场强大小；互不相交。
- 电场线与导体表面**垂直**。
- 平行板之间（远离边缘）电场线平行、等间距 $\Rightarrow$ **匀强电场**；边缘处弯曲（边缘效应）。
- 带电导体球外的电场与球心处同量点电荷的电场相同（径向）。

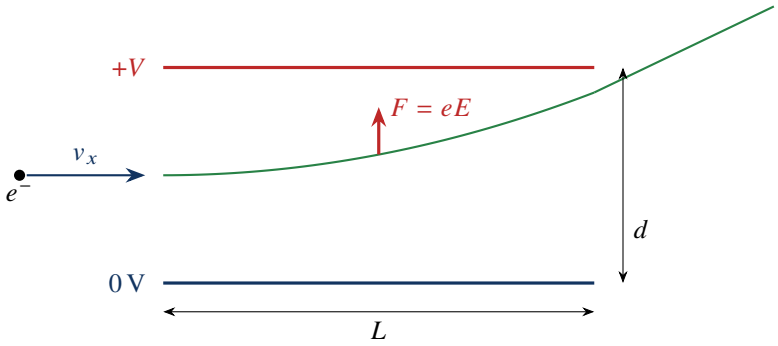
18.2 匀强电场 Uniform electric fields

公式 18.1 (平行板间的场强　*Field between parallel plates*)

$$E = \frac{\Delta V}{\Delta d}$$

ΔV 为两板间电势差，Δd 为板间距。由此 $\mathrm{V\,m^{-1}}$ 与 $\mathrm{N\,C^{-1}}$ 等价。

18.2.1 带电粒子在匀强电场中的运动



水平：v_x 不变
竖直：$a = \dfrac{eE}{m} = \dfrac{eV}{md}$
轨迹为抛物线

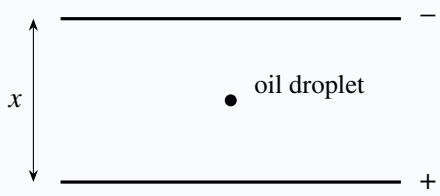
- 粒子初速度平行于场方向：做匀加速（或匀减速）直线运动。电场力做功 qV 转化为动能：$\frac{1}{2}mv^2 = qV$。
- 粒子初速度垂直于场方向：与抛体运动类似——垂直场方向匀速，平行场方向匀加速，轨迹为**抛物线**。离开电场后做直线运动。

- 电子的重力（约 10^{-29} N）远小于电场力，通常忽略；油滴等宏观带电微粒则要考虑重力（Millikan 实验：$qE = mg$ 时悬浮）。
- 偏转程度：电荷越大、质量越小、速度越小，偏转越大。α 粒子与 β 粒子偏转方向相反，β 偏转大得多（质量小）。

例题 18.1 9702/42/F/M/26 Q2

(a) Define electric field at a point. [1]

(b) An oil droplet has charge of magnitude Q. The droplet is held in equilibrium by the electric field between two horizontal parallel plates, as shown in Figure 2.1.



The electric field is created by a potential difference (p.d.) V between the plates, with the lower plate at the higher electric potential. The separation of the plates is x.

(i) Draw four field lines to represent the electric field between the plates. [3]

(ii) Explain, with reference to forces, whether the sign of Q is positive or negative. [2]

(c) For the oil droplet in (b), state expressions, in terms of some or all of Q, V and x, for: (i) the electric field strength E; (ii) the weight W of the droplet. [2]

(d) The plates in (b) have a p.d. across them of 4800 V and a separation of 0.024 m. The magnitude of the charge on the oil droplet is $3e$, where e is the elementary charge. Calculate the mass of the oil droplet. [2]

(e) The polarity of the p.d. across the plates in (b) is reversed. Describe and explain the subsequent motion of the oil droplet. [2]

解答 Solution

(a) 单位正电荷所受的力。 **[B1]**

(b)(i) 四条平行直线，垂直于两板、从一板画到另一板； **[B1]**

大致均匀分布； **[B1]**

箭头**向上**（从高电势的下板指向上板）。 **[B1]**

(ii) 重力向下，所以电场力必须向上； **[B1]**

电场方向向上、电场力也向上，所以 Q 为**正**。 **[B1]**

(c)(i) $E = V/x$ **[A1]**

(ii) $W = QE = \dfrac{QV}{x}$ **[A1]**

(d) $W = mg$ **[C1]**

$m = \dfrac{4800 \times 3 \times 1.60 \times 10^{-19}}{0.024 \times 9.81} = 9.8 \times 10^{-15}\,\text{kg}$ **[A1]**

(e) 电场力和重力都向下； **[B1]**

油滴向下加速运动。 [B1]

18.3 点电荷的电场

定律 18.1 (库仑定律 *Coulomb's law*)

The (electric) force between two point charges is proportional to the product of their charges and inversely proportional to the square of their separation:

$$F = \frac{Q_1 Q_2}{4\pi\varepsilon_0 r^2}$$

$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$ 为真空介电常数，$\dfrac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\mathrm{m\,F^{-1}}$。同号相斥（$F > 0$），异号相吸（$F < 0$）。带电导体球外，电荷可看作集中在球心。

点电荷场强：由 $E = F/q$，

公式 18.2 (点电荷的场强 *Field of a point charge*)

$$E = \frac{Q}{4\pi\varepsilon_0 r^2}$$

方向：正电荷沿半径向外，负电荷沿半径向内。多个电荷时按矢量叠加。

18.4 电势 Electric potential

定义 18.2 (电势 *Electric potential*)

The electric potential at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point.

电势：把单位正的小试探电荷从无穷远移到该点所做的功。单位 V（$= \mathrm{J\,C^{-1}}$），标量。无穷远处电势为零。

公式 18.3 (点电荷的电势与电势能 *Potential and potential energy*)

$$V = \frac{Q}{4\pi\varepsilon_0 r}, \qquad E_\mathrm{P} = qV = \frac{Qq}{4\pi\varepsilon_0 r}$$

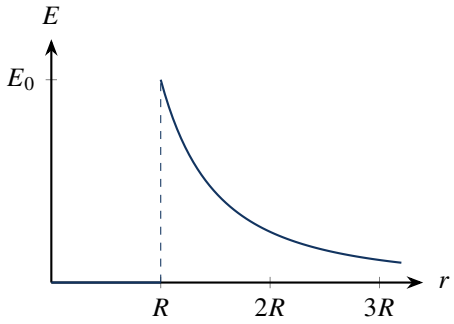
正电荷周围电势为正，负电荷周围电势为负。多个电荷时电势按代数和叠加（标量）。

公式 18.4 (场强与电势梯度 *Field strength and potential gradient*)

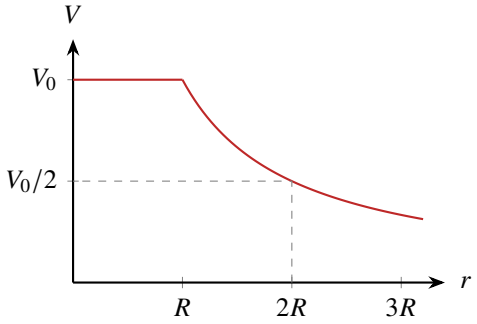
$$E = -\frac{\Delta V}{\Delta x}$$

场强等于电势梯度的负值：V–x 图切线斜率的大小等于该点场强大小；电场方向指向电势降低的方向。

带正电导体球：E–r



带正电导体球：V–r



导体球内部场强为零、电势处处等于表面电势 V_0；球外 $E \propto 1/r^2$，$V \propto 1/r$。

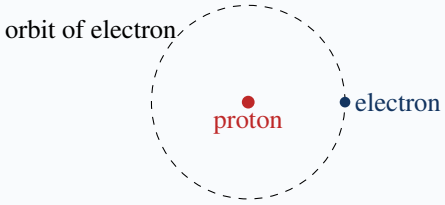
18.4.1 电场与引力场的比较

	引力场	电场
来源	质量 M	电荷 Q
场强定义	单位质量受力 $g = F/m$	单位正电荷受力 $E = F/q$
力的规律	$F = \dfrac{Gm_1m_2}{r^2}$（只有吸引）	$F = \dfrac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$（吸引或排斥）
点源场强	$g = \dfrac{GM}{r^2}$	$E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$
势	$\phi = -\dfrac{GM}{r}$（总为负）	$V = \dfrac{Q}{4\pi\varepsilon_0 r}$（可正可负）
势能	$E_P = -\dfrac{GMm}{r}$	$E_P = \dfrac{Qq}{4\pi\varepsilon_0 r}$
场强与势	$g = -\dfrac{\Delta\phi}{\Delta r}$	$E = -\dfrac{\Delta V}{\Delta r}$

例题 18.2 9702/44/M/J/26 Q1

(a) State Coulomb's law. [2]

(b) In a model of an atom, an electron is in a circular orbit around a proton.



The electric potential energy of the electron is -1.09×10^{-18} J.

(i) Show that the radius of the orbit is 2.11×10^{-10} m. [1]

(ii) Use the information in (b)(i) to determine the magnitude of the electric force between the electron and the proton. [1]

(iii) Calculate the centripetal acceleration of the electron. [1]

(iv) Use your answer in (b)(iii) to determine the period of the orbit of the electron. [2]

(c) The electron in (b) emits a photon so that the electric potential energy of the electron decreases. The

electron remains in a circular orbit around the proton. State how the radius of orbit, centripetal acceleration and angular speed change (decreases / stays the same / increases). [3]

解答 Solution

(a) 两点电荷间的（电）力与它们电荷量的乘积成正比， **[B1]**

与它们之间距离的平方成反比。 **[B1]**

(b)(i) $E_{\mathrm{P}} = \dfrac{Qq}{4\pi\varepsilon_0 r}$：$-1.09\times10^{-18} = -\dfrac{(1.60\times10^{-19})^2}{4\pi\times8.85\times10^{-12}\times r}$，$r = 2.11\times10^{-10}\,\mathrm{m}$ **[A1]**

(ii) $F = \dfrac{(1.60\times10^{-19})^2}{4\pi\times8.85\times10^{-12}\times(2.11\times10^{-10})^2} = 5.17\times10^{-9}\,\mathrm{N}$ **[A1]**

(iii) $a = \dfrac{F}{m} = \dfrac{5.17\times10^{-9}}{9.11\times10^{-31}} = 5.68\times10^{21}\,\mathrm{m\,s^{-2}}$ **[A1]**

(iv) $a = r\omega^2$，$\omega = 2\pi/T$ **[C1]**

$T = 2\pi\sqrt{\dfrac{r}{a}} = 2\pi\sqrt{\dfrac{2.11\times10^{-10}}{5.68\times10^{21}}} = 1.21\times10^{-15}\,\mathrm{s}$ **[A1]**

(c) 半径：减小（势能更负 $\Rightarrow$ r 更小）； **[B1]**

向心加速度：增大（$a \propto 1/r^2$）； **[B1]**

角速度：增大（$\omega = \sqrt{a/r}$）。 **[B1]**

教学提示 Teacher's note 建议 5 课时。可与第 13 章对照教学（比较表）。易错：(1) 场强是矢量（叠加看方向），电势是标量（代数相加，注意正负）；(2) 定义题必须写“positive charge”和“from infinity”；(3) 匀强电场 $E = V/d$ 中 d 是板间距；(4) 带电粒子偏转题与抛体运动完全类比。2026 M/J 41 卷的“比较质子与电子在中点产生的场强和电势”是很好的概念辨析题。

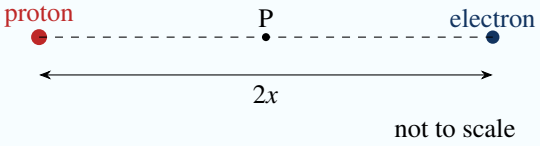
18.5 本章小结 Summary

物理量/关系	公式	说明
场强定义	$E = F/q$	单位正电荷
匀强电场	$E = \Delta V/\Delta d$	$\mathrm{V\,m^{-1}}$
库仑定律	$F = \dfrac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$	
点电荷场强	$E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$	矢量叠加
点电荷电势	$V = \dfrac{Q}{4\pi\varepsilon_0 r}$	标量叠加
电势能	$E_{\mathrm{P}} = \dfrac{Qq}{4\pi\varepsilon_0 r}$	
场强与电势	$E = -\Delta V/\Delta x$	电势梯度

18.6 本章练习 Practice questions

练习 18.1 9702/41/M/J/26 Q4

A hydrogen atom consists of a proton and an electron separated by distance $2x$, as shown in Figure 4.1.



not to scale

Point P is at the midpoint of the line joining the centres of the proton and the electron.

(a) (i) For point P, compare the electric field due to the proton with the electric field due to the electron. [3]

(ii) For point P, compare the electric potential due to the proton with the electric potential due to the electron. [3]

(b) (i) The elementary charge is e and the permittivity of free space is ε_0. State an expression, in terms of x, e and ε_0, for the magnitude of the electric force F between the proton and the electron. [2]

(ii) Determine an expression, in terms of F and x, for the electric potential energy E_P of the proton and the electron. [1]

解答 Solution

(a)(i) 两者场强**大小相等**； [B1]

质子在 P 的场强指向右（背离质子）；电子在 P 的场强也指向右（指向电子）——方向相同。 [B2]

(ii) 两者电势**大小相等**； [B1]

质子产生的电势为正，电子产生的电势为负（符号相反，P 点总电势为零）。 [B2]

(b)(i) $F = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 R^2}$ [C1]

$F = \dfrac{e^2}{16\pi\varepsilon_0 x^2}$ [A1]

(ii) $E_P = \dfrac{e^2}{4\pi\varepsilon_0(2x)} = F \times 2x$，即 $E_P = -2Fx$（大小 $2Fx$）。 [A1]

练习 18.2 原创仿真题 Exam-style

Two parallel metal plates are 4.0 mm apart. The potential difference between them is 200 V. What is the magnitude of the force on an electron between the plates?

A 8.0×10^{-18} N　　**B** 3.2×10^{-17} N

C 8.0×10^{-15} N　　**D** 3.2×10^{-14} N

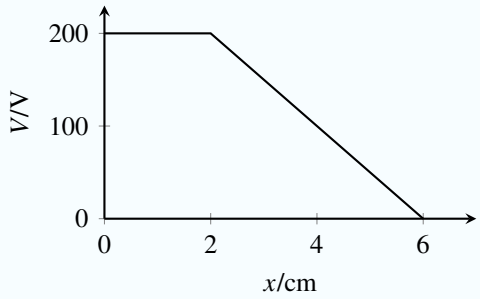
解答 Solution

答案 Answer：C

$E = V/d = 200/(4.0 \times 10^{-3}) = 5.0 \times 10^4$ V m^{-1}；$F = eE = 1.60 \times 10^{-19} \times 5.0 \times 10^4 = 8.0 \times 10^{-15}$ N。

练习 18.3 原创仿真题 Exam-style

The graph shows the variation of electric potential V with distance x along a line.



What is the magnitude of the electric field strength at $x = 4.0\,\text{cm}$?

A $25\,\text{V m}^{-1}$ **B** $50\,\text{V m}^{-1}$ **C** $2500\,\text{V m}^{-1}$ **D** $5000\,\text{V m}^{-1}$

解答 Solution

答案 Answer：**D**

x = 2–6 cm 段电势梯度 $= \dfrac{200}{0.040} = 5000\,\text{V m}^{-1}$，$E$ 等于其大小。（B 是忘了把 cm 换成 m。）

练习 18.4 原创仿真题 Exam-style

Two point charges $+Q$ and $-Q$ are separated by a distance d. What is the electric potential at the midpoint between them, and the electric field strength there?

	potential	field strength
A	zero	zero
B	zero	$\dfrac{2Q}{\pi\varepsilon_0 d^2}$
C	$\dfrac{Q}{\pi\varepsilon_0 d}$	zero
D	$\dfrac{Q}{\pi\varepsilon_0 d}$	$\dfrac{2Q}{\pi\varepsilon_0 d^2}$

解答 Solution

答案 Answer：**B**

电势（标量）：$\dfrac{Q}{4\pi\varepsilon_0(d/2)} + \dfrac{-Q}{4\pi\varepsilon_0(d/2)} = 0$。场强（矢量）：两者方向相同（从 $+Q$ 指向 $-Q$），$2 \times \dfrac{Q}{4\pi\varepsilon_0(d/2)^2} = \dfrac{2Q}{\pi\varepsilon_0 d^2}$。

练习 18.5 原创仿真题 Exam-style

An electron enters the region between two horizontal parallel plates with a horizontal velocity of $3.0 \times 10^7\,\text{m s}^{-1}$, midway between the plates. The plates are 5.0 cm long and 2.0 cm apart, with a potential difference of 400 V between them.

(a) Calculate the acceleration of the electron between the plates. [2]

(b) Calculate the time the electron spends between the plates. [1]

(c) Determine the vertical deflection of the electron as it leaves the plates, and state whether it hits a plate. [2]

(d) Describe the shape of the path of the electron (i) between the plates and (ii) after it leaves the plates. [2]

解答 Solution

(a) $E = V/d = 400/0.020 = 2.0 \times 10^4\ \text{V m}^{-1}$ **[C1]**

$a = \dfrac{eE}{m} = \dfrac{1.60 \times 10^{-19} \times 2.0 \times 10^4}{9.11 \times 10^{-31}} = 3.5 \times 10^{15}\ \text{m s}^{-2}$ **[A1]**

(b) $t = \dfrac{L}{v_x} = \dfrac{0.050}{3.0 \times 10^7} = 1.67 \times 10^{-9}\ \text{s}$ **[A1]**

(c) $y = \frac{1}{2}at^2 = \frac{1}{2} \times 3.5 \times 10^{15} \times (1.67 \times 10^{-9})^2$ **[C1]**

$= 4.9 \times 10^{-3}\ \text{m} = 0.49\ \text{cm} < 1.0\ \text{cm}$（到板的距离），所以不打到板。 **[A1]**

(d)(i) 抛物线（向正极板弯曲）； **[B1]**

(ii) 直线（离开电场后不受力）。 **[B1]**

第十九章 电容 Capacitance

考纲要求 Syllabus learning outcomes

19.1.1 define capacitance, as applied to both isolated spherical conductors and to parallel plate capacitors
定义电容（适用于孤立球形导体和平行板电容器）。

19.1.2 recall and use $C = Q/V$
记住并使用 $C = Q/V$。

19.1.3 derive, using $C = Q/V$, formulae for the combined capacitance of capacitors in series and in parallel
用 $C = Q/V$ 推导电容器串联、并联的等效电容公式。

19.1.4 use the capacitance formulae for capacitors in series and in parallel
使用串联、并联电容公式。

19.2.1 determine the electric potential energy stored in a capacitor from the area under the potential–charge graph
由电势差–电荷图下的面积求电容器储存的电势能。

19.2.2 recall and use $W = \frac{1}{2}QV = \frac{1}{2}CV^2$
记住并使用 $W = \frac{1}{2}QV = \frac{1}{2}CV^2$。

19.3.1 analyse graphs of the variation with time of potential difference, charge and current for a capacitor discharging through a resistor
分析电容器通过电阻放电时电势差、电荷、电流随时间变化的图像。

19.3.2 recall and use $\tau = RC$ for the time constant for a capacitor discharging through a resistor
记住并使用放电时间常数 $\tau = RC$。

19.3.3 use equations of the form $x = x_0\,\mathrm{e}^{-t/RC}$ where x could represent current, charge or potential difference for a capacitor discharging through a resistor
使用 $x = x_0\mathrm{e}^{-t/RC}$（$x$ 可为电流、电荷或电势差）。

19.1 电容 Capacitance

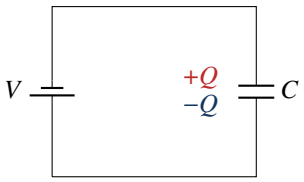
电容器 (*capacitor*) 是储存电荷（和电能）的元件，最简单的是两块相互靠近、彼此绝缘的平行金属板。充电时两板带等量异号电荷 $+Q$ 和 $-Q$，电容器“所带电荷”指**一块板**上电荷的大小 Q。

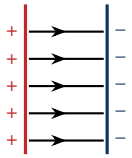
定义 19.1 (电容 *Capacitance*)

Capacitance is the charge stored per unit potential difference: $C = \dfrac{Q}{V}$. For a parallel plate capacitor, Q is the charge on one plate and V is the p.d. between the plates. For an isolated spherical conductor, Q is the charge on the sphere and V is its potential.

电容 = 电荷 / 电势差。单位 farad，$1\,\mathrm{F} = 1\,\mathrm{C\,V^{-1}}$；常用 μF、nF、pF。

孤立导体球的电容：半径 R 的导体球带电 Q 时电势 $V = \dfrac{Q}{4\pi\varepsilon_0 R}$，所以 $C = \dfrac{Q}{V} = 4\pi\varepsilon_0 R$（另一“极板”可看作在无穷远处）。





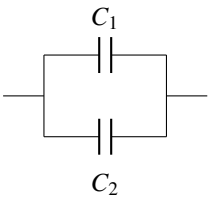
电场储存能量

19.2 电容器的串联与并联



串联：电荷相同

$$\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}$$



并联：电压相同

$$C=C_1+C_2$$

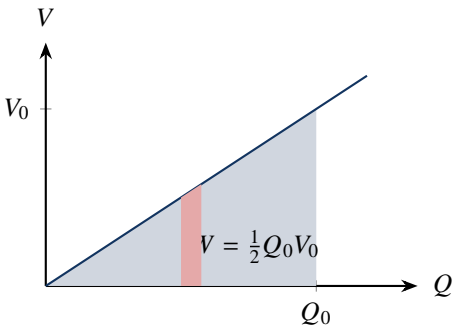
并联推导：各电容器电压都为 V，总电荷 $Q=Q_1+Q_2$（电荷守恒），即 $CV=C_1V+C_2V$，所以 $C=C_1+C_2$。

串联推导：各电容器带相同电荷 Q（中间相连的两极板原来不带电，感应出等量异号电荷），总电压 $V=V_1+V_2$（基尔霍夫第二定律），即 $\frac{Q}{C}=\frac{Q}{C_1}+\frac{Q}{C_2}$，所以 $\frac{1}{C}=\frac{1}{C_1}+\frac{1}{C_2}$。

> **易错提醒 Common mistakes** 电容的串并联公式与电阻**恰好相反**：电容并联相加，串联取倒数和。两个相同电容 C 串联得 $C/2$，并联得 $2C$。

19.3 电容器储存的能量

充电过程中，每移动一小份电荷 ΔQ 需要克服已有电压做功 $V\Delta Q$。V 随 Q 线性增大（$V=Q/C$），所以总功为 V–Q 图下的三角形面积：

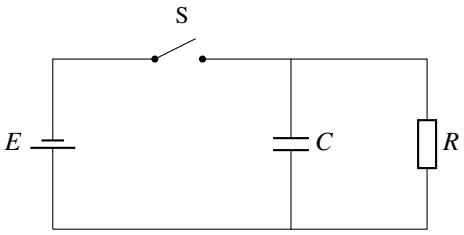


公式 19.1 (电容器储能 *Energy stored in a capacitor*)

$$W=\tfrac{1}{2}QV=\tfrac{1}{2}CV^2=\frac{Q^2}{2C}$$

考试技巧 Exam tip 电池给电容器充电时，电池提供的能量为 QV，电容器只储存了 $\frac{1}{2}QV$，另一半在充电过程中消耗在电阻（导线）上转化为内能。

19.4 电容器的放电 Discharging a capacitor



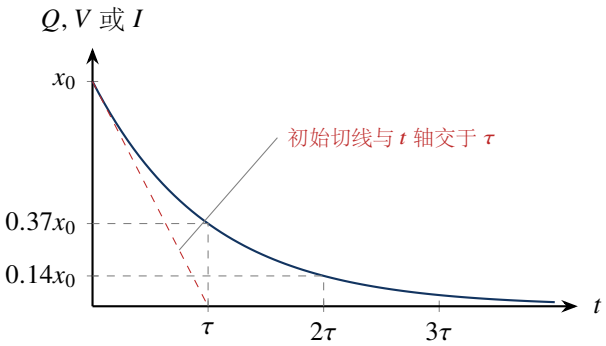
S 闭合：电容器充电至 $V_0 = E$
S 断开：电容器通过 R 放电

放电时，电流 $I = V/R$，而 $V = Q/C$。电荷越少，电压越小，电流越小，电荷减少得越慢——这是**指数衰减**（exponential decay）：

公式 19.2 (放电规律 *Discharge equations*)

$$Q = Q_0\,\mathrm{e}^{-t/RC}, \qquad V = V_0\,\mathrm{e}^{-t/RC}, \qquad I = I_0\,\mathrm{e}^{-t/RC}$$

$$\tau = RC \quad (\text{时间常数 time constant，单位 s})$$



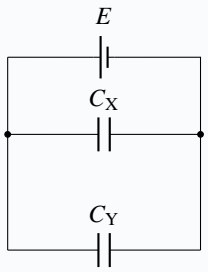
- 经过一个时间常数 $\tau = RC$，Q、V、I 都减为初值的 $1/\mathrm{e} \approx 37\%$。
- 减为一半所需时间：$t_{1/2} = RC\ln 2 \approx 0.69RC$；每经过相同时间，都减少相同的比例。
- ln 图：$\ln V = \ln V_0 - \dfrac{t}{RC}$，$\ln V$–$t$ 图是斜率为 $-\dfrac{1}{RC}$ 的直线（实验中常用来求 RC）。
- 放电电流方向与充电时相反；I–t 图下面积等于放出的总电荷 Q_0。
- τ 越大放电越慢：增大 R 或 C 都会使放电变慢。

易错提醒 Common mistakes 储存的**能量** $W \propto V^2$，所以能量衰减的时间常数是 $RC/2$：经过 τ 后能量减为初值的 $\mathrm{e}^{-2} \approx 14\%$。

例题 19.1 9702/41/M/J/26 Q5

(a) Define the capacitance of a parallel plate capacitor. [2]

(b) Two capacitors X and Y have capacitances C_X and C_Y respectively. The capacitors are connected in parallel to a supply that has electromotive force (e.m.f.) E.



Derive the expression for the relationship between C_X, C_Y and the combined capacitance C_T of the circuit. [3]

(c) Two capacitors are connected in series to a supply that has e.m.f. 6.00 V. One of the capacitors has capacitance C and the other has capacitance $2C$. The total energy stored by the two capacitors is 1.30 mJ. Calculate C in μF. [3]

解答 Solution

(a) 电荷与电势差之比， [M1]

其中电荷指一块板上的电荷，电势差指两板之间的电势差。 [A1]

(b) 两电容器电压相同：$E = V_X = V_Y$； [B1]

总电荷 $Q_T = Q_X + Q_Y$； [M1]

$C_T V = C_X V + C_Y V$，所以 $C_T = C_X + C_Y$。 [A1]

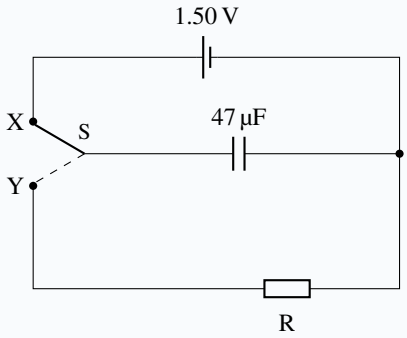
(c) $E = \frac{1}{2}CV^2$ [C1]

串联：$\dfrac{1}{C_T} = \dfrac{1}{C} + \dfrac{1}{2C}$，$C_T = \frac{2}{3}C$ [C1]

$1.30 \times 10^{-3} = \frac{1}{2} \times \frac{2}{3}C \times 6.00^2$，$C = 1.08 \times 10^{-4}\,\text{F} = 108\,\mu\text{F}$ [A1]

例题 19.2 9702/44/M/J/26 Q6

A capacitor of capacitance 47 μF, a cell of electromotive force (e.m.f.) 1.50 V and a resistor R are connected into the circuit of Figure 6.1.



The two-way switch S alternates between terminals X and Y with a frequency of 0.20 Hz: it is at X from $t = 0$ to 2.5 s, at Y from 2.5 s to 5.0 s, at X from 5.0 s to 7.5 s and at Y from 7.5 s to 10.0 s. The time constant of the discharge circuit is 2.4 s.

(a) (i) Calculate the resistance, in kΩ, of R. [2]

(ii) Show that the minimum potential difference (p.d.) across the capacitor is 0.53 V. [2]

(b) Sketch the variation of the p.d. V_C across the capacitor with t between $t = 0$ and $t = 10.0$ s. [3]

(c) The frequency at which the switch alternates is now doubled. Explain, without calculation, whether the minimum p.d. across the capacitor is now less than, equal to or greater than 0.53 V. [2]

解答 Solution

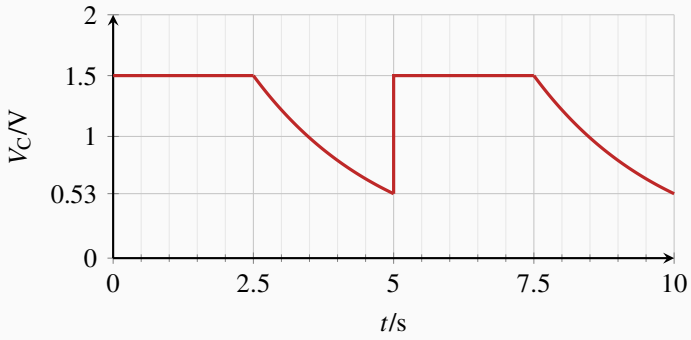
(a)(i) $\tau = RC$ **[C1]**

$R = \dfrac{2.4}{47 \times 10^{-6}} = 5.1 \times 10^4\,\Omega = 51\,\text{k}\Omega$ **[A1]**

(ii) 每次放电时间 2.5 s： **[C1]**

$V_{\text{min}} = 1.50\,\text{e}^{-2.5/2.4} = 0.53\,\text{V}$ **[A1]**

(b)（开关在 X 时电容器与电池直接相连，瞬间充满到 1.50 V）



0–2.5 s、5.0–7.5 s 为 1.50 V 水平线； **[B1]**

2.5–5.0 s、7.5–10.0 s 为斜率逐渐减小的放电曲线； **[B1]**

放电曲线最低点为 0.53 V。 **[B1]**

(c) 每次放电时间缩短（1.25 s）； **[B1]**

所以最小电压**大于** 0.53 V。 **[B1]**

教学提示 Teacher's note 建议 4 课时。易错：(1) 电容串并联公式与电阻相反；(2) 指数衰减中把 t 与 τ 单位统一（ms/μF 组合）；(3) 能量是 $\frac{1}{2}CV^2$ 而不是 CV^2。建议做数据记录器放电实验，作 $\ln V$–t 图求 RC——这也是 Paper 3/5 的常见题材。

19.5 本章小结 Summary

物理量/关系	公式	说明
电容定义	$C = Q/V$	$\mathrm{F} = \mathrm{C\,V^{-1}}$
孤立导体球	$C = 4\pi\varepsilon_0 R$	
并联 / 串联	$C = C_1 + C_2$；$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$	与电阻相反
储能	$W = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	V–Q 图面积
放电	$x = x_0 \mathrm{e}^{-t/RC}$	x：Q, V, I
时间常数	$\tau = RC$	$t_{1/2} = 0.69RC$

19.6 本章练习 Practice questions

练习 19.1 9702/42/M/J/26 Q6

(a) Define electric potential at a point. [2]

(b) An isolated conducting sphere of radius R is in a vacuum. The sphere has a charge of $+Q$ distributed around its surface.

(i) State an expression, in terms of R and Q, for the magnitude V_0 of the electric potential at the surface of the sphere. Identify any other symbols that you use. [1]

(ii) Sketch the variation of the electric potential V at distance x from the centre of the sphere between $x = R$ and $x = 3R$. [3]

(c) The dome of an electrostatic generator, called a Van de Graaff generator, may be considered to be a conducting sphere of radius 11.9 cm. The dome is charged so that the electric potential at its surface is -4.80×10^4 V.

(i) Calculate the charge on the surface of the dome. [2]

(ii) Explain why the dome acts as a capacitor. [1]

(iii) Calculate the capacitance of the dome. Give a unit with your answer. [2]

(iv) Determine the energy stored by the dome. [2]

解答 Solution

(a) 单位（正）电荷所做的功， **[B1]**

把正电荷从无穷远移到该点。 **[B1]**

(b)(i) $V_0 = \dfrac{Q}{4\pi\varepsilon_0 R}$，$\varepsilon_0$ 为真空介电常数。 **[B1]**

(ii) 从 (R, V_0) 开始、数值与斜率都逐渐减小的曲线； **[B1]**

经过 $(2R, \frac{1}{2}V_0)$ 和 $(3R, \frac{1}{3}V_0)$； **[B1]**

曲线全部在 $V > 0$ 区域。 **[B1]**

(c)(i) $Q = 4\pi\varepsilon_0 RV = -4.80 \times 10^4 \times 4\pi \times 8.85 \times 10^{-12} \times 0.119$ **[C1]**

$= -6.35 \times 10^{-7}\,\text{C}$ [A1]

(ii) 它能储存电荷（和能量）。 [B1]

(iii) $C = Q/V = \dfrac{6.35 \times 10^{-7}}{4.80 \times 10^{4}}$ [C1]

$= 1.32 \times 10^{-11}\,\text{F}$ [A1]

(iv) $W = \frac{1}{2}QV = \frac{1}{2} \times 6.35 \times 10^{-7} \times 4.80 \times 10^{4}$ [C1]

$= 0.015\,\text{J}$ [A1]

练习 19.2 原创仿真题 Exam-style

Three capacitors, each of capacitance 6.0 μF, are connected so that two of them are in parallel and this combination is in series with the third. What is the total capacitance?

A 2.0 μF **B** 4.0 μF **C** 9.0 μF **D** 18 μF

解答 Solution

答案 Answer：**B**

并联部分 12 μF，与 6.0 μF 串联：$\dfrac{1}{C} = \dfrac{1}{12} + \dfrac{1}{6} = \dfrac{3}{12}$，$C = 4.0\,\mu\text{F}$。

练习 19.3 原创仿真题 Exam-style

The potential difference across a capacitor is increased from 2.0 V to 6.0 V. By what factor does the energy stored in the capacitor increase?

A 3 **B** 6 **C** 8 **D** 9

解答 Solution

答案 Answer：**D**

$W \propto V^2$：$(6.0/2.0)^2 = 9$。

练习 19.4 原创仿真题 Exam-style

A capacitor of capacitance 220 μF is charged to 12 V and then discharged through a resistor of resistance 15 kΩ.

(a) Calculate the time constant of the circuit. [1]

(b) Calculate the initial discharge current. [1]

(c) Calculate the p.d. across the capacitor 5.0 s after the discharge begins. [2]

(d) Calculate the time taken for the energy stored in the capacitor to fall to one quarter of its initial value. [3]

解答 Solution

(a) $\tau = RC = 15 \times 10^{3} \times 220 \times 10^{-6} = 3.3\,\text{s}$ [A1]

(b) $I_0 = V_0/R = 12/15\,000 = 8.0 \times 10^{-4}\,\text{A}$（0.80 mA） [A1]

(c) $V = 12\,\text{e}^{-5.0/3.3}$ [C1]

$= 2.6\,\text{V}$ **[A1]**

(d) 能量变为 $1/4 \Rightarrow V$ 变为 $1/2$（$W \propto V^2$） **[C1]**

$\frac{1}{2} = \text{e}^{-t/3.3}$，$t = 3.3 \ln 2$ **[C1]**

$= 2.3\,\text{s}$ **[A1]**

第二十章 磁场 Magnetic fields

考纲要求 Syllabus learning outcomes

20.1.1 understand that a magnetic field is an example of a field of force produced either by moving charges or by permanent magnets
理解磁场是由运动电荷或永磁体产生的一种力场。

20.1.2 represent magnetic fields by field lines
用磁感线表示磁场。

20.2.1 understand that a force might act on a current-carrying conductor placed in a magnetic field
理解磁场中的载流导体可能受力。

20.2.2 recall and use the equation $F = BIL\sin\theta$, with directions as interpreted by Fleming's left-hand rule
记住并使用 $F = BIL\sin\theta$，方向由弗莱明左手定则判断。

20.2.3 define magnetic flux density as the force acting per unit current per unit length on a wire placed at right-angles to the magnetic field
把磁感应强度定义为垂直于磁场放置的导线单位电流、单位长度所受的力。

20.3.1 determine the direction of the force on a charge moving in a magnetic field
判断磁场中运动电荷所受力的方向。

20.3.2 recall and use $F = BQv\sin\theta$
记住并使用 $F = BQv\sin\theta$。

20.3.3 understand the origin of the Hall voltage and derive and use the expression $V_H = BI/(ntq)$, where t = thickness
理解霍尔电压的成因，推导并使用 $V_H = BI/(ntq)$（t 为厚度）。

20.3.4 understand the use of a Hall probe to measure magnetic flux density
理解用霍尔探头测量磁感应强度。

20.3.5 describe the motion of a charged particle moving in a uniform magnetic field perpendicular to the direction of motion of the particle
描述带电粒子垂直于匀强磁场运动时的运动情况。

20.3.6 explain how electric and magnetic fields can be used in velocity selection
解释如何用电场和磁场进行速度选择。

20.4.1 sketch magnetic field patterns due to the currents in a long straight wire, a flat circular coil and a long solenoid
画出长直导线、平面圆线圈和长螺线管中电流产生的磁场分布。

20.4.2 understand that the magnetic field due to the current in a solenoid is increased by a ferrous core
理解铁磁性铁芯能增强螺线管电流的磁场。

20.4.3 explain the origin of the forces between current-carrying conductors and determine the direction of the forces
解释载流导体之间作用力的来源，并判断其方向。

20.5.1 define magnetic flux as the product of the magnetic flux density and the cross-sectional area perpendicular to the direction of the magnetic flux density
把磁通量定义为磁感应强度与垂直于磁感应强度方向的截面积的乘积。

20.5.2 recall and use $\Phi = BA$
记住并使用 $\Phi = BA$。

20.5.3 understand and use the concept of magnetic flux linkage

理解并使用磁链的概念。

20.5.4 understand and explain experiments that demonstrate: that a changing magnetic flux can induce an e.m.f. in a circuit; that the induced e.m.f. is in such a direction as to oppose the change producing it; the factors affecting the magnitude of the induced e.m.f.

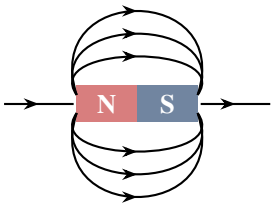
理解并解释以下实验：变化的磁通量能在电路中感应出电动势；感应电动势的方向总是阻碍引起它的变化；影响感应电动势大小的因素。

20.5.5 recall and use Faraday's and Lenz's laws of electromagnetic induction

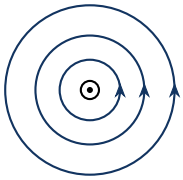
记住并使用法拉第电磁感应定律和楞次定律。

20.1 磁场与磁感线

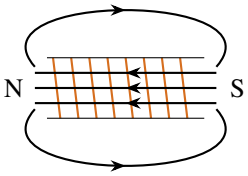
磁场 (*magnetic field*) 是运动电荷（电流）或永磁体周围存在的、对其他运动电荷或磁性物质施加力的区域。



条形磁铁 bar magnet



长直导线（电流流出纸面 ⊙）
同心圆，越远越稀



长螺线管 solenoid
内部为匀强磁场

- 磁感线：在磁体外部从 N 极指向 S 极，在内部从 S 指向 N，形成闭合曲线；疏密表示磁场强弱；互不相交。
- **长直导线**：以导线为圆心的同心圆，方向由**右手螺旋定则**（握住导线，拇指指电流方向，四指弯曲方向即磁场方向）。
- **平面圆线圈**：中心附近磁感线近似平行、垂直于线圈平面；外部类似短条形磁铁。
- **长螺线管**：内部为平行、等间距的匀强磁场，外部类似条形磁铁。用右手握住螺线管，四指沿电流方向，拇指指向 N 极。
- 在螺线管中插入**铁芯**（ferrous core，如软铁）会使磁场大大增强（电磁铁原理）。



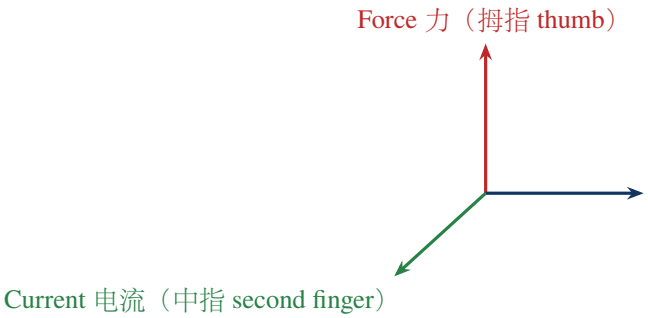
⊗ 电流流入纸面　　⊙ 电流流出纸面　　磁场垂直纸面向里　　磁场垂直纸面向外

20.2 磁场对电流的作用力

公式 20.1 (磁场对载流导线的力 *Force on a current-carrying conductor*)

$$F = BIL\sin\theta$$

θ 为导线（电流方向）与磁场方向的夹角。导线平行于磁场时不受力；垂直时力最大 $F = BIL$。



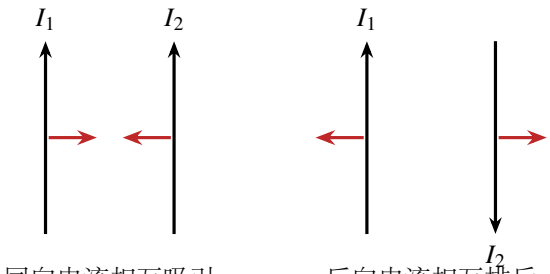
弗莱明左手定则 Fleming's left-hand rule：
三指互相垂直
"FBI"：拇指 F，食指 B，中指 I

定义 20.1 (磁感应强度 *Magnetic flux density*)

Magnetic flux density is the force acting per unit current per unit length on a wire placed at right angles to the magnetic field: $B = \dfrac{F}{IL}$.

磁感应强度（磁通密度）：垂直于磁场放置的导线，单位电流、单位长度所受的力。单位 tesla (T)，$1\,\mathrm{T} = 1\,\mathrm{N\,A^{-1}\,m^{-1}}$。

20.2.1 平行载流导线间的作用力



同向电流相互吸引　　反向电流相互排斥

解释：导线 1 的电流在导线 2 处产生磁场，导线 2 的电流在此磁场中受力（左手定则）；反之亦然。两力大小相等、方向相反（第三定律力对）。

20.3 磁场对运动电荷的作用力

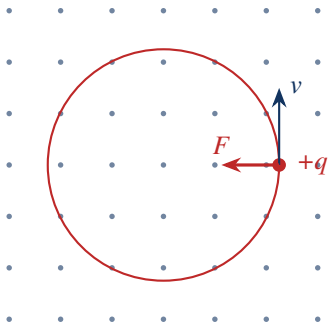
公式 20.2 (洛伦兹力 *Force on a moving charge*)

$$F = BQv\sin\theta$$

方向用左手定则判断：中指指正电荷的运动方向（负电荷则指向其运动的反方向，因为约定电流方向与电子运动方向相反）。

推导：导线中 $I = nAvq$，长度 L 内有 $N = nAL$ 个载流子，$F = BIL = B(nAvq)L = (nAL)\,Bqv$，每个载流子受力 Bqv。

20.3.1 带电粒子在匀强磁场中的圆周运动



B 垂直纸面向外 ⊙

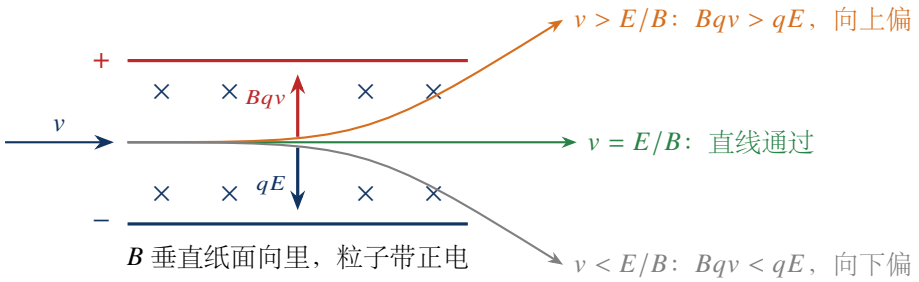
磁场力始终垂直于速度 ⇒ 不做功

速率不变，做匀速圆周运动

$$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{Bq}$$

- 磁场力始终**垂直于速度**，只改变速度方向，不改变速率，不做功，动能不变。
- 半径 $r = \dfrac{mv}{Bq} = \dfrac{p}{Bq}$：动量越大半径越大；荷质比 q/m 越大半径越小。
- 周期 $T = \dfrac{2\pi m}{Bq}$，与速度无关。
- 若速度与磁场不垂直，粒子做螺旋运动（不要求定量）。
- 应用：质谱仪（由半径求质量）、粒子探测器中的径迹（判断电荷正负与动量）。

20.3.2 速度选择器 Velocity selector

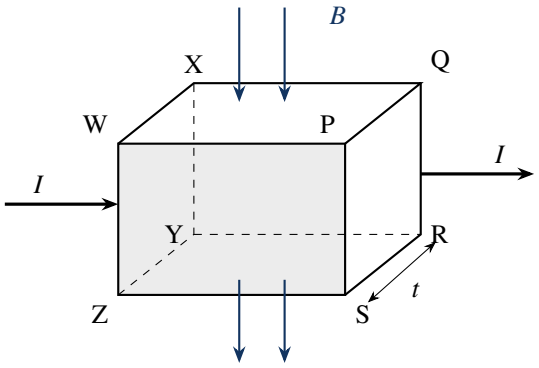


电场与磁场互相垂直，并都垂直于粒子速度。电场力 qE 与磁场力 Bqv 方向相反，只有满足

$$qE = Bqv \quad \Rightarrow \quad v = \frac{E}{B}$$

的粒子不偏转，从出口小孔射出。选择的速度与粒子的电荷、质量都**无关**。

20.3.3 霍尔效应 Hall effect



电子向左运动，受磁场力向后
在背面 QRYX 聚集负电荷
前面 PSZW 带正电（电势较高）
霍尔电压 V_H 在前后两面之间

霍尔电压的成因：

(1) 导体（半导体）片中通有电流，载流子在磁场中受到垂直于 I 和 B 的磁场力 Bqv；

(2) 载流子偏向一侧，在两个相对的面上积累异号电荷，产生横向电场和电势差；

(3) 电场力 qE 随电荷积累增大，直到 $qE = Bqv$ 时达到稳定，此时的电势差即为**霍尔电压** V_H。

推导 $V_H = \dfrac{BI}{ntq}$：设片宽（两霍尔面之间的距离）为 d，厚度（沿 B 方向）为 t，截面积 $A = dt$。

$$qE = Bqv, \quad E = \frac{V_H}{d} \Rightarrow V_H = Bvd; \qquad I = nAvq = n(dt)vq \Rightarrow v = \frac{I}{ndtq}$$

$$V_H = B \cdot \frac{I}{ndtq} \cdot d = \frac{BI}{ntq}.$$

公式 20.3 (霍尔电压 *Hall voltage*)

$$V_H = \frac{BI}{ntq}$$

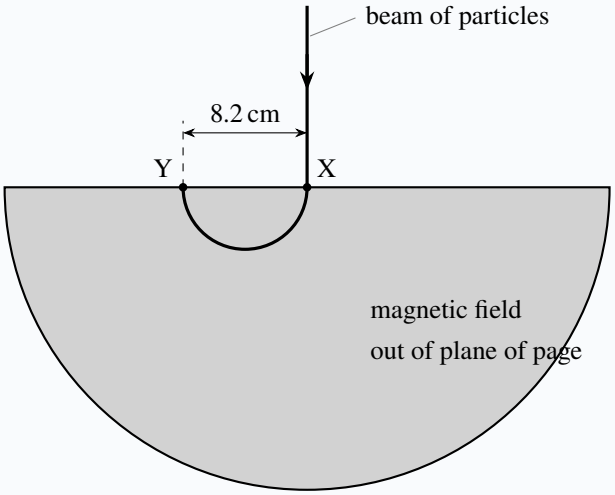
n：载流子数密度；t：片的厚度（沿磁场方向）；q：每个载流子的电荷。

霍尔探头：用**半导体**薄片（n 小，t 小 $\Rightarrow V_H$ 大，易测量）。保持 I 恒定，$V_H \propto B$，可用于测量磁感应强度。使用时必须使探头平面**垂直于磁场**（转动探头找到读数最大的位置）。

例题 20.1 9702/41/M/J/26 Q6

(a) Define magnetic flux density. [2]

(b) Figure 6.1 shows a beam of charged particles in a device known as a mass spectrometer.



The shaded region contains a uniform magnetic field that is out of the plane of the page. The particles enter the region of the magnetic field at point X and strike a detector at point Y.

(i) Explain whether the sign of the charge on the particles is positive or negative. [2]

(ii) Explain why the path followed by the beam of particles inside the magnetic field is circular. [2]

(c) Each of the particles has a mass of 2.0×10^{-26} kg and a charge of magnitude 3.2×10^{-19} C. All of the particles have a speed of $7.3 \times 10^{4}\ \mathrm{m\,s^{-1}}$. The distance XY is 8.2 cm.

(i) Show that the centripetal acceleration of one of the particles is $1.3 \times 10^{11}\ \mathrm{m\,s^{-2}}$. [2]

(ii) Use the information in (c)(i) to determine the flux density B of the uniform magnetic field. Give a unit with your answer. [2]

(d) The particles are replaced with particles that have half of the charge and a quarter of the mass of the particles in (b). The sign of the charge and the speed of the particles are both unchanged. Describe the path of this beam of particles. [2]

解答 Solution

(a) 单位长度、单位电流所受的力；导线垂直于磁场。（三点全对 2 分，两点 1 分） **[B2]**

(b)(i) 在 X 处粒子所受磁场力向左（指向圆心）； **[B1]**

由左手定则（B 向外、F 向左），等效电流方向与粒子运动方向（向下）相同，所以带**正**电。 **[B1]**

(ii) 磁场力始终垂直于粒子运动方向； **[B1]**

只改变速度方向不改变速率（提供向心加速度）。 **[B1]**

(c)(i) $a = \dfrac{v^2}{r}$，$r = 8.2/2 = 4.1$ cm **[C1]**

$$a = \frac{(7.3 \times 10^4)^2}{0.041} = 1.3 \times 10^{11}\ \mathrm{m\,s^{-2}}$$ **[A1]**

(ii) $BQv = ma$ **[C1]**

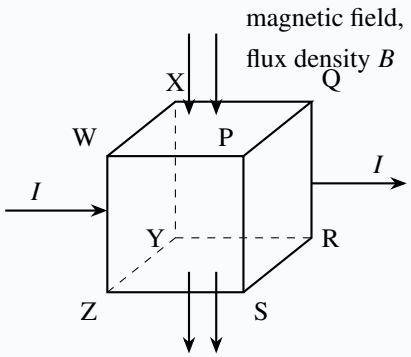
$$B = \frac{2.0 \times 10^{-26} \times 1.3 \times 10^{11}}{3.2 \times 10^{-19} \times 7.3 \times 10^{4}} = 0.11\ \mathrm{T}$$ **[A1]**

(d) $r = \dfrac{mv}{BQ} \propto \dfrac{m}{Q}$：$\dfrac{1/4}{1/2} = \dfrac{1}{2}$。从 X 沿同一方向进入、向同侧偏转， **[B1]**

半圆半径减半（打在 XY 中点）。 **[B1]**

例题 20.2 9702/42/M/J/26 Q7

A Hall probe consists of a slice of conducting material, as shown in Figure 7.1.



Current I enters the slice at face WXYZ and leaves it at face PQRS. The slice is in a uniform magnetic field of flux density B. This magnetic field is perpendicular to faces WXQP and ZYRS. A Hall voltage V_H is induced between two of the faces of the slice.

(a) (i) Using the letters P, Q, R, S, W, X, Y and Z, identify the faces of the slice between which the Hall voltage is induced. [1]

(ii) The magnitude of the Hall voltage is given by $V_H = \dfrac{BI}{nqt}$. State the meanings of the symbols n, q and t in this equation. [3]

(b) (i) Explain the origin of the Hall voltage. [3]

(ii) The charge carriers are electrons. Identify which of the faces in (a)(i) is at the higher potential. Explain your reasoning. [1]

解答 Solution

(a)(i) PSZW 与 QRYX（前后两面）。 **[B1]**

(ii) n：载流子数密度（单位体积内载流子数）； **[B1]**

q：每个载流子的电荷； **[B1]**

t：片的厚度（PS、QR、XY 或 WZ 的长度）。 **[B1]**

(b)(i) 任三点：载流子在磁场中受力；该力垂直于 I 和 B；载流子偏向一个面；两面之间的电荷分离产生电势差；当磁场力与由此产生的电场力大小相等、方向相反时达到稳定。 **[B3]**

(ii) 电子偏向 QRYX 面聚集，所以 PSZW 面电势较高。 **[B1]**

20.4 电磁感应 Electromagnetic induction

20.4.1 磁通量与磁链

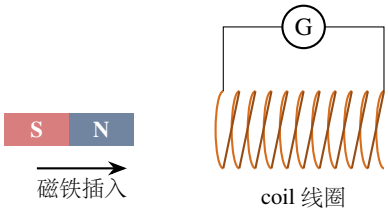
定义 20.2 (磁通量 *Magnetic flux*)

Magnetic flux is the product of the magnetic flux density and the cross-sectional area perpendicular to the direction of the magnetic flux density: $\Phi = BA$.

磁通量 = 磁感应强度 × 垂直于磁场方向的面积。单位 weber (Wb)，1 Wb = 1 T m^2。面积与磁场成角时 $\Phi = BA\cos\theta$（θ 为磁场与线圈平面**法线**的夹角）。

磁链 (magnetic flux linkage)：N 匝线圈中每匝的磁通量为 Φ，则磁链 $= N\Phi = NBA$，单位 Wb（或 Wb-turns）。

20.4.2 电磁感应现象



磁铁相对线圈运动 ⇒ 磁链变化 ⇒ 检流计偏转
磁铁静止 ⇒ 无偏转
反向运动 ⇒ 反向偏转
运动越快、磁铁越强、匝数越多 ⇒ 偏转越大

定律 20.1 (法拉第电磁感应定律 *Faraday's law*)

The magnitude of the induced e.m.f. is proportional to (equal to) the rate of change of magnetic flux linkage.

$$E = -\frac{\Delta(N\Phi)}{\Delta t}$$

感应电动势的大小等于磁链的变化率。负号表示楞次定律。

定律 20.2 (楞次定律 *Lenz's law*)

The direction of the induced e.m.f. (or current) is such as to oppose the change (in flux linkage) that produces it.

感应电动势（感应电流）的方向总是**阻碍**引起它的磁通量变化。这是**能量守恒**的结果：要维持相对运动，外界必须克服阻力做功，这些功转化为电能。

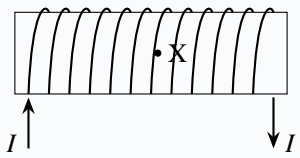
楞次定律的演示：把磁铁 N 极插入线圈，线圈靠近磁铁一端产生 N 极（相互排斥，阻碍靠近）；拔出时产生 S 极（相互吸引，阻碍离开）。磁铁穿过铜管下落明显变慢（铜管中的涡流阻碍下落）。

影响感应电动势大小的因素：磁场的强弱 B、线圈匝数 N、线圈面积 A、相对运动的速度（磁通变化的快慢）。

导线切割磁感线：长度 L 的导线以速度 v 垂直切割磁感线，Δt 内扫过面积 $Lv\Delta t$，$E = \dfrac{B\cdot Lv\Delta t}{\Delta t} = BLv$。

例题 20.3 9702/44/M/J/26 Q7(b)(d)

A coil of wire is wound onto a hollow insulating tube to form a solenoid, as shown in Figure 7.1.



There is a constant current I in the solenoid in the direction indicated (entering at the left-hand end, leaving at the right-hand end, with the turns passing upwards at the front of the tube).

(i) Draw five field lines to represent the magnetic field inside and outside the solenoid, and indicate with the letters N and S the magnetic poles of the solenoid. [4]

The current in the solenoid now varies sinusoidally so that the flux density B of the magnetic field at point X varies with time t according to $B = 2.0\cos 200\pi t$ where B is in mT and t is in s. A small coil is placed at point X inside the solenoid, so that its axis is parallel to the axis of the solenoid. The small coil has 45 turns, each with a radius of 2.8×10^{-3} m.

(ii) Calculate the maximum magnetic flux linkage $N\Phi$ of the small coil. Give a unit with your answer.[3]

(iii) Describe and explain how the variation of B with t may be used to determine the maximum electromotive force (e.m.f.) induced in the small coil. No calculations are required. [3]

解答 Solution

(i) 内部为平行、等间距的水平直线；外部为类似条形磁铁的闭合曲线，与内部连续； **[B1+B1]**

内部箭头从右向左； **[B1]**

左端为 N 极，右端为 S 极。 **[B1]**

(ii) $N\Phi = NBA$ **[C1]**

$= 45 \times 2.0 \times 10^{-3} \times \pi \times (2.8 \times 10^{-3})^2$ **[C1]**

$= 2.2 \times 10^{-6}$ Wb **[A1]**

(iii) 在 B–t 图最陡处作切线，求最大斜率（最大 $\Delta B/\Delta t$）； **[B1]**

最大斜率 × 线圈面积 × 匝数 = 磁链的最大变化率； **[B1]**

由法拉第定律，感应电动势的最大值等于磁链的最大变化率。 **[B1]**

教学提示 Teacher's note 建议 8 课时（本章内容最多）。几个必须熟练的方向判断：右手螺旋定则（磁场方向）、左手定则（受力方向，注意负电荷）、楞次定律（感应电流方向）。三个推导：$F = BQv$、$V_H = BI/ntq$、$r = mv/Bq$。定义题：magnetic flux density、magnetic flux、Faraday's law、Lenz's law。实验：用磁铁和线圈 + 检流计演示电磁感应；用霍尔探头测螺线管内磁场。

20.5 本章小结 Summary

物理量/关系	公式	说明
导线受力	$F = BIL\sin\theta$	左手定则
磁感应强度	$B = F/IL$	$\mathrm{T} = \mathrm{N\,A^{-1}\,m^{-1}}$
运动电荷受力	$F = BQv\sin\theta$	不做功
圆周运动	$r = mv/BQ$	$T = 2\pi m/BQ$
速度选择器	$v = E/B$	与 q、m 无关
霍尔电压	$V_\mathrm{H} = BI/ntq$	半导体 n 小
磁通量 / 磁链	$\Phi = BA$；$N\Phi$	Wb
法拉第定律	$E = -\Delta(N\Phi)/\Delta t$	楞次定律：阻碍变化

20.6 本章练习 Practice questions

练习 20.1 9702/42/F/M/26 Q6(c)

A rectangular wire frame with sides 6.0 cm (vertical) and 4.0 cm (horizontal) oscillates about its vertical axis with simple harmonic motion. The displacement s of each vertical edge of the frame from its equilibrium position varies with time t according to $s = 3.2\sin 6.9t$, where s is in mm and t is in s. The maximum speed of the edges is 22 mm s^{-1}. There is a small gap XY at the top of the frame. The frame is in a uniform horizontal magnetic field of flux density 0.047 T that is parallel to the equilibrium position of the frame.

(i) Calculate the electromotive force (e.m.f.) E_0 induced across XY when the frame is moving at its maximum speed. [3]

(ii) Sketch the variation of the induced e.m.f. E across XY with t between $t = 0$ and $t = 2T$. [2]

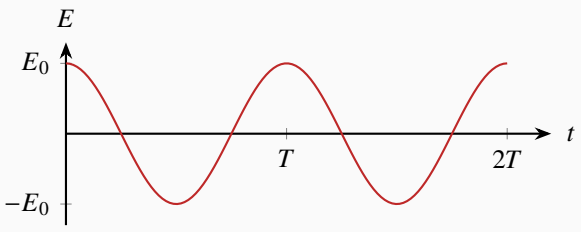
解答 Solution

(i) 两条竖直边（各长 6.0 cm）都切割磁感线，$E = \dfrac{B\Delta A}{\Delta t}$ **[C1]**

$E_0 = 0.047 \times 2 \times 0.060 \times 22 \times 10^{-3}$ **[C1]**

$= 1.2 \times 10^{-4}$ V **[A1]**

(ii) 感应电动势 ∝ 速度，$v = v_0\cos\omega t$，所以 $E = E_0\cos\omega t$：两个周期、振幅 E_0 的余弦曲线，$t = 0$ 时为最大值，在 $0.25T$、$0.75T$、$1.25T$、$1.75T$ 处为零。 **[B1+B1]**



练习 20.2 原创仿真题 Exam-style

Two long parallel wires carry currents in opposite directions. Which statement describes the forces between the wires?

A They attract each other with forces of equal magnitude.

B They repel each other with forces of equal magnitude.

C They attract each other; the wire with the larger current exerts the larger force.

D They repel each other; the wire with the larger current exerts the larger force.

解答 Solution

答案 Answer：**B**

反向电流相互排斥；两力是第三定律力对，大小相等。

练习 20.3 原创仿真题 Exam-style

A proton and an α-particle enter the same uniform magnetic field with the same speed, perpendicular to the field. What is the ratio $\dfrac{\text{radius of path of } \alpha\text{-particle}}{\text{radius of path of proton}}$?

A $\frac{1}{2}$ **B** 1 **C** 2 **D** 4

解答 Solution

答案 Answer：**C**

$r = \dfrac{mv}{Bq} \propto \dfrac{m}{q}$：$\alpha$ 粒子 $\dfrac{4}{2} = 2$，质子 $\dfrac{1}{1} = 1$，比值为 2。

练习 20.4 原创仿真题 Exam-style

In a velocity selector, the electric field strength is $3.0 \times 10^4\ \text{V m}^{-1}$ and the magnetic flux density is 0.15 T. The fields are perpendicular to each other and to the velocity of the particles. What is the speed of the particles that pass through undeflected?

A $4.5 \times 10^3\ \text{m s}^{-1}$ **B** $2.0 \times 10^5\ \text{m s}^{-1}$

C $4.5 \times 10^5\ \text{m s}^{-1}$ **D** $5.0 \times 10^{-6}\ \text{m s}^{-1}$

解答 Solution

答案 Answer：**B**

$v = E/B = 3.0 \times 10^4 / 0.15 = 2.0 \times 10^5\ \text{m s}^{-1}$。

练习 20.5 原创仿真题 Exam-style

A square coil of side 5.0 cm has 200 turns. It is placed with its plane perpendicular to a uniform magnetic field of flux density 0.40 T. The coil is pulled completely out of the field in 0.25 s.

(a) Calculate the initial magnetic flux linkage of the coil. [2]

(b) Calculate the average e.m.f. induced in the coil. [2]

(c) Explain, with reference to Lenz's law, why work must be done to pull the coil out of the field. [2]

解答 Solution

(a) $N\Phi = NBA = 200 \times 0.40 \times (0.050)^2$ **[C1]**

$= 0.20\,\text{Wb}$ **[A1]**

(b) $E = \dfrac{\Delta(N\Phi)}{\Delta t} = \dfrac{0.20}{0.25}$ **[C1]**

$= 0.80\,\text{V}$ **[A1]**

(c) 感应电流的方向阻碍磁链减少，即感应电流在磁场中受力阻碍线圈离开； **[B1]**

必须施加外力克服这一阻力做功，外界做的功转化为电能（能量守恒）。 **[B1]**

第二十一章 交流电 Alternating currents

考纲要求 Syllabus learning outcomes

21.1.1 understand and use the terms period, frequency and peak value as applied to an alternating current or voltage
理解并使用交流电流或电压的周期、频率和峰值。

21.1.2 use equations of the form $x = x_0 \sin \omega t$ representing a sinusoidally alternating current or voltage
使用 $x = x_0 \sin \omega t$ 形式的方程表示正弦交流电流或电压。

21.1.3 recall and use the fact that the mean power in a resistive load is half the maximum power for a sinusoidal alternating current
记住并使用：正弦交流电在纯电阻负载中的平均功率是最大功率的一半。

21.1.4 distinguish between root-mean-square (r.m.s.) and peak values and recall and use $I_{\text{r.m.s.}} = I_0/\sqrt{2}$ and $V_{\text{r.m.s.}} = V_0/\sqrt{2}$ for a sinusoidal alternating current
区分方均根值与峰值，记住并使用 $I_{\text{r.m.s.}} = I_0/\sqrt{2}$，$V_{\text{r.m.s.}} = V_0/\sqrt{2}$。

21.2.1 distinguish graphically between half-wave and full-wave rectification
从图像上区分半波整流和全波整流。

21.2.2 explain the use of a single diode for the half-wave rectification of an alternating current
解释用单个二极管实现交流电的半波整流。

21.2.3 explain the use of four diodes (bridge rectifier) for the full-wave rectification of an alternating current
解释用四个二极管（桥式整流器）实现全波整流。

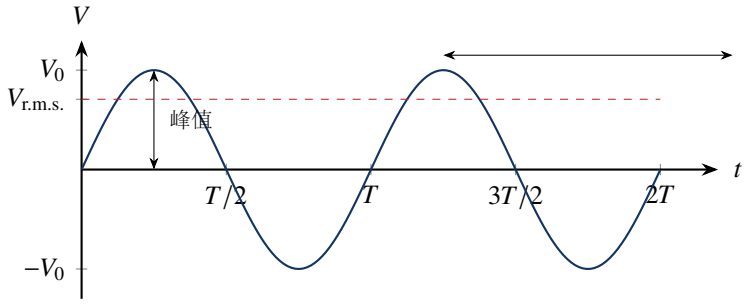
21.2.4 analyse the effect of a single capacitor in smoothing, including the effect of the values of capacitance and the load resistance
分析单个电容器的平滑作用，包括电容和负载电阻大小的影响。

21.1 交流电的特征

定义 21.1 (交流电 *Alternating current*)

An alternating current is a current whose direction reverses regularly (periodically).

交流电：方向周期性改变的电流。正弦交流电 $I = I_0 \sin \omega t$，$V = V_0 \sin \omega t$，其中 I_0、V_0 为**峰值** (peak value)，$\omega = 2\pi f = 2\pi/T$。



- **峰值** (peak value)：最大值 V_0；**峰峰值** (peak-to-peak) $= 2V_0$。

- 英国电网 $V_{\text{r.m.s.}} = 230\,\text{V}$，$f = 50\,\text{Hz}$，峰值约 325 V。

21.1.1 方均根值 Root-mean-square values

定义 21.2 (方均根值（有效值） *Root-mean-square value*)

The r.m.s. value of an alternating current is the value of the direct (steady) current that would produce the same mean power (heating effect) in a resistor as the alternating current.

方均根值（有效值）：使同一电阻产生相同平均功率（相同热效应）的直流电的值。

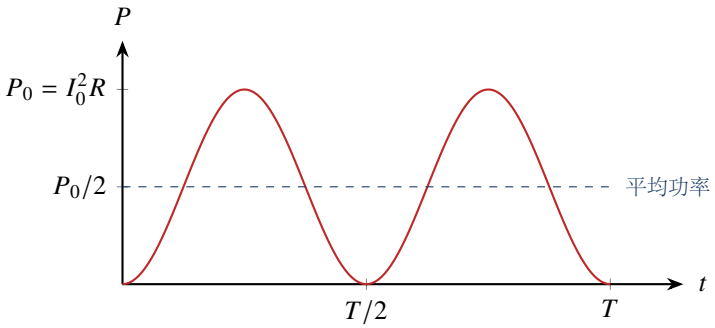
推导：电阻中瞬时功率 $P = I^2R = I_0^2R\sin^2\omega t$。$\sin^2\omega t$ 在 0 与 1 之间对称变化，平均值为 $\frac{1}{2}$，所以

$$\langle P\rangle = \tfrac{1}{2}I_0^2R = \tfrac{1}{2}P_{\max}.$$

令 $I_{\text{r.m.s.}}^2R = \frac{1}{2}I_0^2R$，得

公式 21.1 (正弦交流电的有效值 *r.m.s. values for sinusoidal a.c.*)

$$I_{\text{r.m.s.}} = \frac{I_0}{\sqrt{2}},\qquad V_{\text{r.m.s.}} = \frac{V_0}{\sqrt{2}},\qquad \langle P\rangle = I_{\text{r.m.s.}}V_{\text{r.m.s.}} = \tfrac{1}{2}I_0V_0 = \tfrac{1}{2}P_{\max}$$

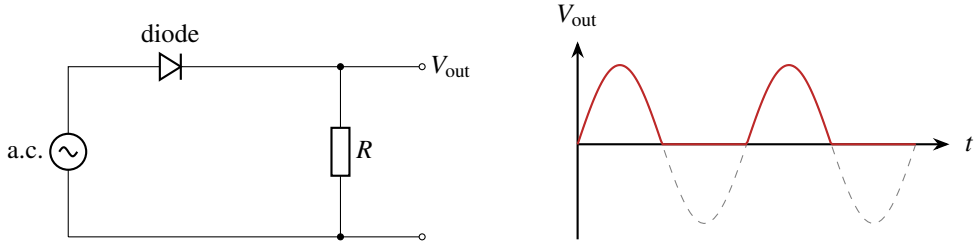


注意：功率随时间变化的频率是电流频率的 2 倍，且功率始终 ≥ 0。

易错提醒 Common mistakes $I_0/\sqrt{2}$ 只适用于**正弦**交流电。方波交流电（幅值 V_0）的有效值就是 V_0；半波整流的正弦交流电有效值为 $V_0/2$（平均功率减半：$V_{\text{r.m.s.}}^2 = \frac{1}{2}\cdot\frac{1}{2}V_0^2$）。

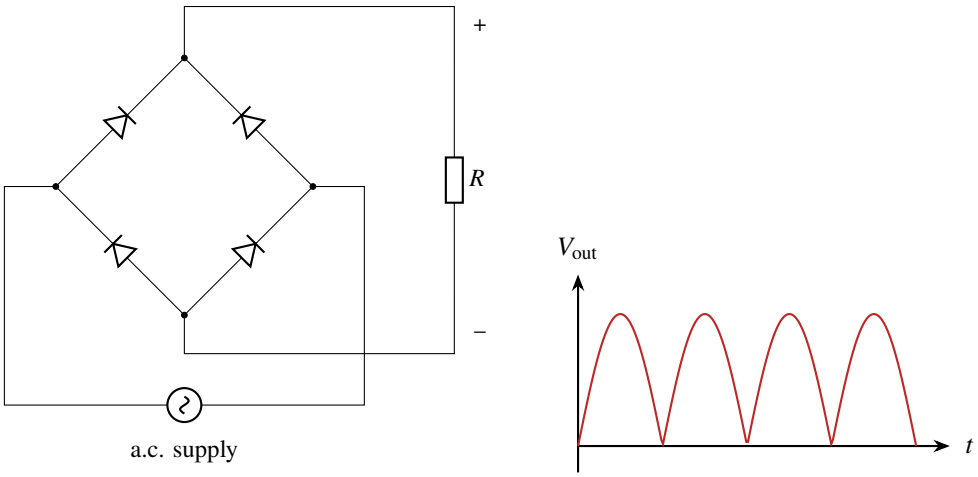
21.2 整流与平滑 Rectification and smoothing

21.2.1 半波整流



二极管只允许电流单向通过：输入电压正半周时二极管正向导通，负载上有电流；负半周时二极管反向截止，电流为零。输出为单向（直流）但大小变化的“脉动直流”。

21.2.2 全波整流（桥式整流）

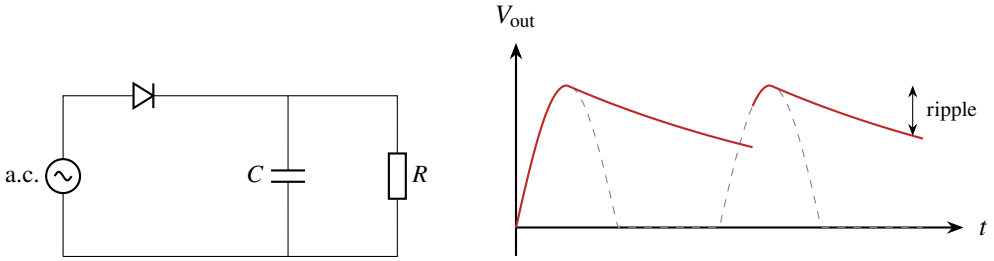


桥式整流器由四个二极管组成。无论输入哪一端为正，总有一对对角的二极管导通，使通过负载 R 的电流**方向始终相同**：

- 正半周：电流经一对二极管通过负载；
- 负半周：电流经另一对二极管通过负载，方向仍与正半周相同。

输出为全波整流波形，频率是输入的 2 倍（每个周期两个“峰”）。

21.2.3 电容器平滑 Smoothing



在负载两端并联一个电容器：

- 输入电压上升时，电容器被充电至峰值；
- 输入电压下降到低于电容器电压时，二极管截止，电容器通过负载**放电**，维持负载上的电压（指数下降）；
- 下一个峰到来时电容器再次充电。输出电压的起伏称为**纹波** (ripple)。

平滑效果取决于时间常数 RC：RC 远大于交流电周期时，放电很少，纹波很小。增大电容 C 或增大负载电阻 R（减小负载电流）都能使输出更平滑；减小 R（负载电流增大）则纹波变大。全波整流加平滑比半波整流加平滑纹波更小（放电时间只有一半）。

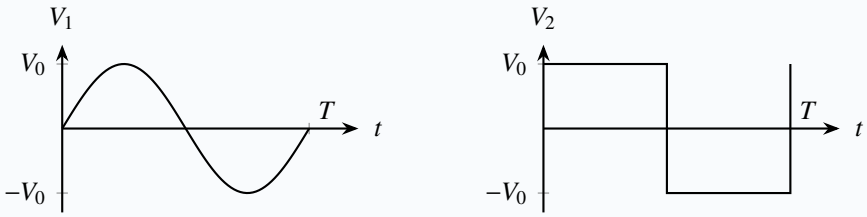
例题 21.1 9702/41/M/J/26 Q7

(a) (i) State what is meant by half-wave rectification of an alternating current. [2]

(ii) Figure 7.1 shows an incomplete diagram of the circuit used to achieve half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT} across a resistor. Complete the circuit diagram. [2]



(b) Two alternating voltages V_1 and V_2 have the same peak voltage V_0 and the same period T. V_1 is sinusoidal and V_2 is a square wave (alternating between $+V_0$ and $-V_0$).



The alternating voltages are applied in turn to the input terminals in the completed circuit. Complete the table to show, in terms of V_0, expressions for the root-mean-square (r.m.s.) values of the two unrectified and the two half-wave rectified voltages.

r.m.s. value	V_1	V_2
unrectified		
half-wave rectified		

[4]

解答 Solution

(a)(i) 允许电流单方向通过， **[B1]**

完全阻止反方向的电流（只保留每个周期的半个周期）。 **[B1]**

(ii) 在一条导线的缺口处接入二极管，另一缺口直接用导线连通； **[B1]**

二极管符号正确。 **[B1]**

(b) V_1 未整流：$V_0/\sqrt{2}$； **[B1]**

V_2 未整流：V_0（方波任何时刻 $V^2 = V_0^2$）； **[B1]**

V_1 半波整流：$V_0/2$（平均功率减半，$V_{\text{r.m.s.}} = \sqrt{V_0^2/4}$）； **[B1]**

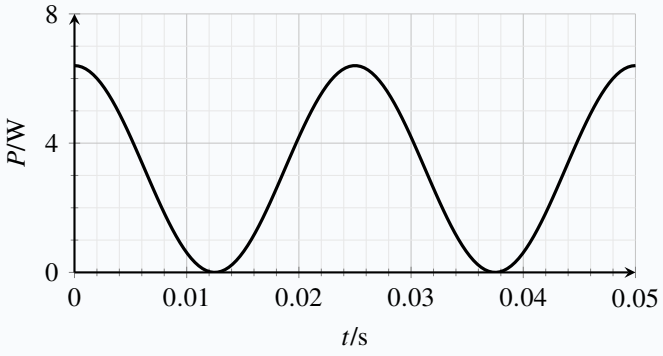
V_2 半波整流：$V_0/\sqrt{2}$（一半时间 V_0、一半时间 0）。 **[B1]**

例题 21.2 9702/42/F/M/26 Q7

(a) State what is meant by the root-mean-square (r.m.s.) value of an alternating current. [2]

(b) There is a sinusoidal alternating current in a resistor of resistance 26 Ω. The power P dissipated in the

resistor varies with time t as shown in Figure 7.1.



(i) Determine the mean power dissipated in the resistor. [1]

(ii) Calculate the r.m.s. current in the resistor. [2]

(c) The current in the resistor in (b) is now half-wave rectified.

(i) State the name of the electric circuit component that is used to achieve this rectification. [1]

(ii) Sketch the variation of the power P dissipated in the resistor between $t = 0$ and $t = 0.050\,\text{s}$. [2]

(iii) Determine the r.m.s. current in the resistor. Explain your reasoning. [2]

解答 Solution

(a) 一个恒定（直流）电流， **[M1]**

它在电阻中产生与该交流电相同的平均功率。 **[A1]**

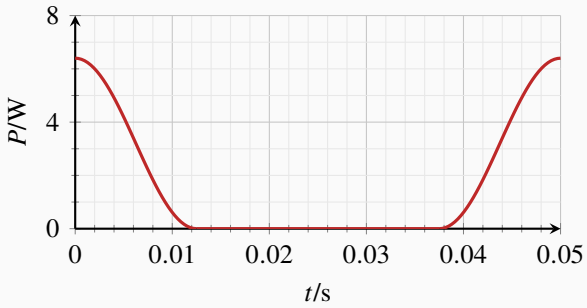
(b)(i) 平均功率 $= P_{\text{max}}/2 = 6.4/2 = 3.2\,\text{W}$ **[A1]**

(ii) $P = I^2R$ **[C1]**

$I_{\text{r.m.s.}} = \sqrt{3.2/26} = 0.35\,\text{A}$ **[A1]**

(c)(i) 二极管 diode。 **[B1]**

(ii) 在 $t = 0.0125\,\text{s}$ 到 $0.0375\,\text{s}$ 之间 $P = 0$，其余时间与原曲线相同（$\sin^2$ 形状，最大值 $6.4\,\text{W}$）。（或另一半周为零。） **[M1+A1]**



(iii) 平均功率减为原来的一半，即 $1.6\,\text{W}$； **[B1]**

$I_{\text{r.m.s.}} = \sqrt{1.6/26} = 0.25\,\text{A}$ **[A1]**

教学提示 Teacher's note 建议 4 课时。r.m.s. 定义要说"steady/direct current"与"same mean power (in a resistor)"。整流电路要求能画出单二极管和桥式整流电路，并说明桥式整流在正、负半周各是哪对二极管导通。平滑题注意分析 C 与 R 对纹波的影响（时间常数），也常结合第 19 章的指数放电计算。

21.3 本章小结 Summary

物理量/关系	公式	说明
正弦交流电	$x = x_0 \sin \omega t$	$\omega = 2\pi f$
有效值	$I_{\text{r.m.s.}} = I_0/\sqrt{2}$，$V_{\text{r.m.s.}} = V_0/\sqrt{2}$	仅正弦
平均功率	$\langle P \rangle = \frac{1}{2} P_{\max} = I_{\text{r.m.s.}} V_{\text{r.m.s.}}$	纯电阻
半波整流	1 个二极管	只保留半周
全波整流	4 个二极管（桥式）	两个半周都利用
平滑	并联电容 C	$RC \gg T$ 时纹波小

21.4 本章练习 Practice questions

练习 21.1 9702/42/M/J/26 Q8

A sinusoidal alternating input voltage V_{IN} is supplied to a circuit that produces a rectified and smoothed output voltage V_{OUT}. The variations with time t of V_{IN} and V_{OUT} are shown in Figure 8.1 and Figure 8.2 respectively.

Figure 8.1

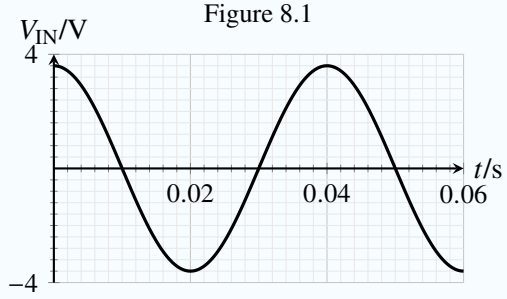
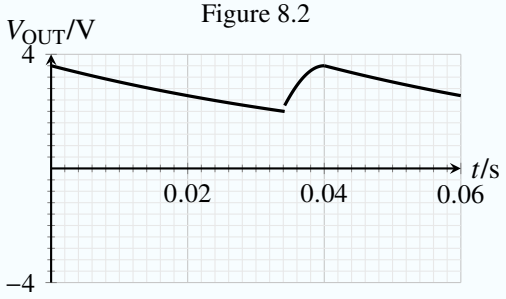


Figure 8.2



(a) State the name of the component in the circuit that produces (i) the rectification; (ii) the smoothing. [2]

(b) Determine (i) the peak value of V_{IN}; (ii) the period of V_{IN}. [2]

(c) Apart from the peak value and period of V_{IN}, determine three other conclusions that can be drawn from Figure 8.1 and Figure 8.2 about the input and output voltages and about the circuit. The conclusions may be qualitative or quantitative. [3]

解答 Solution

(a)(i) 二极管； [B1]

(ii) 电容器。 [B1]

(b)(i) 3.6 V [A1]

(ii) 0.040 s [A1]

(c) 任三点（数值需带单位，至少 2 位有效数字）： [B3]

- 是半波整流（每个周期只充电一次）；
- 输入频率 $f = 25\,\text{Hz}$（或 $\omega = 160\,\text{rad s}^{-1}$）；
- 输入电压有效值 $= 3.6/\sqrt{2} = 2.5\,\text{V}$；
- 平滑电路的时间常数：由 V 在约 0.034 s 内从 3.6 V 降到 2.0 V，$RC = \dfrac{0.034}{\ln(3.6/2.0)} \approx 0.058\,\text{s}$；
- 输出电压纹波（变化量）约 1.6 V（约 44%）。

练习 21.2 原创仿真题 Exam-style

The peak value of a sinusoidal alternating voltage is 325 V. What is the r.m.s. value of the voltage?

A 163 V **B** 230 V **C** 325 V **D** 460 V

解答 Solution

答案 Answer：**B**

$V_{\text{r.m.s.}} = 325/\sqrt{2} = 230\,\text{V}$。

练习 21.3 原创仿真题 Exam-style

A sinusoidal alternating current of peak value 2.0 A passes through a resistor of resistance 50 Ω. What is the mean power dissipated in the resistor?

A 50 W **B** 100 W **C** 141 W **D** 200 W

解答 Solution

答案 Answer：**B**

$\langle P \rangle = \frac{1}{2} I_0^2 R = \frac{1}{2} \times 2.0^2 \times 50 = 100\,\text{W}$。

练习 21.4 原创仿真题 Exam-style

A capacitor is connected across the load resistor in a full-wave rectifier circuit to smooth the output. Which change reduces the ripple of the output voltage?

A using a capacitor of smaller capacitance

B using a load resistor of smaller resistance

C using a capacitor of larger capacitance

D using a half-wave rectifier instead

解答 Solution

答案 Answer：C

增大 C 使时间常数 RC 增大，放电变慢，纹波减小。

练习 21.5 原创仿真题 Exam-style

A bridge rectifier is connected to a sinusoidal a.c. supply of frequency 50 Hz and to a load resistor.

(a) Draw the circuit diagram of the bridge rectifier connected to the supply and the load resistor. [3]

(b) Explain how the bridge rectifier produces a current in the load resistor that is always in the same direction. [2]

(c) State the frequency of the variation of the rectified output voltage. [1]

解答 Solution

(a) 四个二极管组成菱形（桥）；交流电源接在桥的一对对角上；负载电阻接在另一对对角上；二极管方向一致（使输出一端始终为正）。 **[B3]**

(b) 输入的每个半周中，只有一对对角的二极管正向导通； **[B1]**

两个半周中导通的二极管对不同，但都使电流从同一方向流过负载。 **[B1]**

(c) 100 Hz（每个输入周期出现两个输出峰）。 **[B1]**

第二十二章 量子物理 Quantum physics

考纲要求 Syllabus learning outcomes

22.1.1 understand that electromagnetic radiation has a particulate nature
理解电磁辐射具有粒子性。

22.1.2 understand that a photon is a quantum of electromagnetic energy
理解光子是电磁能量的量子。

22.1.3 recall and use $E = hf$
记住并使用 $E = hf$。

22.1.4 use the electronvolt (eV) as a unit of energy
使用电子伏特 (eV) 作为能量单位。

22.1.5 understand that a photon has momentum and that the momentum is given by $p = E/c$
理解光子具有动量 $p = E/c$。

22.2.1 understand that photoelectrons may be emitted from a metal surface when it is illuminated by electromagnetic radiation
理解金属表面受电磁辐射照射时可能发射光电子。

22.2.2 understand and use the terms threshold frequency and threshold wavelength
理解并使用截止频率和截止波长。

22.2.3 explain photoelectric emission in terms of photon energy and work function energy
用光子能量和逸出功解释光电发射。

22.2.4 recall and use $hf = \Phi + \frac{1}{2}mv_{\max}^2$
记住并使用 $hf = \Phi + \frac{1}{2}mv_{\max}^2$。

22.2.5 explain why the maximum kinetic energy of photoelectrons is independent of intensity, whereas the photoelectric current is proportional to intensity
解释光电子最大动能与光强无关，而光电流与光强成正比。

22.3.1 understand that the photoelectric effect provides evidence for a particulate nature of electromagnetic radiation while phenomena such as interference and diffraction provide evidence for a wave nature
理解光电效应证明电磁辐射的粒子性，而干涉和衍射证明其波动性。

22.3.2 describe and interpret qualitatively the evidence provided by electron diffraction for the wave nature of particles
定性描述并解释电子衍射为粒子波动性提供的证据。

22.3.3 understand the de Broglie wavelength as the wavelength associated with a moving particle
理解德布罗意波长是与运动粒子相联系的波长。

22.3.4 recall and use $\lambda = h/p$
记住并使用 $\lambda = h/p$。

22.4.1 understand that there are discrete electron energy levels in isolated atoms (e.g. atomic hydrogen)
理解孤立原子（如氢原子）中电子具有分立的能级。

22.4.2 understand the appearance and formation of emission and absorption line spectra
理解发射线状谱和吸收线状谱的形态与成因。

22.4.3 recall and use $hf = E_1 - E_2$
记住并使用 $hf = E_1 - E_2$。

22.1 光子 Photons

定义 22.1 (光子 *Photon*)

A photon is a quantum (discrete packet) of electromagnetic energy.

光子：电磁辐射的能量量子（一份不可分割的能量）。电磁辐射以光子形式被发射和吸收，表现出**粒子性**。

公式 22.1 (光子的能量与动量 *Energy and momentum of a photon*)

$$E = hf = \frac{hc}{\lambda}, \qquad p = \frac{E}{c} = \frac{h}{\lambda}$$

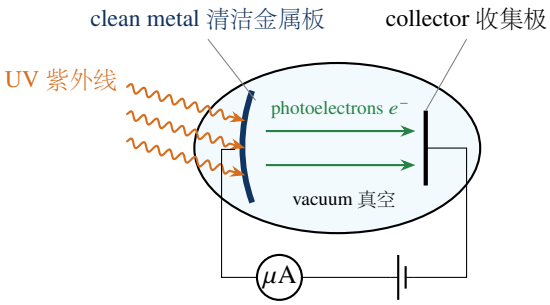
$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$（普朗克常量）。光子没有静止质量，但有动量；光照射到物体表面会产生压力（辐射压）。

电子伏特 (electronvolt)：一个电子通过 1 V 电势差加速所获得的动能。

$$1\,\mathrm{eV} = 1.60 \times 10^{-19}\,\mathrm{J}$$

电子在电势差 V 下加速：$\frac{1}{2}mv^2 = eV$。可见光光子能量约 1.8–3.1 eV。

22.2 光电效应 The photoelectric effect



紫外线照射清洁的锌板：
立即有光电子发射
（可见光照射：无论多强都没有）

当频率足够高的电磁辐射照射清洁的金属表面时，金属表面会发射电子，称为**光电效应** (*photoelectric effect*)，发射出的电子称为**光电子** (*photoelectrons*)。

22.2.1 实验事实（波动理论无法解释）

(1) 存在**截止频率** (*threshold frequency*) f_0：低于 f_0 的辐射无论多强、照射多久都不能产生光电子；
(2) 光电子的**最大动能只取决于入射光频率**，与光强无关（频率越高，最大动能越大）；
(3) 光电流（单位时间发射的光电子数）与光强成正比；
(4) 光电子的发射几乎是**瞬时**的（无延迟）。

按波动理论，能量连续地分布在波前上，任何频率的光只要足够强、照射足够久，电子都能积累到足够能量逸出——这与 (1)(2)(4) 矛盾。

22.2.2 光子理论解释

定义 22.2 (逸出功与截止频率 *Work function and threshold frequency*)

Work function Φ: the minimum energy required to remove an electron from the surface of a metal.

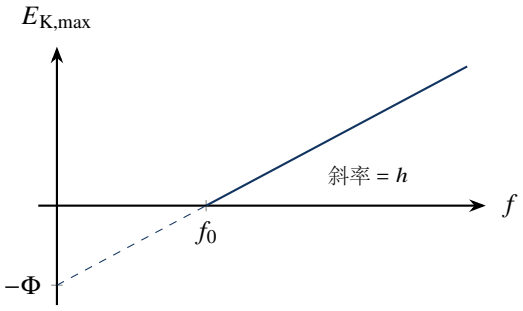
Threshold frequency f_0: the minimum frequency of electromagnetic radiation required to cause the emission of electrons from the surface. $\Phi = hf_0 = \dfrac{hc}{\lambda_0}$ (λ_0 = threshold wavelength, the maximum wavelength).

逸出功：使电子从金属表面逸出所需的最小能量。**截止频率**：能使金属表面发射电子的最低频率；对应的**截止波长**是能发射电子的最长波长。

- 每个光子的能量 hf 被**一个**电子全部吸收（一对一，瞬时）。
- 若 $hf < \Phi$，电子不能逸出；光强增大只是光子数增多，每个光子能量不变，仍不能发射——解释了截止频率。
- 若 $hf > \Phi$，多余能量成为电子的动能。表面附近的电子逸出需要的能量最少，获得**最大动能**：

公式 22.2 (爱因斯坦光电效应方程 *Photoelectric equation*)

$$hf = \Phi + \tfrac{1}{2}mv_{\max}^2$$



$E_{K,max}$–f 图是斜率为 h 的直线（与金属种类无关），横截距为 f_0，纵截距为 $-\Phi$。

光强的影响：光强增大 ⇒ 单位时间入射光子数增多 ⇒ 单位时间发射的光电子数增多 ⇒ 光电流增大；但每个光子能量不变，所以最大动能不变。

例题 22.1 9702/44/M/J/26 Q8

(a) When a clean metal surface is illuminated by electromagnetic radiation, electrons may be emitted from the surface.

(i) State the name of this phenomenon. [1]

(ii) State what is meant by the threshold frequency of the metal. [1]

(iii) Explain why the existence of the threshold frequency provides evidence for the photon model of electromagnetic radiation. [3]

(b) When a clean metal sheet is illuminated by radiation of wavelength 254 nm, electrons are emitted with a maximum kinetic energy of 1.91×10^{-19} J. When the same metal sheet is illuminated by radiation

of wavelength 178 nm, the maximum kinetic energy of the emitted electrons is 5.27×10^{-19} J.

(i) Use these data to calculate a value, to three significant figures, for the Planck constant. [3]

(ii) Determine a value, in eV, for the work function energy of the metal. Give your answer to two significant figures. [3]

解答 Solution

(a)(i) 光电效应（光电发射）。 **[B1]**

(ii) 能使金属表面发射电子的最低频率。 **[B1]**

(iii) 任三点：电子离开表面需要一个最小能量；被电子吸收的一个光子必须有足够能量；光子是能量包，能量与频率成正比；所以光子能量的最小值对应辐射的最小频率。 **[B3]**

(b)(i) $\dfrac{hc}{\lambda} = \Phi + E_{\mathrm{K}}$，两式相减消去 Φ： **[C1]**

$$h \times 3.00 \times 10^{8} \left(\frac{1}{178 \times 10^{-9}} - \frac{1}{254 \times 10^{-9}} \right) = (5.27 - 1.91) \times 10^{-19} \quad \textbf{[C1]}$$

$$h = 6.66 \times 10^{-34}\ \mathrm{J\,s} \quad \textbf{[A1]}$$

(ii) $\Phi = \dfrac{6.66 \times 10^{-34} \times 3.00 \times 10^{8}}{178 \times 10^{-9}} - 5.27 \times 10^{-19} = 5.95 \times 10^{-19}\ \mathrm{J}$ **[C1]**

除以 1.60×10^{-19} **[C1]**

$\Phi = 3.7\ \mathrm{eV}$ **[A1]**

22.3 波粒二象性 Wave–particle duality

	波动性的证据	粒子性的证据
电磁辐射	干涉、衍射、偏振	光电效应、线状谱（光子发射与吸收）
物质（电子等）	电子衍射	电荷量子化、Rutherford 散射、碰撞、径迹

22.3.1 德布罗意波长

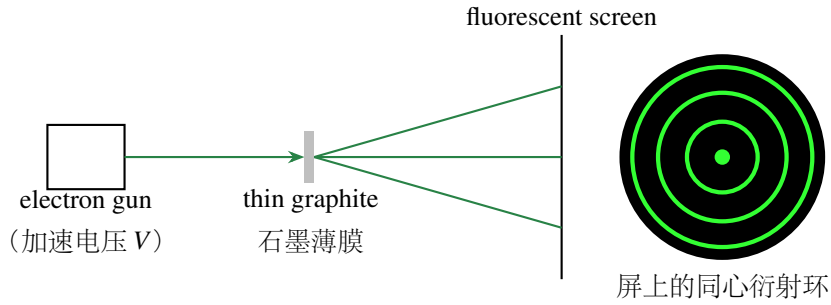
定义 22.3 (德布罗意波长 *de Broglie wavelength*)

The de Broglie wavelength is the wavelength associated with a moving particle: $\lambda = \dfrac{h}{p} = \dfrac{h}{mv}$.

德布罗意波长：与运动粒子相联系的波长。动量越大，波长越短。

被电势差 V 加速的电子：$\frac{1}{2}mv^2 = eV$，$p = \sqrt{2meV}$，$\lambda = \dfrac{h}{\sqrt{2meV}}$。加速电压越高，波长越短。

22.3.2 电子衍射



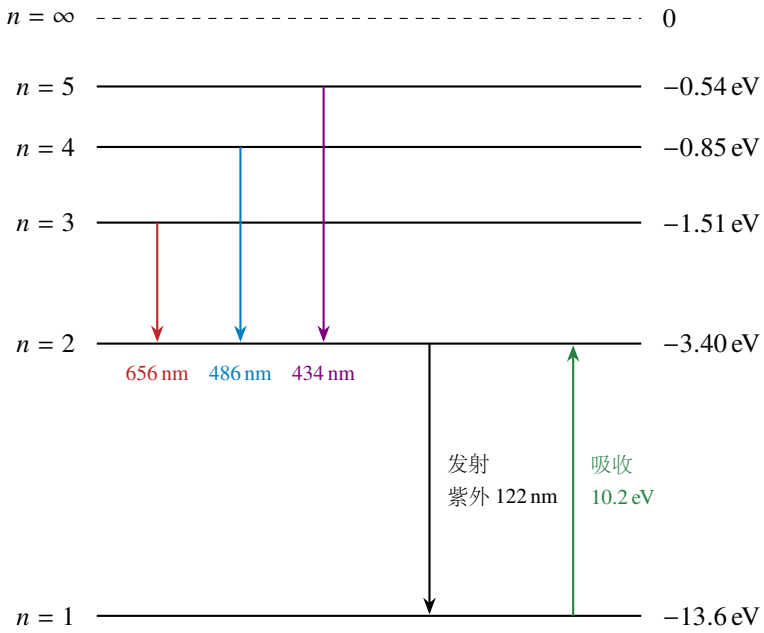
电子束通过石墨薄膜后，在荧光屏上形成**同心明暗圆环**——这是衍射图样，证明电子具有**波动性**：

- 石墨中原子层间距（约 10^{-10} m）与电子的德布罗意波长相当，起到衍射光栅的作用；
- 增大加速电压 ⇒ 电子速度、动量增大 ⇒ 波长减小 ⇒ 衍射角减小，圆环**半径变小**；
- 用磁铁靠近，圆环移动——说明形成圆环的是带电粒子（电子），而不是 X 射线。

22.4 能级与线状谱

22.4.1 分立能级

孤立原子中的电子只能处于某些特定的能量状态，称为**能级**(*energy levels*)。能级是**分立的**（量子化的），通常以负值表示（电子完全脱离原子时能量为零）。最低能级称为**基态**。



氢原子能级图（不按比例）：向下箭头表示发射光子，向上箭头表示吸收光子

公式 22.3 (能级跃迁 *Transitions between energy levels*)

$$hf = \frac{hc}{\lambda} = E_1 - E_2$$

电子从高能级 E_1 跃迁到低能级 E_2 时，发射一个能量恰好等于能级差的光子；吸收一个能量恰好等于能级差的光子，可从低能级跃迁到高能级。

22.4.2 发射谱与吸收谱



发射线状谱：黑背景上的亮线

吸收线状谱：连续谱上的暗线

- **发射谱**：高温低压气体中被激发的原子，电子从高能级向低能级跃迁，发射特定频率的光子——只出现某些离散波长的亮线。
- **吸收谱**：连续光谱（白光）通过较冷的气体时，原子吸收能量恰好等于能级差的光子跃迁到高能级，这些波长的光被吸收（随后向各方向重新发射，在原方向上强度减弱）——连续谱上出现暗线。同一元素的吸收线与发射线位置相同。
- 线状谱只有特定波长 ⇒ 光子能量只有特定值 ⇒ 原子能级是**分立**的。每种元素有自己独特的谱线，可用于鉴别元素（恒星成分）。

教学提示 Teacher's note 建议 6 课时。光电效应四个实验事实与光子解释必须一一对应。三条常考定义：photon、work function、threshold frequency、de Broglie wavelength。“为何最大动能与光强无关”的标准答案：intensity determines the number of photons per unit time, not the energy of each photon; one photon is absorbed by one electron。能级题注意 eV 与 J 的换算，以及“发射光子数目最多可能的谱线数”等组合计数问题。

22.5 本章小结 Summary

物理量/关系	公式	说明
光子能量	$E = hf = hc/\lambda$	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$
光子动量	$p = E/c = h/\lambda$	
电子伏特	$1\,\mathrm{eV} = 1.60 \times 10^{-19}\,\mathrm{J}$	
光电方程	$hf = \Phi + \frac{1}{2}mv_{\max}^2$	$\Phi = hf_0$
德布罗意波长	$\lambda = h/p = h/mv$	电子衍射证明
能级跃迁	$hf = E_1 - E_2$	线状谱

22.6 本章练习 Practice questions

练习 22.1 9702/42/M/J/26 Q9

(a) State what is meant by the de Broglie wavelength. [1]

(b) Complete the table to identify experimental evidence for the wave nature and the particulate nature of matter and of electromagnetic radiation.

	wave nature	particle nature
matter		
radiation		

[4]

(c) An electron in a vacuum is accelerated from rest by a potential difference (p.d.) of 3.5 kV. For the accelerated electron, calculate (i) the speed; (ii) the de Broglie wavelength. [5]

解答 Solution

(a) 与运动粒子相联系的波长。 **[B1]**

(b) 物质的波动性：电子衍射； **[B1]**

物质的粒子性：如 Brownian 运动、电荷量子化、Rutherford 散射、放射性衰变等； **[B1]**

辐射的波动性：衍射 / 干涉 / 偏振； **[B1]**

辐射的粒子性：光电效应（或线状谱）。 **[B1]**

(c)(i) $qV = \frac{1}{2}mv^2$ **[C1]**

$1.60 \times 10^{-19} \times 3500 = \frac{1}{2} \times 9.11 \times 10^{-31} \times v^2$ **[C1]**

$v = 3.5 \times 10^7\ \text{m s}^{-1}$ **[A1]**

(ii) $\lambda = \dfrac{h}{mv} = \dfrac{6.63 \times 10^{-34}}{9.11 \times 10^{-31} \times 3.5 \times 10^7}$ **[C1]**

$= 2.1 \times 10^{-11}\ \text{m}$ **[A1]**

练习 22.2 9702/42/F/M/26 Q8(a)

A star in a distant galaxy consists mostly of hydrogen. Visible light emitted by the star consists of discrete spectral absorption lines at wavelengths 410 nm, 434 nm, 486 nm and 656 nm. Explain, with reference to photons, how the existence of these spectral lines provides evidence for discrete electron energy levels in the hydrogen atom. [3]

解答 Solution

任三点： **[B3]**

- 光子具有分立的能量，光子能量取决于波长（$E = hc/\lambda$）；
- 光子能量等于两个能级之差；
- 谱线波长是分立的，说明电子跃迁只涉及分立的能量变化；
- 分立的能量变化说明能级必定是分立的。

练习 22.3 原创仿真题 Exam-style

What is the energy, in eV, of a photon of wavelength 500 nm?

A 0.40 eV **B** 2.5 eV **C** 4.0 eV **D** 4.0×10^{-19} eV

解答 Solution

答案 Answer：**B**

$E = \dfrac{hc}{\lambda} = \dfrac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{500 \times 10^{-9}} = 3.98 \times 10^{-19}\,\text{J} = \dfrac{3.98 \times 10^{-19}}{1.60 \times 10^{-19}} = 2.5\,\text{eV}$。

练习 22.4 原创仿真题 Exam-style

The intensity of the light incident on a metal surface is doubled, without changing its frequency. Photoelectrons are emitted. What happens to the maximum kinetic energy of the photoelectrons and to the photoelectric current?

	maximum kinetic energy	photoelectric current
A	doubles	doubles
B	doubles	unchanged
C	unchanged	doubles
D	unchanged	unchanged

解答 Solution

答案 Answer：**C**

频率不变，每个光子能量不变，最大动能不变；光强加倍，单位时间光子数加倍，光电流加倍。

练习 22.5 原创仿真题 Exam-style

What is the momentum of a photon of energy 3.0 eV?

A $1.6 \times 10^{-27}\,\text{kg m s}^{-1}$ **B** $1.0 \times 10^{-8}\,\text{kg m s}^{-1}$

C $4.8 \times 10^{-19}\,\text{kg m s}^{-1}$ **D** $9.0 \times 10^{-11}\,\text{kg m s}^{-1}$

解答 Solution

答案 Answer：**A**

$p = \dfrac{E}{c} = \dfrac{3.0 \times 1.60 \times 10^{-19}}{3.00 \times 10^{8}} = 1.6 \times 10^{-27}\,\text{kg m s}^{-1}$。

练习 22.6 原创仿真题 Exam-style

Some of the energy levels of the hydrogen atom are −13.6 eV ($n = 1$), −3.40 eV ($n = 2$) and −1.51 eV ($n = 3$).

(a) Explain why the energy levels are given negative values. [1]

(b) Calculate the wavelength of the photon emitted when an electron moves from the $n = 3$ level to the

$n = 2$ level. State the region of the electromagnetic spectrum of this photon. [3]

(c) State the number of different photon energies that could be emitted when electrons in atoms excited to the $n = 3$ level return to the ground state. [1]

(d) Explain why light of wavelength 500 nm passing through cool hydrogen gas is not absorbed. [2]

解答 Solution

(a) 规定电子完全脱离原子（在无穷远处）时能量为零；电子被原子束缚，必须外界提供能量才能使其脱离，所以束缚态能量为负。 **[B1]**

(b) $\Delta E = -1.51 - (-3.40) = 1.89\,\text{eV} = 3.02 \times 10^{-19}\,\text{J}$ **[C1]**

$\lambda = \dfrac{hc}{\Delta E} = \dfrac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{3.02 \times 10^{-19}} = 6.58 \times 10^{-7}\,\text{m}$（约 658 nm） **[A1]**

可见光（红光）。 **[B1]**

(c) 3 种（$3 \to 2$、$2 \to 1$、$3 \to 1$）。 **[B1]**

(d) 500 nm 光子的能量 = 2.49 eV，不等于冷氢气中基态（$n = 1$）电子到任何更高能级的能级差（最小为 10.2 eV）； **[B1]**

原子只能吸收能量恰好等于能级差的光子，所以不吸收。 **[B1]**

第二十三章 核物理 Nuclear physics

考纲要求 Syllabus learning outcomes

23.1.1 understand the equivalence between energy and mass as represented by $E = mc^2$ and recall and use this equation
理解 $E = mc^2$ 所表示的质能等价，记住并使用该方程。

23.1.2 represent simple nuclear reactions by nuclear equations
用核反应方程表示简单的核反应。

23.1.3 define and use the terms mass defect and binding energy
定义并使用质量亏损和结合能。

23.1.4 sketch the variation of binding energy per nucleon with nucleon number
画出比结合能随核子数变化的曲线。

23.1.5 explain what is meant by nuclear fusion and nuclear fission
解释核聚变与核裂变。

23.1.6 explain the relevance of binding energy per nucleon to nuclear reactions, including nuclear fusion and nuclear fission
解释比结合能与核反应（含聚变、裂变）的关系。

23.1.7 calculate the energy released in nuclear reactions using $E = c^2\Delta m$
用 $E = c^2\Delta m$ 计算核反应释放的能量。

23.2.1 understand that fluctuations in count rate provide evidence for the random nature of radioactive decay
理解计数率的涨落证明放射性衰变是随机的。

23.2.2 understand that radioactive decay is both spontaneous and random
理解放射性衰变是自发且随机的。

23.2.3 define activity and decay constant, and recall and use $A = \lambda N$
定义活度和衰变常量，记住并使用 $A = \lambda N$。

23.2.4 define half-life
定义半衰期。

23.2.5 use $\lambda = 0.693/t_{1/2}$
使用 $\lambda = 0.693/t_{1/2}$。

23.2.6 understand the exponential nature of radioactive decay, and sketch and use the relationship $x = x_0\mathrm{e}^{-\lambda t}$, where x could represent activity, number of undecayed nuclei or received count rate
理解放射性衰变的指数规律，画出并使用 $x = x_0\mathrm{e}^{-\lambda t}$（$x$ 可为活度、未衰变核数或接收到的计数率）。

23.1 质能方程与结合能

23.1.1 质能等价

公式 23.1 (质能方程　*Mass–energy equivalence*)

$$E = mc^2, \qquad \Delta E = c^2 \Delta m$$

质量与能量是等价的：系统能量变化 ΔE 时，质量随之变化 $\Delta m = \Delta E / c^2$。

常用换算：$1\,\mathrm{u} = 1.66 \times 10^{-27}\,\mathrm{kg}$，对应能量 $1.66 \times 10^{-27} \times (3.00 \times 10^8)^2 = 1.49 \times 10^{-10}\,\mathrm{J} \approx 931\,\mathrm{MeV}$（$1\,\mathrm{MeV} = 1.60 \times 10^{-13}\,\mathrm{J}$）。

23.1.2 质量亏损与结合能

定义 23.1 (质量亏损　*Mass defect*)

The mass defect of a nucleus is the difference between the total mass of the separate (individual) nucleons and the mass of the nucleus.

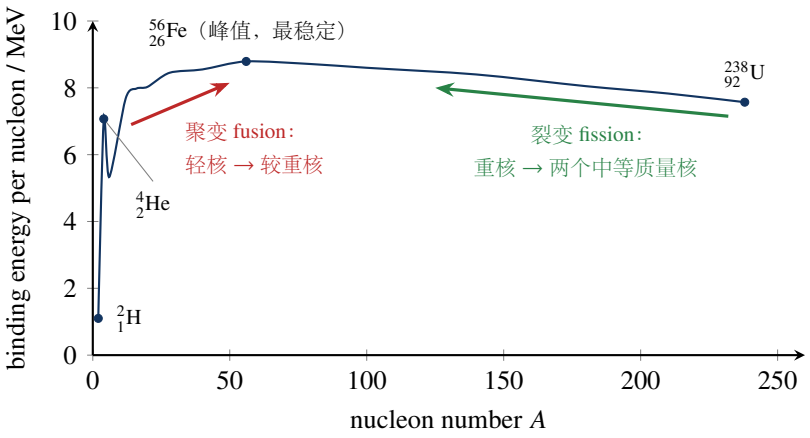
质量亏损：组成原子核的全部核子单独存在（彼此相距无穷远）时的总质量与该原子核质量之差：$\Delta m = Z m_\mathrm{p} + (A - Z) m_\mathrm{n} - m_\mathrm{nucleus}$。原子核的质量总是**小于**组成它的核子质量之和。

定义 23.2 (结合能　*Binding energy*)

The binding energy of a nucleus is the minimum energy required to separate all the nucleons of the nucleus to infinity (or the energy released when the nucleus is formed from its separate nucleons).

结合能：把原子核的全部核子完全分开到无穷远所需的最小能量（等于由自由核子结合成该核时释放的能量）。结合能 $= \Delta m\, c^2$。

比结合能（每核子结合能 binding energy per nucleon）= 结合能 / 核子数，衡量原子核的**稳定程度**：比结合能越大，核越稳定。



- 曲线在 $A \approx 56$（铁、镍附近）达到最大值约 8.8 MeV。

- **核聚变** (fusion)：两个轻核结合成一个较重的核。产物比结合能更大，释放能量。需要极高温度（使原子核获得足够动能克服静电斥力），发生在恒星内部。
- **核裂变** (fission)：一个重核（如 $^{235}_{92}\mathrm{U}$）吸收一个中子后分裂为两个中等质量的核并放出几个中子。产物比结合能更大，释放能量。用于核电站。
- 判断规则：反应后**总结合能增大**（总质量减小）$\Rightarrow$ 释放能量。释放的能量 = 产物总结合能 − 反应物总结合能 = $\Delta m\,c^2$（Δm 为反应前后质量之差）。

考试技巧 Exam tip 计算核反应能量的两种方法：(1) 质量法：Δm = 反应物总质量 − 产物总质量；(2) 质量亏损（结合能）法：Δm = 产物质量亏损之和 − 反应物质量亏损之和。核反应中**质量亏损增大**就释放能量。注意 u、kg、J、MeV 之间的换算。

例题 23.1 9702/41/M/J/26 Q9

(a) Define the mass defect of a nucleus. [2]

(b) Thorium-228 ($^{228}_{90}\mathrm{Th}$) is radioactive. Table 9.1 shows the mass defects of thorium-228 and some other nuclides.

nuclide	mass defect / u
$^{4}_{2}\mathrm{He}$	0.030 377
$^{224}_{88}\mathrm{Ra}$	1.846 840
$^{228}_{89}\mathrm{Ac}$	1.869 820
$^{228}_{90}\mathrm{Th}$	1.871 290
$^{228}_{91}\mathrm{Pa}$	1.866 129

(i) Use the data in Table 9.1 to explain why thorium-228 is an α-emitter and not a β^- emitter or a β^+ emitter. [4]

(ii) Calculate the energy, in MeV, released by the α-decay of one thorium-228 nucleus. Give your answer to three significant figures. [3]

解答 Solution

(a) 原子核质量与组成它的核子（质量之和）之差， **[M1]**

核子彼此分开到无穷远时。 **[A1]**

(b)(i) α 衰变产物为 $^{224}_{88}\mathrm{Ra} + {}^{4}_{2}\mathrm{He}$； **[B1]**

β^+ 衰变产物为 $^{228}_{89}\mathrm{Ac}$，β^- 衰变产物为 $^{228}_{91}\mathrm{Pa}$； **[B1]**

α 衰变产物总质量亏损 = 1.846 840 + 0.030 377 = 1.877 217 u； **[B1]**

只有产物的质量亏损大于母核（1.871 290 u）时衰变才能发生：α 衰变满足，而 Ac、Pa 的质量亏损都小于 Th，所以不发生 β 衰变。 **[B1]**

(ii) $E = \Delta mc^2$ **[C1]**

$= (1.877\,217 - 1.871\,290) \times 1.66 \times 10^{-27} \times (3.00 \times 10^{8})^2 = 8.85 \times 10^{-13}\,\mathrm{J}$ **[C1]**

$= 5.53\,\mathrm{MeV}$ **[A1]**

23.2 放射性衰变 Radioactive decay

23.2.1 随机性与自发性

- **自发** (spontaneous)：衰变不受外界条件（温度、压强、化学状态等）影响。
- **随机** (random)：无法预测某一个特定原子核**何时**衰变，也无法预测下一个衰变的是哪个核；每个核在单位时间内衰变的概率相同（等于衰变常量 λ）。
- **证据**：用 GM 管测量放射源的计数率，读数在平均值附近**无规则涨落**——这说明衰变是随机的。

定义 23.3 (活度与衰变常量 *Activity and decay constant*)

Activity: the number of nuclei that decay per unit time.

Decay constant: the probability that a nucleus will decay per unit time.

活度 A：单位时间内衰变的原子核数，单位 becquerel，1 Bq = 1 次衰变/秒。**衰变常量** λ：单个原子核在单位时间内发生衰变的概率，单位 s^{-1}。

公式 23.2 (活度 *Activity*)

$$A = \lambda N$$

N 为尚未衰变的原子核数。活度与剩余核数成正比——这正是指数衰变的原因。

23.2.2 半衰期与指数衰减

定义 23.4 (半衰期 *Half-life*)

The half-life of a radioactive isotope is the (mean) time taken for the number of undecayed nuclei (or the activity) to be reduced to half of its initial value.

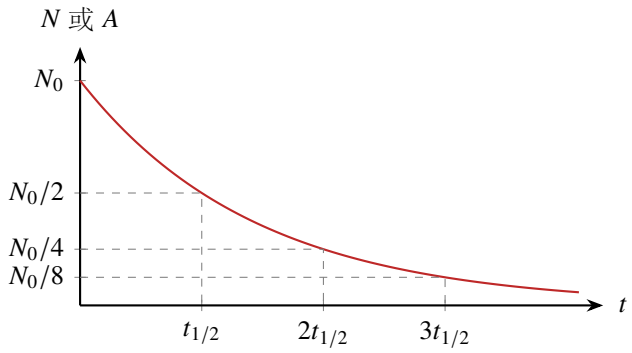
半衰期：放射性同位素中未衰变原子核数（或活度）减少到原来一半所需的（平均）时间。

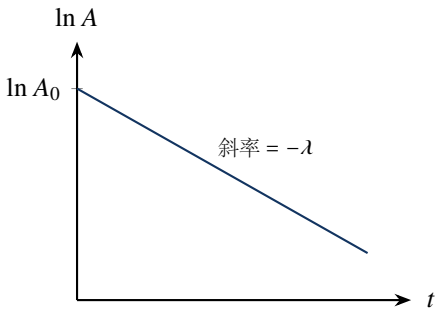
公式 23.3 (指数衰变规律 *Exponential decay*)

$$x = x_0\,\mathrm{e}^{-\lambda t}, \qquad \lambda = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{t_{1/2}}$$

x 可以是活度 A、未衰变核数 N 或接收到的计数率 C（扣除本底后）。经过 n 个半衰期，剩余 $\left(\frac{1}{2}\right)^n$。

推导 $\lambda t_{1/2} = \ln 2$：$t = t_{1/2}$ 时 $N = \frac{1}{2}N_0$，$\frac{1}{2} = \mathrm{e}^{-\lambda t_{1/2}}$，两边取对数 $\lambda t_{1/2} = \ln 2$。





实验测半衰期：先测**本底计数率**（background count rate），再测放射源的计数率并减去本底（corrected count rate）。作 $\ln C$–t 图，斜率 $= -\lambda$。

计数率小于活度的原因：放射源向各个方向发射辐射，只有一部分射向探测器；部分辐射在源内部、空气中或探测器窗口被吸收；进入探测器的辐射也不一定都被记录（探测效率不足 100%）。

例题 23.2 9702/42/F/M/26 Q9

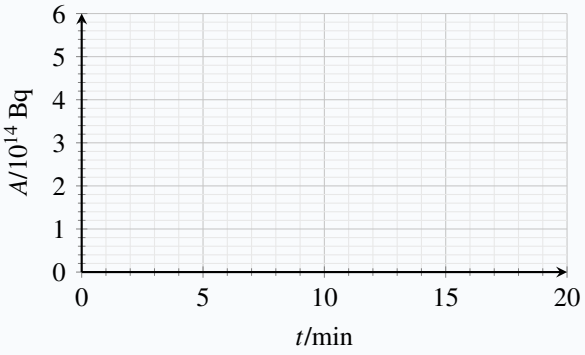
(a) State what is meant by radioactive decay. [2]

(b) An isotope X is radioactive and decays to a stable product with a half-life of 12.0 minutes. A sample of X has an activity of 5.80×10^{14} Bq at time $t = 0$.

(i) Calculate the decay constant of X. Give a unit with your answer. [2]

(ii) Calculate the activity of the sample at $t = 20.0$ minutes. [2]

(iii) Sketch the variation of the activity A of the sample with t between $t = 0$ and $t = 20.0$ minutes. [3]



(c) A scientist uses a radiation detector to determine the measured count rate of radiation emitted by the sample in (b). Explain why the measured count rate at any given time is less than the activity of the sample. [3]

解答 Solution

(a) （电离）辐射的自发发射， [B1]

从不稳定原子核发射。 [B1]

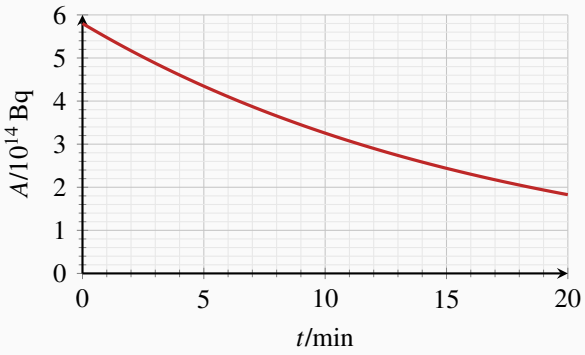
(b)(i) $\lambda = \dfrac{\ln 2}{12.0 \times 60}$ [C1]

$= 9.63 \times 10^{-4}\ \text{s}^{-1}$ [A1]

(ii) $A = 5.80 \times 10^{14} \exp\left(-\dfrac{\ln 2 \times 20.0}{12.0}\right)$ [C1]

$= 1.83 \times 10^{14}\,\text{Bq}$ **[A1]**

(iii)



起点 $(0, 5.80)$； **[B1]**

斜率为负且大小逐渐减小； **[B1]**

经过 $(12, 2.90)$ 和 $(20, 1.83)$。 **[B1]**

(c) 任三点：辐射向各方向发射，只有一部分射向探测器；部分辐射在样品内被吸收；部分在空气中被吸收；部分被探测器窗口吸收；部分到达探测器但未被记录。 **[B3]**

教学提示 Teacher's note 建议 5 课时。质量亏损/结合能的定义中"separated to infinity"是得分点。比结合能曲线必须能画出并标出峰值位置（$A \approx 56$）及纵坐标数量级（约 8 MeV），解释聚变和裂变为何都释放能量。衰变计算注意时间单位统一（λ 的单位与 t 一致）。可用骰子模拟放射性衰变的随机性，帮助理解"概率"的意义。

23.3 本章小结 Summary

物理量/关系	公式	说明
质能方程	$E = mc^2$，$\Delta E = c^2 \Delta m$	$1\,\text{u} \approx 931\,\text{MeV}$
质量亏损	$\Delta m = \sum m_{\text{核子}} - m_{\text{核}}$	
结合能	$E_\text{B} = \Delta m\, c^2$	分开到无穷远
活度	$A = \lambda N$	Bq
半衰期	$\lambda = 0.693/t_{1/2}$	
衰变规律	$x = x_0 \text{e}^{-\lambda t}$	x：A, N, 计数率

23.4 本章练习 Practice questions

练习 23.1 9702/42/M/J/26 Q10

(a) Define the binding energy of a nucleus. [2]

(b) A star emits radiation that is produced in a series of nuclear reactions that form helium from hydrogen. The reactions may be summarised by the overall equation

$$4\,{}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{0}_{+1}\mathrm{X}$$

where X is a particle that has no binding energy. All of the radiation emitted by the star is produced by this series of nuclear reactions. The mass defect of the ${}^{4}_{2}\mathrm{He}$ nuclide is 0.030 377 u.

(i) State the name of particle X. [1]

(ii) Show that the binding energy of the ${}^{4}_{2}\mathrm{He}$ nuclide is 4.54×10^{-12} J. [2]

(c) The star has a luminosity (total power output) of 7.98×10^{27} W. The mass of hydrogen in the star decreases continuously as a result of the series of nuclear reactions. Determine the rate of decrease of mass of hydrogen in the star. [4]

解答 Solution

(a) 把原子核的全部核子分开所需的能量， [M1]

分开到无穷远。 [A1]

(b)(i) 正电子 positron（β^+）。 [B1]

(ii) $E = \Delta mc^2 = 0.030\,377 \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ [C1]

$= 4.54 \times 10^{-12}$ J [A1]

(c) 每秒形成的 He 核数 $= \dfrac{7.98 \times 10^{27}}{4.54 \times 10^{-12}} = 1.76 \times 10^{39}\ \mathrm{s}^{-1}$ [C1]

每秒聚变的 H 核数 $= 4 \times 1.76 \times 10^{39} = 7.03 \times 10^{39}\ \mathrm{s}^{-1}$ [C1]

H 质量 $= 1.67 \times 10^{-27} \times$ 核数 [C1]

质量减少率 $= 1.17 \times 10^{13}\ \mathrm{kg\,s}^{-1}$ [A1]

练习 23.2 原创仿真题 Exam-style

The graph of binding energy per nucleon against nucleon number has a maximum near nucleon number 56. Which statement is correct?

A Energy is released when a nucleus with nucleon number 56 splits into two smaller nuclei.

B Energy is released when two nuclei of nucleon number 2 fuse together.

C Nuclei with nucleon number about 56 have the smallest mass defect per nucleon.

D The fission of a uranium nucleus decreases the total binding energy.

解答 Solution

答案 Answer：B

轻核聚变后比结合能增大，释放能量。A：铁附近核分裂后比结合能减小，需要吸收能量；C：比

结合能最大即每核子质量亏损最大；D：裂变使总结合能增大。

练习 23.3 原创仿真题 Exam-style

A radioactive sample has a half-life of 8.0 days. What fraction of the original radioactive nuclei remains undecayed after 24 days?

A $\frac{1}{8}$ **B** $\frac{1}{4}$ **C** $\frac{1}{3}$ **D** $\frac{7}{8}$

解答 Solution

答案 Answer：A

24 天 = 3 个半衰期，剩余 $(1/2)^3 = 1/8$。（D 是已衰变的比例。）

练习 23.4 原创仿真题 Exam-style

Which observation provides evidence that radioactive decay is random?

A The activity of a source decreases exponentially with time.

B The count rate from a source fluctuates about a mean value.

C The half-life of an isotope does not depend on temperature.

D α-particles are emitted with discrete energies.

解答 Solution

答案 Answer：B

计数率的涨落是随机性的证据；C 说明衰变是自发的（不受外界影响）。

练习 23.5 原创仿真题 Exam-style

A sample of iodine-131 has a half-life of 8.02 days. At a certain time the sample contains 4.0×10^{15} undecayed nuclei.

(a) Calculate the decay constant of iodine-131 in s^{-1}. [2]

(b) Calculate the activity of the sample at this time. [1]

(c) Calculate the number of undecayed nuclei remaining 30.0 days later. [2]

(d) A detector placed near the sample records a count rate that is much less than the activity. The background count rate is $0.40\,\mathrm{s}^{-1}$. State how the background count rate should be taken into account when determining the half-life experimentally. [1]

解答 Solution

(a) $\lambda = \dfrac{0.693}{8.02 \times 24 \times 3600}$ **[C1]**

$= 1.0 \times 10^{-6}\,\mathrm{s}^{-1}$ **[A1]**

(b) $A = \lambda N = 1.0 \times 10^{-6} \times 4.0 \times 10^{15} = 4.0 \times 10^{9}\,\mathrm{Bq}$ **[A1]**

(c) $N = 4.0 \times 10^{15}\,\mathrm{e}^{-0.693 \times 30.0/8.02}$ **[C1]**

$= 3.0 \times 10^{14}$ **[A1]**

(d) 从每次测得的计数率中减去本底计数率，用修正后的计数率作图（如 $\ln C$–t）。 **[B1]**

第二十四章 医学物理 Medical physics

考纲要求 Syllabus learning outcomes

24.1.1 understand that a piezo-electric transducer can be used to generate and detect ultrasound
理解压电换能器可用来产生和接收超声波。

24.1.2 explain how ultrasound is used to obtain diagnostic information about internal structures
解释如何利用超声波获得体内结构的诊断信息。

24.1.3 understand the meaning of specific acoustic impedance and its importance to the intensity reflection coefficient at a boundary
理解特性声阻抗的含义及其对界面强度反射系数的重要性。

24.1.4 recall and use $Z = \rho c$
记住并使用 $Z = \rho c$。

24.1.5 recall and use $I_R/I_0 = (Z_1 - Z_2)^2/(Z_1 + Z_2)^2$
记住并使用强度反射系数公式。

24.1.6 recall and use $I = I_0 e^{-\mu x}$ for the attenuation of ultrasound in matter
记住并使用超声波在物质中的衰减 $I = I_0 e^{-\mu x}$。

24.2.1 explain that X-rays are produced by electron bombardment of a metal target and calculate the minimum wavelength of X-rays produced from the accelerating p.d.
解释 X 射线由电子轰击金属靶产生，并由加速电压计算最短波长。

24.2.2 understand the use of X-rays in imaging internal body structures, including an understanding of the term contrast in X-ray imaging
理解用 X 射线对体内结构成像，包括“对比度”的含义。

24.2.3 recall and use $I = I_0 e^{-\mu x}$ for the attenuation of X-rays in matter
记住并使用 X 射线在物质中的衰减公式。

24.2.4 understand that computed tomography (CT) scanning produces a 3D image of an internal structure by first combining multiple X-ray images taken in the same section from different angles to obtain a 2D image of the section, then repeating this process along an axis and combining 2D images of multiple sections
理解 CT 扫描：先把同一断面从不同角度拍摄的多幅 X 射线图像合成为该断面的二维图像，再沿轴向逐层重复并把多个断面合成为三维图像。

24.3.1 understand that a tracer is a substance containing radioactive nuclei that can be introduced into the body and is then absorbed by the tissue being studied
理解示踪剂是含放射性核、可引入体内并被被研究组织吸收的物质。

24.3.2 recall that a tracer that decays by β^+ decay is used in positron emission tomography (PET scanning)
记住 PET 扫描使用 β^+ 衰变的示踪剂。

24.3.3 understand that annihilation occurs when a particle interacts with its antiparticle and that mass–energy and momentum are conserved in the process
理解粒子与其反粒子相互作用时发生湮灭，过程中质能与动量守恒。

24.3.4 explain that, in PET scanning, positrons emitted by the decay of the tracer annihilate when they interact with electrons in the tissue, producing a pair of gamma-ray photons travelling in opposite directions
解释 PET 中示踪剂放出的正电子与组织中的电子湮灭，产生一对方向相反的 γ 光子。

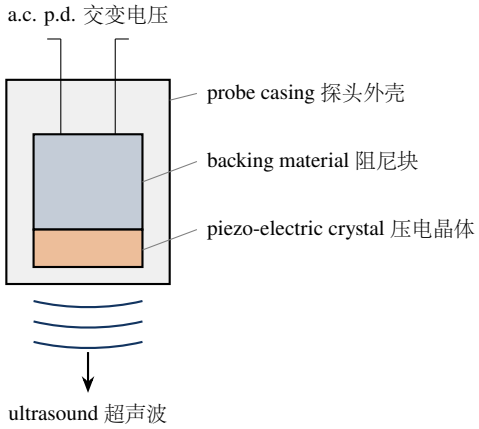
24.3.5 calculate the energy of the gamma-ray photons emitted during the annihilation of an electron-positron pair
计算电子–正电子湮灭产生的 γ 光子能量。

24.3.6 understand that the gamma-ray photons from an annihilation event travel outside the body and can be detected, and an image of the tracer concentration in the tissue can be created by processing the arrival times of the gamma-ray photons
理解湮灭产生的 γ 光子射出体外被探测，通过处理光子到达时间可得到组织中示踪剂浓度的图像。

24.1 超声波 Ultrasound

超声波是频率高于人耳听觉上限（约 20 kHz）的声波（纵波）。医学诊断常用 1–15 MHz，在软组织中波长约 0.1–1.5 mm，可以分辨毫米级结构。

24.1.1 压电换能器 Piezo-electric transducer



产生：在晶体两面加交变电压，晶体反复伸缩（逆压电效应），以超声频率振动，产生超声波；电压频率等于晶体固有频率时共振，振幅最大。

接收：反射回来的超声波使晶体振动（形变），晶体两面产生交变电压（压电效应）。

同一探头交替作为发射器和接收器：发射一个短脉冲，然后“静听”回波。阻尼块使晶体振动迅速停止，从而产生短脉冲（提高分辨率）。

24.1.2 特性声阻抗与反射

定义 24.1 (特性声阻抗 *Specific acoustic impedance*)

The specific acoustic impedance of a medium is the product of the density of the medium and the speed of sound in the medium: $Z = \rho c$.

特性声阻抗 = 介质密度 × 介质中的声速，单位 $\mathrm{kg\,m^{-2}\,s^{-1}}$。

♣

公式 24.1 (强度反射系数 *Intensity reflection coefficient*)

$$\frac{I_\mathrm{R}}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$$

（垂直入射）透射比例 $= 1 - \dfrac{I_R}{I_0}$。两种介质声阻抗相差越大，反射越强。

界面	声阻抗差异	反射
空气–软组织	极大（4×10^2 对 1.6×10^6）	几乎全反射（约 99.9%）
软组织–骨骼	较大	部分反射（约 30–40%）
软组织–软组织（不同器官）	很小	反射很少（<1%），但足以成像

耦合剂（凝胶）的作用：探头与皮肤之间若有空气，几乎所有超声都在空气–皮肤界面被反射。涂抹声阻抗与皮肤接近的凝胶，排除空气，使超声波能进入人体。

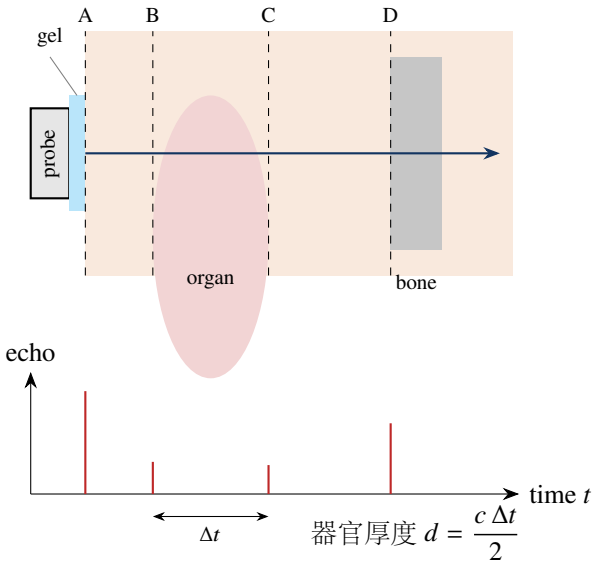
24.1.3 衰减

公式 24.2 (衰减 *Attenuation*)

$$I = I_0\,\mathrm{e}^{-\mu x}$$

μ 为线性衰减（吸收）系数，单位 m^{-1} 或 cm^{-1}；x 为穿过的厚度。$\ln(I/I_0) = -\mu x$，$\ln(I/I_0)$–x 图斜率为 $-\mu$。

24.1.4 超声诊断成像 (A-scan / B-scan)



- 探头发出超声脉冲，在各组织界面部分反射。测量回波返回的时间 t，由 $2d = ct$ 得界面深度；回波强弱给出界面两侧声阻抗差异等信息。
- A 扫描（A-scan）：屏幕上显示回波强度随时间变化的尖峰，用于测量深度、厚度。B 扫描（B-scan）：探头在体表移动（或用探头阵列），把各方向回波合成二维图像（如胎儿检查）。

- 优点：无电离辐射、相对安全、便宜、可实时成像；缺点：不能穿过骨骼和含气器官（肺）成像，分辨率有限。

例题 24.1 9702/42/F/M/26 Q10

(a) (i) Explain how ultrasound waves are detected by a piezoelectric transducer. [2]

(ii) Define the specific acoustic impedance of a medium. [2]

(b) Table 10.1 lists the specific acoustic impedances Z of air, water and glass.

medium	Z / kg m^{-2} s^{-1}
air	4.39×10^2
water	1.49×10^6
glass	9.98×10^6

(i) The density of water is 998 kg m^{-3}. Determine the speed of sound in water. [2]

(ii) Calculate, to three significant figures, the intensity reflection coefficient for sound incident on a water–glass boundary. [2]

(iii) Explain, without calculation, what happens when sound is incident on an air–glass boundary. [3]

解答 Solution

(a)(i) 超声波使换能器中的晶体振动（形变）； **[B1]**

晶体的振动产生（感应出）交变电动势，被检测到。 **[B1]**

(ii) 介质密度与声速的乘积， **[M1]**

声速指声音在该介质中的速度。 **[A1]**

(b)(i) $c = Z/\rho = \dfrac{1.49 \times 10^6}{998}$ **[C1]**

$= 1490\ \text{m s}^{-1}$ **[A1]**

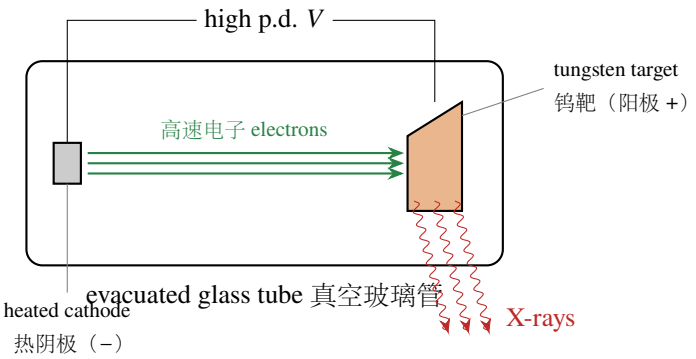
(ii) $\dfrac{I_R}{I_0} = \left(\dfrac{9.98 - 1.49}{9.98 + 1.49}\right)^2$ **[C1]**

$= 0.548$ **[A1]**

(iii) 任三点：两者声阻抗相差极大；强度反射系数约等于 1；（几乎）全部声强被反射；（几乎）没有声波透射进入玻璃。 **[B3]**

24.2 X 射线 X-rays

24.2.1 X 射线的产生

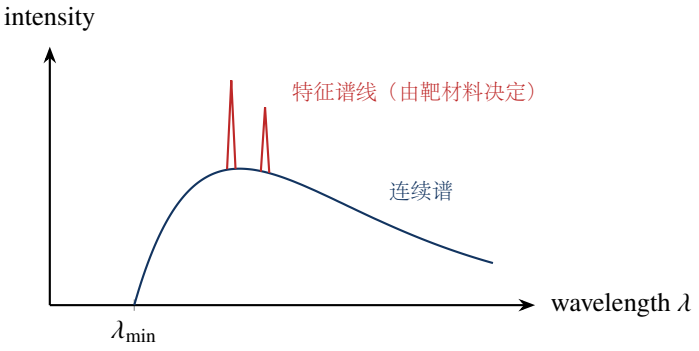


热阴极发射的电子被高电压 V（几十 kV）加速，高速轰击金属靶（如钨），在靶中急剧减速（大部分动能转化为内能，约 99%），一部分动能转化为 X 射线光子的能量。

连续谱与最短波长：每个电子在靶中可能经多次减速，每次产生的光子能量各不相同，所以 X 射线有连续的波长范围。当一个电子的**全部**动能 eV 一次转化为**一个**光子时，光子能量最大、波长最短：

公式 24.3 (X 射线最短波长　*Minimum X-ray wavelength*)

$$eV = \frac{hc}{\lambda_{\min}} \quad \Rightarrow \quad \lambda_{\min} = \frac{hc}{eV}$$

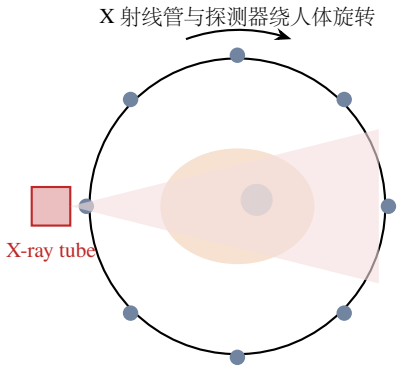


24.2.2 X 射线成像与对比度

X 射线穿过人体时被不同组织不同程度地吸收（衰减），$I = I_0 e^{-\mu x}$。透射的 X 射线在探测器（或胶片）上形成“阴影图”：

- 骨骼 μ 大（含钙，原子序数高），吸收多，图像中呈白色；软组织 μ 小，吸收少。
- **对比度** (contrast)：图像中不同区域亮暗的差异。两种组织的衰减系数相差越大，透射强度差别越大，对比度越好。软组织之间 μ 相近，对比度差——可服用钡餐等“造影剂”（μ 大）来增强对比度。
- **半值厚度** (half-value thickness) $x_{1/2}$：使强度减半的厚度，$\mu x_{1/2} = \ln 2$。

24.2.3 CT 扫描 Computed tomography



(1) 从很多不同角度拍摄同一薄层（断面）的 X 射线图像；
(2) 计算机合成得到该断面的二维图像；
(3) 沿身体轴线移动，逐层重复；
(4) 把多个断面的二维图像组合成三维图像。

CT 能分辨普通 X 光片中重叠的结构，软组织对比度更好，但辐射剂量比普通 X 光片大得多。

例题 24.2 9702/41/M/J/26 Q8

(a) State what is meant by a photon. [2]

(b) X-rays for use in medical diagnosis are produced by accelerating charged particles towards a metal target.

(i) State the name of the charged particles. [1]

(ii) Describe how X-rays are formed at the target. [2]

(iii) For any given accelerating voltage, there is a range of wavelengths of X-rays produced that has a minimum value. Explain, with reference to photons, why there is a range of wavelengths and why this range has a minimum value. [3]

(iv) Calculate the minimum wavelength of X-rays produced when the accelerating voltage is 52.0 kV. [3]

(c) X-rays are used to create an image of the internal structure of an object that is made from two types of material of similar thickness. Describe and explain how a property of the materials determines whether or not the image produced has good contrast. [2]

解答 Solution

(a) 一份（量子化的）能量， [M1]
电磁辐射的能量。 [A1]
(b)(i) 电子。 [B1]
(ii) 电子被靶减速； [B1]
电子（部分）动能转化为光子（X 射线）能量。 [B1]
(iii) 电子的部分动能在靶中转化为内能（热），各电子转化为光子的能量不同； [B1]
X 射线波长与光子能量成反比； [B1]
当电子的全部动能转化为一个光子的能量时，光子能量最大、波长最短。 [B1]
(iv) $\dfrac{hc}{\lambda} = eV$ [C1]
$\lambda = \dfrac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{1.60 \times 10^{-19} \times 52.0 \times 10^{3}}$ [C1]

$= 2.39 \times 10^{-11}\,\mathrm{m}$ [A1]

(c) 讨论两种材料（线性）衰减系数的差别； [B1]

两种材料透射强度差别大时对比度好（衰减系数相近则对比度差）。 [B1]

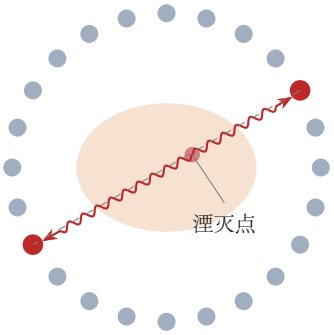
24.3 PET 扫描 Positron emission tomography

- **示踪剂** (*tracer*)：含放射性核的物质，引入体内后被被研究的组织吸收（如含 $^{18}_{9}\mathrm{F}$ 的葡萄糖类似物，被代谢旺盛的肿瘤细胞大量吸收）。PET 使用发生 β^+ **衰变**的示踪剂。
- 示踪剂放出的正电子在组织中运动很短距离（约 1 mm）后，与电子相遇发生**湮灭** (*annihilation*)。
- 湮灭中**质能守恒**和**动量守恒**：正负电子的总动量近似为零，所以产生**两个**能量相等、**方向相反**的 γ 光子（一个光子无法同时满足动量守恒和能量守恒）。

公式 24.4 (湮灭光子能量　*Annihilation photon energy*)

$$2m_e c^2 = 2E_\gamma \quad \Rightarrow \quad E_\gamma = m_e c^2 = 8.2 \times 10^{-14}\,\mathrm{J} = 0.511\,\mathrm{MeV}$$

（忽略正负电子湮灭前的动能。）每个光子的波长 $\lambda = h/(m_e c) = 2.4 \times 10^{-12}\,\mathrm{m}$。



两个 0.511 MeV 光子沿相反方向射出，
被相对的两个探测器（红色）同时探测；
湮灭点在两探测器的连线（虚线）上，
由两光子到达**时间差**确定其在连线上的位置；
大量事件经计算机处理 ⇒ 示踪剂浓度分布图像

PET 成像过程：γ 光子穿出人体，被环绕病人的探测器环探测。只有在极短时间窗内被两个相对探测器“同时”探测到的光子对才被记录；湮灭位置在两探测器连线上，并由到达时间差确定。计算机处理大量事件，得到组织中示踪剂浓度的图像——显示的是器官的**功能（代谢活动）**，而不仅是结构。

教学提示 Teacher's note　建议 5 课时。本章公式均在数据表中给出，但物理过程的文字描述是主要得分点：压电换能器的产生与接收、为什么要用凝胶、X 射线最短波长的原因、对比度、CT 步骤、PET 中为什么产生两个反向光子。开放式“三个结论”题（如本章 2026 M/J 44 卷超声衰减图）要能从斜率求 μ、从图像读厚度、从跳变求透射系数。

24.4 本章小结 Summary

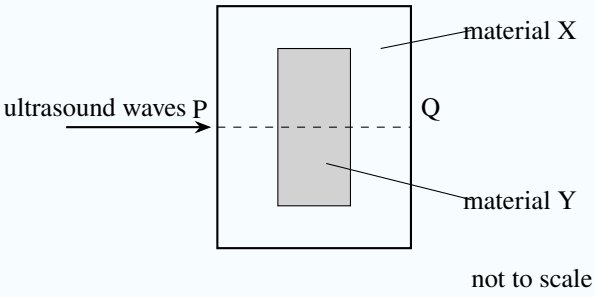
物理量/关系	公式	说明
声阻抗	$Z = \rho c$	$\mathrm{kg\,m^{-2}\,s^{-1}}$
强度反射系数	$\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$	差异大则反射强
衰减	$I = I_0 e^{-\mu x}$	超声与 X 射线
X 射线最短波长	$\lambda_{min} = hc/eV$	
回波深度	$2d = ct$	
湮灭光子	$E_\gamma = m_e c^2 = 0.511\,\mathrm{MeV}$	两个反向光子

24.5 本章练习 Practice questions

练习 24.1 9702/44/M/J/26 Q9

(a) Explain how ultrasound waves are generated by a piezoelectric transducer. [3]

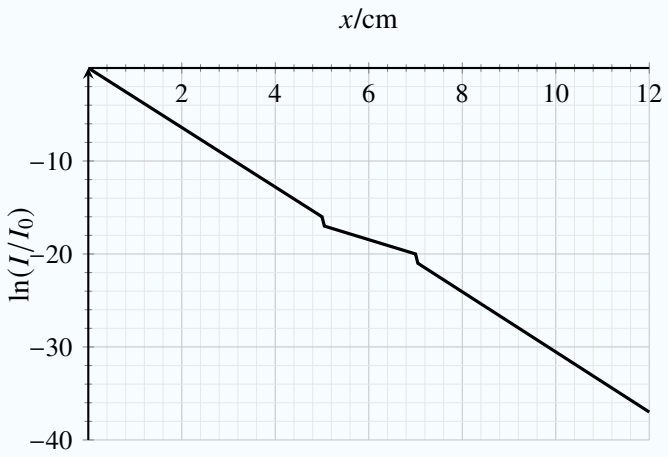
(b) Ultrasound waves are used to obtain information about a structure that consists of two materials, as shown in Figure 9.1.



The waves enter the structure at point P and travel through the structure to point Q.

(i) Material Y has a density of $8960\,\mathrm{kg\,m^{-3}}$. The speed of sound in material Y is $4720\,\mathrm{m\,s^{-1}}$. Calculate the specific acoustic impedance Z of material Y. Give a unit with your answer. [2]

(ii) The intensity of ultrasound entering the structure at point P is I_0. The transmitted intensity between P and Q at distance x from P is I. Figure 9.2 shows the variation with x of the value of $\ln(I/I_0)$.



Determine three conclusions about the structure and the materials from which it is made that may be drawn from Figure 9.2. The conclusions may be qualitative or quantitative. [3]

解答 Solution

(a) 给晶体加电压使其形变； **[B1]**

加**交变**电压使晶体（以超声频率）振动； **[B1]**

晶体振动使周围介质（空气）产生振动，即发出超声波。 **[B1]**

(b)(i) $Z = \rho c = 8960 \times 4720$ **[C1]**

$= 4.23 \times 10^7\ \text{kg m}^{-2}\ \text{s}^{-1}$ **[A1]**

(ii) 任三点（数值带单位）： **[B3]**

- X 的总厚度（沿 PQ）为 10.0 cm（两段各 5.0 cm），Y 的厚度为 2.0 cm；
- X 的衰减系数 = 16/5.0 = 3.2 cm^{-1}；Y 的衰减系数 = 3/2.0 = 1.5 cm^{-1}，X 的更大；
- 在 X–Y 界面 $\ln(I/I_0)$ 突降约 1，强度透射系数约 $\text{e}^{-1} = 0.37$（反射系数约 0.63）。

练习 24.2 原创仿真题 Exam-style

Why is a gel applied between an ultrasound transducer and the skin?

A to increase the speed of the ultrasound in the skin

B to reduce the attenuation of the ultrasound in the body

C to reduce the reflection of ultrasound at the skin surface

D to increase the frequency of the ultrasound

解答 Solution

答案 Answer：C

凝胶的声阻抗与皮肤接近，排除空气，大大减少皮肤表面的反射。

练习 24.3 原创仿真题 Exam-style

A beam of X-rays passes through a material with a linear attenuation coefficient of $0.35\,\text{cm}^{-1}$. What thickness of the material reduces the intensity of the beam to 25% of its initial value?

A 0.71 cm **B** 2.0 cm **C** 4.0 cm **D** 7.1 cm

解答 Solution

答案 Answer：C

$0.25 = e^{-0.35x}$，$x = \dfrac{\ln 4}{0.35} = 4.0\,\text{cm}$（即两个半值厚度，每个 1.98 cm）。

练习 24.4 原创仿真题 Exam-style

Which statement about the formation of a CT scan image is correct?

A A single X-ray image is taken and processed by a computer to give a 3D image.

B X-ray images of one slice are taken from many different angles and combined to form a 2D image of that slice; images of many slices are then combined to form a 3D image.

C Ultrasound images from different angles are combined to form a 3D image.

D Gamma-ray photons emitted from a tracer are detected in coincidence to form a 3D image.

解答 Solution

答案 Answer：B

D 描述的是 PET 扫描。

练习 24.5 原创仿真题 Exam-style

In a PET scan, a patient is given a tracer containing fluorine-18, which decays by β^+ emission.

(a) Explain why the emitted positron produces two gamma-ray photons that travel in opposite directions. [3]

(b) Calculate the energy, in MeV, of each gamma-ray photon. Assume that the kinetic energies of the positron and the electron are negligible. [2]

(c) Explain how the position of the annihilation event in the body is determined. [2]

解答 Solution

(a) 正电子与组织中的电子湮灭，质量全部转化为能量； **[B1]**

湮灭前总动量近似为零，动量守恒要求产生的光子总动量为零； **[B1]**

所以产生两个能量相等、方向相反的光子。 **[B1]**

(b) $E_\gamma = m_e c^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 = 8.2 \times 10^{-14}\,\text{J}$ **[C1]**

$= 8.2 \times 10^{-14} / 1.60 \times 10^{-13} = 0.51\,\text{MeV}$ **[A1]**

(c) 两个光子被探测器环上相对的两个探测器（几乎同时）探测到，湮灭点在两探测器连线上； **[B1]**

由两光子到达时间之差确定湮灭点在连线上的位置。 **[B1]**

第二十五章 天体物理与宇宙学 Astronomy and cosmology

考纲要求 Syllabus learning outcomes

25.1.1 understand the term luminosity as the total power of radiation emitted by a star
理解光度是恒星辐射的总功率。

25.1.2 recall and use the inverse square law for radiant flux intensity F in terms of the luminosity L of the source $F = L/(4\pi d^2)$
记住并使用辐射通量强度的平方反比定律 $F = L/(4\pi d^2)$。

25.1.3 understand that an object of known luminosity is called a standard candle
理解光度已知的天体称为标准烛光。

25.1.4 understand the use of standard candles to determine distances to galaxies
理解用标准烛光测定星系距离。

25.2.1 recall and use Wien's displacement law $\lambda_{\max} \propto 1/T$ to estimate the peak surface temperature of a star
记住并用维恩位移定律 $\lambda_{\max} \propto 1/T$ 估算恒星表面温度。

25.2.2 use the Stefan–Boltzmann law $L = 4\pi\sigma r^2 T^4$
使用斯特藩–玻尔兹曼定律 $L = 4\pi\sigma r^2 T^4$。

25.2.3 use Wien's displacement law and the Stefan–Boltzmann law to estimate the radius of a star
用维恩定律和斯特藩–玻尔兹曼定律估算恒星半径。

25.3.1 understand that the lines in the emission spectra from distant objects show an increase in wavelength from their known values
理解遥远天体发射光谱中的谱线波长比已知值增大（红移）。

25.3.2 use $\Delta\lambda/\lambda \approx \Delta f/f \approx v/c$ for the redshift of electromagnetic radiation from a source moving relative to an observer
对相对观察者运动的光源，使用红移公式 $\Delta\lambda/\lambda \approx \Delta f/f \approx v/c$。

25.3.3 explain why redshift leads to the idea that the Universe is expanding
解释为什么红移导致宇宙膨胀的观点。

25.3.4 recall and use Hubble's law $v \approx H_0 d$ and explain how this leads to the Big Bang theory (candidates will only be required to use SI units)
记住并使用哈勃定律 $v \approx H_0 d$，并解释它如何导出大爆炸理论（只需使用 SI 单位）。

25.1 光度与标准烛光

定义 25.1 (光度 *Luminosity*)

The luminosity of a star is the total power of (electromagnetic) radiation emitted by the star.

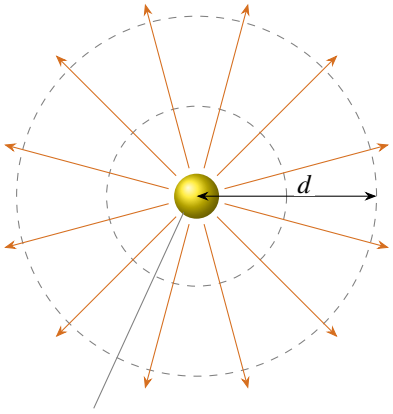
光度 L：恒星向外辐射电磁波的**总功率**，单位 W。太阳光度约 3.8×10^{26} W。

♣

定义 25.2 (辐射通量强度 *Radiant flux intensity*)

The radiant flux intensity is the radiant power passing normally through unit area.

辐射通量强度 F：垂直通过单位面积的辐射功率，单位 $\mathrm{W\,m^{-2}}$（即在观察者处测得的“亮度”）。



star 恒星，光度 L

能量均匀分布在半径为 d 的球面上：

$$F = \frac{L}{4\pi d^2}$$

距离加倍，F 变为 $\frac{1}{4}$

定律 25.1 (平方反比定律 *Inverse square law*)

$$F = \frac{L}{4\pi d^2}$$

（假设辐射在各方向均匀发射、传播途中没有被吸收。）

定义 25.3 (标准烛光 *Standard candle*)

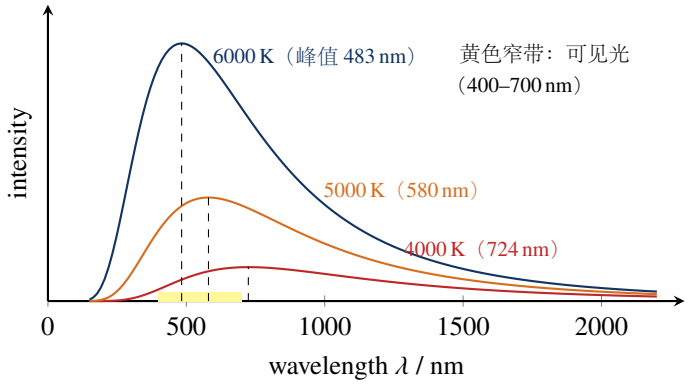
A standard candle is an astronomical object of known luminosity.

标准烛光：光度已知的天体，如造父变星（Cepheid variable stars，光度与脉动周期有确定关系）、Ia 型超新星（峰值光度几乎相同）。

用标准烛光测距：(1) 在目标星系中识别出一个标准烛光，由其类型知道光度 L；(2) 在地球上测量其辐射通量强度 F；(3) 由 $d = \sqrt{\frac{L}{4\pi F}}$ 计算距离。

25.2 恒星的温度与半径

25.2.1 维恩位移定律



定律 25.2 (维恩位移定律 *Wien's displacement law*)

$$\lambda_{\max} \propto \frac{1}{T} \qquad (\lambda_{\max} T = \text{constant} \approx 2.9 \times 10^{-3}\ \text{m K})$$

The wavelength at which the maximum (intensity of) radiation is emitted from a black body is inversely proportional to its thermodynamic (surface) temperature.

黑体辐射强度最大处对应的波长与其热力学温度成反比。温度越高，峰值波长越短（颜色偏蓝）。比较两颗恒星：$\lambda_1 T_1 = \lambda_2 T_2$。

25.2.2 斯特藩–玻尔兹曼定律

定律 25.3 (斯特藩–玻尔兹曼定律 *Stefan–Boltzmann law*)

$$L = 4\pi\sigma r^2 T^4$$

$\sigma = 5.67 \times 10^{-8}\ \text{W m}^{-2}\ \text{K}^{-4}$；$r$ 为恒星半径，$4\pi r^2$ 为表面积；T 为表面热力学温度（恒星近似为黑体）。

估算恒星半径的步骤：(1) 测光谱峰值波长 $\lambda_{\max}$，由维恩定律求表面温度 T；(2) 由标准烛光法等得到距离 d，测 F，求光度 $L = 4\pi d^2 F$；(3) $r = \sqrt{\dfrac{L}{4\pi\sigma T^4}}$。

例题 25.1 9702/41/M/J/26 Q10

(a) State Wien's displacement law. [2]

(b) The Sun has a radius of 6.96×10^8 m and a surface temperature of 5770 K. The Earth is at a distance of 1.52×10^{11} m from the Sun.

(i) Calculate the luminosity of the Sun. [2]

(ii) Determine the radiant flux intensity at the Earth of radiation emitted by the Sun. Give a unit with

your answer. [3]

(iii) Star X has a wavelength of maximum emission of $0.581\lambda_S$, where λ_S is the wavelength of maximum emission from the Sun. Calculate the surface temperature of star X. [1]

解答 Solution

(a) （表面）热力学温度与波长成反比， [M1]

该波长是辐射强度（发射率）最大时的波长。 [A1]

(b)(i) $L = 4\pi \times 5.67 \times 10^{-8} \times (6.96 \times 10^8)^2 \times 5770^4$ [C1]

$= 3.83 \times 10^{26}\,\mathrm{W}$ [A1]

(ii) $F = \dfrac{L}{4\pi d^2}$ [C1]

$= \dfrac{3.83 \times 10^{26}}{4\pi \times (1.52 \times 10^{11})^2}$ [C1]

$= 1320\,\mathrm{W\,m^{-2}}$ [A1]

(iii) $T = \dfrac{5770}{0.581} = 9930\,\mathrm{K}$ [A1]

25.3 红移、哈勃定律与大爆炸

25.3.1 红移



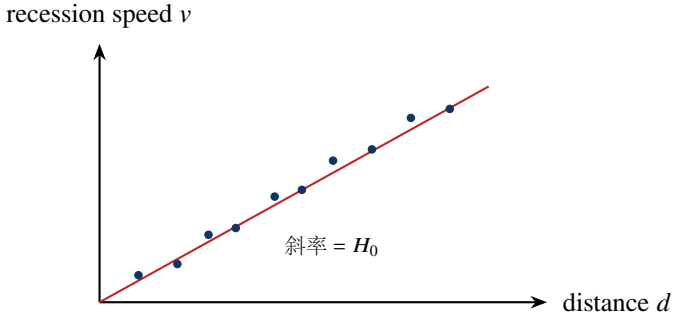
遥远星系光谱中的特征谱线（如氢的吸收线）的波长都比实验室中测得的**增大**，称为**红移** (*redshift*)。这是光源远离观察者时的多普勒效应（对电磁波）：

公式 25.1 (红移 *Redshift*)

$$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$

$\Delta\lambda$ 为波长增量，λ 为原（发射）波长，v 为光源的退行速度（$v \ll c$ 时适用）。

25.3.2 哈勃定律与宇宙膨胀



定律 25.4 (哈勃定律 *Hubble's law*)

$$v \approx H_0 d$$

The recession speed of a galaxy is (approximately) proportional to its distance from the observer.

星系的退行速度与它到观察者的距离（近似）成正比。H_0 为哈勃常量，SI 单位 s^{-1}，约 $2.2 \times 10^{-18}\,\mathrm{s}^{-1}$。

宇宙膨胀与大爆炸：

(1) 几乎所有遥远星系都有红移 ⇒ 它们都在远离我们；

(2) 越远的星系退行越快（哈勃定律）⇒ 宇宙在整体膨胀（任何位置的观察者都会看到同样的规律，并不说明我们位于中心）；

(3) 把膨胀"倒放"，所有物质在过去某一时刻聚集在一个极小、极热、极密的点 ⇒ 宇宙起源于一次"大爆炸"。

(4) 估算宇宙年龄：若退行速度不变，$t = \dfrac{d}{v} = \dfrac{1}{H_0} \approx 4.5 \times 10^{17}\,\mathrm{s} \approx 1.4 \times 10^{10}$ 年。

例题 25.2 9702/44/M/J/26 Q10

(a) (i) State what is meant by a standard candle. [1]

(ii) Explain how standard candles may be used to determine the distance of a galaxy from an observer. [2]

(b) A star in a distant galaxy has luminosity L. The star emits radiation that has maximum intensity at a wavelength that is known to be λ_{star} after correcting for redshift. The Sun has surface temperature T_0 and emits radiation that has maximum intensity at wavelength λ_0.

(i) Explain why the wavelength of the radiation observed from the star has to be corrected for redshift. [2]

(ii) State an expression, in terms of λ_{star}, λ_0 and T_0, for the surface temperature T_{star} of the star. [1]

(iii) Determine an expression, in terms of L and T_{star}, for the radius of the star. Identify any other symbols that you use. [1]

解答 Solution

(a)(i) 光度已知的天体。 **[B1]**

(ii) 测量（地球上接收到的）辐射通量强度 F； **[B1]**

用 $F = \dfrac{L}{4\pi d^2}$ 计算距离 d。 **[B1]**

(b)(i) 恒星（所在星系）正在远离观察者； **[B1]**

所以观测到的波长大于发射波长。 **[B1]**

(ii) $T_{\text{star}} = \dfrac{\lambda_0 T_0}{\lambda_{\text{star}}}$ **[A1]**

(iii) $L = 4\pi\sigma r^2 T_{\text{star}}^4$，$r = \sqrt{\dfrac{L}{4\pi\sigma T_{\text{star}}^4}}$，$\sigma$ 为斯特藩–玻尔兹曼常量。 **[A1]**

教学提示 Teacher's note 建议 4 课时。本章公式都在数据表中，但定义（luminosity、radiant flux intensity、standard candle、Wien's law 中"thermodynamic temperature"与"wavelength of maximum emission"）和"红移 → 膨胀 → 大爆炸"的逻辑链需要准确表述。计算中注意 nm 换 m，以及 T^4 的计算量级。

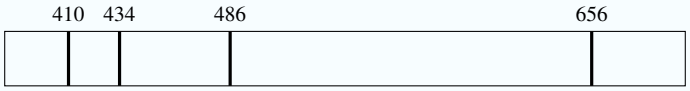
25.4 本章小结 Summary

物理量/关系	公式	说明
平方反比定律	$F = \dfrac{L}{4\pi d^2}$	标准烛光测距
维恩位移定律	$\lambda_{\max} \propto 1/T$	$\lambda_1 T_1 = \lambda_2 T_2$
斯特藩–玻尔兹曼	$L = 4\pi\sigma r^2 T^4$	估算半径
红移	$\Delta\lambda/\lambda \approx \Delta f/f \approx v/c$	远离 ⇒ 波长变长
哈勃定律	$v \approx H_0 d$	$1/H_0 \approx$ 宇宙年龄

25.5 本章练习 Practice questions

练习 25.1 9702/42/F/M/26 Q8(b)

A star in a distant galaxy consists mostly of hydrogen. Visible light emitted by the star consists of discrete spectral absorption lines at wavelengths (as measured in a laboratory) of 410 nm, 434 nm, 486 nm and 656 nm, as represented in Figure 8.1.



The galaxy is at a distance of 6.7×10^{24} m from the Earth.

(i) Explain why the spectral lines observed on the Earth are at different wavelengths from those shown in Figure 8.1. [2]

(ii) State where the four spectral lines observed on the Earth would appear relative to those in Figure 8.1. [2]

(iii) The Hubble constant may be taken as $2.1 \times 10^{-18}\ \text{s}^{-1}$. Determine the observed wavelength, in nm, of the lowest frequency spectral line of visible light from the star. [3]

解答 Solution

(i) 星系远离地球运动，产生红移； **[B1]**

观测到的波长大于发射波长。 **[B1]**

(ii) 四条谱线都向右（长波方向）移动； **[B1]**

各线移动量与原波长成正比（$\Delta\lambda \propto \lambda$，移动比例相同，约 4.7%）。 **[B1]**

(iii) $v = H_0 d = 2.1 \times 10^{-18} \times 6.7 \times 10^{24} = 1.41 \times 10^7\ \text{m s}^{-1}$ **[C1]**

频率最低即波长最长的 656 nm 线：$\Delta\lambda = \dfrac{v}{c}\lambda = \dfrac{1.41 \times 10^7 \times 656}{3.0 \times 10^8} = 31\ \text{nm}$ **[C1]**

观测波长 = 656 + 31 = 687 nm [A1]

练习 25.2 原创仿真题 Exam-style

The distance from a star to an observer is doubled. The luminosity of the star is unchanged. What happens to the radiant flux intensity received by the observer?

A It is halved.

B It is reduced to one quarter.

C It is unchanged.

D It is doubled.

解答 Solution

答案 Answer：**B**

$F \propto 1/d^2$。

练习 25.3 原创仿真题 Exam-style

The Sun has a surface temperature of about 5800 K and a peak emission wavelength of about 500 nm. A star has a peak emission wavelength of 290 nm. What is the approximate surface temperature of the star?

A 3400 K **B** 5800 K **C** 10 000 K **D** 17 000 K

解答 Solution

答案 Answer：**C**

$T = 5800 \times \dfrac{500}{290} = 1.0 \times 10^4$ K。

练习 25.4 原创仿真题 Exam-style

Two stars P and Q have the same surface temperature. The luminosity of P is 100 times the luminosity of Q. What is the ratio $\dfrac{\text{radius of P}}{\text{radius of Q}}$?

A $\sqrt{10}$ **B** 10 **C** 100 **D** 10^4

解答 Solution

答案 Answer：**B**

$L \propto r^2T^4$，T 相同，$r \propto \sqrt{L} = \sqrt{100} = 10$。

练习 25.5 原创仿真题 Exam-style

A spectral line from a distant galaxy has a wavelength of 502 nm. The same line measured in a laboratory has a wavelength of 486 nm.

(a) Calculate the speed of recession of the galaxy. [2]

(b) The Hubble constant is $2.2 \times 10^{-18}\,\mathrm{s}^{-1}$. Calculate the distance to the galaxy. [2]

(c) Use the Hubble constant to estimate the age of the Universe in years. State the assumption you make. [3]

解答 Solution

(a) $v = c\dfrac{\Delta\lambda}{\lambda} = 3.00 \times 10^8 \times \dfrac{16}{486}$ **[C1]**

$= 9.9 \times 10^6\,\mathrm{m\,s}^{-1}$ **[A1]**

(b) $d = \dfrac{v}{H_0} = \dfrac{9.9 \times 10^6}{2.2 \times 10^{-18}}$ **[C1]**

$= 4.5 \times 10^{24}\,\mathrm{m}$ **[A1]**

(c) $t \approx \dfrac{1}{H_0} = \dfrac{1}{2.2 \times 10^{-18}} = 4.5 \times 10^{17}\,\mathrm{s}$ **[C1]**

$= \dfrac{4.5 \times 10^{17}}{3.15 \times 10^7} = 1.4 \times 10^{10}$ 年 **[A1]**

假设：星系自大爆炸以来一直以恒定速度退行（膨胀速率不变）。 **[B1]**

第三部分

实验技能 Practical Skills

第二十六章 实验技能 Practical skills (Papers 3 & 5)

考纲要求 Syllabus learning outcomes

P3 Paper 3 Advanced Practical Skills: manipulation, measurement and observation; presentation of data and observations; analysis, conclusions and evaluation
Paper 3（AS 实验操作）：操作、测量与观察；数据与观察结果的呈现；分析、结论与评价。两道题，各 20 分。

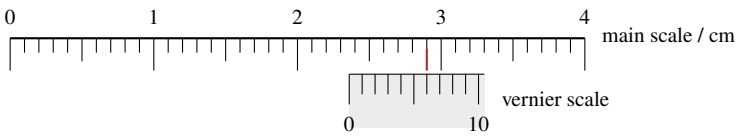
P5 Paper 5 Planning, Analysis and Evaluation: planning an investigation; analysis of data with uncertainties; conclusions and evaluation
Paper 5（A2 规划、分析与评价）：Q1 设计实验（15 分）；Q2 含不确定度的数据分析（15 分）。

1.3 assess the uncertainty in a derived quantity by simple addition of absolute or percentage uncertainties
通过不确定度的简单相加评估导出量的不确定度（与第 1 章衔接）。

26.1 测量与读数 Measurement

26.1.1 常用仪器的精度

仪器	读数精度（记录到）	使用要点
米尺 metre rule	1 mm（记作 0.1 cm）	视线垂直刻度以避免视差；不要从磨损的端点量起
游标卡尺 vernier calipers	0.1 mm 或 0.05/0.02 mm	检查零点误差；测内径、外径、深度
千分尺 micrometer	0.01 mm	用棘轮 (ratchet) 旋紧；检查零点误差；不同位置多测几次
秒表 stopwatch	0.01 s（实际受反应时间约 0.1–0.2 s 限制）	测 10–20 个周期总时间再除
量角器 protractor	1°	
温度计 thermometer	0.5 或 1°C	读数前搅拌；等示数稳定
数字电表 multimeter	最后一位	量程合适；注意正负极
电子天平 top-pan balance	0.1 g 或 0.01 g	先调零（tare）



游标卡尺：主尺读数 2.3 cm（游标 0 刻线左侧），游标第 6 条刻线与主尺对齐（红色），读数 = 2.3 + 6 × 0.01 = 2.36 cm = 23.6 mm。

26.1.2 误差与改进的常见表述

- **随机误差**：读数难以判断的终点（如振动周期的开始/结束）、反应时间、读数时物体在动、视差、温度波动等 ⇒ 重复测量取平均、用视频逐帧回放、用光电门/数据记录器、使用标志线 (fiducial marker)。
- **系统误差**：零点误差、仪器未校准、热量散失等 ⇒ 校准、修正零点、加隔热、改进方法。
- 测量量太小导致百分比不确定度大 ⇒ 测多个周期 / 多张纸的总厚度 / 更长的金属丝等。

26.2 记录数据 Recording data

- 表头写成“物理量 / 单位”，如 l / cm、T / s、$\dfrac{1}{d^2}$ / mm^{-2}；表格内只写数字。
- 同一列原始数据的**小数位数**必须一致，并与仪器精度相符（米尺读数写到 0.1 cm，如 25.0 cm 而不是 25 cm）。
- 计算量的**有效数字**位数与原始数据中有效数字最少的相同（或多一位）。
- 尽量取覆盖较大范围的 6 组以上数据，且均匀分布；能重复的读数要重复，并记录每次读数。

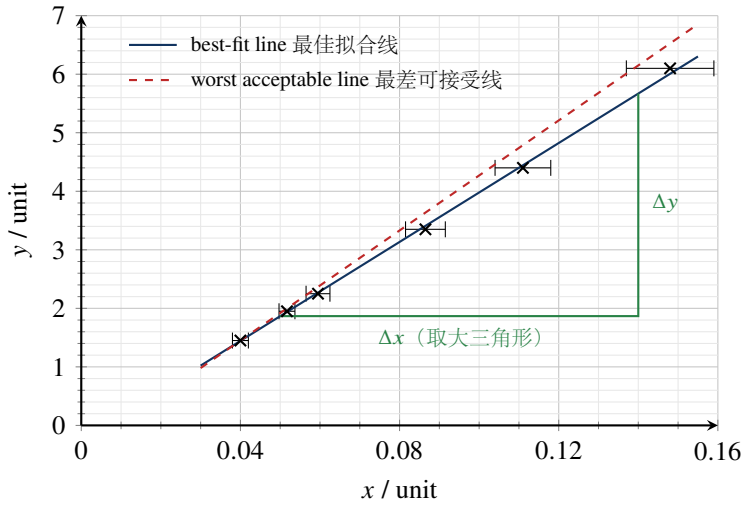
26.3 不确定度 Uncertainties

公式 26.1 (不确定度的估计与合成 *Estimating and combining uncertainties*)

- 单次读数：Δx 取最小分度（或根据读数难度更大）；
- 重复读数：$\Delta x = \dfrac{1}{2}$(最大值 − 最小值)（半极差），测量值取平均；
- 加减：绝对不确定度相加；乘除：百分比不确定度相加；乘方 x^n：乘以 $|n|$；
- 对数：$y = \lg x$ 时 $\Delta y = \lg(x + \Delta x) - \lg x$（或 $\frac{1}{2}[\lg(x + \Delta x) - \lg(x - \Delta x)]$）；
- 最大/最小法：把各量取极值代入计算，$\Delta y = y_{\max} - y$。

判断实验结果是否支持某关系（Paper 3 典型问题）：算出两组数据得到的常数 k_1、k_2，计算百分比差 $\dfrac{|k_1 - k_2|}{k_1} \times 100\%$，与题目给出（或自己估算）的百分比不确定度比较：差值小于不确定度 ⇒ 支持；大于 ⇒ 不支持。结论必须与计算一致。

26.4 作图与图像分析 Graphs



- **坐标轴**：标注“物理量 / 单位”；标度均匀、易读（每格 1、2、5 的倍数，避免 3、7 等）；数据点占据图纸一半以上（不必从原点开始）。
- **描点**：用细小的“×”或“⊙”，精确到半个小格。
- **最佳拟合线**：一条细直线，点均匀分布在线两侧（不是简单连接首末两点）。
- **斜率**：在线上取两个相距较远（大于线长一半）的点（不一定是数据点），写出读数，gradient $= \dfrac{\Delta y}{\Delta x}$。
- **截距**：若横轴不从 0 开始，必须用 $y = mx + c$ 代入线上一点求 c，不能直接读。
- **误差棒**：以数据点为中心、总长 $2\Delta y$（或 $2\Delta x$）的线段。
- **最差可接受直线** (worst acceptable line)：穿过所有误差棒、斜率最大（或最小）的直线。斜率不确定度 = | 最佳线斜率 − 最差线斜率 |；截距不确定度同理。

26.4.1 化曲为直 Linearising relationships

关系式	作图	斜率	截距
$y = kx^2$	y–x^2	k	0
$y = \dfrac{k}{x} + c$	y–$\dfrac{1}{x}$	k	c
$T = 2\pi\sqrt{l/g}$	T^2–l	$4\pi^2/g$	0
$y = ax^n$	$\lg y$–$\lg x$	n	$\lg a$
$y = Ae^{-kx}$	$\ln y$–x	$-k$	$\ln A$
$y = Ae^{kx^2}$	$\ln y$–x^2	k	$\ln A$

易错提醒 Common mistakes 求未知常数时，必须把“斜率/截距”的表达式与实际图像对应好，并给出**正确单位**；lg 图的截距要用 10^c 还原，ln 图用 e^c。lg 图的纵轴表头写成 $\lg(v\,/\,\mathrm{m\,s^{-1}})$。

例题 26.1 9702/51/M/J/26 Q2

A long tube contains a liquid of density σ. Two rubber bands are placed on the tube towards the bottom; the distance between the two rubber bands is z. A small steel ball of diameter d is dropped from the surface of the liquid. The time t for the ball to pass between the two rubber bands is measured. The experiment is repeated with steel balls of different diameters. It is suggested that t and d are related by the equation

$$\frac{z}{t} = \frac{gd^2(\rho - \sigma)}{18\beta}$$

where g is the acceleration of free fall, ρ is the density of steel and β is a constant.

(a) A graph is plotted of t on the y-axis against $\frac{1}{d^2}$ on the x-axis. Determine an expression for the gradient. [1]

(b) Values of d and t are given in Table 2.1. Calculate and record values of $\frac{1}{d^2}$ / mm^{-2}. Include the absolute uncertainties in $\frac{1}{d^2}$. [2]

d / mm	$\frac{1}{d^2}$ / mm^{-2}	t / s
2.6 ± 0.1		6.10
3.0 ± 0.1		4.40
3.4 ± 0.1		3.35
4.1 ± 0.1		2.25
4.4 ± 0.1		1.95
5.0 ± 0.1		1.45

(c) (i) Plot a graph of t/s against $\frac{1}{d^2}$/mm^{-2}. Include error bars for $\frac{1}{d^2}$. [2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines. [2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer. [2]

(d) The density ρ of steel is $(7850 \pm 50)\,\text{kg}\,\text{m}^{-3}$. The density σ of the liquid is $(1250 \pm 50)\,\text{kg}\,\text{m}^{-3}$. Calculate $(\rho - \sigma)$. Include the absolute uncertainty. [1]

(e) (i) Using your answers to (a), (c)(iii) and (d), determine the value of β. Include an appropriate unit.
Data: $g = 9.81\,\text{m}\,\text{s}^{-2}$, $z = (97 \pm 2)$ mm. [2]

(ii) Determine the percentage uncertainty in β. [1]

(f) The experiment is repeated using a steel ball with a diameter of (1.20 ± 0.01) cm. Determine the time t. Include the absolute uncertainty. [2]

解答 Solution

(a) $t = \frac{18\beta z}{g(\rho - \sigma)} \cdot \frac{1}{d^2}$，所以 gradient $= \frac{18\beta z}{g(\rho - \sigma)}$。 **[1]**

(b) $\frac{1}{d^2}$：0.148, 0.111, 0.0865, 0.0595, 0.0517, 0.0400； [1]

不确定度 $\Delta\left(\frac{1}{d^2}\right) = 2\frac{\Delta d}{d}\cdot\frac{1}{d^2}$：±0.011, ±0.007, ±0.005, ±0.003, ±0.002, ±0.002。 [1]

(c)(i)(ii) 见上一节示意图（点 ×、水平误差棒、实线为最佳拟合线、虚线为最差可接受线）。最佳拟合线须经过 (0.065, 2.50) 与 (0.136, 5.50) 附近。 [2+2]

(iii) 最佳拟合线：gradient $= \frac{5.50-2.50}{0.136-0.065} = 42\,\text{s mm}^2$； [1]

最差线（过 (0.038, 1.45) 与 (0.137, 6.10)）gradient ≈ 47，不确定度 = 47 − 42 = 5：gradient = (42 ± 5) s mm^2。 [1]

(d) $\rho-\sigma = (6600\pm100)\,\text{kg m}^{-3}$（绝对不确定度相加）。 [1]

(e)(i) $\beta = \frac{g(\rho-\sigma)\times\text{gradient}}{18z} = \frac{9.81\times6600\times42\times10^{-6}}{18\times0.097}$ [1]

$= 1.6\,\text{kg m}^{-1}\,\text{s}^{-1}$（注意 gradient 换算为 s m^2） [1]

(ii) $\frac{5}{42}+\frac{100}{6600}+\frac{2}{97} = 11.9\%+1.5\%+2.1\%\approx15\%$ [1]

(f) $\frac{1}{d^2} = \frac{1}{12.0^2} = 6.94\times10^{-3}\,\text{mm}^{-2}$，$t = \text{gradient}\times\frac{1}{d^2} = 42\times6.94\times10^{-3} = 0.29\,\text{s}$； [1]

$\frac{\Delta t}{t} = \frac{5}{42}+2\times\frac{0.01}{1.20} = 13.6\%$，$t = (0.29\pm0.04)$ s。 [1]

26.5 评价与改进（Paper 3 第 2 题）

Paper 3 Q2 的最后部分要求写出 **4 个局限性（不确定度来源）**和 **4 个对应的改进**。高分要点：

- 第一条几乎总是：**只取了两组数据，不足以得出有效结论** ⇒ 多取几组数据并作图（或比较多个常数值）。
- 每个困难都要**具体**：说清是测哪个量、为什么困难（如“长度难测，因为弹簧会动 / 视差 / 线圈端点不明确”）。
- 改进要可操作：用视频逐帧回放并在画面中放计时器、标志线、夹具固定、铅垂线、水平仪、游标卡尺代替米尺等；“更仔细”“用计算机”不得分。
- “反应时间”单独写通常不得分；需联系具体测量（如很难判断摆动的开始和结束）。

例题 26.2 9702/31/M/J/26 Q2(f)–(h)（附示例数据）

In an experiment to investigate the oscillations of a spring system, a mass m hangs from a set of springs. The extension of spring A is e. The period of oscillation is T_0 when spring A is slack and T when spring A is extended. It is suggested that the relationship between T_0, T, e and m is

$$(T_0 - T) = \frac{Ce}{m}$$

where C is a constant. A student obtains the following results:

m / g	e / cm	T_0 / s	T / s
200	4.5	0.645	0.590
300	6.5	0.780	0.728

(a) Using the data, calculate two values of C. [1]

(b) It is suggested that the percentage uncertainty in the values of C is 15%. Using this uncertainty, explain whether the results support the relationship. [1]

(c) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment. [4]

(ii) Describe four improvements that could be made to this experiment. [4]

解答 Solution

(a) $C = \dfrac{(T_0 - T)m}{e}$：$C_1 = \dfrac{0.055 \times 200}{4.5} = 2.4\,\mathrm{s\,g\,cm^{-1}}$；$C_2 = \dfrac{0.052 \times 300}{6.5} = 2.4\,\mathrm{s\,g\,cm^{-1}}$。 [1]

(b) 百分比差 $= \dfrac{2.44 - 2.40}{2.44} \times 100\% = 1.6\%$，小于 15%，所以结果**支持**该关系。 [1]

(c)(i) 与 (ii) 一一对应（各 4 分）： [4+4]

局限性 limitation	改进 improvement
A. 只有两组数据，不足以得出有效结论	测更多组（不同 m、l）数据并作图（或比较多个 C 值）
B. l 难以测量：弹簧会动 / 视差 / 线圈端点不清楚	夹住米尺测量；用游标卡尺；在弹簧两端放指针（标记）
C. 难以使弹簧 A 水平、其余弹簧竖直；支架在桌面上滑动	用铅垂线和三角板；支架底座加重物或用 G 形夹固定
D. 弹簧沿夹杆滑动 / 滑落	在杆上刻槽、用夹子或胶泥固定弹簧
E. 难以判断一次完整振动的起止（周期难测）	拍摄视频（画面中有计时器）逐帧回放；在平衡位置设标志线

26.6 设计实验（Paper 5 第 1 题）

考试技巧 Exam tip Paper 5 Q1（15 分）评分结构：**定义问题** 2 分（指出自变量、因变量、需控制的变量）；**数据收集方法** 4 分（可行装置图带标注、如何测量各量、用什么仪器）；**分析方法** 3 分（作什么图、关系成立的判据、如何由斜率/截距求常数）；**附加细节** 6 分（安全措施、具体操作细节、如何减小误差、重复测量等）。

答题模板：

(1) **变量**：自变量 x（如何改变）；因变量 y（如何测量）；控制变量（逐个列出并说明如何保持不变）。

(2) **装置图**：完整、可行、标注关键部件（支架、夹子、仪表等），电路图要能工作。

(3) **步骤**：说明用何仪器测量每个量（含精度考虑），如“用秒表测 10 个周期的时间 t，$T = t/10$”。

(4) **分析**：把关系式化为直线（常用 lg–lg 或 ln 图），写出“若图像为过原点的直线 / 斜率为 n 的直线，则关系成立”，并写出常数 = f(gradient 或 intercept)。

(5) **细节与安全**：护目镜、防止重物坠落、高温/高压/激光/放射源的防护；重复测量取平均；用大的取值范围以减小百分比不确定度；如何确保对齐、水平、平行等。

例题 26.3 9702/51/M/J/26 Q1

A rod is fixed horizontally. A solid bar of mass M is attached to the rod by two vertical strings so that the bar is horizontal. The vertical strings are equal horizontal distances a from the centre of the bar. The separation between the rod and the bar is L. One end of the bar is displaced horizontally and then released. The bar rotates with oscillations about its centre. The period of the oscillations is T. It is suggested that T is related to a by the relationship

$$T = \frac{2\pi P}{a^q}\sqrt{\frac{L}{M}}$$

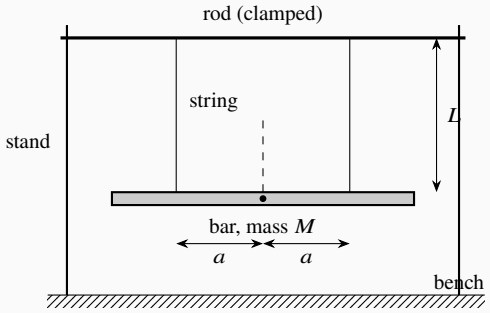
where P and q are constants. Plan a laboratory experiment to test the relationship between T and a. Draw a diagram showing the arrangement of your equipment. Explain how the results could be used to determine values for P and q. In your plan, you should include: the procedure to be followed; the measurements to be taken; the control of variables; the analysis of the data; any safety precautions to be taken. [15]

解答 Solution

定义问题：改变 a，测量 T（a 为自变量，T 为因变量）； [1]

保持 L 不变（以及 M 不变）。 [1]

装置与数据收集：



标注完整的可行装置图（支架固定在桌上、绳悬挂横杆、标出 a、L）； [1]

用秒表测 n（≥ 10）次振动的总时间 t，$T = t/n$； [1+1]

用米尺（或卡尺）测 a 和 L。 [1]

分析：$\lg T = -q\lg a + \lg\left(2\pi P\sqrt{L/M}\right)$。作 $\lg T$–$\lg a$ 图； [1]

若为直线则关系成立，$q = -$gradient； [1]

$P = \dfrac{1}{2\pi}\sqrt{\dfrac{M}{L}} \times 10^{y\text{-intercept}}$。 [1]

附加细节（任 6 点）：用 G 形夹把支架固定在桌上 / 横杆下方放软垫防止坠落；保持 M 不变并用天平测 M；从横杆中点（先找到中点并标记）量 a；用水准仪确保横杆水平、用三角板/铅垂线确保绳竖直；用胶泥防止绳滑动；小角度振动并用平衡位置处的标志线计时；用较大的 L 使 T 较大（减小百分比不确定度）；每个 a 重复测 T 取平均。 [6]

教学提示 Teacher's note 建议 6–8 课时，并贯穿全年的实验课。Paper 3 拿分关键在“规范”：表头单位、数据位数一致、描点精确、取大三角形求斜率、写出读数点坐标。Paper 5 Q2 的误差棒、最差线与不确定度合成每年题型几乎一样，务必让学生按 MS 的步骤反复训练；Q1 规划题背好“附加细节”清单（安全、控制变量的方法、减小不确定度的措施）。

26.7 本章练习 Practice questions

练习 26.1 9702/52/M/J/26 Q2

A bar is fixed between two stands. A rod of mass M is suspended from the bar by two parallel strings separated by L. One end of the rod is displaced horizontally and released; the rod rotates with oscillations about its centre. The time for 10 oscillations is t. The experiment is repeated for different values of L. It is suggested that T and L are related by the equation

$$T = \frac{2\pi}{L}\sqrt{\frac{\alpha D}{Mg}}$$

where g is the acceleration of free fall, D is the distance between the bar and the rod, and α is a constant.

(a) A graph is plotted of T^2 on the y-axis against $\dfrac{1}{L^2}$ on the x-axis. Determine an expression for the gradient. [1]

(b) Values of L and t are given in the table. Calculate and record values of $\dfrac{1}{L^2}/\mathrm{m}^{-2}$, T/s and T^2/s^2. Include the absolute uncertainties in T and T^2. [2]

L / cm	t / s
27.4	15.8 ± 0.2
30.4	14.5 ± 0.2
34.8	12.6 ± 0.2
40.8	10.8 ± 0.2
48.4	9.2 ± 0.2
55.2	8.0 ± 0.2

(c) Plot the graph with error bars, draw the line of best fit and a worst acceptable line, and determine the gradient of the line of best fit with its absolute uncertainty. [6]

(d) The distance between the bar and the rod is measured several times: 0.826 m, 0.829 m, 0.824 m, 0.821 m. Determine the mean distance D. Include the absolute uncertainty. [1]

(e) Using $g = 9.81\,\mathrm{m\,s^{-2}}$ and $M = (97 \pm 2)\,\mathrm{g}$, determine the value of α with an appropriate unit, and its percentage uncertainty. [3]

(f) Determine the separation L of the two strings that gives a value of T of $(2.20 \pm 0.02)\,\mathrm{s}$. Include the absolute uncertainty. [2]

解答 Solution

(a) $T^2 = \dfrac{4\pi^2\alpha D}{Mg} \cdot \dfrac{1}{L^2}$, gradient $= \dfrac{4\pi^2\alpha D}{Mg}$。 [1]

(b)

$1/L^2$ / m^{-2}	T / s	T^2 / s^2
13.3	1.58 ± 0.02	2.50 ± 0.06
10.8	1.45 ± 0.02	2.10 ± 0.06
8.26	1.26 ± 0.02	1.59 ± 0.05
6.01	1.08 ± 0.02	1.17 ± 0.04
4.27	0.92 ± 0.02	0.85 ± 0.04
3.28	0.80 ± 0.02	0.64 ± 0.03

[1+1]

(c) 最佳拟合线经过 (5.0–5.3, 1.0) 与 (11.3–11.6, 2.2)；gradient ≈ 0.19 s^2 m^2，最差线得不确定度约 ±0.01。 [6]

(d) $D = (0.825 \pm 0.004)$ m（平均值；半极差 $= (0.829 - 0.821)/2$）。 [1]

(e) $\alpha = \dfrac{Mg \times \text{gradient}}{4\pi^2 D} = \dfrac{0.097 \times 9.81 \times 0.19}{4\pi^2 \times 0.825} = 5.5 \times 10^{-3}\,\text{kg m}^2$； [2]

百分比不确定度 $= \dfrac{2}{97} + \dfrac{0.01}{0.19} + \dfrac{0.004}{0.825} \approx 2.1\% + 5.3\% + 0.5\% = 8\%$。 [1]

(f) $L = \sqrt{\dfrac{\text{gradient}}{T^2}} = \sqrt{\dfrac{0.19}{2.20^2}} = 0.20\,\text{m}$； [1]

$\dfrac{\Delta L}{L} = \dfrac{1}{2}\left(\dfrac{0.01}{0.19} + 2 \times \dfrac{0.02}{2.20}\right) = 3.5\%$，$L = (0.198 \pm 0.007)$ m。 [1]

练习 26.2 9702/52/F/M/26 Q2

A student places a tube and a loudspeaker into an oven. The frequency of the signal generator is adjusted until the tube produces a loud sound; the speed v of sound in air is determined. The temperature of the air in the oven is t (in °C) and $T = t + 273$. It is suggested that

$$v = T^n \sqrt{\frac{\gamma R}{M}}$$

where R is the molar gas constant, M is the molar mass of air and n and γ are constants.

(a) A graph is plotted of lg v on the y-axis against lg T on the x-axis. Determine expressions for the gradient and y-intercept. [1]

(b) Values of T and v are: 289 K, (339 ± 2); 297 K, (346 ± 2); 306 K, (351 ± 2); 317 K, (357 ± 2); 326 K, (361 ± 3); 337 K, (370 ± 3) (all v in m s^{-1}). Calculate values of lg$(v/\text{m s}^{-1})$ with their absolute uncertainties. [2]

(c) Plot the graph with error bars, draw the best-fit and worst acceptable lines, and determine the gradient and y-intercept of the best-fit line, each with its absolute uncertainty. [6]

(d) Using $M = 0.029\,\text{kg mol}^{-1}$ and $R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$, determine n and γ, and the percentage uncertainty in n. [3]

(e) The experiment is repeated inside a freezer where the temperature of the air is −12 °C. Determine the speed v of sound in air at this temperature. [1]

解答 Solution

(a) $\lg v = n\lg T + \frac{1}{2}\lg\frac{\gamma R}{M}$：gradient $= n$，y-intercept $= \frac{1}{2}\lg\frac{\gamma R}{M}$。 [1]

(b) $\lg v$：2.530, 2.539, 2.545, 2.553, 2.558, 2.568；不确定度约 ±0.003（后两行 ±0.003–0.004）。 [1+1]

(c) 最佳线经过 (2.468, 2.535) 与 (2.5235, 2.565) 附近，gradient ≈ 0.54；截距须用 $c = y - mx$ 求（横轴不从 0 开始）：$c \approx 2.535 - 0.54 \times 2.468 = 1.20$；用最差线求二者的不确定度（gradient 约 ±0.1）。 [6]

(d) n = gradient ≈ 0.54；$\gamma = \frac{M}{R} \times 10^{2c} = \frac{0.029}{8.31} \times 10^{2.42} \approx 0.9$（参考值，随所画直线不同而不同）； [2]

n 的百分比不确定度 $= \frac{\Delta\text{gradient}}{\text{gradient}} \times 100\%$（约 20%）。 [1]

(e) $T = 261\,\text{K}$：$\lg v = 0.54 \times \lg 261 + 1.21 = 2.507$，$v \approx 320\,\text{m s}^{-1}$。 [1]

练习 26.3 9702/52/M/J/26 Q1

A capacitor is connected to a reed switch, a d.c. power supply of negligible internal resistance (e.m.f. E) and a resistor. An alternating current of frequency f supplied to terminals F and G of the reed switch makes switch S vibrate at frequency f: for half of each cycle the capacitor is charged from the d.c. supply, and for the other half it discharges through the resistor. The mean current in the resistor is I. The capacitor is made from two circular metal plates in air, each of area A, separated by a small distance h. It is suggested that

$$h = \frac{KAfE}{I}$$

where K is a constant. Plan a laboratory experiment to test the relationship between I and h. Draw a diagram, and explain how the results could be used to determine a value for K. Include the procedure, measurements, control of variables, analysis and safety precautions. [15]

解答 Solution

定义问题：改变 h，测 I； [1]

保持 f（以及 A、E）不变。 [1]

数据收集：电路图——干簧开关与电容器正确连接、用电流表测 I、用电压表测 E； [1]

干簧开关的 F、G 端接信号发生器（交流电源）； [1]

两块圆形金属板用支架（或绝缘垫片）支撑并相互平行； [1]

用游标卡尺测 h（不能用米尺）。 [1]

分析：作 I–$\frac{1}{h}$ 图（或 lg I–lg h 图）； [1]

若为过原点的直线（或 lg 图斜率为 −1）则关系成立； [1]

$K = \frac{\text{gradient}}{AfE}$（$I$–$1/h$ 图）。 [1]

附加细节（任 6 点）：支架夹紧防止极板倒下 / 改变 h 前断开电源防止电击 / 电阻发热勿触碰；保持 A、E 不变；用卡尺测直径 d，$A = \pi d^2/4$；用示波器测 f（数格数 × 时基得周期，$f = 1/T$，测多个周期取平均）；极板下垫绝缘材料；用大面积极板（或大 E、大 f）使 I 较大；在极板不同位置测 h 取平均；用等厚垫片或三角板确保两板平行对齐。 [6]

练习 26.4 9702/52/F/M/26 Q1

Two identical springs, each with spring constant k, are attached to each other. The spring combination is attached to a wooden block and to a string wound on the spindle of a d.c. motor. The terminals of the motor are connected to a capacitor of capacitance C, which is initially charged to a potential difference V and then discharged through the motor. As the spindle turns, the string wraps around it and the spring combination extends; the maximum extension is e. It is suggested that $\frac{ke^2}{2} = \beta CV^2$ where β is a constant. Plan a laboratory experiment to test the relationship between e and C, and explain how the results could be used to determine a value for β. [15]

解答 Solution

定义问题：改变 C，测 e；保持 V 不变。 **[1+1]**

数据收集：可行装置图（木块和电动机用 G 形夹固定在桌上，标注弹簧、细绳、转轴）；电压表并联在电容器（或充电电源）两端；可行的充放电电路（直流电源、开关，电容器只通过电动机放电，电容器符号正确）；用米尺或卡尺测弹簧长度求 e。 **[4]**

分析：作 e^2–C 图（或 lg e–lg C 图）；过原点的直线（或 lg 图斜率 0.5）则关系成立；$e^2 = \frac{2\beta V^2}{k}C$，所以 $\beta = \frac{k \times \text{gradient}}{2V^2}$（$e^2$–$C$ 图）。 **[3]**

附加细节（任 6 点）：弹簧可能断裂弹出——用防护屏或戴护目镜；保持 k 不变，并用挂砝码测伸长求 $k = mg/x$（或作力–伸长图求斜率）；用电容器经已知电阻放电测时间常数来确定 C；把米尺固定在弹簧旁或在弹簧末端加指针；e = 伸长后长度 − 原长；用较大的 V（或较小的 k）使 e 较大；每个 C 重复测 e 取平均；检查弹簧每次都能恢复原长（未超过弹性极限）。 **[6]**

附录 A 数据、公式与术语 Data, formulae and glossary

A.1 考试数据表 Data

物理量	Quantity	符号与数值	
自由落体加速度	acceleration of free fall	$g = 9.81\,\mathrm{m\,s^{-2}}$	
真空中光速	speed of light in free space	$c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$	
元电荷	elementary charge	$e = 1.60 \times 10^{-19}\,\mathrm{C}$	
原子质量单位	unified atomic mass unit	$1\,\mathrm{u} = 1.66 \times 10^{-27}\,\mathrm{kg}$	
质子静止质量	rest mass of proton	$m_\mathrm{p} = 1.67 \times 10^{-27}\,\mathrm{kg}$	
电子静止质量	rest mass of electron	$m_\mathrm{e} = 9.11 \times 10^{-31}\,\mathrm{kg}$	
阿伏加德罗常数	Avogadro constant	$N_\mathrm{A} = 6.02 \times 10^{23}\,\mathrm{mol^{-1}}$	
摩尔气体常量	molar gas constant	$R = 8.31\,\mathrm{J\,K^{-1}\,mol^{-1}}$	
玻尔兹曼常量	Boltzmann constant	$k = 1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$	
万有引力常量	gravitational constant	$G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$	
真空介电常数	permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$	$\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\mathrm{m\,F^{-1}}$
普朗克常量	Planck constant	$h = 6.63 \times 10^{-34}\,\mathrm{J\,s}$	
斯特藩–玻尔兹曼常量	Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$	

A.2 考试公式表 Formulae

公式 Formula	表达式
匀加速直线运动 uniformly accelerated motion	$s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$
液体压强 hydrostatic pressure	$\Delta p = \rho g \Delta h$
浮力 upthrust	$F = \rho g V$
声波多普勒效应 Doppler effect for sound waves	$f_\mathrm{o} = \dfrac{f_\mathrm{s} v}{v \pm v_\mathrm{s}}$
电流 electric current	$I = Anvq$
串联电阻 resistors in series	$R = R_1 + R_2 + \cdots$
并联电阻 resistors in parallel	$\dfrac{1}{R} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \cdots$

公式 Formula	表达式
引力势 gravitational potential	$\phi = -\dfrac{GM}{r}$
引力势能 gravitational potential energy	$E_\mathrm{P} = -\dfrac{GMm}{r}$
理想气体压强 pressure of an ideal gas	$p = \dfrac{1}{3}\dfrac{Nm}{V}\langle c^2\rangle$
简谐运动 simple harmonic motion	$a = -\omega^2 x$
简谐运动的速度 velocity of particle in s.h.m.	$v = v_0\cos\omega t\,,\quad v = \pm\omega\sqrt{x_0^2 - x^2}$
点电荷电势 electric potential	$V = \dfrac{Q}{4\pi\varepsilon_0 r}$
电势能 electrical potential energy	$E_\mathrm{P} = \dfrac{Qq}{4\pi\varepsilon_0 r}$
串联电容 capacitors in series	$\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \cdots$
并联电容 capacitors in parallel	$C = C_1 + C_2 + \cdots$
电容器储能 energy of charged capacitor	$W = \frac{1}{2}QV$
电容器放电 discharge of a capacitor	$x = x_0\mathrm{e}^{-t/RC}$
霍尔电压 Hall voltage	$V_\mathrm{H} = \dfrac{BI}{ntq}$
交流电 alternating current/voltage	$x = x_0\sin\omega t$
放射性衰变 radioactive decay	$x = x_0\mathrm{e}^{-\lambda t}$
衰变常量 decay constant	$\lambda = \dfrac{0.693}{t_{1/2}}$
强度反射系数 intensity reflection coefficient	$\dfrac{I_\mathrm{R}}{I_0} = \dfrac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
斯特藩–玻尔兹曼定律 Stefan–Boltzmann law	$L = 4\pi\sigma r^2 T^4$
多普勒红移 Doppler redshift	$\dfrac{\Delta\lambda}{\lambda} \approx \dfrac{\Delta f}{f} \approx \dfrac{v}{c}$

易错提醒 Common mistakes 以下常用公式**不在**数据表中，必须记住：$F = ma$、$p = mv$、$W = Fs$、$E_\mathrm{K} = \frac{1}{2}mv^2$、$\Delta E_\mathrm{P} = mg\Delta h$、$P = Fv$、$v = f\lambda$、$I = I_0\cos^2\theta$、$d\sin\theta = n\lambda$、$\lambda = ax/D$、$Q = It$、$V = IR$、$P = VI$、$R = \rho L/A$、$E = V + Ir$、$\omega = 2\pi/T$、$a = v^2/r$、$F = Gm_1m_2/r^2$、$g = GM/r^2$、$pV = nRT = NkT$、$\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$、$\Delta U = q + W$、$E = \frac{1}{2}m\omega^2x_0^2$、$F = Q_1Q_2/4\pi\varepsilon_0 r^2$、$E = V/d$、$C = Q/V$、$\tau = RC$、$F = BIL\sin\theta$、$F = BQv\sin\theta$、$\Phi = BA$、$E = -\Delta(N\Phi)/\Delta t$、$I_\mathrm{r.m.s.} = I_0/\sqrt{2}$、$E = hf$、$p = h/\lambda$、$hf = \Phi + \frac{1}{2}mv_\mathrm{max}^2$、$E = mc^2$、$A = \lambda N$、$Z = \rho c$、$I = I_0\mathrm{e}^{-\mu x}$、$F = L/4\pi d^2$、$\lambda_\mathrm{max} \propto 1/T$、$v = H_0 d$ 等。

A.3 SI 单位与词头 SI units and prefixes

导出单位	符号	物理量	SI 基本单位
newton	N	力	$\mathrm{kg\,m\,s^{-2}}$
pascal	Pa	压强	$\mathrm{kg\,m^{-1}\,s^{-2}}$
joule	J	能量、功	$\mathrm{kg\,m^{2}\,s^{-2}}$
watt	W	功率	$\mathrm{kg\,m^{2}\,s^{-3}}$
coulomb	C	电荷	$\mathrm{A\,s}$
volt	V	电势差	$\mathrm{kg\,m^{2}\,s^{-3}\,A^{-1}}$
ohm	Ω	电阻	$\mathrm{kg\,m^{2}\,s^{-3}\,A^{-2}}$
farad	F	电容	$\mathrm{kg^{-1}\,m^{-2}\,s^{4}\,A^{2}}$
tesla	T	磁感应强度	$\mathrm{kg\,s^{-2}\,A^{-1}}$
weber	Wb	磁通量	$\mathrm{kg\,m^{2}\,s^{-2}\,A^{-1}}$
hertz	Hz	频率	$\mathrm{s^{-1}}$
becquerel	Bq	活度	$\mathrm{s^{-1}}$

A.4 命令词 Command words

命令词	英文含义	中文说明
Calculate	work out from given facts, figures or information	由给定数据计算，写出过程
Compare	identify/comment on similarities and/or differences	指出异同（每一方面都要涉及两者）
Define	give precise meaning	给出精确定义（多用“per unit”“ratio”）
Describe	state the points of a topic / give characteristics and main features	描述要点、特征或步骤
Determine	establish an answer using the information available	由已有信息（常含读图）求出答案
Explain	set out purposes or reasons / make relationships clear	说明原因，给出因果联系
Identify	name/select/recognise	指出、选出
Justify	support a case with evidence/argument	用证据或论证支持结论
Predict	suggest what may happen based on available information	根据已知信息预测
Show (that)	provide structured evidence that leads to a given result	写出完整推导/计算过程得到给定结果（结果通常多写一位有效数字）

命令词	英文含义	中文说明
Sketch	make a simple freehand drawing showing the key features	画出关键特征（形状、截距、渐近线、峰值位置等）
State	express in clear terms	简明陈述，无需解释
Suggest	apply knowledge and understanding to situations where there are a range of valid responses	运用知识提出合理看法（答案不唯一）

A.5 中英术语对照表 Glossary

English	中文	English	中文
absolute zero	绝对零度	accuracy	准确度
activity	活度	alternating current	交流电
amplitude	振幅	angular frequency	角频率
angular speed	角速度	annihilation	湮灭
antinode	波腹	antiparticle	反粒子
attenuation coefficient	衰减系数	baryon	重子
binding energy	结合能	capacitance	电容
centre of gravity	重心	centripetal acceleration	向心加速度
charge carrier	载流子	coherence	相干性
compression	压缩；密部	conservation of momentum	动量守恒
couple	力偶	critical damping	临界阻尼
damping	阻尼	de Broglie wavelength	德布罗意波长
decay constant	衰变常量	density	密度
diffraction	衍射	diffraction grating	衍射光栅
displacement	位移	Doppler effect	多普勒效应
drag	阻力	efficiency	效率
elastic deformation	弹性形变	elastic limit	弹性极限
electric field strength	电场强度	electric potential	电势
electromotive force (e.m.f.)	电动势	electronvolt	电子伏特
energy level	能级	equilibrium	平衡
extension	伸长量	field line	场线
fission	裂变	flux linkage	磁链
forced oscillation	受迫振动	frequency	频率

English	中文	English	中文
fundamental particle	基本粒子	fusion	聚变
geostationary orbit	地球同步静止轨道	gravitational field strength	引力场强度
gravitational potential	引力势	hadron	强子
half-life	半衰期	Hall voltage	霍尔电压
homogeneous (equation)	齐次的（方程）	ideal gas	理想气体
intensity	强度	interference	干涉
internal energy	内能	internal resistance	内阻
isotope	同位素	kinetic energy	动能
lepton	轻子	limit of proportionality	比例极限
longitudinal wave	纵波	luminosity	光度
magnetic flux	磁通量	magnetic flux density	磁感应强度
mass defect	质量亏损	mean-square speed	方均速率
meson	介子	moment	力矩
momentum	动量	natural frequency	固有频率
neutrino	中微子	node	波节
nucleon	核子	nucleon number	核子数（质量数）
path difference	程差	period	周期
phase difference	相位差	photoelectric effect	光电效应
photon	光子	plastic deformation	塑性形变
polarisation	偏振	positron	正电子
potential difference	电势差	potential divider	分压器
power	功率	precision	精密度
pressure	压强	progressive wave	行波
proton number	质子数	quark	夸克
radian	弧度	radiant flux intensity	辐射通量强度
random error	随机误差	rarefaction	疏部
rectification	整流	redshift	红移
resistance	电阻	resistivity	电阻率
resonance	共振	resultant	合（力、矢量）
root-mean-square (r.m.s.)	方均根（有效值）	scalar	标量
simple harmonic motion	简谐运动	smoothing	平滑（滤波）
specific acoustic impedance	特性声阻抗	specific heat capacity	比热容
specific latent heat	比潜热	spring constant	劲度系数
standard candle	标准烛光	stationary wave	驻波

English	中文	English	中文
strain	应变	stress	应力
superposition	叠加	systematic error	系统误差
tension	张力	terminal velocity	终端速度
thermal equilibrium	热平衡	thermistor	热敏电阻
threshold frequency	截止频率	time constant	时间常数
torque	转矩（力偶矩）	tracer	示踪剂
transverse wave	横波	uncertainty	不确定度
upthrust	浮力	vector	矢量
velocity selector	速度选择器	wavelength	波长
weight	重量（重力）	work done	功
work function	逸出功	Young modulus	杨氏模量